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Impact of cancer diagnoses on the outcomes of patients with COVID-19 – a systematic review and meta-analysis

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Impact of cancer diagnoses on the outcomes of patients with COVID-19 – a systematic review and meta-analysis

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ABSTRACT

The coronavirus disease 2019 (COVID-19) has caused significant mortality and morbidity across the globe. Cancer patients are especially vulnerable given their immunocompromised state. We aimed to determine the proportion of cancer patients with COVID-19, their severity and mortality outcomes through a systematic review and meta-analysis (MA).

Methods:

Systematic review was performed through online databases, PubMed, Medline and Google Scholar, with keywords listed in methods section (November 1st 2019 - May 1st 2020). Studies with clinical outcomes of at least 10 COVID-19 patients and at least 1 with a diagnosis of cancer were included. The studies for MA were assessed with PRISMA guidelines and Newcastle-Ottawa Scale (NOS). The data were pooled using a random-effects model. The main outcomes were planned before data

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collection, including proportion of cancer patients among COVID-19 populations, relative risk of severe outcomes and death of cancer patients compared to general COVID-19 patients.

Results:

We identified 47 case series (33,390 patients), with 230 cancer patients with individual patient data (IPD). We found that the pooled proportion of cancer among COVID-19 patients was 0.03 (0.03-0.04, I2 = 94.65%, p<0.01). The pooled relative risk (RR) of death was 1.44 (95% CI 1.19-1.76) between cancer patients and the general population with COVID-19 infection. The pooled RR of severe outcome was 1.49 (95% CI 1.10-2.03) between cancer and general COVID-19 patients. The presence of lung cancer, stage IV cancer and active treatment did not result in significantly increased RR of severe outcome. Among the available IPD, only age and gender were associated with severe outcomes.

Conclusion:

Cancer patients were at a higher risk of severe and death outcomes from COVID-19 infection as compared to general COVID-19 populations. A collaborative effort is required for a more complete database of cancer patients with COVID-19 infection.

BACKGROUND

The coronavirus disease 2019 (COVID-19) pandemic, since its emergence in December 2019, has caused significant mortality and morbidity across the globe. By mid-Aug 2020, more than 20 million people were infected with more than 700,000 deaths. It has disrupted many aspects of healthcare delivery and overwhelmed many hospitals and their intensive care resources.

Cancer patients are an at-risk population as they are generally older, immunocompromised, and have multiple co-morbidities. However the impact of cancer diagnoses on the outcomes of patients with COVID-19 is unclear. A recently published multi-centre study of 105 cancer patients in China suggested that cancer patients with COVID-19 infection had higher risk of severe disease outcomes as compared to 536 age-matched non-cancer patients[1]. Another study from the Chinese Centre for

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Disease Control and Prevention Group reported a case fatality rate (CFR) of 2.3% (1023/44672) among confirmed COVID-19 cases and a more than double CFR of 5.6% (6/107) among the cancer patients with COVID-19[2]. In contrast, a database report of 5,688 COVID-19 positive patients from a tertiary hospital in New York found no significant difference in risk of death between cancer and non-cancer patients (RR 1.15, 95% CI 0.84-1.57) although there was an increased risk of intubation among cancer patients (RR 1.89, 95% CI 1.37-2.61)[3].

Hence, we aimed to determine firstly the proportion of cancer patients among the reported COVID-19 cases, secondly whether cancer patients with COVID-19 had more severe outcomes and higher mortality compared to the general population and lastly, if there were any specific characteristics of patients with cancer that were associated with an increased risk of severe COVID-19 infection.

METHODS

Search Methods and Study Selection

A systematic search of the literature was conducted to identify published clinical trials that include cancer patients with COVID-19 infection. The investigators searched biomedical databases including PubMed, Medline and Google Scholar using the terms *COVID-19, coronavirus, nCoV, SARS-CoV-2, characteristics, comorbidity, mortality, death, risk, oncology, cancer, malignancy, neoplasm, tumor, tumour, and carcinoma*. The search was conducted on May 3, 2020 for a search period from November 1, 2019 through to May 1, 2020. The references of relevant meta-analysis and published reviewers were searched for additional eligible studies. The articles included met the following criteria: (1) COVID-19 diagnosis should be confirmed by Real Time Polymerase Chain Reaction (RT-PCR), or based on 2019 Novel Coronavirus Disease (COVID-19) Diagnostic criteria, if applicable (Appendix 3), (2) studies should contain clinical data of at least ten COVID-19 patients with at least one patient with a diagnosis of cancer, (3) severity outcomes (defined as hypoxaemia, admission to intensive care unit (ICU), and/or mechanical ventilation) and/or death outcomes should be presented and, (4) studies should be available in English and full text. Exclusion criteria included studies that reported only paediatric or pregnant women data. Studies published online ahead of print were eligible. For the

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pooled analysis on severity and death outcomes of cancer and general COVID-19 patients, we mandated that the included studies include at least 10 cancer patients.

The literature search and study selection were performed independently by three of the authors (SH, QZ and VSY), and discrepancies were resolved by consensus. The study followed the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.

Data Extraction

Two investigators (SH and VSY) independently graded the studies based on Newcastle-Ottawa Scale (NOS) for assessing the quality of non-randomised studies and discrepancies were resolved by consensus. Data extraction was performed by SH, and the parameters extracted included the location (hospital and city) of the study, type of study, number of patients, demographics including median or mean age, gender, proportion of patient with co-morbidities including cancer, and the clinical outcomes of severe disease and death (of cancer patients and overall COVID-19 population) were collected for the available studies.

Statistical analysis

All statistical analyses were performed using the STATA V.15 software. Meta-analysis using random-effects models was performed to analyse the pooled proportion of cancer among COVID-19 patients. This was further stratified into subgroups by region (within Hubei Province, China vs outside Hubei Province, including other countries), median/mean age (above or equal to vs below 60 years old) and gender (proportion of male more than or equal to 60% vs less than 60% in population). We further analysed the pooled relative risks (RR) of severe outcomes and death for cancer patients with COVID-19 versus the general population with COVID-19. The subgroup of lung cancer and stage IV patients were also evaluated for relative risk of severe outcomes. The results were presented as proportion (decimal) with 95% confidence interval (95% CI). Heterogeneity among studies was analysed by the I^2 statistics, where $I^2 > 50\%$ was considered moderate heterogeneity and chi-square $p < 0.05$ was considered statistically significant. Chi-square tests were performed to look for

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associations between cancer patient characteristics (age, gender, lung cancers, stage IV cancers, co-morbidities and recent systemic therapy) and severe COVID-19 outcomes in the individual patient data (IPD).

Patient and Public Involvement

There is no patient directly involved in this study.

RESULTS

Eligible studies and characteristics

We identified 2410 relevant publications after literature search and review. An additional 16 studies were identified from the references of published meta-analyses. After screening and eligibility assessment, we included in our meta-analysis a total of 47 case series involving 33,390 patients (Figure 1). Twenty-two studies that did not meet the MA criteria included published meta-analyses on co-morbidities of COVID-19 and database reports were summarised in Appendix 1.

Among the 47 studies, 36 studies were from public hospitals in China (23 from Hubei Province) with the rest from Korea (1), Iran (1), Italy (2), United Kingdom (UK) (1) and USA (6) (Table 1). Most studies were retrospective cohort studies and a few observational cross-sectional studies. The quality of all the included studies ranged from 6-8 (out of 9) using the Newcastle-Ottawa Scale (Appendix 1 Supplementary Table 3). Among the 47 case series, six contained detailed descriptions of COVID-19 patients with cancer (Table 2). Only five studies provided detailed IPD (Appendix 2 Supplementary Table 1). For the pooled analysis on relative risk of severe outcomes among cancer COVID-19 patients, we identified eight case series that contained severity data, and seven studies that contained mortality data for analysis.

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Table 1: Baseline characteristics, comorbidities and severity outcomes of forty-seven COVID-19 case series

No.	Authors	Location	Definition of severity	Total	Death n (%)	Severe n (%)	Gender – Male n (%)	Age	DM	HTN	Pulmonary disease	Smoking	CKD	CVS	CLD	Cancer total n (%)	Cancer – severe n (% of cancer total)	Cancer – death (% of cancer total)
1	Benelli et al.[4]	Crema, Italy	ICU + NIV, death	411	72 (17.5)	140 (34.1)	359 (66.6)	70.5 (1-99)	67 (16.3)	193 (47.0)	48 (11.7)	N.A	22 (5.3)	93 (22.6)	N.A	33 (8.0)	10 (30.3)	9 (27.3)
2	Cai et al.[5]	Shenzhen, China	ICU, death	298	3 (1.0)	58 (19.5)	145 (48.7)	47.5 (33-61)	18 (6.0)	47 (15.8)	N.A	N.A	N.A	25 (8.4)	28 (9.4)	4 (1.3)	2 (50.0)	N.A
3	Cao M et al.[6]	Shanghai, China	ICU, death	198	1 (0.5)	19 (9.6)	101 (51.0)	50.1 (±16.3)	15 (7.6)	42 (21.2)	N.A	11 (5.6)	N.A	12 (6.0)	6 (3.0)	4 (2.0)	0	N.A
4	Cao J et al.[8]	Hubei, China	ICU, death	102	17 (16.7)	18 (17.6)	53 (52.0)	54 (37-67)	11 (10.8)	28 (27.5)	10 (9.8)	N.A	4 (3.9)	11 (10.8)	2 (2.0)	4 (3.9)	N.A	1 (25.0)
5	Chen J et al.[7]	Shanghai, China	ICU, death	249	2 (0.8)	22 (8.8)	126 (50.6)	51 (31-64)	25 (10.0)	N.A	5 (2.0)	N.A	N.A	55 (22.1)	N.A	1 (0.4)	N.A	N.A
6	Chen N et al.[8]	Hubei, China	ICU, death	99	11 (0.11)	23 (23.2)	67 (67.7)	55.5 (±13.1)	12 (12.1)	0	1 (1.0)	N.A	N.A	40 (40.4)	N.A	1 (1.0)	N.A	N.A
7	Colaneri et al.[9]	Pavia, Italy	Require high FiO2	44	2 (4.5)	17 (38.6)	28 (63.6)	67.5 (10-94)	7 (15.9)	15 (34.1)	2 (4.5)	N.A	N.A	11 (25.0)	N.A	6 (13.6)	2 (4.5)	N.A
8	Dai et al.[1]	Hubei, China	NIV and ventilation, death	641	N.A	84 (13.1)	57 (8.9)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	105 (16.4)	20 (19.1)	12 (11.4)
9	Du et al.[10]	Hubei, China	Death	179	21 (11.7)	N.A	97 (54.1)	57.6 (±13.7)	33 (18.4)	58 (32.4)	N.A	N.A	4 (2.2)	29 (16.2)	N.A	4 (2.2)	N.A	1 (25.0)
10	Duanmu et al.[11]	California, USA	Death	100	1 (1.0)	N.A	56 (56.0)	45 (32-65)	10 (10.0)	19 (19.0)	1 (1.0)	2 (2.0)	6 (6.0)	N.A	N.A	3 (3.0)	N.A	N.A
11	Feng et al.[12]	Hubei, China	See below	476	38 (8.0)	124 (26.1)	271 (56.9)	53 (40-64)	17 (3.6)	40 (8.4)	14 (2.9)	44 (9.2)	2 (0.4)	26 (5.5)	N.A	12 (2.5)	7 (58.3)	N.A
12	Guan et al.[13]	China	Composite endpoint[a]	1099	15 (!.4)	67 (6.1)	640 (58.2)	47 (35-58)	81 (7.4)	165 (15.0)	12 (1.1)	158 (14.4)	8 (0.7)	42 (3.8)	23 (2.1)	10 (0.9)	1 (10.0)	N.A
13	Guo et al.[14]	Hubei, China	Mechanical ventilation	187	43 (23.0)	45 (24.1)	91 (48.7)	58.5 (±14.7)	28 (15.0)	61 (32.6)	4 (2.1)	18 (9.6)	6 (3.2)	29 (15.5)	N.A	13 (7.0)	N.A	N.A
14	Huang C et al.[15]	Hubei, China	ICU	41	6 (14.6)	13 (31.7)	30 (73.2)	49 (41-58)	8 (19.5)	6 (14.6)	1 (2.4)	3 (7.3)	N.A	6 (14.6)	1 (2.4)	1 (2.4)	0	N.A
15	Huang Y et al.[16]	Hubei, China	ICU	34	N.A	8 (23.5)	14 (41.2)	56 (26-88)	4 (11.8)	8 (23.5)	3 (8.8)	N.A	N.A	6 (17.6)	1 (2.9)	3 (8.8)	N.A	N.A
16	Kim et al.[17]	Korea	Require high FiO2	28	0	6 (24.1)	15 (53.6)	40 (20-73)	2 (7.1)	0	0	5 (17.9)	0	0	1 (3.6)	1 (3.6)	N.A	N.A
17	Li et al.[18]	Hubei, China	IDSA/ATS	548	90 (16.4)	269 (49.1)	279 (50.9)	60 (48-69)	83 (15.1)	166 (30.3)	17 (3.1)	N.A	10 (1.8)	34 (6.2)	5	24 (0.9)	14	N.A
18	Lian et al.[19]	Zhejiang, China	Not defined	788	N.A	18 (2.3)	407 (51.6)	N.A	57 (7.2)	126 (16.0)	3	54 (6.9)	7 (0.9)	11 (1.4)	31 (3.9)	6 (0.8)	N.A	N.A
19	Liang et al.[20]	China	Composite endpoint	1590	N.A	254 (16.0)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	18 (1.1)	9 (50.0)	9 (50.0)
20	Liu K et al.[21]	Hubei, China	NIV	137	16 (11.7)	34 (24.8)	61 (44.5)	57 (20-83)	14 (10.2)	13 (9.5)	2 (1.5)	N.A	N.A	10	N.A	2 (1.5)	N.A	N.A
21	Ma et al.[22]	Hubei, China	See below[b]	1380	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	37 (2.7)	20 (54.1)	5 (13.5)
22	Mehta et al.[23]	New York, USA	Death	1308	210 (16.1)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	218 (16.7)	N.A	61 (28.0)
23	Miyashita et al.[3]	New York, USA	Intubation, death	5688	555 (9.8)	351 (6.2)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	334 (5.9)	N.A	37 (11.1)
24	Myers et al.[24]	California, USA	ICU	377	100 (26.5)	113 (30.0)	212 (56.2)	61 (50-73)	118 (31.3)	164 (43.5)	28 (7.4)	N.A	48 (12.7)	22 (5.8)	21 (5.6)	18 (4.8)	6 (33.3)	N.A
25	Nikpouraghdam et al.[25]	Iran	Death	2964	239 (8.1)	N.A	1955 (66.0)	56 (46-65)	113 (3.8)	59 (2.0)	60 (2.0)	N.A	18 (0.6)	37 (1.2)	N.A	17 (0.6)	N.A	1 (5.9%)
26	Pan et al[26]	Hubei, China	ICU, death	204	36 (17.6)	16 (7.8)	107 (52.5)	52.9 (±16·0)	44 (21.6)	N.A	9 (4.4)	N.A	N.A	44 (21.6)	N.A	13 (6.4)	N.A	N.A
27	Paranjpe et al.[27]	New York, USA	Death	1078	310 (28.8)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	64 (5.9)	N.A	24 (37.5)
28	Richardson et al.[28]	New York, USA	ICU, death	5700	553 (9.7)	320 (5.6)	3437 (60.3)	63 (52-75)	1808 (31.7)	3026 (53.1)	287 (5.0)	2691 (47.2)	454 (8.0)	966 (16.9)	30 (0.5)	320 (5.6)	N.A	N.A
29	Shi H et al.[29]	Hubei. China	Not defined	81	N.A	N.A	42 (51.9)	49·5 (±11·0)	19 (23.5)	12 (14.8)	9 (11.1)	N.A	3 (3.7)	14 (17.3)	N.A	4 (4.9)	N.A	N.A
30	Shi Y et al.[30]	Zhejiang, China	Not defined	487	0	49 (10.1)	259 (53.2)	46 (±19·0)	29 (6.0)	99 (20.3)	0	40 (8.2)	7 (1.4)	11 (2.3)	22 (4.5)	5 (1.0)	2 (40.0)	N.A
31	Tomlins et al.[31]	UK	NIV + intubation, death	95	20 (21.1)	16 (16.8)	60 (63.2)	75 (59-82)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	20 (21.1)	N.A	3 (15.0)
32	Wan et al.[32]	Chongqing, China	Not defined	135	1 (0.7)	40 (29.6)	72 (53.3)	47 (36-55)	12 (8.9)	13 (9.6)	0	9 (6.7)	N.A	7 (5.2)	2 (1.5)	4 (3.0)	3 (75.0)	N.A
33	Wang D et al.[33]	Hubei, China	ICU	138	6 (4.3)	36 (26.1)	75 (54.3)	56 (42-68)	14 (10.1)	43 (31.2)	4 (2.9)	N.A	2 (1.5)	27 (19.6)	0	10 (7.2)	4 (40.0)	N.A
34	Wang R et al.[34]	Anhui. China	ICU	125	0	19 (15.2)	71 (56.8)	38.8 (±13.8)	10 (8.0)	N.A	2 (1.6)	16 (12.8)	N.A	18 (14.4)	N.A	1 (0.8)	N.A	N.A
35	Wu C[35] et al.	Hubei, China	ARDS, death	201	44 (21.9)	84 (41.8)	N.A	51 (43-60)	22 (10.9)	39 (19.4)	5 (2.5)	N.A	2 (1.0)	8 (4.0)	7 (3.5)	1 (0.5)	N.A	N.A
36	Wu J et al.[36]	Jiangsu, China	See below	80	0	3 (37.5)	39 (48.8)	46.1 (±15.4)	5 (6.3)	N.A	0	N.A	1 (1.3)	25 (31.3)	1 (1.3)	1 (1.3)	N.A	N.A

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37	Xu et al.[37]	Guangzhou, China	Not defined	90	N.A	N.A	39 (43.3)	50 (18-86)	5 (55.6)	17 (18.9)	1 (1.1)	N.A	N.A	3 (3.3)	N.A	2 (2.2)	N.A	N.A
38	Yang et al.[38]	Zhejiang, China	NIV	149	0	2 (1.3)	81 (54.4)	45.1 (±13.4)	9 (6.0)	0	1 (0.7)	N.A	N.A	28 (18.8)	N.A	2 (1.3)	N.A	N.A
39	Yu et al.[39]	Hubei, China	ARDS, death	1524	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	12 (0.8)	3 (25.0)	3 (25.0)
40	Zhang L et al.[40]	Hubei, China	Composite endpoint	1276	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	28 (2.2)	15 (53.6)	8 (28.6)
41	Zhang G et al.[41]	Hubei, China	IDSA/ATS	221	12 (5.4)	55 (24.9)	108 (48.9)	55 (39-66.5)	22 (10.0)	54 (24.4)	6 (2.7)	N.A	6 (2.7)	37 (16.7)	7 (3.2)	9 (4.1)	4	N.A
42	Zhang J et al.[42]	Hubei, China	See below	663	25 (3.8)	409 (61.7)	321 (48.4)	55.6 (44-69)	67 (10.1)	N.A	51 (7.7)	N.A	N.A	164 (24.7)	31 (4.7)	14 (2.1)	11 (78.6)	N.A
43	Zhang N et al.[c]	Hubei, China	See below	1548	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	67 (4.3)	32 (47.8)	18 (26.9)
44	Zhang S et al.[d]	Hubei, China	Death	315	47 (14.9)	178 (56.5)	175 (55.6)	57 (44-66)	41 (13.0)	78 (24.8)	3 (1.0)	N.A	2 (0.6)	42 (13.3)	9 (2.9)	12 (3.8)	N.A	4 (33.3)
45	Zhao et al.[43]	Hubei, China	Not defined	91	2 (2.2)	30 (33.0)	49 (53.8)	46 (Not stated)	3 (3.3)	18 (19.8)	1 (1.1)	N.A	1 (1.1)	N.A	N.A	3 (3.3)	2 (2.2)	N.A
46	Zhou et al.[44]	Hubei, China	Not defined	191	54 (28.3)	119 (62.3)	119 (62.3)	56 (46-67)	36 (18.8)	58 (30.4)	6 (3.1)	11 (5.8)	2 (1.0)	15 (7.9)	0	2 (1.0)	N.A	N.A
47	Zhu et al.[45]	Hefei, China	Not defined	32	N.A	N.A	15 (46.9)	46 (35-52)	4 (12.5)	7 (21.9)	2 (6.3)	6 (18.8)	1 (3.1)	10 (31.3)	2 (6.3)	2 (6.3)	N.A	N.A

[a]composite endpoint: Severe disease composite endpoints (ICU, mechanical ventilation)

[b]see below: Severity as defined by Chinese National Health Commission Definition of Severity (Refer to Appendix 3)

[c]article in press (accepted, peer-reviewed), [d]article not peer-reviewed

All figures presented as n (%) – number (percentage) unless otherwise stated. Age presented as median with interquartile range (IQR): age (range) or as mean: age (± standard deviation (SD))

Abbreviations: ARDS – acute respiratory distress syndrome; CKD – chronic kidney disease; CLD (chronic liver disease); CVS – cardiovascular disease (and included cerebrovascular disease); DM – diabetes mellitus; FiO2 – supplemental oxygen; HTN – hypertension; ICU – intensive care unit; IDSA/ATS – Infectious Disease Society of America/American Thoracic Society Guidelines (Refer to Appendix 3); NIV – non-invasive ventilation; N.A – not applicable

Overall, lung cancer was the most common tumour type in the six case series at 24.0% (64/267) (Table 2). There were more males than females affected, with a median age above 60 in all case series. This median age was notably higher than the non-cancer populations reported in previous studies[1]. Severe disease was documented in 44.6% (119/267) of the patients in the six studies (Table 2). Overall, 230 patients from five case series were identified with oncological history, stage, recent treatment history and clinical outcomes (Appendix 2 Supplementary Table 1). However, among these patients, only 24.3% of the patient received recent systemic therapy, limiting the usefulness of the dataset in predicting the outcomes of COVID-19 in patients with active cancer.

Table 2: Baseline characteristics of 267 cancer patients with COVID-19 infections from six case series

Characteristics	Zhang L et al. 2020 (n=28) n (%) or median (range)		Yu et al. 2020 (n=12) n (%) or median (range)		Liang et al. 2020 (n=18) n (%) or median (range)		Ma et al. 2020 (n=37) n (%) or median (range)		Dai et al. 2020 (n=105) n (%) or median (range)		Zhang H et al. 2020 (n=67) n (%) or median (range)	
Age	65.0 (56.0-70.0)		66 (48-78)		63.1 (51-75)		62 (59-70)		64 (50-78)		66 (37-98)	
Gender												
Male	17 (60.7)		10 (83.3)		12 (66.7)		20 (54.1)		57 (54.7)		41 (61.2)	
Female	11 (39.3)		2 (16.7)		6 (33.3)		17 (45.9)		48 (45.3)		26 (38.8)	
Stage												
I-III	18 (64.3)		5 (41.7)		13 (72.2)		N.A		88 (83.8)		N.A	
IV	10 (35.7)		6 (50.0)		4 (22.2)		N.A		17 (16.2)		N.A	
Unknown	N.A		1 (8.3)		1 (5.6)		N.A		N.A		N.A	
Smoking history												
Yes	N.A		N.A		4 (22.2)		N.A		36 (34.3)		9 (13.4)	
No	N.A		N.A		N.A		N.A		69 (65.7)		N.A	
Unknown	N.A		N.A		N.A		N.A		N.A		N,A	
	Lung ca	7 (25.0)	Lung ca	7 (58.3)	Lung ca	5 (27.8)	CRC	11 (29.7)	Lung ca	22 (20.1)	Lung ca	15 (22.4)
	Esophageal ca	4 (14.3)	Rectal ca	1 (8.3)	CRC	4 (22.2)	Lung ca	8 (21.6)	GI ca	13 (12.4)	CRC	11 (16.4)
	Breast ca	3 (10.7)	Colon ca	1 (8.3)	Breast ca	3 (16.7)	Breast ca	7 (18.9)	Breast ca	11 (10.5)	Thyroid ca	8 (11.9)
	Laryngoca	2 (7.1)	Pancreatic ca	1 (8.3)	Bladder ca	2 (11.1)	Gynae ca	5 (13.5)	Thyroid ca	11 (10.5)	Urinary system	8 (11.9)
	Liver ca	2 (7.1)	Breast ca	1 (8.3)	Lymphoma	1 (0.56)	Other ca	6 (16.2)	Blood ca	9 (8.6)	Breast & gynae	7 (10.4)
	Prostatic ca	2 (7.1)	Urothelial ca	1 (8.3)	Thyroid ca	1 (0.56)			Cervix ca	6 (5.7)	Head and Neck	5 (7.5)
	Cervical ca	1 (3.6)			Adrenal ca	1 (0.56)			Esophageal ca	6 (5.7)	CNS tumour	4 (5.9)

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	Gastric ca	1 (3.6)			RCC	1 (0.56)					Haematological	3 (4.5)
	Colon ca	1 (3.6)									Gastric ca	3 (4.5)
	Rectum ca	1 (3.6)									Esophageal ca	2 (3.0)
	NPC	1 (3.6)									HCC	1 (1.15)
	Endometrial ca	1 (3.6)										
	Ovarian ca	1 (3.6)										
	Ca of testis	1 (3.6)										
Baseline treatment	ChemoRT	25 (89.3)	BSC	4 (33.3)	Surveillance	12 (75.0)	Antitumour therapy	13 (35.1)	Chemotherapy	17 (16.2)	Follow up	44 (65.7)
	Operation	21 (75.0)	Chemo-immuno Therapy	2 (16.7)	Chemo and surgery within 1 month	4 (25.0)			RT	13 (12.4)	On treatment	23 (34.3)
	Targeted therapy/ immunotherapy	6 (21.4)	Surveillance	2 (16.7)					Surgery	8 (7.6)		
	Chemo (<14d)	3 (10.7)	Targeted Therapy/RT	1 (8.3)					Immunotherapy	6 (5.7)		
	Targeted Therapy	2 (7.1)	Chemo/RT	1 (8.3)					Targeted Therapy	4 (3.8)		
	RT (<14d)	1 (3.6)	Adjuvant RT	1 (8.3)								
	Immunotherapy (<14d)	1 (3.6)	Not started	1 (8.3)								
Severe disease	Severe 8/15 Non-invasive ventilation 2/15 Invasive ventilation	15 (53.6)	Severe (25.0)	3	Severe	9 (50.0)	Severe	20 (54.1)	Severe ICU	40 (38.0) 20 (19.1)	Severe Mechanical ventilation	32 (47.8) 28 (41.8)
ARDS	8 (28.6)		N.A		N.A		N.A		N.A		14 (20.9)	
Death	8 (28.6)		3 (25.0)		9 (50.0)		5 (13.5)		12 (11.4)		18 (26.9)	

Abbreviations: ARDS – acute respiratory distress syndrome; BSC – best supportive care; Ca – carcinoma; CNS – central nervous system; CRC – colorectal cancer; GI – gastrointestinal; ICU – intensive care unit; NPC – nasopharyngeal cancer; RCC – renal cell carcinoma; RT – radiotherapy

Proportion of cancer diagnoses

The overall pooled proportion of cancer patients was 0.03 (95% CI 0.02-0.04) with significant heterogeneity among the included studies, I^2 = 94.64%, chi-square p=<0.01 (Figure 2a). The proportion of cancer patients was significantly lower in the studies with median or mean age less than 60 years, at 0.02 (95% CI 0.01-0.02) compared to 0.07 (95% CI 0.05-0.09) for studies with median or mean age above or equal 60 years (Figure 2b). The proportion of cancer was higher in the studies with male pre-dominance (more than or at least 60% population being male), at 0.05 (95% CI 0.02-0.07) than studies with less than 60% population being male, at 0.02 (95% CI 0.02-0.03; Figure 2c). The proportion of cancer in Hubei Province was 0.03 (95% CI 0.02-0.05) and outside Hubei Province 0.04 (95% CI 0.02-0.04; Appendix 2 Supplementary Figure 1).

Mortality outcomes of cancer patients with COVID-19

Overall, the mortality rates of cancer patients with COVID-19 infection ranged widely from 5.9% to 50% (Table 3). The pooled relative risk of death was significantly higher in patients with cancer and COVID-19 patients than in the general population with COVID-19 (RR 1.41; 95% CI 1.15-1.73), I^2 = 26.2%, chi-square p=0.229 (Figure 3). The pooled mortality rate was estimated at 0.22 (95% CI 0.21-0.33) among cancer patients with COVID-19, which was significantly higher than that of the overall COVID-19 patients, at 0.09 (95% CI 0.07-0.11; Appendix 2 Supplementary 2a and 2b).

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Table 3: Mortality rate among cancer patients diagnosed with COVID-19 infection

Case series	Death/cancer population	Mortality rate in cancer cohorts (%)	Death/total COVID-19 population	Mortality rate in general COVID-19 cohorts (%)	Relative risk of death (95% CI)
Liang et al.	9/18	50.0	N.A	N.A.	N.A
Zhang L et al.	8/28	28.6	N.A	N.A	N.A
Yu et al.	3/12	25	N.A	N.A	N.A
Ma et al.	5/37	13.5	N.A	N.A.	N.A
Zhang N et al	18/67	26.9	N.A	N.A	N.A
Dai et al.	12/105	11.4	N.A	N.A	N.A
Benelli et al.	**9/33**	**27.3**	**72/411**	**17.5**	**1.56 (0.86-2.82)**
Mehta et al.	**61/218**	**28.0**	**210/1308**	**16.1**	**1.74 (1.36-2.23)**
Miyashita et al.	**37/334**	**11.1**	**555/5688**	**9.8**	**1.14 (0.83-1.55)**
Nikpouraghdam et al.	**1/17**	**5.9**	**239/2964**	**8.1**	**0.73 (0.11-4.90)**
Paranjpe et al.	**24/64**	**37.5**	**310/1078**	**28.8**	**1.30 (0.94-1.81)**
Tomlins et al	**3/20**	**15.0**	**20/95**	**21.1**	**0.71 (0.23-2.17)**
Zhang S et al.	**4/12**	**33.3**	**47/315**	**14.9**	**2.23 (0.96-5.19)**

Severity outcomes of cancer patients with COVID-19

Further analyses were conducted for the eight case series with available severity outcomes for both cancer and general COVID-19 patient cohorts. We found that cancer patients had higher risk of severe COVID-19 outcomes compared to the general COVID-19 population, with a pooled relative risk of 1.49 (95% CI 1.10-2.03) and moderate heterogeneity, I^2 = 64.0%, chi-square p=0.0007 (Figure 4a). Also, we noted that the pooled proportion of severe outcomes in cancer patients was 0.42 (95% CI 0.30-0.54), which was higher than the proportion of severe cases of all COVID-19 patients at 0.27 (95% CI 0.21-0.33; Appendix 2, Supplementary Figure 2a and 2b). There was no statistically significant difference in relative risk of severe outcomes between those patients with lung cancers versus non lung cancers (RR at 1.13; 95% CI 0.57-2.22, I^2 = 34.6%, chi-square p=0.205; Figure 4b). There was no statistically significant difference in relative risks of severe outcomes between stage IV vs non stage IV cancers (RR 1.48; 95% CI 0.89-2.47; Figure 4c).

The severe outcomes of COVID-19 in cancer patients was also analysed from the available 230 individual patient data and summarised in Appendix 2 Supplementary table 1. Only patient age and gender appeared to be significantly associated with severity outcomes by Chi-Square Test (Appendix 2 Supplementary Table 2).

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DISCUSSION

We performed a systematic review and meta-analysis to find out the proportion of cancer diagnoses among COVID-19 patients, and whether they had more severe outcomes compared to the general population. We also looked at the patient characteristics that may be associated with severe clinical outcomes among these patients. To our knowledge, this is the largest meta-analysis that directly addresses the clinical outcomes of COVID-19 patients with cancer, including the relative risks of severity and death as compared to general COVID-19 patients.

Firstly, we found that the pooled proportion of cancer patients among COVID-19 cases was 3%. This result was similar to the proportion rate from an earlier meta-analysis[46] of 11 studies with a proportion of 2%, but higher than the 0.2% described by the Chinese Centre for Disease Control and Prevention Group[2]. We found that the cancer proportion was higher among older patients and in male patients with COVID-19. In the subsequent IPD analysis, it was also advanced age and male gender that were significantly associated with severity of COVID-19 infection. We also noted that the cancer proportion among the COVID-19 case series from Hubei Province was 3% (Appendix 2 Supplementary Figure 1), which was higher than the estimated cancer prevalence of 0.45% in the region[20]. However, cancer proportion rate should be taken in the context of geographical variations, as well as differences in the oncology patient load across medical centres.

We have shown that COVID-19 patients with cancer had significantly more severe outcomes compared to the general COVID-19 patients, with a pooled RR of 1.41 (95% CI 1.08-1.87, p=0.024). The pooled estimated proportion of severe COVID-19 outcomes was 42% among cancer patients in our meta-analysis, which was higher compared to a recently published cohort of 928 cancer patients from USA, Canada and Spain that reported that 26% of their patients met a composite endpoint of death, ICU admission and/or mechanical ventilation[47]. Overall, the proportion of cancer patients with severe COVID-19 outcomes varied across studies but remained high, highlighting the vulnerability of cancer patients to SARS-CoV-2 infection.

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We also observed that lung cancer was the most common cancer among those affected by COVID-19. However, this may be confounded by the fact that lung cancer is also the most common cancer type in China[48]. A few case series noted that patients with lung cancers had significantly worse outcomes compared to non-lung cancer cohorts[1,40], which was also supported by the early data from TERAVOLT, an international database of thoracic cancers (in press). However, we did not find the relative risk of severity to be significant for lung cancers (vs other tumour types) or stage IV cancers (vs stage I-III) across four studies. This was also echoed by our IPD analysis, where lung cancer and the stage of cancer were not significantly associated with severe outcomes. Separately, haematological cancers also showed severe outcomes with high mortality rates in two studies[1,49], and should also be investigated further in a collaborative effort. The available individual data in our study were not large enough to analyse the influence of modality of treatment (targeted therapy, immunotherapy or chemotherapy) on the outcome of COVID-19 infections. Even the larger studies published recently did not consistently find active systemic therapy to be associated with risk of severe COVID-19 infection[50]. There were also mixed observations from small series, with some suggesting that patients on recent chemotherapy[20] or immunotherapy[1] were at higher risk. These questions remain unanswered and require further collaborative efforts to identify risk factors of severe COVID-19 infection among cancer patients.

Overall, the pooled relative risk of death among the cancer patients was 1.41 (95% CI 1.15-1.73), compared to general COVID-19 patients. However, pooled mortality risk of cancer patients with COVID-19 should be interpreted with caution due to the limited sample size. Some case series also had patients who remained hospitalised at the time of reporting, hence without full mortality outcomes. CFR were lower at 5.6% among cancer COVID-19 patients in a large Chinese report[2] and 11.1% from a tertiary centre in New York, USA[3]. Notably, among selected COVID-19 mortality reports from Italy and USA, around 20-30% of deaths had a history of cancer[23,51].

In light of the severity of COVID-19 in cancer patients, efforts should be undertaken to reduce nosocomial exposure in cancer patients, as nosocomial infections may contribute to incidence of COVID-19 in oncology patients[39]. The decision to attend clinics and continue non-urgent systemic therapy should be made judiciously, especially in regions with high burden of disease transmission.

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This has been echoed by various oncology bodies, including American Society of Clinical Oncology (ASCO) and European Society of Medical Oncology (ESMO)[52]. Given the higher risk of severe outcomes and mortality among cancer patients, it is important that cancer patients with COVID-19 infection should be monitored closely and decisions for ICU support discussed early. ICU outcomes should also be an important question for future retrospective studies in view of the ethical and clinical considerations of putting patients with advanced cancer on mechanical ventilation, especially in countries with significant outbreak and limited ICU resources.

The limitations of this meta-analysis include the heterogeneity and retrospective nature of the case studies, including reporting and selection biases. A few studies were reported from the same Hospitals and hence there may be overlapping patients although efforts were taken to reduce this (Appendix 1 Supplementary table 2). The small numbers and incomplete death outcomes at the time of reporting also restricted our analysis. Due to the limited clinical data, we could not adjust for co-morbidities and other characteristics that may contribute to severe disease. It should also be noted that severity outcomes were defined inconsistently across some studies. A few studies also included clinically diagnosed COVID-19 cases.

Whilst we acknowledge other risk factors such as age[35,53,54], gender[55], cardiovascular risk factors[54,56,57] including diabetes mellitus[58,59] and hypertension[60] remain important in COVID-19 infection, this is the largest meta-analysis to show that malignancy is also a significant risk factor for severe disease and mortality. However, we did not identify other tumour or treatment related factors associated with severe outcomes among cancer patients with COVID-19. As more data becomes available for cancer patients with COVID-19 infection, further analyses may be performed to discern the clinical outcomes of cancer patients with COVID-19 infection, with specific attention to underlying lung cancers, haematological cancers and the effects of various systemic therapies or other treatment modalities. Information on COVID-19 patients with cancer should also be collected in more detail and with consistency across studies, to allow further meta-analyses to ascertain the outcomes of cancer patients with COVID-19 infection.

CONCLUSION

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Given that COVID-19 infection may persist in our population in the foreseeable future, it is imperative to continue active research and data collection on the effect of SARS-CoV-2 on oncology patients. Due to the burden of their disease, these patients will continue to require systemic chemotherapy and urgent care[61]. As such, treating physicians need to be able to make more discerning decisions on the choice of systemic treatment and escalation of care.

Overall, we found through meta-analysis that the severity and mortality rates of COVID-19 in cancer patients were higher than that in the general population. However, the dataset remains small and should be interpreted with caution. We hope that with collaborative efforts, a comprehensive set of patient outcomes of COVID-19 patients, and specifically, of cancer patients, can be collated to better understand the disease trajectory and outcomes of COVID-19.

Word count: 2827 words*

**excluding tables and figures*

Figure legends

Figure 1: Flow chart of study identification and exclusion for data analysis
Figure 2a: Random effects pooled proportion of cancer patients among patients diagnosed with COVID-19. *Yu, Dai and Zhang N et al was excluded in the proportion calculations (did not represent overall COVID-19 populations)
Figure 2b: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by median/mean age ≥ 60 vs < 60 years)
Figure 2c: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by male ≥ 60% or < 60%)
Figure 3: Random effects pooled relative risk of death for cancer COVID-19 patients versus total population of COVID-19 patients
Figure 4a: Random effects pooled relative risk for severe outcome of cancer COVID-19 patients versus total population of COVID-19 patients
Figure 4b: Random effects pooled relative risk for severe outcome of lung cancer vs other cancer patients infected with COVID-19
Figure 4c: Random effects pooled relative risk for severe outcome of stage IV cancer vs non-stage IV cancer patients infected with COVID-19

Declarations

No conflicts of interest to declare.
Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.
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Data sharing not applicable to this article as no datasets were generated or analysed during the current study

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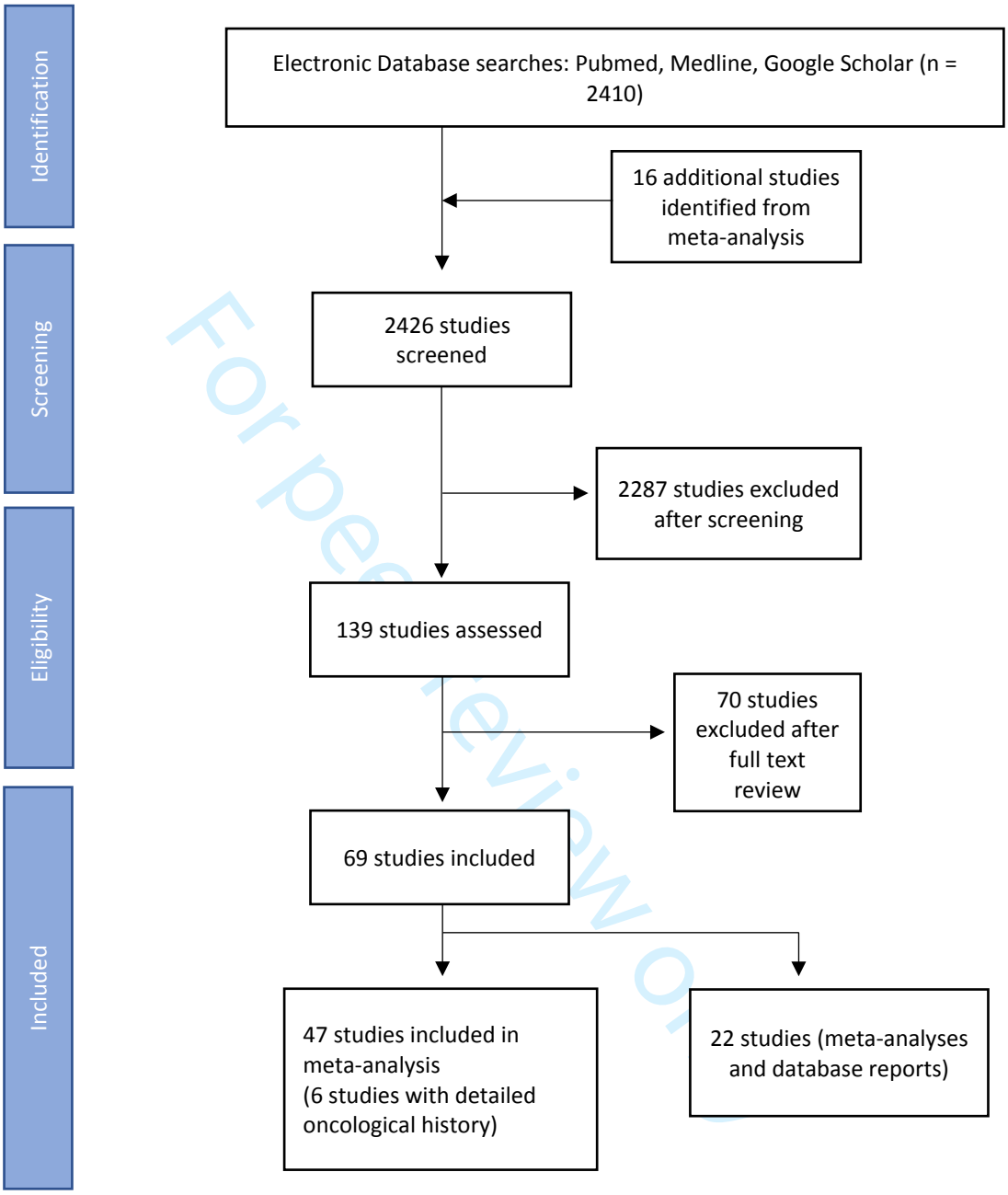
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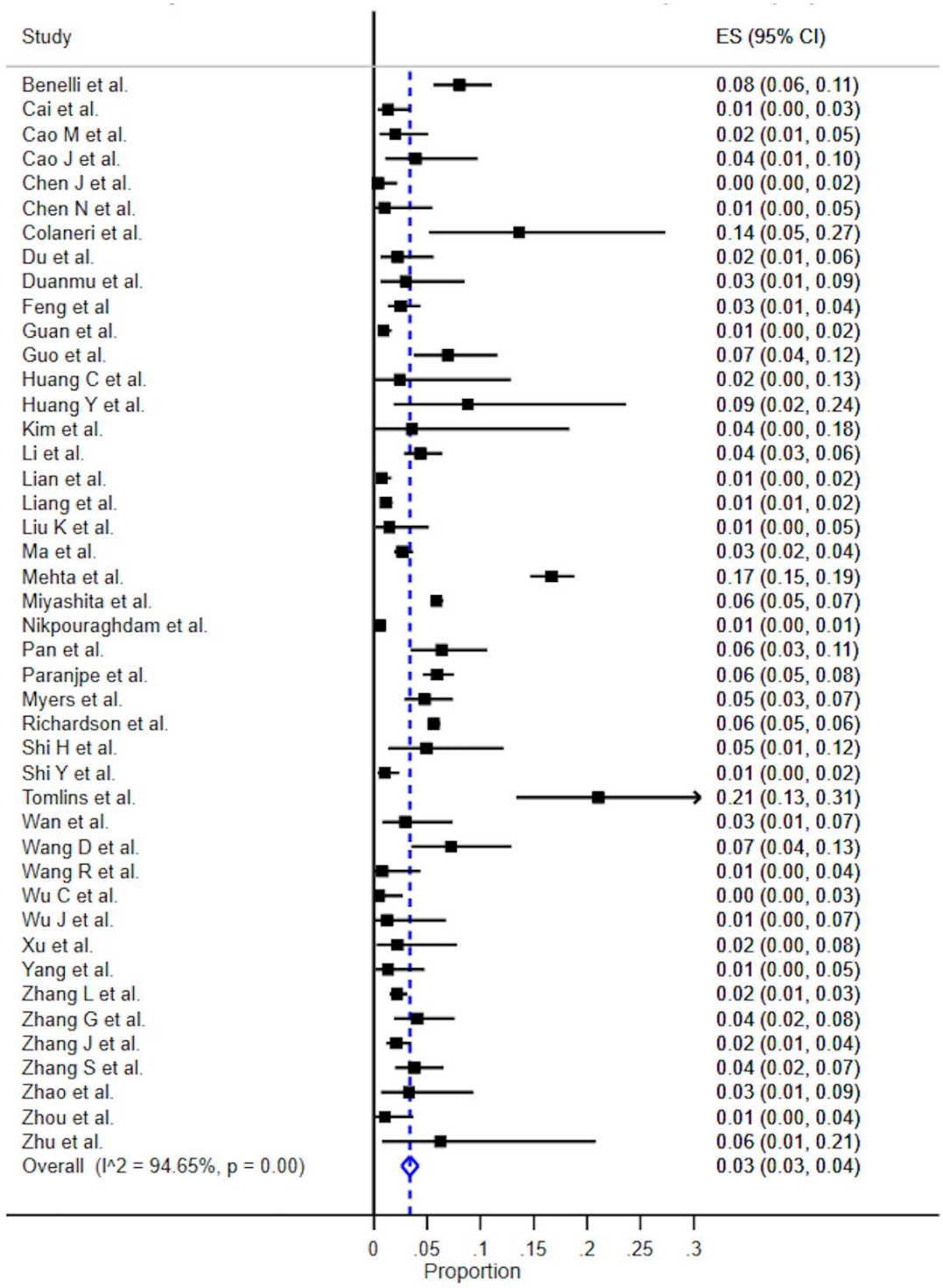
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Figure 1: Flow chart of study identification and exclusion for data analysis



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Figure 2a: Random effects pooled proportion of cancer patients among patients diagnosed with COVID-19



*Yu, Dai and Zhang N et al was excluded in the proportion calculations (did not represent overall COVID-19 populations)

Figure 2b: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by median/mean age ≥ 60 vs < 60 years)

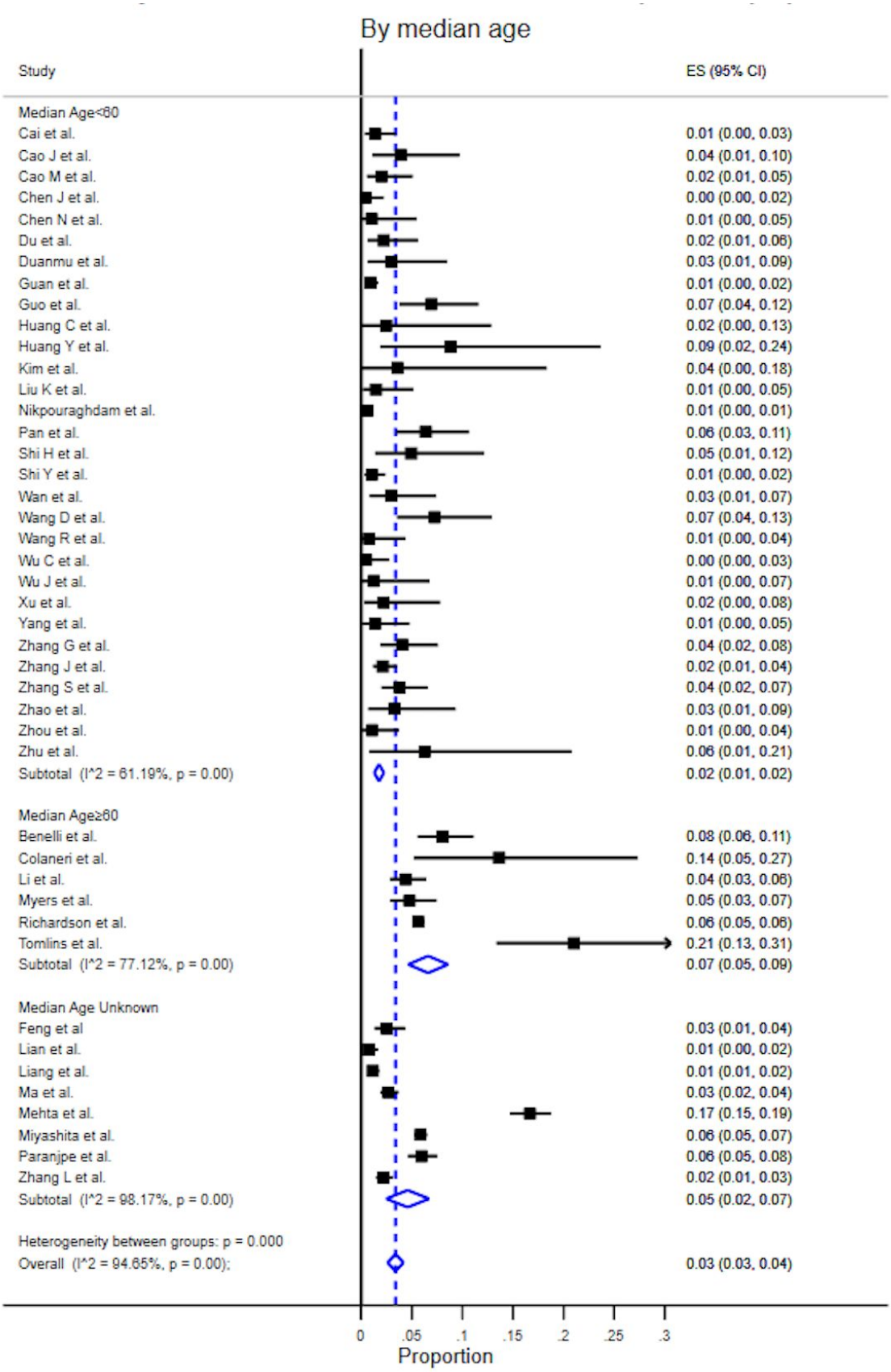


Figure 2c: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by male ≥ 60% or < 60%)

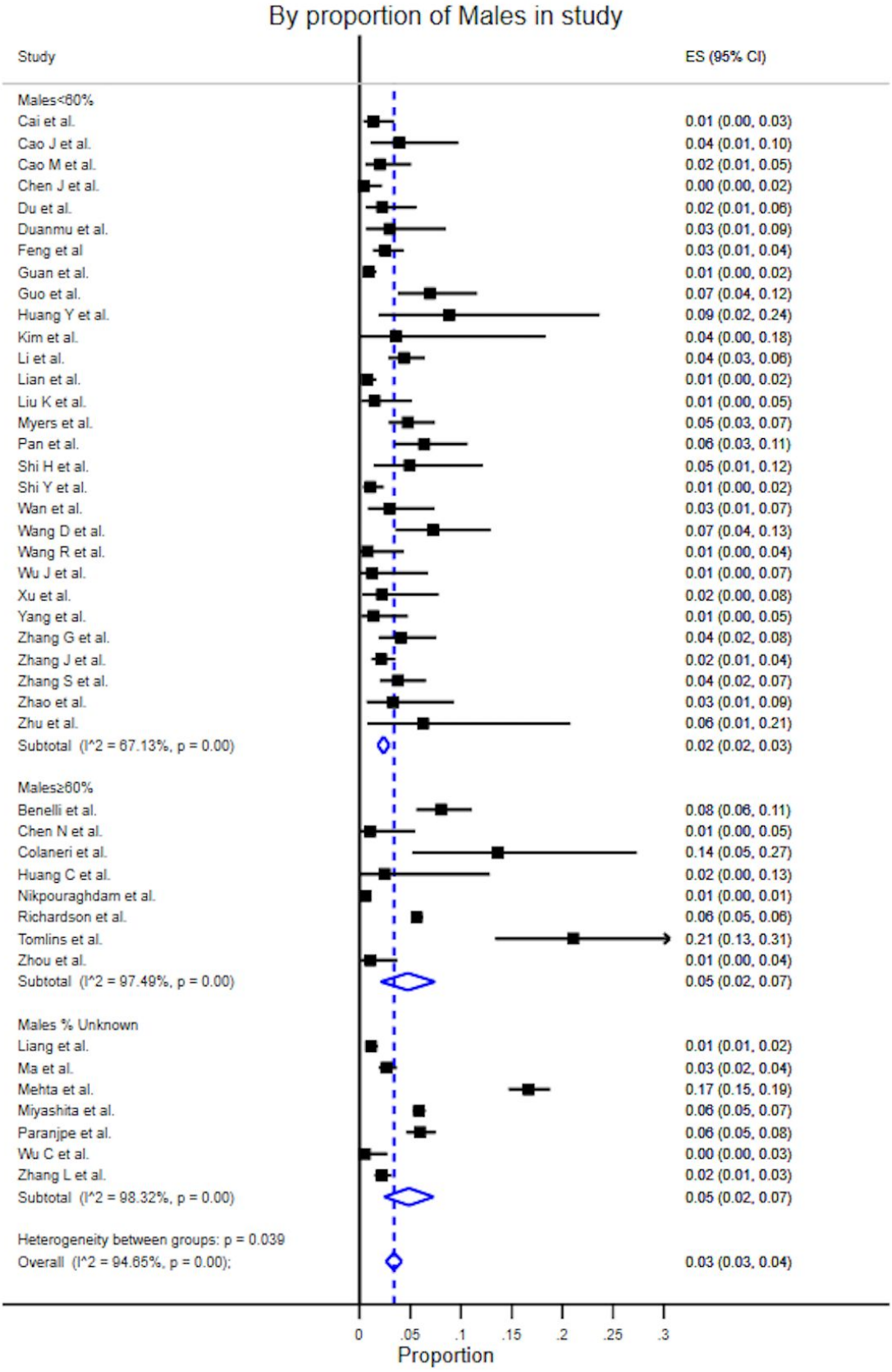
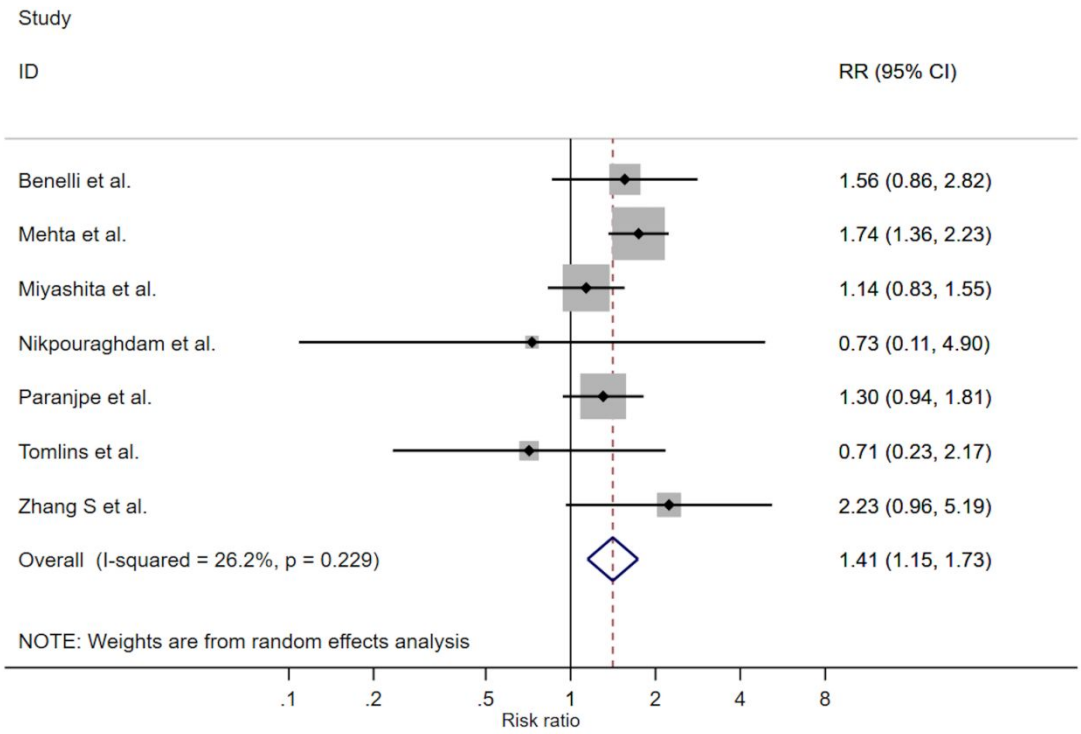
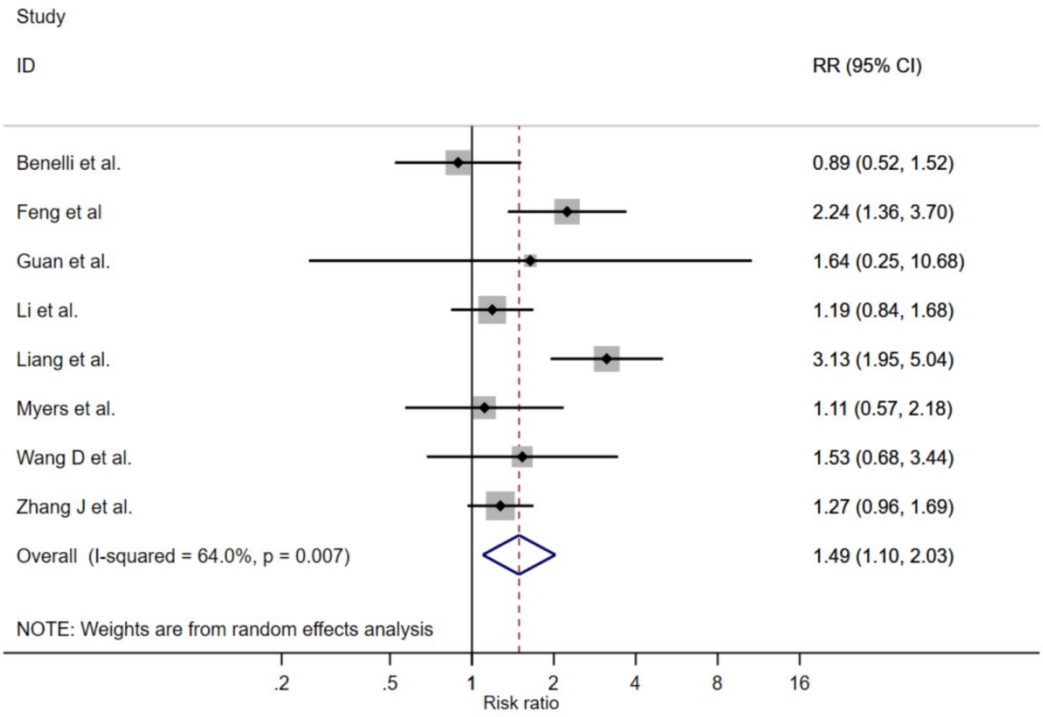


Figure 3: Random effects pooled relative risk of death for cancer COVID-19 patients versus total population of COVID-19 patients



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Figure 4a: Random effects pooled relative risk for severe outcome of cancer COVID-19 patients versus total population of COVID-19 patients



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Figure 4b: Random effects pooled relative risk for severe outcome of lung cancer vs other cancer patients infected with COVID-19

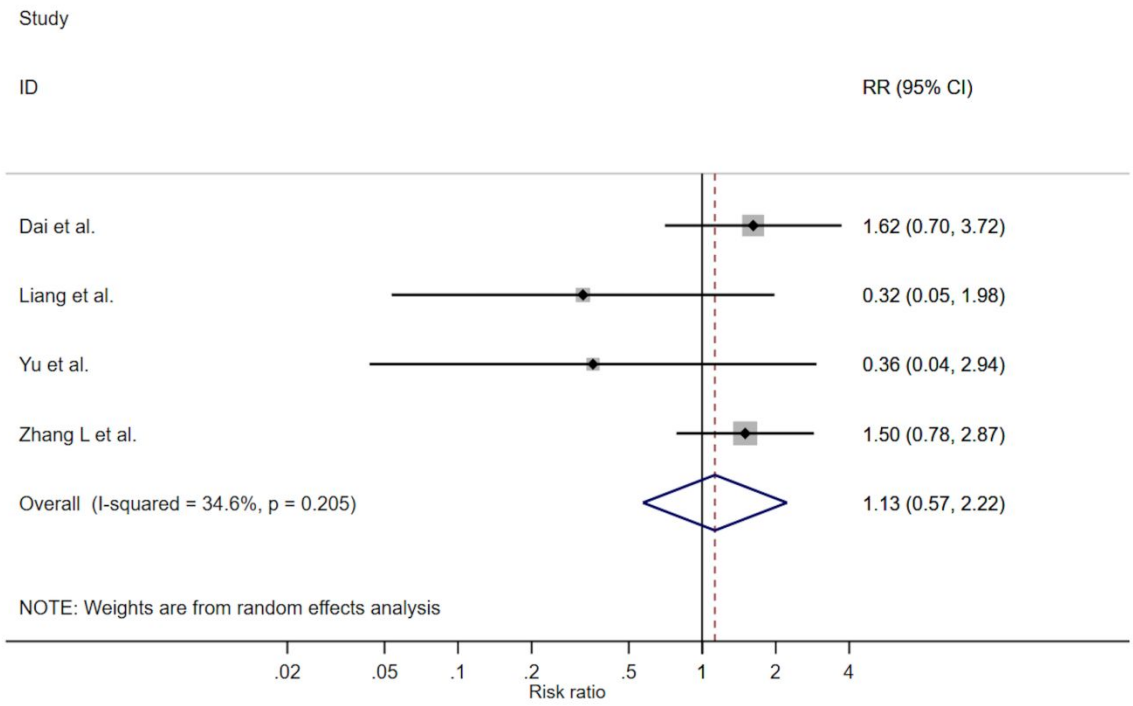
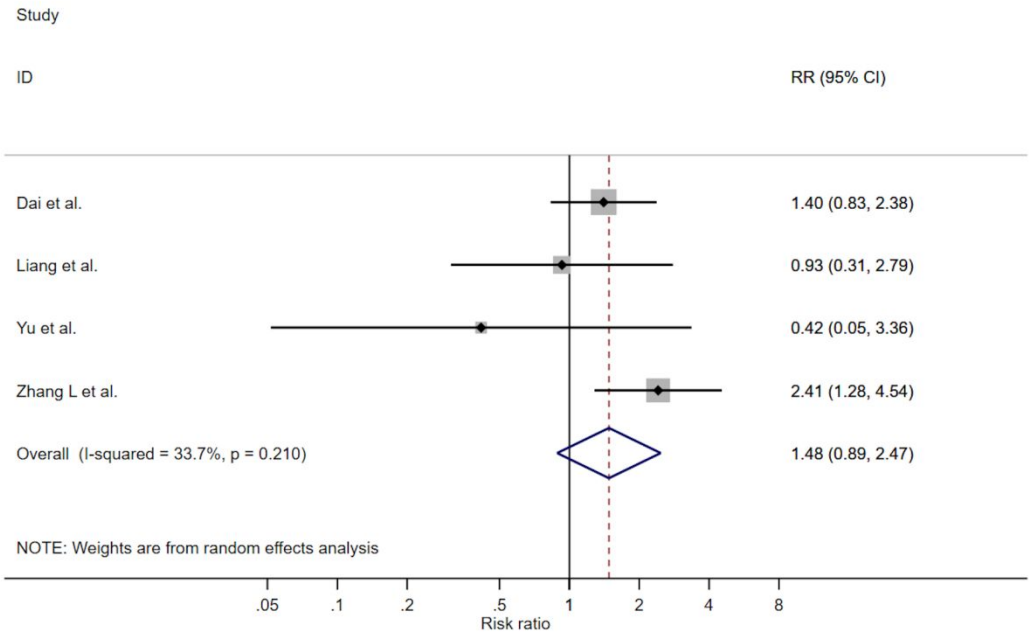


Figure 4c: Random effects pooled relative risk for severe outcome of stage IV cancer vs non-stage IV cancer patients infected with COVID-19



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Appendix 1

Supplementary Table 1: Meta-analyses and database reports on COVID-19

No.	Authors	Title:
1.	Desai et al.	COVID-19 and Cancer: Lessons From a Pooled Meta-Analysis
2.	Emami et al.	Prevalence of Underlying Diseases in Hospitalized Patients with COVID-19: a Systematic Review and Meta-Analysis
3.	Wang B et al.	Does comorbidity increase the risk of patients with COVID-19: evidence from meta-analysis
4.	Cao Y et al.	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2: A systematic review and meta-analysis
5.	Hu Y et al.	Prevalence and severity of corona virus disease 2019 (COVID-19): A systematic review and meta-analysis
6.	Yang J et al.	Prevalence of comorbidities and its effects in patients infected with SARS-CoV-2: a systematic review and meta-analysis
7.	Fu L et al.	Clinical characteristics of coronavirus disease 2019 (COVID-19) in China: A systematic review and meta-analysis
8.	Li LQ et al.	COVID-19 patients' clinical characteristics, discharge rate, and fatality rate of meta-analysis
9.	Zhu J et al.	Clinicopathological Characteristics of 8697 Patients With COVID-19 in China: A Meta-Analysis
10.	Zheng Z et al.	Risk factors of critical & mortal COVID-19 cases: A systematic literature review and meta-analysis
11.	Rodriguez-Morales et al.	Clinical, laboratory and imaging features of COVID-19: A systematic review and meta-analysis
12.	Ofori-asenso et al.	Cancer is associated with severe disease in COVID-19 patients: a systematic review and meta-analysis
13.	Khan et al.	Effects of underlying morbidities on the occurrence of deaths in COVID- 19 patients: A systematic review and meta-analysis (in-press)
14.	Moujaess et al.	Cancer patients and research during COVID-19 pandemic: A systematic review of current evidence
15.	Sun P et al.	Clinical characteristics of hospitalized patients with SARS-CoV-2 infection: A single arm meta-analysis
16.	Jin JM et al.	Gender Differences in Patients with COVID-19 Focus on Severity and Mortality
17.	Deng G et al	Clinical determinants for fatality of 44,672 patients with COVID-19
18.	Wu and Mcgoogan	Characteristics of and Important Lessons From the Coronavirus Disease 2019 (COVID-19) Outbreak in China: Summary of a Report of 72 314 Cases From the Chinese Center for Disease Control and Prevention
19.	Onder G, Rezza G, Brusaferro S.	Case-Fatality Rate and Characteristics of Patients Dying in Relation to COVID-19 in Italy
20.	Guan et al	Comorbidity and its impact on 1590 patients with Covid-19 in China: A Nationwide Analysis
21.	Adams ML, Katz DL, Grandpre J.	Population-Based Estimates of Chronic Conditions Affecting Risk for Complications from Coronavirus Disease, United States
22.	Kang YJ et al	Mortality Rate of Infection With COVID-19 in Korea From the Perspective of Underlying Disease

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Supplementary Table 2: Case series included in meta-analysis

No.	Author	Title	No.	Type of study	Outcome measured	Quality of study	Hospital	Location
1	Benelli et al. 2020	SARS-CoV-2 comorbidity and outcome in hospitalized patients in Crema, Italy.	539	Single centre prospective cohort study	Severity	7	General Hospital of Crema	Crema, Italy
2	Cai et al. 2020	COVID-19 in a designated infectious diseases hospital outside Hubei Province, China.	298	Single centre retrospective cohort study	Severity	7	Third People's Hospital of Shenzhen	Shenzhen, China
3	Cao M et al. 2020	Clinical Features of Patients Infected with the 2019 Novel Coronavirus (COVID-19) in Shanghai, China.	198	Single centre retrospective cohort study (not-peer reviewed, pre-print)	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 15 Feb 2020	Shanghai, China
4	Cao J et al. 2020	Clinical features and short-term outcomes of 102 patients with coronavirus disease 2019 in Wuhan, China.	102	Single centre retrospective cohort study	Death	7	Wuhan University Zhongnan Hospital in Wuhan, China, 3 Jan to 1 Feb 2020	Wuhan, Hubei, China
5	Chen J et al. 2020	Clinical progression of patients with COVID-19 in Shanghai, China.	249	Single centre retrospective cohort study	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 6 Feb, 2020	Shanghai, China
6	Chen N et al. 2020	Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study.	99	Single centre retrospective cohort study	Severity	7	Jinyintan Hospital in Wuhan, 1 to 20 Jan 2020	Wuhan, Hubei, China
7	Colaneri et al. 2020	Clinical characteristics of coronavirus disease (COVID-19) early findings from a teaching hospital in Pavia, North Italy, 21 to 28 February 2020.	44	Single centre retrospective cohort study	Severity	6	Fondazione IRCCS Policlinico San Matteo, 21 to 28 Feb 2020	Pavia, Italy
8	Dai et al. 2020	Patients with cancer appear more vulnerable to SARS-COV-2: a multi-center study during the COVID-19 outbreak.	641	Multicentre retrospective matched cohort study	Severity	8	14 hospitals in Wuhan, China, 1 Jan to 24 Feb 2020 (Zhongnan, Tongji, Renmin Hospital etc)	Wuhan, Hubei, China
9	Du et al. 2020	Predictors of Mortality for Patients with COVID-19 Pneumonia Caused by SARS-CoV-2: A Prospective Cohort Study.	179	Single centre prospective study	Death	7	Wuhan Pulmonary Hospital, 25 Dec 2019 to 7 Feb 2020	Wuhan, Hubei, China
10	Duanmu et al. 2020	Characteristics of Emergency Department Patients With COVID-19 at a Single Site in Northern California: Clinical Observations and Public Health Implications.	100	Single centre observational cross-section study	Admission from ED, Death	6	Stanford University School of Medicine, 4 to 23 March 2020	California, USA
11	Feng et al. 2020	COVID-19 with Different Severity: A Multi-center Study of Clinical Features	476	Single centre retrospective cohort study	Severity and death	7	3 hospitals in Wuhan, Shanghai and Anhui, 1 Jan to 15 Feb 2020	China
12	Guan et al. 2020	Clinical Characteristics of Coronavirus Disease 2019 in China.	1099	Multicentre retrospective cohort study	Severity and death	7	552 hospitals across China, December 11, 2019, and January 29 2020	China
13	Guo et al. 2020	Cardiovascular Implications of Fatal Outcomes of Patients With Coronavirus Disease 2019 (COVID-19)	187	Single centre retrospective cohort study	Severity and death	7	Seventh Hospital of Wuhan City, from 23 Jan to 23 Feb 2020	Wuhan, Hubei, China
14	Huang C et al. 2020	Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China.	41	Single centre prospective cohort study	Severity	7	Jinyintan Hospital till 2 Jan 2020	Wuhan, Hubei, China
15	Huang Y et al. 2020	Clinical characteristics of laboratory confirmed positive cases of SARS-CoV-2 infection in Wuhan, China: A retrospective single center analysis.	34	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital, 21 Dec to 28 Jan 2020	Wuhan, Hubei, China
16	Kim et al. 2020	Clinical Course and Outcomes of Patients with Severe Acute Respiratory Syndrome Coronavirus 2 Infection: a Preliminary Report of the First 28 Patients from the Korean Cohort Study on COVID-19.	28	Multicentre retrospective cohort study	Severity	6	11 Hospitals in Korea	Korea
17	Li et al. 2020	Risk factors for severity and mortality in adult COVID-19 inpatients in Wuhan.	548	Single centre retrospective cohort study	Severity and death	7	Tongji Hospital, 26 Jan to 5 Feb 2020	Wuhan, Hubei, China
18	Lian et al. 2020	Analysis of Epidemiological and Clinical features in older patients with Corona Virus Disease 2019 (COVID-19) out of Wuhan.	788	Multicentre retrospective cohort study	Severity	8	Zhejiang Province, 17 Jan to 12 Feb 2020	Wuhan. Hubei, China
19	Liang et al. 2020	Cancer patients in SARS-CoV-2 infection: a nationwide analysis in China.	1590	Multicentre retrospective cohort study	Severity and death	8	575 hospitals in China as of 31 Jan 2020	China
20	Liu et al. 2020	Clinical characteristics of novel coronavirus cases in tertiary hospitals in Hubei Province.	137	Multicentre retrospective cohort study	Death	7	9 hospitals in Hubei, 30 Dec 2019 to 24 Jan 2020 (Tongji Hosp, other hospitals outside Wuhan)	Hubei, China
21	Ma et al. 2020	Clinical characteristics and prognosis in cancer patients with COVID-19: A single center's retrospective study.	1380	Single centre retrospective cohort study	Severity and death	8	Renmin Hospital of Wuhan University, 1 Jan to 30 March 2020	Wuhan, Hubei, China
22	Mehta et al. 2020	Case Fatality Rate of Cancer Patients with COVID-19 in a New York Hospital System.	218	Single centre retrospective cohort study (peer-reviewed, pre-print)	Death	7	Montefiore Medical Center and Albert Einstein College of Medicine, 18 March to 8 April 2020	New York, USA
23	Miyashita et al. 2020	Do Patients with Cancer Have a Poorer Prognosis of COVID-19? An Experience in New York City.	5688	Single centre retrospective cohort study	Severity and death	8	Mount Sinai Health System, 1 March to 6 April 2020	New York, USA
24	Myers et al. 2020	Characteristics of Hospitalized Adults With COVID-19 in an Integrated Health Care System in California.	377	Multicentre retrospective cohort study	Severity	7	Kaiser Permanente Northern California (KPNC) health system, 1 to 31 March 2020	California, USA
25	Nikpouraghdam et al. 2020	Epidemiological characteristics of coronavirus disease 2019 (COVID-19) patients in IRAN: A single center study.	2964	Single centre retrospective cohort study	Death	7	Baqiyatallah Hospital, 19 Feb to 15 April 2020	Tehran, Iran

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26	Pan et al. 2020	Clinical Characteristics of COVID-19 Patients With Digestive Symptoms in Hubei, China: A Descriptive, Cross-Sectional, Multicenter Study	204	Multicentre retrospective cross sectional study	Severity and death	7	3 hospitals in Hubei province, including Wuhan Hanan Hospital, Wuhan Union Hospital, and Huanggang Central Hospital, 18 Jan to 28 Feb 2020	Hubei, China
27	Paranjpe et al. 2020	Clinical Characteristics of Hospitalized Covid-19 Patients in New York City.	1078	Multicentre retrospective cohort study (not peer reviewed, pre-print)	Death	8	Mount Sinai Health System Hospitals, 27 Feb to 2 April 2020	New York, USA
28	Richardson et al. 2020	Presenting Characteristics, Comorbidities, and Outcomes Among 5700 Patients Hospitalized With COVID-19 in the New York City Area.	5700	Multicentre retrospective cohort study	Severity and death	8	Northwell Health Hospitals, 1 March to 4 April 2020	New York, USA
29	Shi H et al. 2020	Radiological findings from 81 patients with COVID-19 pneumonia in Wuhan, China: a descriptive study.	81	Multicentre retrospective cohort study	Not defined	6	Wuhan Jinyintan hospital or Union Hospital of Tongji Medical College, 20 Dec to 23 Jan 2020	Wuhan, Hubei, China
30	Shi Y et al. 2020	Host susceptibility to severe COVID-19 and establishment of a host risk score: findings of 487 cases outside Wuhan.	487	Multicentre retrospective cohort study	Not defined	7	Hospitals in Zhejiang Province, until 17 Feb 2020	Zhejiang, China
31	Tomlins et al. 2020	Clinical features of 95 sequential hospitalised patients with novel coronavirus 2019 disease (COVID-19), the first UK cohort	95	Single centre retrospective cohort study	Severity and death	7	North Bristol NHS Trust, 10 to 30 March 2020	UK
32	Wan et al. 2020	Clinical features and treatment of COVID-19 patients in northeast Chongqing.	135	Single centre retrospective cohort study	Not defined	7	Chongqing University Three Gorges Hospital, 23 Jan to 8 Feb 2020	Chongqing, Sichuan, China
33	Wang D et al. 2020	Clinical Characteristics of 138 Hospitalized Patients With 2019 Novel Coronavirus-Infected Pneumonia in Wuhan, China.	138	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital of Wuhan University, 1 to 28 Jan 2020	Wuhan, Hubei, China
34	Wang R et al. 2020	Epidemiological and clinical features of 125 Hospitalized Patients with COVID-19 in Fuyang, Anhui, China.	125	Single centre retrospective cohort study	Severity	7	NO.2 People's Hospital of Fuyang City 20 Jan to 9 Feb 2020.	Anhui, China
35	Wu C et al. 2020	Risk Factors Associated With Acute Respiratory Distress Syndrome and Death in Patients With Coronavirus Disease 2019 Pneumonia in Wuhan, China.	201	Single centre retrospective study	Severity and death	8	Jinyintan Hospital in Wuhan, China, 25 Dec to 26 Jan 2020	Wuhan, Hubei, China
36	Wu J et al. 2020	Clinical Characteristics of Imported Cases of COVID-19 in Jiangsu Province: A Multicenter Descriptive Study.	80	Multicentre retrospective cohort study	Severity	7	Three Grade IIIA hospitals of Jiangsu, 22 Jan to 14 Feb 2020	Jiangsu, China
37	Xu et al. 2020	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2.	90	Single centre retrospective study	Severity	7	Guangzhou Eighth People's Hospital, 23 Jan to 4 Feb 2020	Guangzhou, China
38	Yang et al. 2020	Clinical characteristics and imaging manifestations of the 2019 novel coronavirus disease (COVID-19):A multi-center study in Wenzhou city, Zhejiang, China.	149	Multicentre retrospective cohort study	Severity	7	Three tertiary hospitals of Wenzhou, 17 Jan to 10 Feb 2020	Wenzhou, Zhejiang, China
39	Yu et al. 2020	SARS-CoV-2 Transmission in Patients With Cancer at a Tertiary Care Hospital in Wuhan, China.	1524	Single centre retrospective study	Severity and death	6	Zhongnan Hospital, 30 Dec to 17 Feb 2020	Wuhan, Hubei, China
40	Zhang G et al. 2020	Clinical features and short-term outcomes of 221 patients with COVID-19 in Wuhan. China.	221	Single centre retrospective cohort study	Severity	7	Zhongnan Hospital of Wuhan, 2 Jan to 10 Feb 2020	Wuhan, Hubei, China
41	Zhang J et al. 2020	Risk factors for disease severity, unimprovement, and mortality in COVID-19 patients in Wuhan, China.	663	Single centre retrospective cohort study	Severity	7	Renmin Hospital of Wuhan University, 11 Jan to 2 Feb 2020	Wuhan, Hubei, China
42	Zhang L et al. 2020	Clinical characteristics of COVID-19-infected cancer patients: a retrospective case study in three hospitals within Wuhan, China.	1276	Single centre retrospective cohort study	Severity and death	8	Tongji Sino-French New Town Hospital, Union Red Cross Hospital, and Union West Hospital, 13 Jan to 26 Feb 2020	Wuhan, Hubei, China
43	Zhang N et al. 2020	A multicentre study of coronavirus disease 2019 outcomes of cancer patients in Wuhan, China	1548	Multicentre retrospective observational study (peer reviewed, pre-print)	Severity, death	7	Zhongnan Hospital of Wuhan University, the Fifth hospital of Wuhamn, the seventh hospital of Wuhan and Wuhan Hankou hospital, 5 Jan to 18 Feb 2020	Wuhan, Hubei, China
44	Zhang S et al. 2020	Short Term Outcomes and Risk Factors for Mortality in Patients with COVID-19 in Wuhan, China: A Retrospective Study.	315	Single centre retrospective cohort study	Death	7	Wuhan Union Hospital West Area, 12 Jan to 3 Feb 2020	Wuhan, Hubei, China
45	Zhao et al. 2020	Clinical characteristics of patients with 2019 coronavirus disease in a non-Wuhan area of Hubei Province, China: a retrospective study.	91	Single centre retrospective cohort study	Severity	7	Jingzhou Central Hospital, 16 Jan to 10 Feb 2020	Jingzhou, Hubei, China
46	Zhou et al. 2020	Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study.	191	Multicentre retrospective cohort study	Death	7	Jinyintan Hospital and Wuhan Pulmonary Hospital, 29 Dec to 31 Jan 2020	Wuhan, Hubei, China
47	Zhu et al. 2020	Initial clinical features of suspected coronavirus disease 2019 in two emergency departments outside of Hubei, China.	32	Single centre retrospective cohort study	Not defined	6	First Affiliated Hospital of USTC, 24 Jan to 20 Feb 2020	Hefei, China

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The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort
   a) truly representative of the average cancer COVID-19 cases in the community *
   b) somewhat representative of the average cancer COVID-19 cases in the community *
   c) selected group of users eg nurses, doctors
   d) no description of the derivation of the cohort

2) Selection of the non-exposed cohort
   a) drawn from the same community as the exposed cohort *
   b) drawn from a different source
   c) no description of the derivation of the non-exposed cohort

3) Ascertainment of exposure
   a) secure record (eg medical records) *
   b) structured interview *
   c) written self-report
   d) no description

4) Demonstration that outcome of interest was not present at start of study
   a) yes *
   b) no

Comparability

5) Comparability of cohorts on the basis of the design or analysis
   a) study controls for median age *
   b) study controls for any additional factors such as other co-morbidities *

Outcome

6) Assessment of outcome
   a) independent blind assessment *
   b) record linkage *
   c) self-report
   d) no description

7) Was follow-up long enough for outcomes to occur
   a) yes (until discharge or death or at least 1 month, whichever come first) *
   b) no

8) Adequacy of follow up of cohorts
   a) complete follow up - all subjects accounted for *
   b) subjects lost to follow up unlikely to introduce bias - small number lost - > 80 % follow up, or description provided of those lost) *
   c) follow up rate < 60 % and no description of those lost
   d) no statement

Case-control studies

1) Is the case definition adequate?
   a) yes, with independent validation *
   b) yes, eg record linkage or based on self reports
   c) no description

2) Representativeness of the cases
   a) consecutive or obviously representative series of cases *
   b) potential for selection biases or not stated

3) Selection of Controls
   a) community controls *
   b) hospital controls
   c) no description

4) Definition of Controls
   a) no history of disease (endpoint) *
   b) no description of source

Comparability

5) Comparability of cases and controls on the basis of the design or analysis
   a) study controls for median age *
   b) study controls for any additional factor * (co-morbidities)

Exposure

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6) Ascertainment of exposure
- a) secure record (eg surgical records) *
- b) structured interview where blind to case/control status *
- c) interview not blinded to case/control status
- d) written self report or medical record only
- e) no description

7) Same method of ascertainment for cases and controls
- a) yes *
- b) no

8) Non-Response rate
- a) same rate for both groups *
- b) non respondents described
- c) rate different and no designation

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Supplementary Table 3: The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

N	Authors	Selection				Comparability		Outcome			Study quality
	Cohort studies	1 Exposed cohort truly representative	2 Non-exposed cohort drawn from same community	3 Ascertainment of exposure	4 Outcome of interest not present at start	5A Cohorts comparable ie. Median age	5B Cohorts comparable on other co-morbidities	6 Quality of outcome assessment	7 Follow-up long enough for outcomes to occur	8 Complete accounting for cohorts	
	Case control studies	1 Case definition adequate	2 Representativeness with the cases	3 Selection of controls	4 Definition of controls	5A Case and control comparable on the basis of analysis ie median age	5B Case and control comparable on other co-morbidities	6 Ascertainment of exposure	7 Same method of ascertainment for cases and controls	8 Non-Response rate	
1	Benelli et al.	*	*	*	*			*	*	*	7
2	Cai et al.	*	*	*	*			*	*	*	7
3	Cao M et al.	*	*	*	*			*	*	*	7
4	Cao J et al.	*	*	*	*			*	*	*	7
5	Chen J et al	*	*	*	*			*	*	*	7
6	Chen N et al.	*	*	*	*			*	*	*	7
7	Colaneri et al.	*	*	*	*			*		*	6
8	Dai et al.	*	*	*	*	*		*	*	*	8
9	Du et al.	*	*	*	*			*	*	*	7
10	Duanmu et al.	*	*	*	*			*		*	6
11	Feng et al.	*	*	*	*			*	*	*	7
12	Guan et al.	*	*	*	*			*	*	*	7
13	Guo et al.	*	*	*	*			*	*	*	7
14	Huang C et al.	*	*	*	*			*	*	*	7
15	Huang Y et al.	*	*	*	*			*		*	6
16	Kim et al.	*	*	*	*			*		*	6
17	Li et al.	*	*	*	*			*	*	*	7
18	Lian et al.	*	*	*	*	*		*	*	*	8
19	Liang et al.	*	*	*	*		*	*	*	*	8
20	Liu et al.	*	*	*	*			*	*	*	7
21	Ma et al.	*	*	*	*	*		*	*	*	8
22	Mehta et al.	*	*	*	*			*	*	*	7
23	Miyashita et al.	*	*	*	*	*		*	*	*	8
24	Myers et al.	*	*	*	*			*	*	*	7
25	Nikpouraghdam et al.	*	*	*	*			*	*	*	7
26	Pan et al.	*	*	*	*			*	*	*	7
27	Paranjpe et al.	*	*	*	*		*	*	*	*	8
28	Richardson et al.	*	*	*	*	*		*	*	*	8
29	Shi H et al.	*	*	*	*			*		*	6
30	Shi Y et al.	*	*	*	*	*		*	*		7
31	Tomlins et al	*	*	*	*			*	*	*	7
32	Wan et al.	*	*	*	*			*	*	*	7
33	Wang D et al.	*	*	*	*			*	*		6
34	Wang R et al.	*	*	*	*			*	*	*	7
35	Wu C et al.	*	*	*	*	*		*	*	*	8
36	Wu J et al	*	*	*	*			*	*	*	7
37	Xu et al.	*	*	*	*			*	*	*	7
38	Yang et al.	*	*	*	*			*	*	*	7
39	Yu et al.	*		*	*			*	*	*	6
40	Zhang G et al.	*	*	*	*			*	*	*	7
41	Zhang J et al.	*	*	*	*			*	*	*	7
42	Zhang L et al.	*	*	*	*		*	*	*	*	8
43	Zhang N et al.	*	*	*	*			*	*	*	7
44	Zhang S et al.	*	*	*	*			*	*	*	7
45	Zhao et al.	*	*	*	*			*	*	*	7
46	Zhou et al.	*	*	*	*			*	*	*	7
47	Zhu et al.	*	*	*	*				*	*	6

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	1-3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	1-3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N.A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	2-4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	2-4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	2-3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Appendix 1
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	2-3
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	2-3

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	2-3
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	3
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	3-4, Figures
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	3-5, Appendix 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8-9, Appendix 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	4-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	4-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	4-8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	4-8, Appendix 2
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7-9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	8
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	7-9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	9

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Appendix 2

Supplementary Table 1: List of 230 patients with available individual patient data and oncological history

Liang et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer in years**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	83	No	4	Bladder ca	Postoperative routine follow-up	Y	NA	N
2	M	87	DM, HTN	2	Colon ca	Postoperative routine follow-up	Y	NA	N
3	M	67	No	0	Adrenal neoplasm	3 weeks after operation	Y	NA	N
4	F	68	No	8	Breast ca	Surgical resection plus adjuvant chemotherapy 8 years ago	Y	NA	N
5	M	58	No	7	CRCC	Recurrence, on immunotherapy	Y	NA	Y
6	F	62	No	4	Breast ca	Surgical resection plus adjuvant chemotherapy 4 years ago	Y	NA	N
7	M	56	No	5	Rectal ca	Surgical resection plus adjuvant chemotherapy 5 years ago	N	NA	N
8	M	53	No	16	Colon ca	Surgical resection plus adjuvant chemotherapy 16 years ago	N	NA	N
9	F	63	No	1	Papillary thyroid ca	Postoperative, in TSH inhibition therapy	N	NA	N
10	M	47	No	1	Lung adenoca	In chemotherapy for advanced tumor	N	NA	Y
11	F	52	No	0.5	Breast ca	Postoperative, loss of chemotherapy information	N	NA	N
12	M	47	No	1	Lymphoma	N.A.	N	NA	NA
13	M	80	DM, HTN, CVS	4	Bladder ca	Postoperative, no chemotherapy information	Y	NA	N
14	M	79	COPD	5	Colorectal ca	Surgical resection plus adjuvant radiotherapy 5 years ago	Y	NA	N
15	M	63	No	1	Lung adenoca	In chemotherapy for advanced tumour	Y	NA	Y
16	F	58	CKD	2	Lung carcinoma in situ	Postoperative routine follow-up	N	NA	N
17	M	58	No	2	Lung adenoca	Postoperative, on targeted therapy	N	NA	N
18	F	55	No	6	Lung adenoca	Advanced, on targeted therapy	N	NA	Y
Yu et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	NA	NA	NA	NSCLC	Concurrent TKI with RT	N	Y	N
2	M	NA	NA	NA	NSCLC	Week 5 of chemoradiotherapy	Y	N	N
3	F	NA	NA	NA	NSCLC	Pleural effusion, BSC	N	Y	Y
4	M	NA	NA	NA	NSCLC	Follow up at 2 year post chemoRT	Y	N	N
5	M	NA	NA	NA	NSCLC	After first cycle of doce, cis and sintilimab	N	Y	Y
6	M	NA	NA	NA	NSCLC	After third cycle of pem/carbo and pembro	N	Y	Y
7	M	NA	NA	NA	NSCLC	Surgery (April 2019) followed by adjuvant doce/cis	N	Y	N
8	M	NA	NA	NA	Rectal ca	BSC	N	Y	Y
9	M	NA	NA	NA	Colon ca	Newly diagnosed	N	Y	NA
10	M	NA	NA	NA	Pancreatic ca	BSC	Y	N	Y
11	F	NA	NA	NA	Breast ca	Adjuvant radiotherapy	N	Y	N
12	M	NA	NA	NA	Urothelial ca	BSC	N	Y	Y
Zhang L et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer in months**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	56	CHB, cirrhosis	26	Liver ca	Stage IV, recurrence with bone mets, bone RT with 14 days	Y	NA	Y
2	M	74	DM	3	Esophageal ca	Stage IV, chemotherapy 1 month ago	Y	NA	Y
3	M	66	HTN, CHD	11	Lung SCC	Surgical resection plus adjuvant chemoRT 10m ago, immune therapy 3m ago, gem and nivo within 14 days	Y	NA	Y
4	M	65	No	23	Lung adenoca	Stage IV, chemo for advanced tumour. Last cycle of pemetrexed was within 14 days	Y	NA	Y
5	M	65	No	4	Esophageal ca	Stage III, surgical resection plus adjuvant chemo and RT	N	NA	N
6	F	48	DM, HTN	5	Serous ca of ovary	Surgical resection plus adjuvant chemo 4m ago, 3 weeks after THBSO	N	NA	N
7	M	54	No	13	Laryngocarcinoma	4 weeks after operation	N	NA	N
8	M	80	HTN, CHD	1	Colorectal ca	3 weeks after operation and neoadjuvant chemoRT	N	NA	N
9	M	65	No	13	Esophageal ca	Surgical resection plus adjuvant chemo and RT 11m ago	Y	NA	0
10	F	70	No	124	Lung adenoca	Postoperative, recurrence. On targeted therapy	Y	NA	Y
11	F	64	No	26	Lung adenoca	Surgical resection plus adjuvant chemo, on EGFR TKI for advanced tumour	N	NA	Y
12	F	24	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 1 year ago	Y	NA	N
13	F	70	DM, HTN	148	Breast ca	Surgical resection plus adjuvant chemo and RT 12 year ago	Y	NA	N
14	F	59	No	62	Lung adenoca	Stage IV, on EGFR TKI treatment	Y	NA	Y
15	F	68	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 20m ago	N	NA	N
16	F	60	No	9	Endometrial ca	Surgical resection 7m ago, on adjuvant chemo 3 weeks ago	N	NA	N
17	M	65	No	26	Small cell lung ca	Maintenance chemo for advanced tumour	N	NA	Y
18	M	57	No	15	Lung SCC	Surgical resection plus adjuvant chemoRT 1 year ago	N	NA	N
19	F	65	No	15	Cervical ca	Surgical resection and adjuvant chemo 1 week ago	N	NA	N
20	M	47	No	51	NPC	Concurrent chemoRT 4m ago	Y	NA	N
21	M	77	No	62	Prostate ca	Postop and adjuvant RT 5 years ago, on follow up	Y	NA	N
22	M	71	DM	75	Esophageal ca	Surgical resection plus adjuvant chemo and RT 6 years ago	Y	NA	N
23	M	65	CHD	26	Prostate ca	Postoperative and adjuvant RT 2 years ago, on follow up	Y	NA	N
24	M	75	COPD	148	Rectal ca	Post-operative routine follow up	Y	NA	N
25	M	56	CHB	73	Liver ca	Surgical resection plus adjuvant chemo and RT 6 years ago, recurrence on adjuvant chemo within 4 weeks	N	NA	Y
26	M	69	No	124	Laryngocarcinoma	Postoperative and adjuvant RT 10 years ago, routine follow up	N	NA	N
27	F	52	No	8	Gastric ca	Postoperative, 4 cycles of chemo and adjuvant RT, routine follow up	N	NA	N
28	M	60	No	13	Testicular ca	Sequential chemo and RT, last cycle of chemo was within 4 weeks	Y	NA	N
Dai et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Date of diagnosed cancer**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	69	Intestinal obstruction	2019.12	Peritoneal cancer	N.A	Y	Y	N
2	M	66	Jaundice, Abdominal infection, chest infection	2019.11	Gastric cancer	N.A	N	Y	Y
3	M	49	No	2020.01	Esophageal SCC	N.A	Y	Y	NA
4	M	65	No	2018.01	Gastric cancer	N.A	N	Y	N
5	M	72	Asthma	2016.09	Lung cancer	Targeted therapy (EGFR-TKI) and PD-1 immunotherapy	N	Y	Y
6	M	73	Chronic bronchitis	2019.11	Laryngeal SCC	Radiotherapy	Y	Y	Y
7	M	69	CHD, HTN	2019.08	Lung cancer	Radiotherapy	N	Y	N

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8	F	55	No	2019.11	Glioblastoma	Radiotherapy and chemotherapy	N	Y	Y
9	M	74	Duodenal ulcer, COPD, HTN, CHD	2019.08	Esophageal cancer	Radiotherapy	N	Y	Y
10	M	55	Hepatic encephalopathy, GI bleeding, Chronic viral hepatitis, Anemia	2014.01	Hepatocellular carcinoma	Radiotherapy	N	Y	Y
11	M	66	HTN, DM	2019.03	Lung SCC	Chemotherapy	Y	Y	Y
12	M	67	Chronic bronchitis, HTN, Atrial fibrillation, Peptic ulcer, CKD	2019.12	Esophageal SCC	Surgery	Y	N	N
13	M	64	Cerebral infarction	2020.01	Myelodysplastic syndrome	N.A	Y	Y	NA
14	M	83	HTN	2016	Meningioma	N.A	Y	Y	NA
15	M	73	HTN	2020.01	Lung cancer	Surgery	Y	Y	N
16	F	48	No	2019.06	Breast cancer	N.A	N	Y	N
17	M	57	Prostatic hyperplasia	2016.02	Lung cancer	Radiotherapy and Targeted therapy (EGFR-TKI)	N	Y	Y
18	M	61	No	2019.12	Lung SCC	DP chemotherapy and PD-1 immunotherapy	Y	N	Y
19	F	84	HTN, DM, lumbar compression fracture	2019.06	Lung cancer	Surgery	Y	N	N
20	M	69	Abdominal aortic aneurysm	2019.09	Lung adenocarcinoma	Radiotherapy and chemotherapy	N	Y	N
21	M	69	Prosopalgia	2019.11	Lung SCC	Chemotherapy and PD-1 immunotherapy	N	Y	Y
22	M	65	Chronic viral hepatitis	2019.04	Lung cancer	N.A	N	Y	N
23	F	76	HTN, CKD, hypothyroidism, cardiac insufficiency	Unknown	Multiple myeloma	N.A	Y	N	NA
24	M	54	Incomplete ileus	2020.01	Rectal cancer	Surgery	N	Y	N
25	F	78	HTN,, COPD, CHD, CKD	2014.06	Chronic granulocytic leukemia	N.A	Y	N	N
26	M	61	HTN, DM	2018.05	Bladder cancer	N.A	Y	Y	N
27	F	50	No	2017.05	Lung cancer	N.A	N	Y	N
28	F	87	HTN	2009	Gastric cancer	N.A	Y	Y	N
29	F	55	No	2018	Cervical cancer	N.A	N	Y	N
30	F	70	HTN	Unknown	Colon cancer	N.A	N	Y	N
31	F	69	No	Unknown	Gastric cancer	N.A	N	Y	NA
32	F	59	No	2009	Breast cancer	N.A	N	Y	N
33	F	48	No	2019.06	Breast cancer	Radiotherapy	N	Y	N
34	M	57	No	2015.12	Lung adenocarcinoma	Concurrent targeted therapy（EGFR-TKI）with radiotherapy	N	Y	Y
35	M	69	Prosopalgia	2019.11	Lung SCC	Radiotherapy and Chemotherapy	N	Y	Y
36	M	69	Chronic viral hepatitis	2019.04	Lung cancer	Chemotherapy and PD-1 immunotherapy	Y	Y	Y
37	M	69	CKD	2019.09	Lung adenocarcinoma	Chemotherapy and Radiotherapy	N	Y	N
38	M	66	No	2019.08	Lung adenocarcinoma	PD-1 immunotherapy	Y	Y	Y
39	M	65	CHD	2019.04	Bladder cancer	Radiotherapy and Chemotherapy	Y	N	Y
40	M	81	HTN	2019.09	Cancer of unknown primary	PD-1 immunotherapy	Y	N	NA
41	F	46	Bacteremia	2019.01	Rectal cancer	N.A	N	Y	N
42	M	76	Intestinal obstruction	2018.11	Rectal cancer	N.A	Y	Y	Y
43	F	58	No	2020.1.2	Ovarian cancer	Chemotherapy	N	Y	N
44	F	74	DM	2020.1.12	Cervical cancer	Chemotherapy	N	Y	N
45	M	26	No	Unknown	Testicular cancer	N.A	N	Y	N
46	F	66	HTN, HLD	Unknown	Breast cancer	N.A	Y	Y	N
47	F	52	DM	Unknown	Thyroid carcinoma	N.A	N	Y	N
48	M	50	Pulmonary tuberculosis, Abdominal infection, Biliary infection, Liver abscess	2015	Thyroid carcinoma	N.A	N	Y	N
49	F	66	No	2019	Thyroid carcinoma	N.A	Y	Y	NA
50	M	83	No	2019.05	Prostate cancer	N.A	N	Y	Y
51	F	29	No	2013.05	Thyroid carcinoma	N.A	N	Y	N
52	F	32	No	2018.01	Thyroid carcinoma	N.A	N	Y	N
53	M	76	No	2018	Prostate cancer	N.A	N	Y	NA
54	F	80	HTN, HLD, Phlebothrombosis, DM	2019.01	Lung adenocarcinoma	N.A	Y	Y	N
55	M	72	HTN	2019	Lung cancer	N.A	Y	N	Y
56	F	57	No	2019.12	Breast cancer	Chemotherapy	N	Y	N
57	F	69	Chronic Hepatitis	2018	Thyroid cancer	N.A	N	Y	N
58	F	60	No	2012	Cervical cancer	N.A	N	Y	N
59	F	72	HTN,,heart disease	2013.06	Breast cancer	N.A	N	Y	NA
60	M	75	HTN	2009	Colon cancer	N.A	N	Y	NA
61	M	67	Hypothyroidism	Unknown	Liver cancer	N.A	N	Y	NA
62	F	52	HTN, CKD	2016.03	Endometrial cancer	N.A	N	Y	N
63	M	64	No	2019.08	Gallbladder carcinoma	N.A	N	Y	NA
64	M	67	No	1994	Nasopharynx cancer	N.A	N	Y	NA
65	M	63	HTN	2010	Kidney cancer	N.A	N	Y	NA
66	M	34	No	2017	Thyroid cancer	N.A	N	Y	NA
67	F	59	No	2019.07	Colon cancer	Chemotherapy	N	Y	NA
68	M	57	Cirrhosis,diabetes	2020.01	Nasopharynx cancer	Chemotherapy	N	Y	NA
69	M	62	No	2015	Bladder cancer	N.A	N	Y	NA
70	F	66	No	2015	Thyroid cancer	N.A	N	Y	NA
71	M	60	No	2015	Multiple myeloma	N.A	N	Y	NA
72	F	64	No	Unknown	Gastric stromal tumor	N.A	N	Y	NA
73	F	66	No	2004	Ocular malignancy	N.A	N	Y	NA
74	M	61	CHD, HTN	2020.01	Multiple myeloma	N.A	N	Y	NA
75	F	53	No	2019.9	Multiple myeloma	N.A	N	Y	NA
76	M	61	DM, HTN	Unknown	Malignant lymphoma	N.A	Y	N	Y
77	F	54	No	2019.03	Lung carcinoma	N.A	Y	N	N
78	F	55	No	2016	Breast cancer	N.A	N	Y	N
79	F	38	No	2019	Breast cancer	N.A	N	Y	N
80	F	50	No	2019	Rectal carcinoma	N.A	Y	N	N
81	M	67	Bronchitis	2017.12	Esophageal carcinoma	N.A	Y	Y	N
82	F	37	No	2018	Thyroid carcinoma	N.A	N	Y	N
83	F	45	No	2019.11	Cervical carcinoma	Surgery and Radiotherapy	Y	Y	N
84	M	35	No	2019.07	Glioma	N.A	Y	Y	NA
85	F	57	No	2019.11	Cervical carcinoma	Surgery	N	Y	N
86	F	49	No	2019.11	Renal carcinoma	N.A	N	Y	N
87	M	62	HTN, Tuberculosis	2018.08	Esophageal carcinoma	N.A	N	Y	N
88	M	70	CHD, Choledochectasia	2019.06	Gastric carcinoma	N.A	N	Y	N
89	M	44	No	2019.12	Thyroid carcinoma	N.A	N	Y	N
90	F	72	Cerebrovascular disease	2017.06	Breast cancer	N.A	N	Y	N
91	F	57	HTN	2019.02	Breast cancer	N.A	Y	Y	N
92	F	56	No	2019.04	Cervical carcinoma	N.A	Y	Y	N
93	M	58	No	2019.06	Rectal carcinoma	Chemotherapy	Y	Y	Y
94	M	71	CHD, HTN	2019.1	Lung SCC	Chemotherapy	Y	Y	Y
95	F	67	DM HTN	2019.12	Breast cancer	Surgery	Y	Y	N
96	M	70	No	2019.11	gastric carcinoma	Chemotherapy	Y	Y	N
97	M	70	HTN	2014	Esophageal carcinoma	N.A	Y	Y	N
98	F	42	HTN	2020.01	Pancreatic cancer	Surgery	N	Y	N

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99	M	43	No	2011	gastric carcinoma	N.A	Y	Y	N
100	M	69	No	2015	Lung cancer	N.A	Y	Y	N
101	M	62	Cerebral hemorrhage, Uremia, DM, HTN	2020.01	Lung cancer	N.A	N	Y	N
102	M	37	HLD	2018.06	Sigmoid colon cancer	N.A	Y	Y	N
103	F	70	CHD, HTN	2002	Thyroid cancer	N.A	Y	Y	N
104	M	51	No	2015	Leukemia	Targeted therapy (Imatinib Mesylate)	N	Y	N
105	F	55	HTN	2019.07	Nasopharyngeal lymphoma	N.A	N	Y	N

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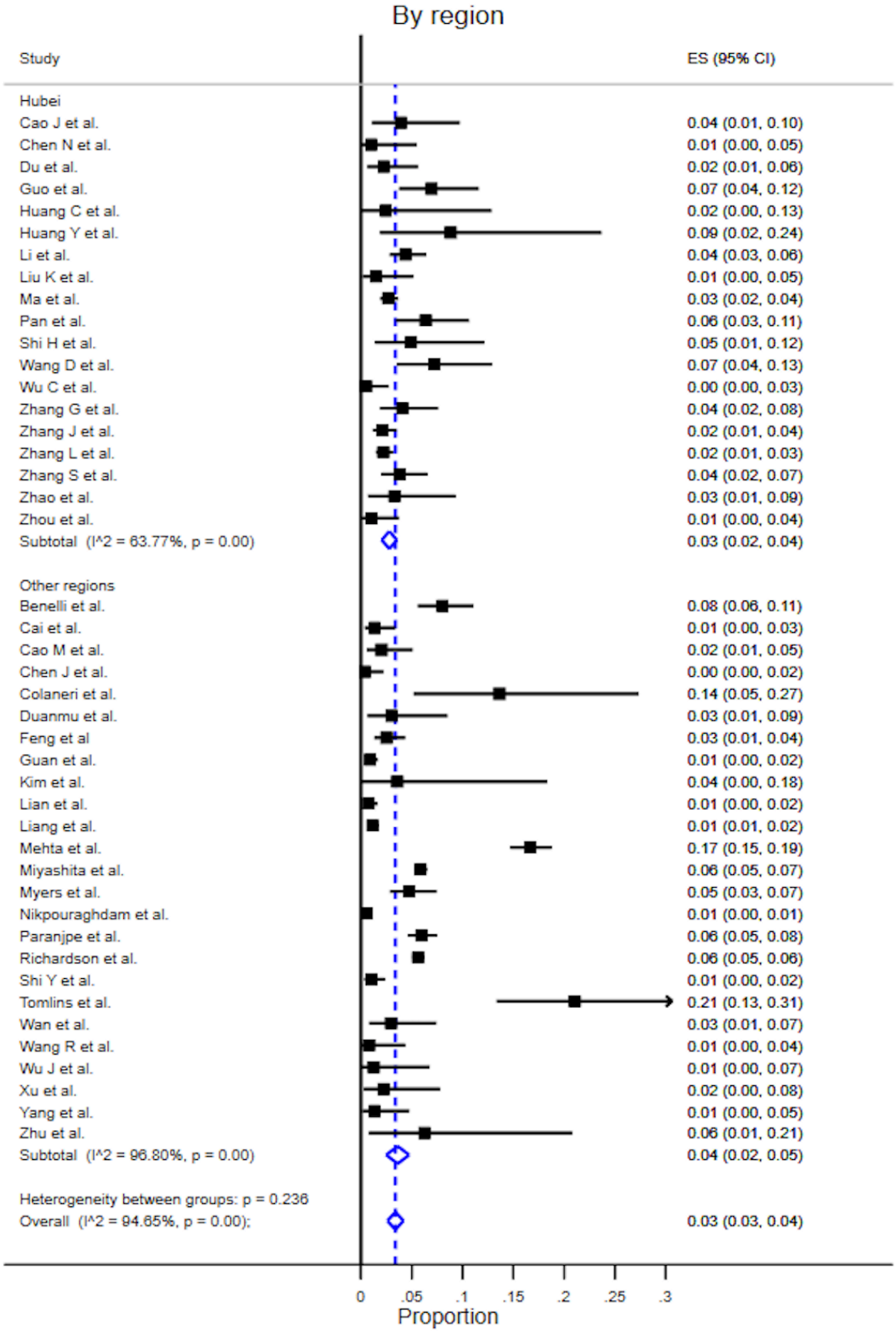
No	Gender	Age	Co-morbidities	Length of dx of cancer in months	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	F	48	No	46	Breast	RT+endocrinotherapy	N	Y	NA
2	M	57	No	53	Lung	targeted therapy/RT	N	Y	NA
3	M	65	Liver cirrhosis	50	Lung	chemotherapy+immunotherapy	N	Y	NA
4	M	66	No	53	Lung	chemotherapy+immunotherapy+targeted therapy	N	Y	NA
5	M	69	No	48	Lung	chemotherapy+RT+immunotherapy	N	Y	NA
6	M	69	No	32	Lung	chemotherapy+RT	Y	N	NA
7	M	61	No	36	Urinary	chemotherapy	Y	N	NA
8	M	37	No	36	liver	surgery+chemotherapy	N	Y	NA
9	M	87	COPD, HTN, CHD, cerebrovascular disease/chronic nephrosis/immunological diseases	44	Gastric	follow-up	Y	N	NA
10	M	61	Disturbance of consciousness	16	Lung	chemotherapy+immunotherapy	Y	N	NA
11	F	67	DM, HTN, liver cirrhosis	8	Esophagus	perioperative	Y	N	NA
12	M	65	HTN, cerebrovascular disease	48	Lung	follow-up	N	Y	NA
13	F	50	No	54	HN	follow-up	N	Y	NA
14	F	46	HTN	50	Thyroid	follow-up	N	Y	NA
15	M	77	HTN	42	Colorectal	follow-up	Y	Y	NA
16	F	70	No	34	Breast	follow-up	Y	Y	NA
17	F	75	No	51	Colorectal	follow-up	N	Y	NA
18	F	53	No	57	HN	follow-up	N	Y	NA
19	M	48	HTN	49	CNS	follow-up	N	Y	NA
20	F	58	HTN, HLD, CHD	45	Thyroid	follow-up	N	Y	NA
21	M	54	No	44	Colorectal	perioperative	N	Y	NA
22	F	78	HLD	18	Hematology	supportive treatment	Y	N	NA
23	M	60	HTN	64	Hematology	supportive treatment	N	Y	NA
24	F	83	HTN	64	CNS	follow-up	Y	Y	NA
25	M	68	HTN	47	Urinary	follow-up	N	Y	NA
26	M	66	COPD, HTN, CHD, liver cirrhosis	48	Urinary	follow-up	N	Y	NA
27	M	63	HTN	29	Urinary	follow-up	Y	N	NA
28	M	77	No	55	Urinary	radiotherapy	Y	Y	NA
29	M	73	HTN/HLD/immune disease	53	Lung	perioperative	N	Y	NA
30	M	66	HD/DM/CHD/immune disease	22	Duodenal	supporting treatment	Y	Y	NA
31	F	84	HTN, immune disease	11	Lung	perioperative	Y	N	NA
32	M	67	DM, HTN, CHD, chronic nephrosis	42	Colorectal	follow-up	Y	Y	NA
33	F	65	DM, HTN, liver cirrhosis	41	CNS	follow-up	N	Y	NA
34	M	78	HTN	40	Colorectal	follow-up	N	Y	NA
35	M	65	No	38	Colorectal	perioperative	Y	Y	NA
36	F	71	Cerebrovascular disease	37	Colorectal	follow-up	Y	Y	NA
37	F	65	HTN, tuberculosis	44	Breast	endocrinotherapy	Y	Y	NA
38	F	40	No	41	Thyroid	follow-up	N	Y	NA
39	F	61	HTN	47	Thyroid	follow-up	N	Y	NA
40	F	70	COPD, DM, HTN, cerebrovascular disease	6	Breast	follow-up	Y	N	NA
41	M	98	No	21	Colorectal	follow-up	Y	N	NA
42	M	64	HTN	53	Esophagus	chemotherapy	N	Y	NA
43	M	67	No	14	Urinary	follow-up	Y	N	NA
44	F	76	COPD, CHD, HTN	52	Hematology	follow-up	N	Y	NA
45	F	68	Tuberculosis	53	Thyroid	follow-up	N	Y	NA
46	M	66	HTN	50	Lung	follow-up	N	Y	NA
47	M	69	No	47	Urinary	follow-up	Y	Y	NA
48	M	69	Tuberculosis	44	Colorectal	follow-up	N	Y	NA
49	M	70	DM, HTN	48	Gastric	follow-up	Y	Y	NA
50	M	72	No	19	Lung	chemotherapy	Y	N	NA
51	F	55	HTN	48	Lung	follow-up	N	Y	NA
52	F	58	No	50	CNS	follow-up	N	Y	NA
53	M	54	HTN	50	HN	follow-up	N	Y	NA
54	F	64	No	40	Breast	follow-up	N	Y	NA
55	F	51	DM, CHD	52	Thyroid	follow-up	N	Y	NA
56	M	56	HTN	50	Thyroid	follow-up	Y	Y	NA
57	M	71	No	13	HN	follow-up	Y	N	NA
58	F	65	DM, HTN	36	Ovary	follow-up	N	Y	NA
59	M	90	HTN	9	Colorectal	perioperative	Y	N	NA
60	F	64	DM, HTN	54	Breast	follow-up	N	Y	NA
61	F	69	No	54	Lung	follow-up	Y	Y	NA
62	M	52	No	39	Thyroid	follow-up	Y	N	NA
63	M	74	HTN	22	Colorectal	follow-up	Y	N	NA
64	M	63	HTN	23	HN	follow-up	Y	N	NA
65	M	70	DM, HTN	37	Lung	follow-up	N	Y	NA
66	M	67	HTN	21	Lung	immunotherapy	Y	N	NA
67	M	64	HTN	57	Urinary	follow-up	Y	Y	NA

Abbreviations: BSC: best supportive care; ca – carcinoma, CHB: chronic hepatitis b; CHD: coronary heart disease; CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; DM: diabetes mellitus; EGFR: epidermal growth factor receptor; HLD: hyperlipidemia; HTN: hypertension; NPC: nasopharyngeal cancer; NSCLC: non-small cell lung cancer; PD1: programmed death 1; RCC: renal cell cancer; RT: radiotherapy; TKI: tyrosine kinase inhibitor; SCC: squamous cell cancer; THBSO: total hysterectomy and bilateral salpingo-oophrectomy
Y: yes; N: no; NA: not applicable/available

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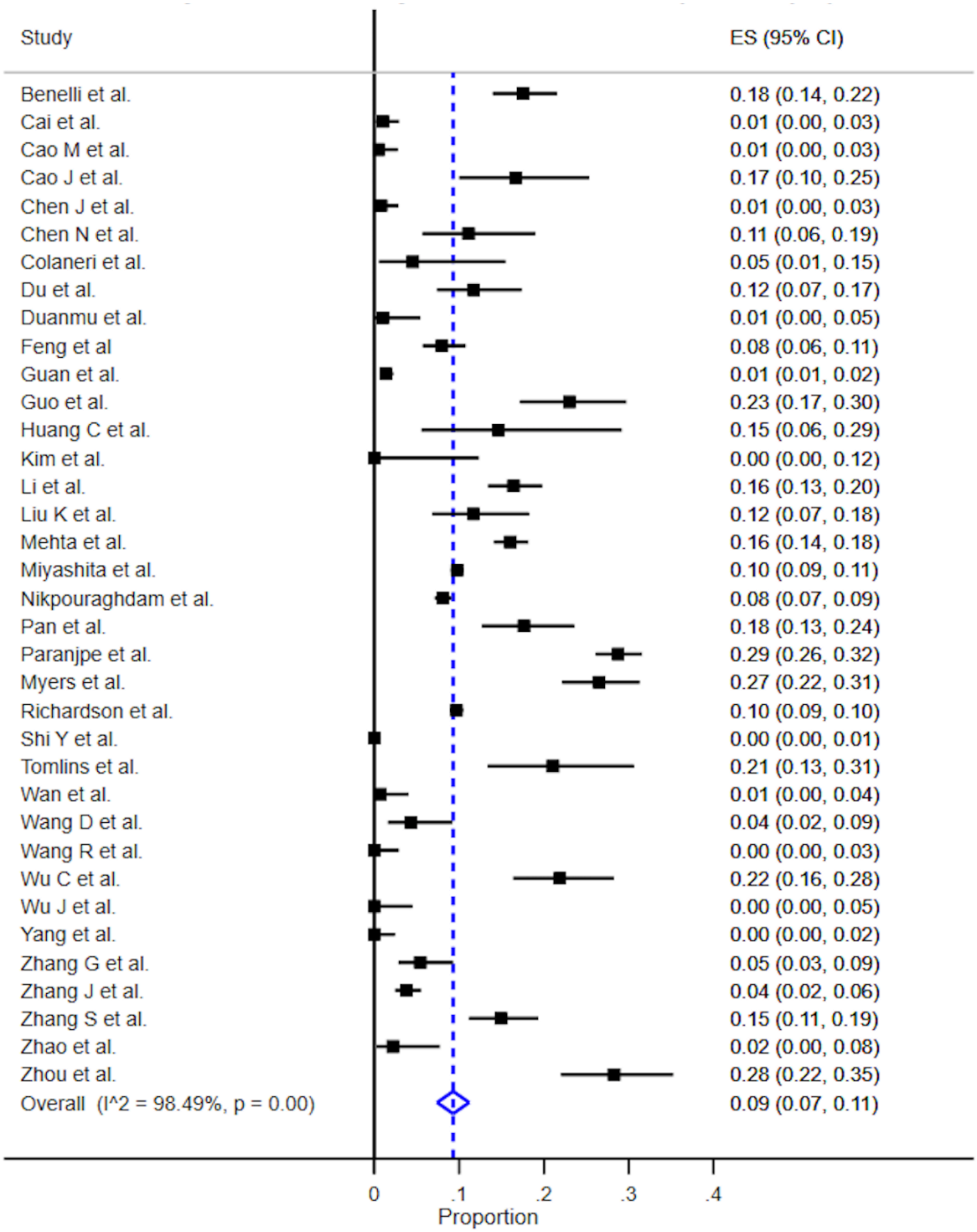
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Supplementary Figure 1: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by region – within Hubei province and outside Hubei province)



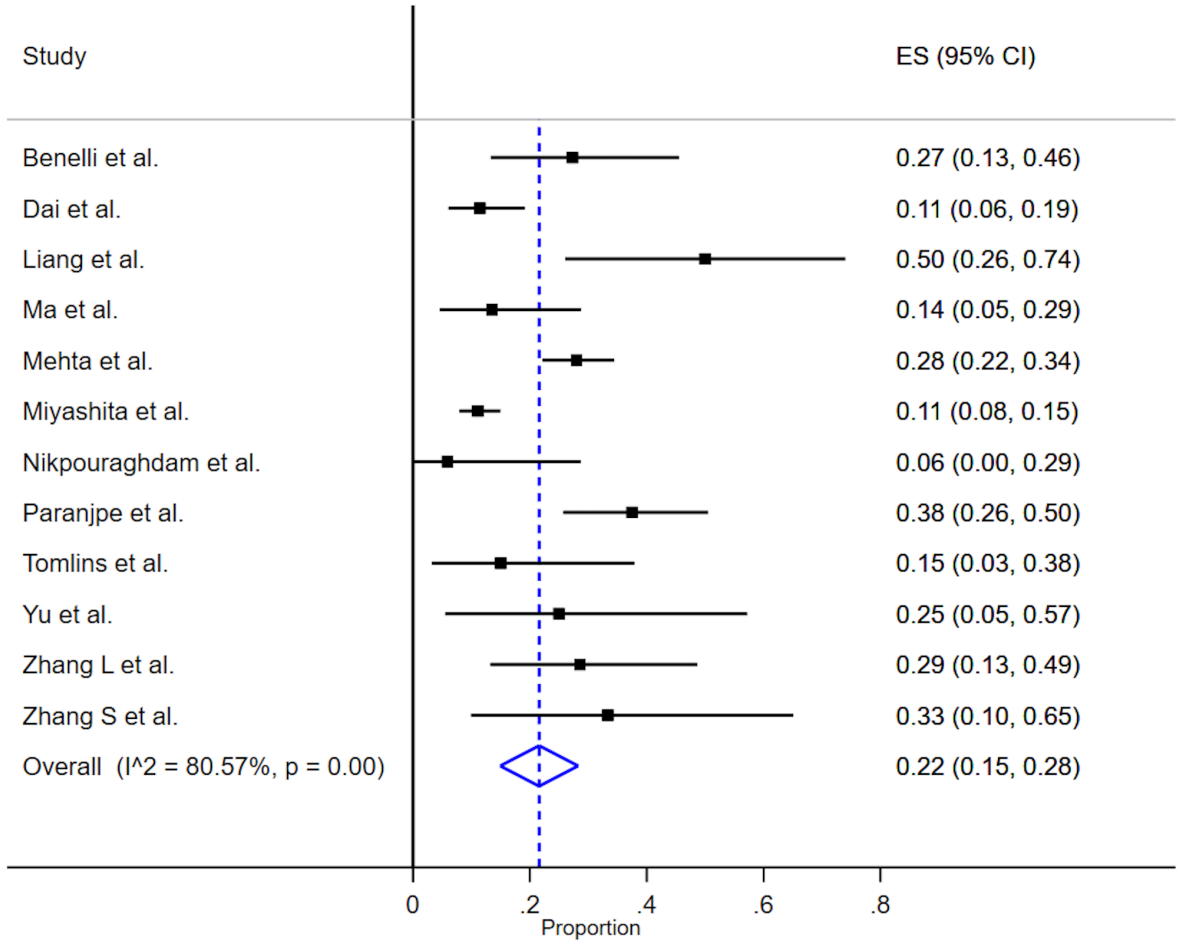
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Supplementary Figure 2a: Random effects pooled proportion of deaths among patients diagnosed with COVID-19



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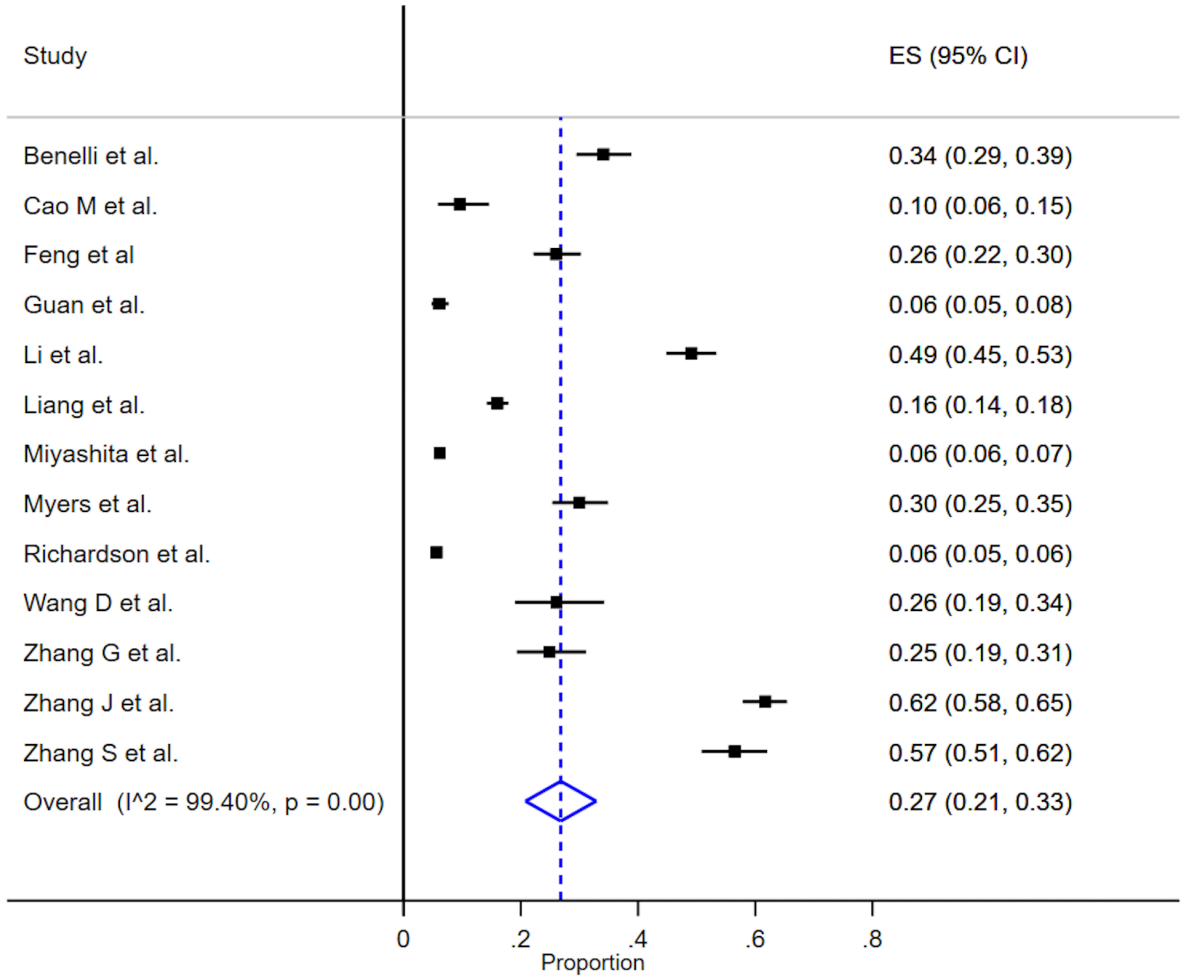
Supplementary Figure 2b: Random effects pooled proportion of deaths among cancer patients diagnosed with COVID-19



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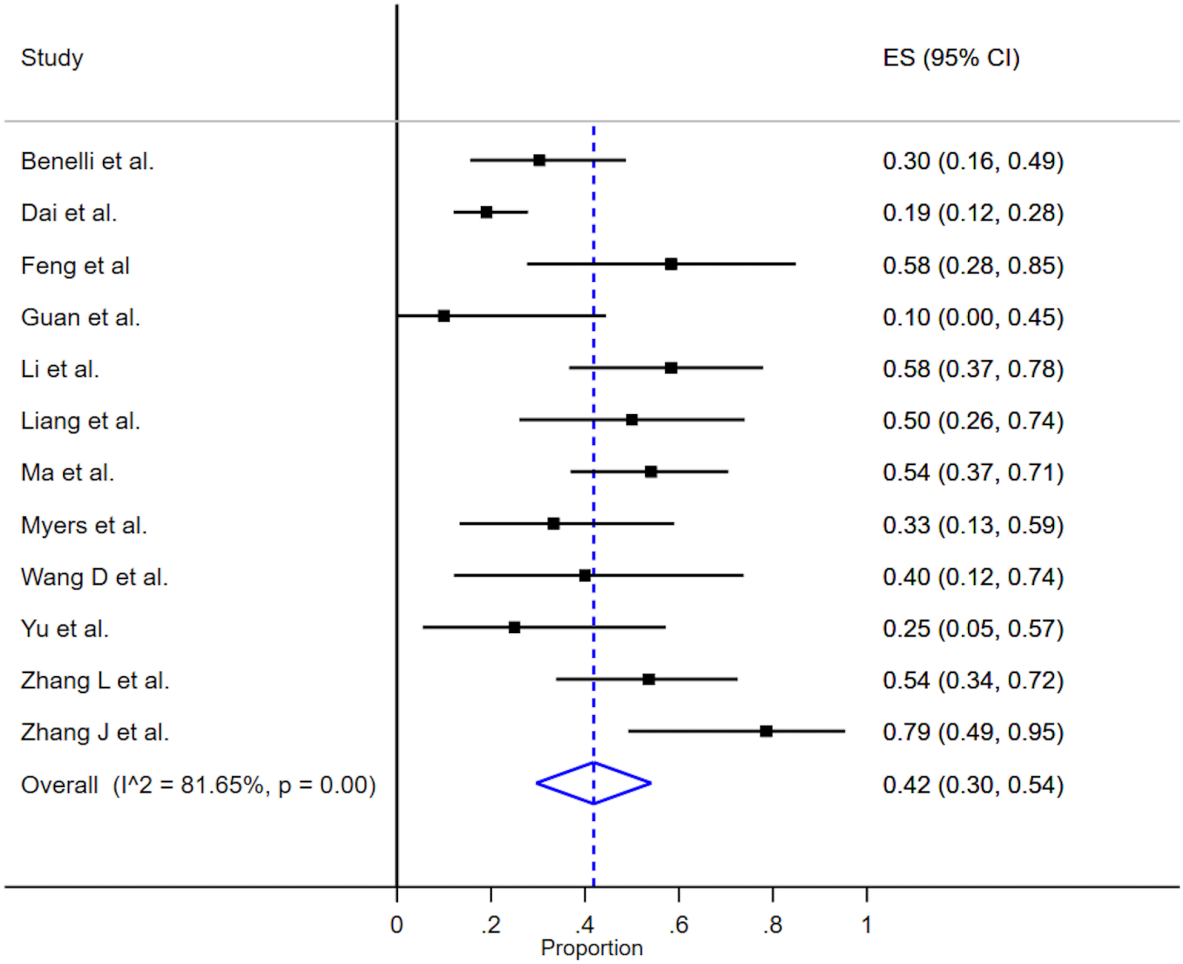
Supplementary Figure 3a: Random effects pooled proportion of severe symptoms among patients diagnosed with COVID-19



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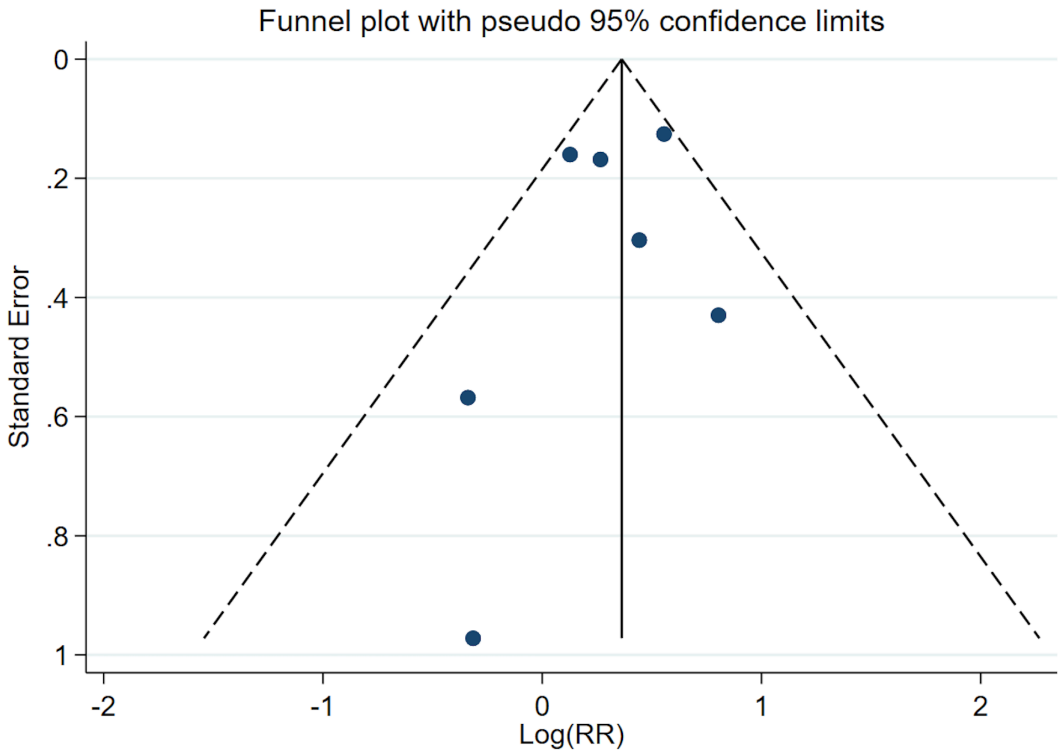
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Supplementary Figure 3b: Random effects pooled proportion of severe symptoms among cancer patients diagnosed with COVID-19

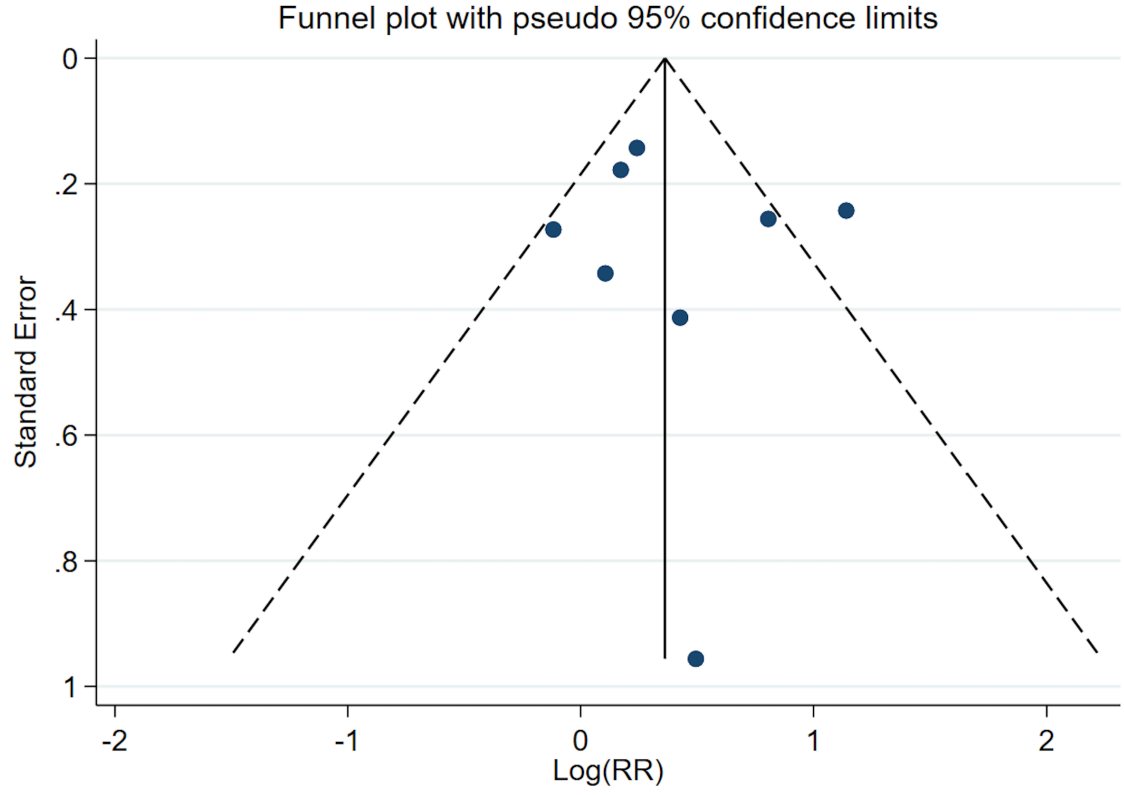


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Supplementary Figure 4a: Funnel plot of seven case series included in pooled relative risk analysis of death outcomes between cancer and general COVID-19 patients

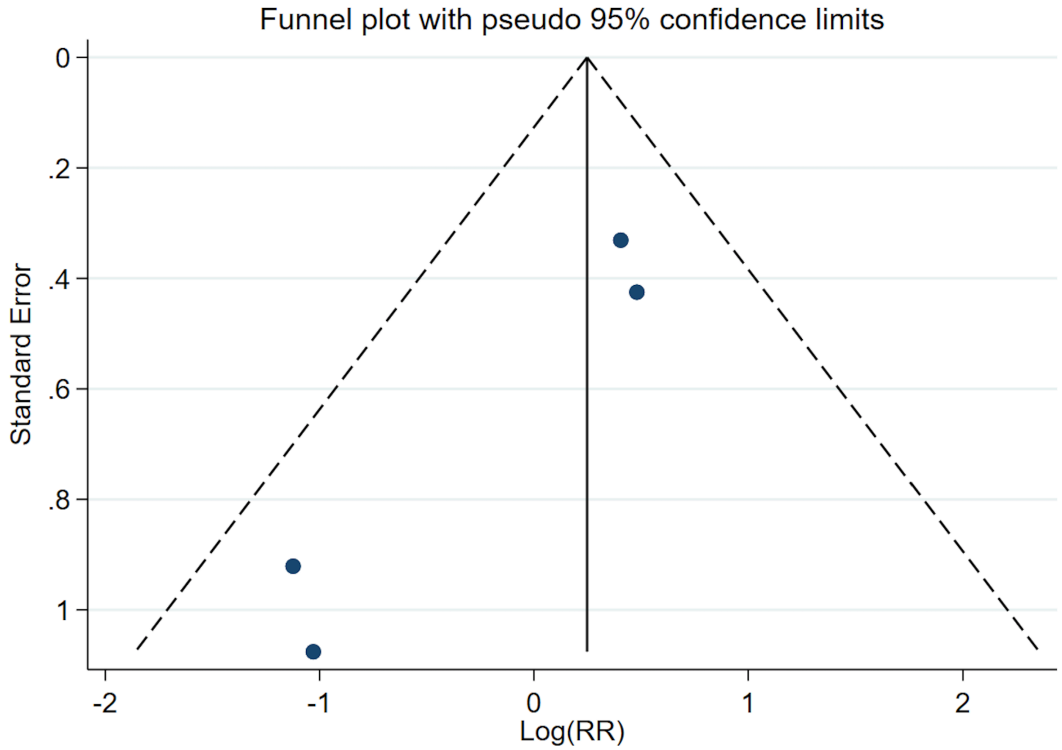


Supplementary Figure 4b: Funnel plot of eight case series included in pooled relative risk analysis of severe outcomes between cancer and general COVID-19 patients



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Supplementary Figure 4c: Funnel plot of four case series included in pooled relative risk analysis of lung cancer vs non-lung cancer and stage IV vs non stage IV cancers



Supplementary Table 2: Characteristics of 230 cancer patients with COVID-19 infection with severe and non-severe outcomes

COVID-19 severity	Non-severe (N = 131)	Severe (N = 99)	P-value (Chi-square)
Gender			
Male	70 (50.0)	70 (50.0)	0.010
Female	61 (67.8)	29 (32.2)	
Co-morbidities			
≥2	20 (8.7)	25 (10.9)	0.093
0-1	102 (44.3)	71 (30.9)	
Stage			
Stage IV	20 (8.7)	20 (8.7)	0.635
Stage I-III	56 (24.3)	40 (17.4)	
Unknown	55 (23.9)	39 (17.0)	
Tumour Type			
Lung cancers	32 (13.9)	24 (10.4)	0.974
Non-lung cancers	99 (43.0)	75 (32.6)	
Age			
≥ 60	63 (27.4)	79 (34.3)	<0.001
< 60	59 (25.7)	17 (7.4)	
Treatment			
Active systemic therapy	33 (14.3)	23 (10.0)	0.758
No systemic therapy	97 (42.2)	76 (33.0)	

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Appendix 3

A. 2019 Novel Coronavirus Disease (COVID-19) Diagnostic criteria 5th edition (5 Feb 2020)

Other provinces except Hubei province

1. Suspected cases by clinical criteria

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Computed tomography (CT) changes for viral pneumonia

c) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease. Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor all of the above clinical manifestations.

2. Laboratory confirmed cases

Suspected cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by real-time reverse transcriptase polymerase chain reaction (RT-PCR) assay.

2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

Hubei province

1. Suspected cases

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease.

Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor any two of clinical manifestations

2. Clinically diagnosed cases

CT chest changes that are consistent with atypical viral pneumonia.

3. Laboratory confirmed cases

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Suspected cases or clinically diagnosed cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by RT-PCR assay.
2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

B. National Health Commission Guideline on Severity Classification of COVID-19 (China)

Type	Criteria
Mild	the clinical symptoms were light, and there was no sign of pneumonia on imaging
Moderate	with fever, respiratory tract and other symptoms, imaging suggests pneumonia
Severe	coincide with any of the following: (1) respiratory distress, respiration rate (RR) ≥ 30 times / min; (2) the oxygen saturation ≤ 93% in the resting state; (3) PaO_2 / FiO_2 ≤ 300 mmHg (1mmHg = 0.133 kPa)
Critically ill	coincide with any of the following: (1) respiratory failure occurs and mechanical ventilation is required; (2) shock; (3) the patient develops other organ failure and needs ICU monitoring and treatment

(A and B released in The Diagnosis and Treatment Protocol for Novel Coronavirus Pneumonia, Version 5, National Health Commission, China)

C. IDSA/ATS Guidelines on Severe Pneumonia 2007

2007 Infectious Diseases Society of America/American Thoracic Society Criteria for Defining Severe Community-acquired Pneumonia
Validated definition includes either one major criterion or three or more minor criteria
Minor criteria
Respiratory rate ≥ 30 breaths/min
Pa_{O2}/Fi_{O2} ratio ≤ 250
Multilobar infiltrates
Confusion/disorientation
Uremia (blood urea nitrogen level ≥ 20 mg/dl)
Leukopenia* (white blood cell count < 4,000 cells/μl)
Thrombocytopenia (platelet count < 100,000/μl)
Hypothermia (core temperature < 36°C)
Hypotension requiring aggressive fluid resuscitation
Major criteria
Septic shock with need for vasopressors
Respiratory failure requiring mechanical ventilation

*Due to infection alone (i.e., not chemotherapy induced).

Metlay et al. Diagnosis and Treatment of Adults with Community-acquired Pneumonia. An Official Clnical Practice Guideline of the American Thoracic Society and Infectious Diseases Society of America. AJRCCM 2019.

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Impact of cancer diagnoses on the outcomes of patients with COVID-19 – a systematic review and meta-analysis

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ABSTRACT

Background:

The coronavirus disease 2019 (COVID-19) has caused significant mortality and morbidity across the globe. Cancer patients are especially vulnerable given their immunocompromised state. We aimed to determine the proportion of COVID-19 patients with cancer, their severity and mortality outcomes through a systematic review and meta-analysis (MA).

Methods:

Systematic review was performed through online databases, PubMed, Medline and Google Scholar, with keywords listed in methods section (November 1st 2019 - Dec 31st 2020). Studies with clinical outcomes of at least 10 COVID-19 patients and at least 1 with a diagnosis of cancer were included. The studies for MA were assessed with PRISMA guidelines and appraised with Newcastle-Ottawa

Scale (NOS). The data were pooled using a random-effects model using STATA software. The main outcomes were planned before data collection, including proportion of cancer patients among COVID-19 populations, relative risk of severe outcomes and death of cancer patients compared to general COVID-19 patients.

Results:

We identified 57 case series (63 413 patients), with 230 cancer patients with individual patient data (IPD). We found that the pooled proportion of cancer among COVID-19 patients was 0.04 (0.03-0.05, I2 = 97.69%, p<0.001). The pooled relative risk (RR) of death was 1.44 (95% CI 1.19-1.76) between cancer patients and the general population with COVID-19 infection. The pooled RR of severe outcome was 1.49 (95% CI 1.18-1.87) between cancer and general COVID-19 patients. The presence of lung cancer and stage IV cancer did not result in significantly increased RR of severe outcome. Among the available IPD, only age and gender were associated with severe outcomes.

Conclusion:

Cancer patients were at a higher risk of severe and death outcomes from COVID-19 infection as compared to general COVID-19 populations. Limitations of this study include publication bias, A collaborative effort is required for a more complete database.

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STRENGTHS AND LIMITATIONS

- Largest meta-analysis on cancer and COVID-19 to date
- Specific objectives looking at proportion of cancer patients among COVID-19 cases reported, with severity and mortality outcomes presented as relative risk against general population with COVID-19
- Detailed cancer cohorts presented with breakdown of tumour types, recent systemic anti-cancer therapy and other characteristics, with analysis of lung cancer vs non-lung cancer patients and stage IV vs non-stage IV patients
- Limitations include publication biases and incomplete reporting of mortality outcomes by selected papers
- Lack of stratification for age and co-morbidities for specific analysis due to limited clinical data

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BACKGROUND

The coronavirus disease 2019 (COVID-19) pandemic, since its emergence in December 2019, has caused significant mortality and morbidity across the globe. In early 2021, more than 100 million people had been infected with more than 2,000,000 deaths. It has disrupted many aspects of healthcare delivery and overwhelmed many hospitals and their intensive care resources.

Cancer patients are an at-risk population as they are generally older, immunocompromised, and have multiple co-morbidities. However, the outcomes of cancer patients with COVID-19 remains conflicted despite multiple cohort studies and meta-analyses. A multi-centre study of 105 cancer patients in China suggested that cancer patients with COVID-19 infection had higher risk of severe disease outcomes as compared to 536 age-matched non-cancer patients[1]. Another study from the Chinese Centre for Disease Control and Prevention Group reported a case fatality rate (CFR) of 2.3% (1023/44672) among confirmed COVID-19 cases and a more than double CFR of 5.6% (6/107) among the cancer patients with COVID-19[2]. In contrast, a database report of 5,688 COVID-19 positive patients from a tertiary hospital in New York found no significant difference in risk of death between cancer and non-cancer patients (RR 1.15, 95% CI 0.84-1.57) although there was an increased risk of intubation among cancer patients (RR 1.89, 95% CI 1.37-2.61)[3], and other cross sectional studies also showed similar risk of morbidity and mortality as the general population[4,5] though with a small number of patients. A few meta-analyses had recently addressed this as well and found that there was increased risk of severe COVID-19 infection among cancer patients on active oncological treatment[6], especially chemotherapy[7].

Hence, we aimed to review the current literature and to determine firstly the proportion of cancer patients among the reported COVID-19 cases, secondly whether cancer patients with COVID-19 had more severe outcomes and higher mortality compared to the general population and lastly, if there were any specific characteristics of patients with cancer that were associated with an increased risk of severe COVID-19 infection.

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METHODS

Search Methods and Study Selection

A systematic search of the literature was conducted to identify studies that include cancer patients with COVID-19 infection. The investigators searched biomedical databases including PubMed, Medline and Google Scholar using the terms *COVID-19, coronavirus, nCoV, SARS-CoV-2, characteristics, comorbidity, mortality, death, risk, oncology, cancer, malignancy, neoplasm, tumor, tumour, and carcinoma* (Appendix 1). The search was conducted on Jan 29, 2021 for a search period from November 1, 2019 through to December 31, 2020. The references of relevant meta-analysis and published reviewers were searched for additional eligible studies. The articles included met the following criteria: (1) COVID-19 diagnosis should be confirmed by Real Time Polymerase Chain Reaction (RT-PCR), or based on 2019 Novel Coronavirus Disease (COVID-19) Diagnostic criteria, if applicable (Appendix 1), (2) studies should contain clinical data of at least ten COVID-19 patients with at least one patient with a diagnosis of cancer (including haematological malignancies), (3) severity outcomes (defined as hypoxaemia, admission to intensive care unit (ICU), and/or mechanical ventilation) and/or death outcomes should be presented and, (4) studies should be available in English and full text. Exclusion criteria included studies that reported only specific patient populations, i.e. intensive care setting, paediatric population or single tumour type cohorts. Studies published online ahead of print and selected pre-print papers were included at the discretion of the authors. For the pooled analysis on severity and death outcomes of cancer and general COVID-19 patients, we mandated that the studies included at least 10 cancer patients.

The literature search and study selection were performed independently by three of the authors (SH, QZ and VSY), and discrepancies were resolved by consensus. The study followed the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines (Appendix 1).

Data Extraction

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Two investigators (SH and VSY) independently graded the studies based on Newcastle-Ottawa Scale (NOS) for assessing the quality of non-randomised studies and discrepancies were resolved by consensus (Appendix 2). Data extraction was performed by SH, and the parameters extracted included the location (hospital and city) of the study, type of study, number of patients, demographics including median or mean age, gender, proportion of patient with co-morbidities including cancer, and the clinical outcomes of severe disease and death (of cancer patients and overall COVID-19 population) were collected for the available studies.

Statistical analysis

All statistical analyses were performed using the STATA V.15 software. Meta-analysis using random-effects models was performed to analyse the pooled proportion of cancer among COVID-19 patients. The included studies were stratified into subgroups by region (within Hubei Province, China vs outside Hubei Province, including other countries), median/mean age (above or equal to vs below 60 years old) and gender (proportion of male more than or equal to 60% vs less than 60% in population). We further analysed the pooled relative risks (RR) of severe COVID-19 infection outcomes and related deaths for cancer patients with COVID-19 versus the general population with COVID-19. The subgroup of lung cancer and stage IV patients were also evaluated for relative risk of severe COVID-19 infection outcomes. The results were presented as proportions (decimal) with 95% confidence interval (95% CI). Heterogeneity between studies was analysed by the I^2 statistics, where $I^2 > 50\%$ was considered moderate heterogeneity and chi-square $p < 0.05$ was considered statistically significant. Chi-square tests were performed to look for associations between cancer patient characteristics (age, gender, lung cancers, stage IV cancers, co-morbidities and recent systemic therapy) and severe COVID-19 outcomes in the individual patient data (IPD).

Patient and Public Involvement

The research question was formed based on the urgent need to understand the effect of COVID-19 on cancer patients and to address the clinical decisions that follows this understanding. No patient

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was directly involved in this systematic review and meta-analysis. We would like to thank the authors and also the patients that were involved in the publications that we had included in our meta-analysis.

RESULTS

Eligible studies and characteristics

We identified 3083 relevant publications after literature search and review. Additional studies were identified from the references of published meta-analyses. After screening and eligibility assessment, we included in our meta-analysis a total of 57 case series involving 63 413 patients (Figure 1; Appendix 2 Supplementary Table 1). Studies that did not meet the MA criteria were summarised in Appendix 2 Supplementary Table 2.

Among the 57 studies, 37 studies were from public hospitals in China (24 from Hubei Province) with the rest from worldwide (USA, Canada and Spain in CCC19 study) (1), Europe (1), Korea (1), Iran (1), France (1), Spain (1), Italy (2), United Kingdom (UK) (3) and USA (9) (Table 1). Most studies were retrospective cohort studies and a few observational cross-sectional studies. The quality of all the included studies ranged from 6-8 (out of 9) using the Newcastle-Ottawa Scale (Appendix 2 Supplementary Table 3). Among the 57 case series, 13 contained detailed descriptions of COVID-19 patients with cancer (Table 2). Only 5 studies provided detailed IPD (Appendix 3 Supplementary Table 1). Forty-seven studies were used for proportion analyses for cancer diagnoses among COVID-19 patients. For the pooled analysis on relative risk of severe outcomes among cancer COVID-19 patients, we identified 10 case series that contained severity data, and 7 studies that contained mortality data for analysis (studies with at least 10 cancer patients). Five studies contained severity outcomes of lung cancer versus other cancer patients with COVID-19 infection, and four studies contained severity outcomes of stage IV vs non-stage IV cancer patients.

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Table 1: Baseline characteristics, comorbidities and severity outcomes of 57 COVID-19 case series

No.	Authors	Location	Definition of severity	Total	Death n (%)	Severe n (%)	Gender – Male n (%)	Age	DM	HTN	Pulmonary disease	Smoking	CKD	CVS	CLD	Cancer total n (%)	Cancer – severe n (% of cancer total)	Cancer – death (% of cancer total)
1	Argenziano et al[8]	New York, USA	ICU, death	1000	211 (21.1)	236 (23.6)	596 (59.6)	63 (50-75	372 (37.2)	601 (60.1)	66 (6.6)	49 (4.9)	137 (13.7)	131 (13.1)	15 (1.5)	67 (6.7)	17 (25.4)	N.A
2	Assaad et al[9]	Lyon, France	Death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	302	N.A	30 (9.9)
3	Benelli et al.[10]	Crema, Italy	ICU + NIV, death	411	72 (17.5)	140 (34.1)	359 (66.6)	70.5 (1-99)	67 (16.3)	193 (47.0)	48 (11.7)	N.A	22 (5.3)	93 (22.6)	N.A	33 (8.0)	10 (30.3)	9 (27.3)
4	Cai et al.[11]	Shenzhen, China	ICU, death	298	3 (1.0)	58 (19.5)	145 (48.7)	47.5 (33-61)	18 (6.0)	47 (15.8)	N.A	N.A	N.A	25 (8.4)	28 (9.4)	4 (1.3)	2 (50.0)	N.A
5	Cao M et al.[12]	Shanghai, China	ICU, death	198	1 (0.5)	19 (9.6)	101 (51.0)	50.1 (±16.3)	15 (7.6)	42 (21.2)	N.A	11 (5.6)	N.A	12 (6.0)	6 (3.0)	4 (2.0)	0	N.A
6	Cao J et al.[12]	Hubei, China	ICU, death	102	17 (16.7)	18 (17.6)	53 (52.0)	54 (37-67)	11 (10.8)	28 (27.5)	10 (9.8)	N.A	4 (3.9)	11 (10.8)	2 (2.0)	4 (3.9)	N.A	1 (25.0)
7	Chen J et al.[13]	Shanghai, China	ICU, death	249	2 (0.8)	22 (8.8)	126 (50.6)	51 (31-64)	25 (10.0)	N.A	5 (2.0)	N.A	N.A	55 (22.1)	N.A	1 (0.4)	N.A	N.A
8	Chen N et al.[14]	Hubei, China	ICU, death	99	11 (0.11)	23 (23.2)	67 (67.7)	55.5 (±13.1)	12 (12.1)	0	1 (1.0)	N.A	N.A	40 (40.4)	N.A	1 (1.0)	N.A	N.A
9	Colaneri et al.[15]	Pavia, Italy	Require high FiO2	44	2 (4.5)	17 (38.6)	28 (63.6)	67.5 (10-94)	7 (15.9)	15 (34.1)	2 (4.5)	N.A	N.A	11 (25.0)	N.A	6 (13.6)	2 (4.5)	N.A
10	Dai et al.[16]	Hubei, China	NIV and ventilation, death	641	N.A	84 (13.1)	57 (8.9)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	105 (16.4)	20 (19.1)	12 (11.4)
11	Docherty et al[17]	UK	HD/ICU, death	20133	5165 (25.7)	3001 (14.9)	12068 (59.9)	73 (58-82)	3650 (18.1)	N.A	3128 (15.5)	852 (4.2)	2830 (14.1)	5469 (27.2)	281 (1.4)	1743 (8.7)	N.A	N.A
12	Du et al.[18]	Hubei, China	Death	179	21 (11.7)	N.A	97 (54.1)	57.6 (±13.7)	33 (18.4)	58 (32.4)	N.A	N.A	4 (2.2)	29 (16.2)	N.A	4 (2.2)	N.A	1 (25.0)
13	Duanmu et al.[19]	California, USA	Death	100	1 (1.0)	N.A	56 (56.0)	45 (32-65)	10 (10.0)	19 (19.0)	1 (1.0)	2 (2.0)	6 (6.0)	N.A	N.A	3 (3.0)	N.A	N.A
14	Feng et al.[20]	Hubei, China	See below	476	38 (8.0)	124 (26.1)	271 (56.9)	53 (40-64)	17 (3.6)	40 (8.4)	14 (2.9)	44 (9.2)	2 (0.4)	26 (5.5)	N.A	12 (2.5)	7 (58.3)	N.A
15	Guan et al.[21]	China	Composite endpoint[a]	1099	15 (!.4)	67 (6.1)	640 (58.2)	47 (35-58)	81 (7.4)	165 (15.0)	12 (1.1)	158 (14.4)	8 (0.7)	42 (3.8)	23 (2.1)	10 (0.9)	1 (10.0)	N.A
16	Guo et al.[22]	Hubei, China	Mechanical ventilation	187	43 (23.0)	45 (24.1)	91 (48.7)	58.5 (±14.7)	28 (15.0)	61 (32.6)	4 (2.1)	18 (9.6)	6 (3.2)	29 (15.5)	N.A	13 (7.0)	N.A	N.A
17	Huang C et al.[23]	Hubei, China	ICU	41	6 (14.6)	13 (31.7)	30 (73.2)	49 (41-58)	8 (19.5)	6 (14.6)	1 (2.4)	3 (7.3)	N.A	6 (14.6)	1 (2.4)	1 (2.4)	0	N.A
18	Huang Y et al.[24]	Hubei, China	ICU	34	N.A	8 (23.5)	14 (41.2)	56 (26-88)	4 (11.8)	8 (23.5)	3 (8.8)	N.A	N.A	6 (17.6)	1 (2.9)	3 (8.8)	N.A	N.A
19	Kim et al. [25]	Korea	Require high FiO2	28	0	6 (24.1)	15 (53.6)	40 (20-73)	2 (7.1)	0	0	5 (17.9)	0	0	1 (3.6)	1 (3.6)	N.A	N.A
20	Kudderer et al.[26]	Worldwide	Composite endpoint	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	928	242 (26.0)	121 (13.0)
21	Lee LY et al.[27]	UK	ICU, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	800	53 (7.0)	226 (28.3)
22	Li et al.[28]	Hubei, China	IDSA/ATS[b]	548	90 (16.4)	269 (49.1)	279 (50.9)	60 (48-69)	83 (15.1)	166 (30.3)	17 (3.1)	N.A	10 (1.8)	34 (6.2)	5	24 (0.9)	14	N.A
23	Lian et al.[29]	Zhejiang, China	Not defined	788	N.A	18 (2.3)	407 (51.6)	N.A	57 (7.2)	126 (16.0)	3	54 (6.9)	7 (0.9)	11 (1.4)	31 (3.9)	6 (0.8)	N.A	N.A
24	Liang et al.[30]	China	Composite endpoint	1590	N.A	254 (16.0)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	18 (1.1)	9 (50.0)	9 (50.0)
25	Liu K et al.[31]	Hubei, China	NIV	137	16 (11.7)	34 (24.8)	61 (44.5)	57 (20-83)	14 (10.2)	13 (9.5)	2 (1.5)	N.A	N.A	10	N.A	2 (1.5)	N.A	N.A
26	Ma et al.[32]	Hubei, China	See below[c]	1380	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	37 (2.7)	20 (54.1)	5 (13.5)
27	Mehta et al.[33]	New York, USA	Death	1308	210 (16.1)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	218 (16.7)	N.A	61 (28.0)
28	Miyashita et al.[3]	New York, USA	Intubation, death	5688	555 (9.8)	351 (6.2)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	334 (5.9)	N.A	37 (11.1)
29	Myers et al.[34]	California, USA	ICU	377	100 (26.5)	113 (30.0)	212 (56.2)	61 (50-73)	118 (31.3)	164 (43.5)	28 (7.4)	N.A	48 (12.7)	22 (5.8)	21 (5.6)	18 (4.8)	6 (33.3)	N.A
30	Nikpouraghdam et al.[35]	Iran	Death	2964	239 (8.1)	N.A	1955 (66.0)	56 (46-65)	113 (3.8)	59 (2.0)	60 (2.0)	N.A	18 (0.6)	37 (1.2)	N.A	17 (0.6)	N.A	1 (5.9%)
31	Pan et al[36]	Hubei, China	ICU, death	204	36 (17.6)	16 (7.8)	107 (52.5)	52.9 (±16·0)	44 (21.6)	N.A	9 (4.4)	N.A	N.A	44 (21.6)	N.A	13 (6.4)	N.A	N.A
32	Paranjpe et al.[37]	New York, USA	Death	1078	310 (28.8)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	64 (5.9)	N.A	24 (37.5)
33	Petrilli et al[38]	New York, USA	ICU, death	5279	665 (12.6)	990 (18.8)	2615 (49.5)	54 (38-66)	1195 (22.6)	2256 (42.7)	786 (14.9)	288 (5.5)	647 (12.3)	704 (13.3)	N.A	403 (7.6)	138 (34.2)	N.A
34	Pinato et al[39]	Europe	HD/ICU, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	890	110 (12.4)	299 (33.6)
35	Robilotti et al[40]	New York, USA	Mechanical ventilation/high flow, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	423	87 (20.6)	51 (12.1)

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36	Richardson et al.[41]	New York, USA	ICU, death	5700	553 (9.7)	320 (5.6)	3437 (60.3)	63 (52-75)	1808 (31.7)	3026 (53.1)	287 (5.0)	2691 (47.2)	454 (8.0)	966 (16.9)	30 (0.5)	320 (5.6)	N.A	N.A
37	Shi H et al.[42]	Hubei. China	Not defined	81	N.A	N.A	42 (51.9)	49·5 (±11·0)	19 (23.5)	12 (14.8)	9 (11.1)	N.A	3 (3.7)	14 (17.3)	N.A	4 (4.9)	N.A	N.A
38	Shi Y et al.[43]	Zhejiang, China	Not defined	487	0	49 (10.1)	259 (53.2)	46 (±19·0)	29 (6.0)	99 (20.3)	0	40 (8.2)	7 (1.4)	11 (2.3)	22 (4.5)	5 (1.0)	2 (40.0)	N.A
39	Tomlins et al.[44]	UK	NIV + intubation, death	95	20 (21.1)	16 (16.8)	60 (63.2)	75 (59-82)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	20 (21.1)	N.A	3 (15.0)
40	Wan et al.[45]	Chongqing, China	Not defined	135	1 (0.7)	40 (29.6)	72 (53.3)	47 (36-55)	12 (8.9)	13 (9.6)	0	9 (6.7)	N.A	7 (5.2)	2 (1.5)	4 (3.0)	3 (75.0)	N.A
41	Wang D et al.[46]	Hubei, China	ICU	138	6 (4.3)	36 (26.1)	75 (54.3)	56 (42-68)	14 (10.1)	43 (31.2)	4 (2.9)	N.A	2 (1.5)	27 (19.6)	0	10 (7.2)	4 (40.0)	N.A
42	Wang R et al.[47]	Anhui. China	ICU	125	0	19 (15.2)	71 (56.8)	38.8 (±13.8)	10 (8.0)	N.A	2 (1.6)	16 (12.8)	N.A	18 (14.4)	N.A	1 (0.8)	N.A	N.A
43	Wu C[48] et al.	Hubei, China	ARDS, death	201	44 (21.9)	84 (41.8)	N.A	51 (43-60)	22 (10.9)	39 (19.4)	5 (2.5)	N.A	2 (1.0)	8 (4.0)	7 (3.5)	1 (0.5)	N.A	N.A
44	Wu J et al.[49]	Jiangsu, China	See below	80	0	3 (37.5)	39 (48.8)	46.1 (±15.4)	5 (6.3)	N.A	0	N.A	1 (1.3)	25 (31.3)	1 (1.3)	1 (1.3)	N.A	N.A
45	Xu et al.[50]	Guangzhou, China	Not defined	90	N.A	N.A	39 (43.3)	50 (18-86)	5 (55.6)	17 (18.9)	1 (1.1)	N.A	N.A	3 (3.3)	N.A	2 (2.2)	N.A	N.A
46	Yang et al.[51]	Zhejiang, China	NIV	149	0	2 (1.3)	81 (54.4)	45.1 (±13.4)	9 (6.0)	0	1 (0.7)	N.A	N.A	28 (18.8)	N.A	2 (1.3)	N.A	N.A
47	Yang K et al.[52]	Hubei, China	ICU, mechanical ventilation, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	205	30 (14.6)	40 (19.5)
48	Yarza et al[53]	Spain	Respi failure	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	63	34 (54.0)	16 (25.4)
49	Yu et al.[54]	Hubei, China	ARDS, death	1524	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	12 (0.8)	3 (25.0)	3 (25.0)
50	Zhang L et al.[55]	Hubei, China	Composite endpoint	1276	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	28 (2.2)	15 (53.6)	8 (28.6)
51	Zhang G et al.[56]	Hubei, China	IDSA/ATS	221	12 (5.4)	55 (24.9)	108 (48.9)	55 (39-66.5)	22 (10.0)	54 (24.4)	6 (2.7)	N.A	6 (2.7)	37 (16.7)	7 (3.2)	9 (4.1)	4 (44.4)	N.A
52	Zhang J et al.[57]	Hubei, China	See below	663	25 (3.8)	409 (61.7)	321 (48.4)	55.6 (44-69)	67 (10.1)	N.A	51 (7.7)	N.A	N.A	164 (24.7)	31 (4.7)	14 (2.1)	11 (78.6)	N.A
53	Zhang N et al.[d]	Hubei, China	See below	1548	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	67 (4.3)	32 (47.8)	18 (26.9)
54	Zhang S et al.[e]	Hubei, China	Death	315	47 (14.9)	178 (56.5)	175 (55.6)	57 (44-66)	41 (13.0)	78 (24.8)	3 (1.0)	N.A	2 (0.6)	42 (13.3)	9 (2.9)	12 (3.8)	N.A	4 (33.3)
55	Zhao et al.[58]	Hubei, China	Not defined	91	2 (2.2)	30 (33.0)	49 (53.8)	46 (Not stated)	3 (3.3)	18 (19.8)	1 (1.1)	N.A	1 (1.1)	N.A	N.A	3 (3.3)	2 (2.2)	N.A
56	Zhou et al.[59]	Hubei, China	Not defined	191	54 (28.3)	119 (62.3)	119 (62.3)	56 (46-67)	36 (18.8)	58 (30.4)	6 (3.1)	11 (5.8)	2 (1.0)	15 (7.9)	0	2 (1.0)	N.A	N.A
57	Zhu et al.[60]	Hefei, China	Not defined	32	N.A	N.A	15 (46.9)	46 (35-52)	4 (12.5)	7 (21.9)	2 (6.3)	6 (18.8)	1 (3.1)	10 (31.3)	2 (6.3)	2 (6.3)	N.A	N.A

[a]composite endpoint: Severe disease composite endpoints (ICU, mechanical ventilation and/or death)

[b]see below: Severity as defined by Chinese National Health Commission Definition of Severity (Refer to Appendix 1)

[c]IDSA/ATS: Infectious Diseases Society of America/American Thoracic Society Criteria for Defining Severe Community-acquired Pneumonia 2007 (Appendix 1)

[d]article in press (accepted, peer-reviewed), [e]article not peer-reviewed

All figures presented as n (%) – number (percentage) unless otherwise stated. Age presented as median with interquartile range (IQR): age (range) or as mean: age (± standard deviation (SD))

Abbreviations: ARDS – acute respiratory distress syndrome; CKD – chronic kidney disease; CLD (chronic liver disease); CVS – cardiovascular disease (and included cerebrovascular disease); DM – diabetes mellitus; FiO2 – supplemental oxygen; HTN – hypertension; ICU – intensive care unit; IDSA/ATS – Infectious Disease Society of America/American Thoracic Society Guidelines (Refer to Appendix 1); NIV – non-invasive ventilation; N.A – not applicable

Overall, there were more males than females affected, with a median age above 60 in all case series. This median age was notably higher than the non-cancer populations reported in previous studies[16]. Severe COVID-19 disease among cancer patients ranged from 9 to 54% in thirteen eligible studies (Table 2). Overall, 230 patients from five case series were identified with oncological history, stage, recent treatment history and clinical outcomes (Appendix 3 Supplementary Table 1). However, among these patients, there was inconsistent reporting in those on active systemic chemotherapy and some

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clinical data on stage of cancer were missing, limiting the usefulness of the dataset in predicting the outcomes of COVID-19 in patients with active cancer.

Table 2: Baseline characteristics of cancer patients with COVID-19 infections from 13 case series

Characteristics	Zhang L et al. 2020 (n=28) n (%) or median (range)		Yu et al. 2020 (n=12) n (%) or median (range)		Liang et al. 2020 (n=18) n (%) or median (range)		Ma et al. 2020 (n=37) n (%) or median (range)		Dai et al. 2020 (n=105) n (%) or median (range)		Zhang N et al. 2020 (n=67) n (%) or median (range)	
Age	65.0 (56.0-70.0)		66 (48-78)		63.1 (51-75)		62 (59-70)		64 (50-78)		66 (37-98)	
Gender												
Male	17 (60.7)		10 (83.3)		12 (66.7)		20 (54.1)		57 (54.7)		41 (61.2)	
Female	11 (39.3)		2 (16.7)		6 (33.3)		17 (45.9)		48 (45.3)		26 (38.8)	
Stage												
I-III	18 (64.3)		5 (41.7)		13 (72.2)		N.A		88 (83.8)		N.A	
IV	10 (35.7)		6 (50.0)		4 (22.2)		N.A		17 (16.2)		N.A	
Unknown	N.A		1 (8.3)		1 (5.6)		N.A		N.A		N.A	
Smoking history												
Yes	N.A		N.A		4 (22.2)		N.A		36 (34.3)		9 (13.4)	
No	N.A		N.A		N.A		N.A		69 (65.7)		N.A	
Unknown	N.A		N.A		N.A		N.A		N.A		N.A	
	Lung ca	7 (25.0)	Lung ca	7 (58.3)	Lung ca	5 (27.8)	CRC	11 (29.7)	Lung ca	22 (20.1)	Lung ca	15 (22.4)
	Esophageal ca	4 (14.3)	Rectal ca	1 (8.3)	CRC	4 (22.2)	Lung ca	8 (21.6)	GI ca	13 (12.4)	CRC	11 (16.4)
	Breast ca	3 (10.7)	Colon ca	1 (8.3)	Breast ca	3 (16.7)	Breast ca	7 (18.9)	Breast ca	11 (10.5)	Thyroid ca	8 (11.9)
	Laryngoca	2 (7.1)	Pancreatic ca	1 (8.3)	Bladder ca	2 (11.1)	Gynae ca	5 (13.5)	Thyroid ca	11 (10.5)	Urinary system	8 (11.9)
	Liver ca	2 (7.1)	Breast ca	1 (8.3)	Lymphoma	1 (0.56)	Other ca	6 (16.2)	Blood ca	9 (8.6)	Breast & gynae	7 (10.4)
	Prostatic ca	2 (7.1)	Urothelial ca	1 (8.3)	Thyroid ca	1 (0.56)			Cervix ca	6 (5.7)	Head and Neck	5 (7.5)
	Cervical ca	1 (3.6)			Adrenal ca	1 (0.56)			Esophageal ca	6 (5.7)	CNS tumour	4 (5.9)
	Gastric ca	1 (3.6)			RCC	1 (0.56)					Haematological	3 (4.5)
	Colon ca	1 (3.6)									Gastric ca	3 (4.5)
	Rectum ca	1 (3.6)									Esophageal ca	2 (3.0)
	NPC	1 (3.6)									HCC	1 (1.15)
	Endometrial ca	1 (3.6)										
	Ovarian ca	1 (3.6)										
	Ca of testis	1 (3.6)										
Baseline treatment	ChemoRT	25 (89.3)	BSC	4 (33.3)	Surveillance	12 (75.0)	Antitumour therapy	13 (35.1)	Chemotherapy	17 (16.2)	Follow up	44 (65.7)
	Operation	21 (75.0)	Chemo-immuno Therapy	2 (16.7)	Chemo and surgery within 1 month	4 (25.0)			RT	13 (12.4)	On treatment	23 (34.3)
	Targeted therapy/ immunotherapy	6 (21.4)	Surveillance	2 (16.7)					Surgery	8 (7.6)		
	Chemo (<14d)	3 (10.7)	Targeted Therapy/RT	1 (8.3)					Immunotherapy	6 (5.7)		
	Targeted Therapy	2 (7.1)	Chemo/RT	1 (8.3)					Targeted Therapy	4 (3.8)		
	RT (<14d)	1 (3.6)	Adjuvant RT	1 (8.3)								
	Immunotherapy (<14d)	1 (3.6)	Not started	1 (8.3)								
Severe disease	Severe 8/15 Non-invasive ventilation 2/15 Invasive ventilation	15 (53.6)	Severe (25.0)	3	Severe	9 (50.0)	Severe	20 (54.1)	Severe ICU	40 (38.0) 20 (19.1)	Severe Mechanical ventilation	32 (47.8) 28 (41.8)
ARDS	8 (28.6)		N.A		N.A		N.A		N.A		14 (20.9)	
Death	8 (28.6)		3 (25.0)		9 (50.0)		5 (13.5)		12 (11.4)		18 (26.9)	

Kuderer et al (CCC19) (n=928) n (%) or median (range)		Lee LY et al (UKCCMP) (n=800) n (%) or median (range)		Robilotti et al (MSKCC) (n=423) n (%) or median (range)		Pinato et al (n=890) n (%) or median (range)		Assaad et al (n=302) n (%) or median (range)		Yang et al (n=205) n (%) or median (range)		Yarza et al (n=63) n (%) or median (range)	
66 (57–76)		69 (59–76)		N.A		Mean: 68.0±12.8		Mean: 58.2±1.1		63 (56–70)		66 (63 - 69)	
468 (50)		449 (56)		212 (50.1)		503 (56.5)		144 (47.7)		96 (46.8)		34 (54)	
459 (49)		349 (44)		211 (49.9)		387 (43.5)		158 (52.3)		109 (53.2)		29 (46)	
N.A		227 (29)		N.A		539 (60.6)		N.A		N.A		52 (82.5)	
N.A		347 (43)		238 (56.3)		351 (39.4)		161 (53.3)		N.A		11 (17.5)	
N.A		205 (25)		N.A		0		N.A		N.A		N.A	
369 (40)		N.A		167 (39.5)		380 (42.7)		N.A		N.A		34 (54.0)	
469 (51)		N.A		249 (58.9)		343 (38.5)		N.A		N.A		29 (46.0)	
90 (10)		N.A		7 (1.7)		167 (18.8)		N.A		N.A		0	
Haematological ca	204 (22.0)	Digestive organs	150 (18.8)	Others	137 (32.4)	Breast	162 (18.2)	Solid tumour	234 (77.5)	Solid tumour	183 (89.3)	Lung	17 (27.0)
Breast ca	191 (20.6)	Other haematological	109 (13.6)	Breast	86 (20.3)	Haematological	137 (15.4)	Haematological	68 (22.5)	Breast	40 (19.5)		
Prostate ca	152 (16.4)	Breast	102 (12.8)	Lymphoma	48 (11.3)	Genito-Urinary	132 (14.8)	Lung	42 (13.9)	Colorectal	28 (13.6)		
Gastrointestinal ca	108 (11.6)	Respiratory and intrathoracic organs	90 (11.3)	Colorectal ca	37 (8.7)	Lung	119 (13.4)			Lung	24 (11.7)		
Lung ca	91 (9.8)	Male genital organs	78 (9.8)	Lung	35 (8.3)	Gastrointestinal	105 (11.8)			Haematological	22 (10.7)		
Gynaecological ca	49 (5.3)	Lymphoma	60 (7.5)	Leukemia	32 (7.6)	Gynaecological	41 (4.6)			Thyroid	16 (7.8)		
RCC	45 (4.8)	Female genital organs	45 (5.6)	Prostate	26 (6.1)	Gastro-esophageal	40 (4.5)			Stomach	12 (5.9)		
Endocrine ca	39 (4.2)	Urinary tract	50 (6.3)	Melanoma	22 (5.2)	Hepatobiliary	45 (5.1)			Cervical	9 (4.4)		
Melanoma	38 (4.1)	Other or unspecified ‡	47 (5.9)			Head and Neck	29 (3.3)			Lymphoma	7 (3.4)		
Head and neck ca	30 (3.2)	Lip, oral cavity, and pharynx	27 (3.4)			Skin	28 (3.1)			Acute lymphoblastic leukaemia	5 (2.4)		

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Sarcoma	24 (2.6)	Melanoma (skin)	27 (3.4)			Other	52 (5.8)			Chronic lymphoblastic leukaemia	4 (2.0)		
CNS	12 (1.3)	CNS	15 (1.9)							Multiple myeloma	3 (1.5)		
Solid tumour, not otherwise specified	43 (4.6)									Acute myelogenous leukaemia	2 (1.0)		
										MDS	1 (<1)		
None	553 (59.6)	Chemotherapy	281 (35.1)	Chemo in last 30 days	191 (45.2)	Chemotherapy	164 (69.8)	Any cancer treatment	194 (64.2)	All	54/182 (29.7)	Chemotherapy	36 (57.1)
Chemotherapy	160 (17.2)	None	272 (34.0)	Surgery in last 30 days	31 (7.3)	Ongoing anticancer therapy at Covid-19 diagnosis	479 (53.8)	Cytotoxics	137 (45.4)	Chemotherapy	31/182 (17.0)	Endocrine therapy	10 (15.9)
Endocrine	85 (9.2)	RT	76 (9.5)	ICI	31 (7.3)	Surgery	417 (46.8)	No cancer treatment	108 (35.8)	Targeted therapy	12/182 (6.6)	Immunotherapy	8 (12.7)
Targeted therapy	75 (8.1)	Targeted treatment	72 (9)			Adjuvant/Neo adjuvant chemotherapy	210 (23.6)	Immunotherapy	26 (8.6)	Others	10/182 (5.5)	Targeted therapy	7 (11.1)
Immunotherapy	38 (4.1)	Hormone therapy	64 (8.0)			Endocrine Therapy	48 (20.4)	Antiangiogenic	18 (6.0)	RT	9/182 (5.0)	No active tx	2 (3.2)
RT	12 (1.3)	Other	60 (7.5)			Target Therapy	41 (17.4)	Anti-CD20	14 (4.6)	Surgery	4/182 (2.2)		
		Immunotherapy	44 (5.5)			Immunotherapy	38 (16.2)	Anti-HER2	12 (4.0)	Immunotherapy	4/182 (2.2)		
		Surgery	29 (3.6)					Antiproteasomes	8 (2.6)				
		No information	10 (1.3)					Everolimus (mTORi)	4 (1.3)				
Severe (composite) 242 (26.1) ICU 132 (14.2) Mechanical ventilation 116 (12.5)		Severe/critical 360 (45) ICU 53 (6.6)		Mechanical ventilation 40 (9.5) High flow 47 (11.1)		Admission to ICU/sub-intensive 110/760 (14.5)		N.A		ICU 30 (14.6) Mechanical ventilation 21(10.2)		Respi failure 34 (54.0)	
NA		NA		NA		127 (22.5)		N.A		23 (11.2)		23 (36.5)	
121 (13.0)		226 (28.3)		51 (12.1)		299 (33.6)		30 (9.9)		40 (19.5)		16 (25.4)	

Abbreviations: ARDS – acute respiratory distress syndrome; BSC – best supportive care; Ca – carcinoma; CNS – central nervous system; CRC – colorectal cancer; GI – gastrointestinal; ICU – intensive care unit; NPC – nasopharyngeal cancer; RCC – renal cell carcinoma; RT – radiotherapy

Proportion of cancer diagnoses

The overall pooled proportion of cancer diagnoses among COVID-19 patients from 47 studies was 0.04 (95% CI 0.03-0.05) with significant heterogeneity among the included studies, I^2 = 97.69%, chi-square p=<0.001 (Figure 2). The proportion of cancer patients was significantly lower in the studies with median or mean age less than 60 years, at 0.03 (95% CI 0.02-0.03) compared to 0.07 (95% CI 0.05-0.09) for studies with median or mean age above or equal 60 years (Figure 3). The proportion of cancer was higher in the studies with male pre-dominance (more than or at least 60% population being male), at 0.05 (95% CI 0.02-0.07) than studies with less than 60% population being male, at 0.03 (95% CI 0.02-0.05; Figure 4). The proportion of cancer in Hubei Province was 0.03 (95% CI 0.02-0.05) and outside Hubei Province 0.04 (95% CI 0.02-0.04; Appendix 3 Supplementary Figure 1).

Mortality outcomes of cancer patients with COVID-19

Overall, the mortality rates of cancer patients with COVID-19 infection ranged widely from 5.9% to 50% across 20 studies (Table 3). The pooled relative risk of death was significantly higher in cancer

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patients with COVID-19 infection than in the general population with COVID-19 infection based on seven studies as shown in table 3 (RR 1.41; 95% CI 1.15-1.73), I^2 = 26.2%, chi-square p=0.229 (Figure 5a). The pooled mortality rate was estimated at 0.22 (95% CI 0.21-0.33) among cancer patients with COVID-19, which was significantly higher than that of the overall COVID-19 patients, at 0.09 (95% CI 0.07-0.11; Appendix 3 Supplementary Figure 2a and 2b).

Table 3: Mortality rate among cancer patients diagnosed with COVID-19 infection*

Case series	Death/cancer population	Mortality rate in cancer cohorts (%)	Death/total COVID-19 population	Mortality rate in general COVID-19 cohorts (%)	Relative risk of death (95% CI)
Liang et al.	9/18	50.0	N.A	N.A.	N.A
Zhang L et al.	8/28	28.6	N.A	N.A	N.A
Yu et al.	3/12	25	N.A	N.A	N.A
Ma et al.	5/37	13.5	N.A	N.A.	N.A
Zhang N et al.	18/67	26.9	N.A	N.A	N.A
Dai et al.	12/105	11.4	N.A	N.A	N.A
Kuderer et al.	121/928	13.0	N.A	N.A	N.A
Lee LY et al.	226/800	28.3	N.A	N.A	N.A
Robilotti et al.	51/423	12.1	N.A	N.A	N.A
Assaad et al.	30/302	9.9	N.A	N.A	N.A
Pinato et al.	299/890	33.6	N.A	N.A	N.A
Yang et al.	40/205	19.5	N.A	N.A	N.A
Yarza et al.	16/63	25.4	N.A	N.A	N.A
Benelli et al.	**9/33**	**27.3**	**72/411**	**17.5**	**1.56 (0.86-2.82)**
Mehta et al.	**61/218**	**28.0**	**210/1308**	**16.1**	**1.74 (1.36-2.23)**
Miyashita et al.	**37/334**	**11.1**	**555/5688**	**9.8**	**1.14 (0.83-1.55)**
Nikpouraghdam et al.	**1/17**	**5.9**	**239/2964**	**8.1**	**0.73 (0.11-4.90)**
Paranjpe et al.	**24/64**	**37.5**	**310/1078**	**28.8**	**1.30 (0.94-1.81)**
Tomlins et al	**3/20**	**15.0**	**20/95**	**21.1**	**0.71 (0.23-2.17)**
Zhang S et al.	**4/12**	**33.3**	**47/315**	**14.9**	**2.23 (0.96-5.19)**

***Only papers with at least 10 cancer patients were included**

Severity outcomes of cancer patients with COVID-19

Further analyses were conducted for the ten case series with available severity outcomes for both cancer and general COVID-19 patient cohorts. We found that cancer patients had higher risk of severe COVID-19 outcomes compared to the general COVID-19 population, with a pooled relative risk of 1.49 (95% CI 1.18-1.87) and moderate heterogeneity, I^2 = 66.7%, chi-square p=0.001 (Figure 5b). Also, we noted that the pooled proportion of severe outcomes in cancer patients was 0.42 (95% CI 0.30-0.54), which was higher than the proportion of severe cases of all COVID-19 patients at 0.27 (95% CI 0.21-0.33; Appendix 3, Supplementary Figure 3a and 3b). There was no statistically significant difference in relative risk of severe outcomes between those patients (available data from 5

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studies identified) with lung cancers versus non lung cancers (RR at 1.46; 95% CI 1.84-2.52, I^2 = 48.1%, chi-square p=0.7; Figure 6a). There was no statistically significant difference in relative risks of severe outcomes between stage IV vs non stage IV cancers from four studies identified (RR 1.48; 95% CI 0.89-2.47; Figure 6b).

The severe outcomes of COVID-19 in cancer patients was also analysed from the available 230 individual patient data and summarised in Appendix 3 Supplementary table 1. Only patient age and gender appeared to be significantly associated with severity outcomes by Chi-Square Test (Appendix 3 Supplementary Table 2).

DISCUSSION

We performed a systematic review and meta-analysis to elucidate the proportion of cancer diagnoses among COVID-19 patients, and whether cancer patients had more severe outcomes compared to the general population. We also looked at the patient characteristics that may be associated with severe clinical outcomes among these patients. To our knowledge, this is the largest meta-analysis that directly addresses the clinical outcomes of COVID-19 patients with cancer, including the relative risks of severity and death as compared to general COVID-19 patients.

Firstly, we found that the pooled proportion of cancer patients among COVID-19 cases was 4%. This result was similar to the proportion rate from an earlier meta-analysis[61] of 11 studies with a proportion of 2%, but higher than the 0.2% described by the Chinese Centre for Disease Control and Prevention Group[2]. We found that the cancer proportion was higher among older patients and in male patients with COVID-19. In the subsequent IPD analysis, it was also advanced age and male gender that were significantly associated with severity of COVID-19 infection. We also noted that the cancer proportion among the COVID-19 case series from Hubei Province was 3% (Appendix 3 Supplementary Figure 1), which was higher than the estimated cancer prevalence of 0.45% in the region[30]. However, cancer proportion can vary across geographical locations and specialist medical centres.

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We have shown that COVID-19 patients with cancer had significantly more severe outcomes compared to the general COVID-19 patients, with a pooled RR of 1.49 (95% CI 1.18-1.87, p=0.001). The pooled estimated proportion of severe COVID-19 outcomes was higher among cancer patients in our meta-analysis compared to the largest cancer cohort data (CCC19)[62]. Overall, the proportion of cancer patients with severe COVID-19 outcomes varied across studies but remained high, highlighting the vulnerability of cancer patients to SARS-CoV-2 infection.

We also observed that lung cancer was the most common cancer among those affected by COVID-19 in Chinese studies (table 2) and this may be due to the fact that lung cancer is the most common cancer in China, rather than a direct relation to the risk of COVID-19 infection[63]. A few case series noted that patients with lung cancers had significantly worse outcomes compared to non-lung cancer cohorts[16,55], which was also supported by data from TERAVOLT[64], an international database of thoracic cancers (not included in our meta-analysis as it is a single tumour registry). TERAVOLT demonstrated 33% mortality and high morbidity among patients with thoracic malignancies, but with low ICU admission rates, presumably partly due to resource allocations and patient prognosis. However, in our meta-analysis, we did not find the relative risk of severity to be significant for lung cancers (vs other tumour types) or stage IV cancers (vs stage I-III) across four studies. This was also echoed by our IPD analysis, where lung cancer and the stage of cancer were not significantly associated with severe outcomes. This could reflect that lung cancer or stage IV cancer did not necessarily confer a higher risk of severe COVID-19 infection, but given the small number of included studies for this part of the analysis, this conclusion should be cautiously interpreted.

Separately, haematological cancers also showed severe outcomes with high mortality rates in multiple cohort studies[16,65].The available individual data in our study were not large enough to analyse the influence of modality of systemic treatment (targeted therapy, immunotherapy or chemotherapy), surgery or radiotherapy on the outcome of COVID-19 infections. Recent studies did not consistently find active systemic therapy to be associated with risk of severe COVID-19 infection[66] though some studies suggest that patients on recent chemotherapy[30] or immunotherapy[16] were at higher risk. These questions remain unanswered and require further collaborative efforts to identify risk factors of severe COVID-19 infection among cancer patients. One suggestion is to have standardised reporting

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format for cancer COVID-19 cohorts including specific tumour type, stage, date of last chemotherapy or systemic treatment, co-morbidities, active vs past cancer diagnosis and also the severity and mortality data for each subgroup.

Overall, the pooled relative risk of death among the cancer patients was 1.41 (95% CI 1.15-1.73), compared to general COVID-19 patients. However, pooled mortality risk of cancer patients with COVID-19 should be interpreted with caution due to the limited sample size. Some case series also had patients who remained hospitalised or with incomplete patient outcome at the time of reporting. CFR were lower at 5.6% among cancer COVID-19 patients in a large Chinese report[2] and 11.1% from a tertiary centre in New York, USA[3]. Notably, among selected COVID-19 mortality reports from Italy and USA, around 20-30% of deaths had a history of cancer[33,67]. We would encourage mortality data to be taken from database reports instead of small case series from single centres.

In light of the severity of COVID-19 in cancer patients, efforts should be undertaken to reduce nosocomial exposure in cancer patients, as nosocomial infections may contribute to incidence of COVID-19 in oncology patients[54]. The decision to attend clinics and continue non-urgent systemic therapy should be made judiciously, especially in regions with high burden of disease transmission. This has been echoed by various oncology bodies, including American Society of Clinical Oncology (ASCO) and European Society of Medical Oncology (ESMO)[68]. Given the higher risk of severe outcomes and mortality among cancer patients, it is important that cancer patients with COVID-19 infection should be monitored closely and decisions for ICU support discussed early. ICU outcomes should also be an important question for future retrospective studies in view of the ethical and clinical considerations of putting patients with advanced cancer on mechanical ventilation, especially in countries with significant outbreak and limited ICU resources. In light of the recent advances in COVID-19 vaccine efforts, another area of active discussion is that of the risk and benefit of vaccination for cancer patients, especially those who are deemed to be significantly immunocompromised. Our data shows that there is risk of severe COVID-19 infection and risk of death among cancer patients compared to general population. Hence, this factor should be strongly considered in the individualised decision for vaccination.

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The limitations of this meta-analysis include the heterogeneity and retrospective nature of the case studies, including reporting and selection biases. A few studies were reported from the same Hospitals and hence there may be overlapping patients although efforts were taken to reduce this. During screening of papers by the authors, we have ensured that the analysed studies were not reported from the same hospital during the same period of admission (Appendix 2 Supplementary table 1). The small numbers and incomplete death outcomes at the time of reporting also restricted our mortality analysis. Due to the limited clinical data, we could not adjust for co-morbidities and other characteristics that may contribute to severe disease. It should also be noted that severity outcomes were defined inconsistently across some studies but efforts were taken to indicate the different definitions in table 1. Of note, some earlier studies also included clinically diagnosed COVID-19 cases.

Whilst we acknowledge other risk factors such as age[48,69,70], gender[71], cardiovascular risk factors[70,72,73] including diabetes mellitus[74,75] and hypertension[76] remain important in COVID-19 infection, this is the largest meta-analysis to show that malignancy is also a significant risk factor for severe disease and mortality. However, we did not identify other tumour or treatment-related factors associated with severe outcomes among cancer patients with COVID-19. We also did not stratify for gender, age and co-morbidities in our analysis due to limited available data for pooled analysis on severity and mortality. As more data becomes available in the literature, further analyses may be performed to discern the clinical outcomes of cancer patients with COVID-19 infection, with specific attention to underlying lung cancers, haematological cancers and the effects of various systemic therapies or other treatment modalities. CCC19, UKCCMP and TERAVOLT are some examples of collaborative efforts on the outcomes of cancer patients with COVID-19 infection, though a limitation is that there is a lack of comparison against general population. Information on COVID-19 patients with cancer should also be collected in more detail and with consistency across studies, to allow further meta-analyses to ascertain the outcomes of cancer patients with COVID-19 infection.

CONCLUSION

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Given that COVID-19 infection may persist in the foreseeable future, it is imperative to continue active research and data collection on the effect of SARS-CoV-2 on oncology patients. Due to the clinical burden of their disease, oncology patients will continue to require systemic chemotherapy and urgent care at medical centres[77]. As such, treating physicians need to be able to make more discerning decisions on the choice of systemic treatment and escalation of care.

Overall, we found through meta-analysis that the severity and mortality rates of COVID-19 in cancer patients were higher than that in the general population. However, the dataset remains small and should be interpreted with caution. We hope that with collaborative efforts, comprehensive patient outcomes of COVID-19 patients, and specifically, of cancer patients, can be collated to better understand the disease trajectory and outcomes of COVID-19.

Figure legends

Figure 1: Flow chart of study identification and exclusion for data analysis
Figure 2: Random effects pooled proportion of cancer patients among patients diagnosed with COVID-19
Figure 3: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by median/mean age ≥ 60 vs < 60 years)
Figure 4: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by male ≥ 60% or < 60%)
Figure 5a: Random effects pooled relative risk of death for cancer COVID-19 patients versus total population of COVID-19 patients
Figure 5b: Random effects pooled relative risk for severe outcome of cancer COVID-19 patients versus total population of COVID-19 patients
Figure 6a: Random effects pooled relative risk for severe outcome of lung cancer vs other cancer patients infected with COVID-19
Figure 6b: Random effects pooled relative risk for severe outcome of stage IV cancer vs non-stage IV cancer patients infected with COVID-19

Declarations

a. Contributorship statement
SH and VSY conceived the idea of the study. SH, VSY and QZ contributed to data collection. ST, SH and YYS contributed to data analysis. SH, QZ, JC, GWYC and VSY contributed to the manuscript writing. MLKC and CX contributed to clinical data. MLKC, YYS and VSY provided guidance.

b. Competing interests
No conflicts of interest to declare.
Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

c. Funding
No funding or acknowledgement to declare.

d. Data sharing statement
Data are available in a public, open access repository.

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e. Ethics statement
Ethics approval was not required as no human or animal subjects were involved. No consent taking was required.

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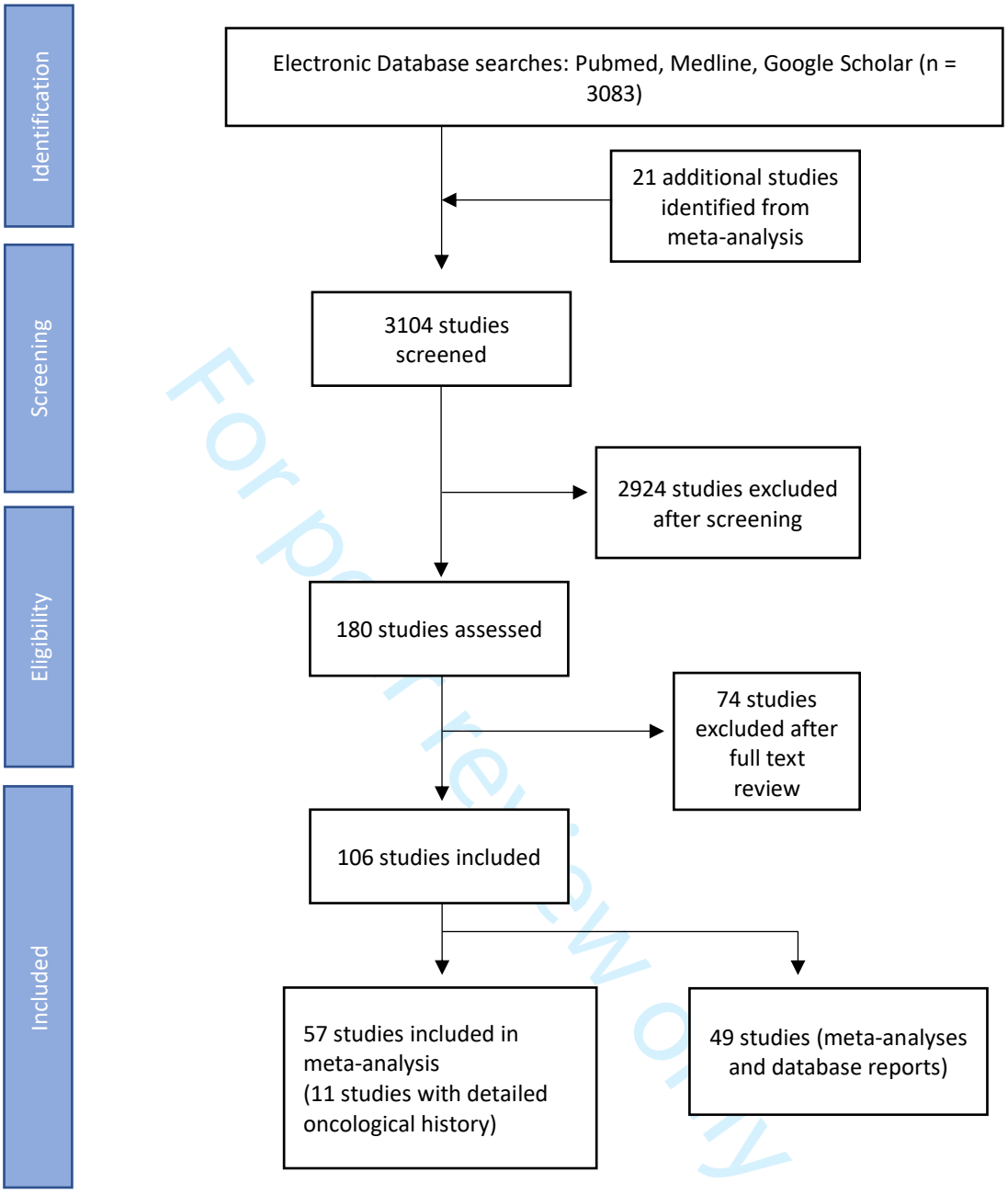
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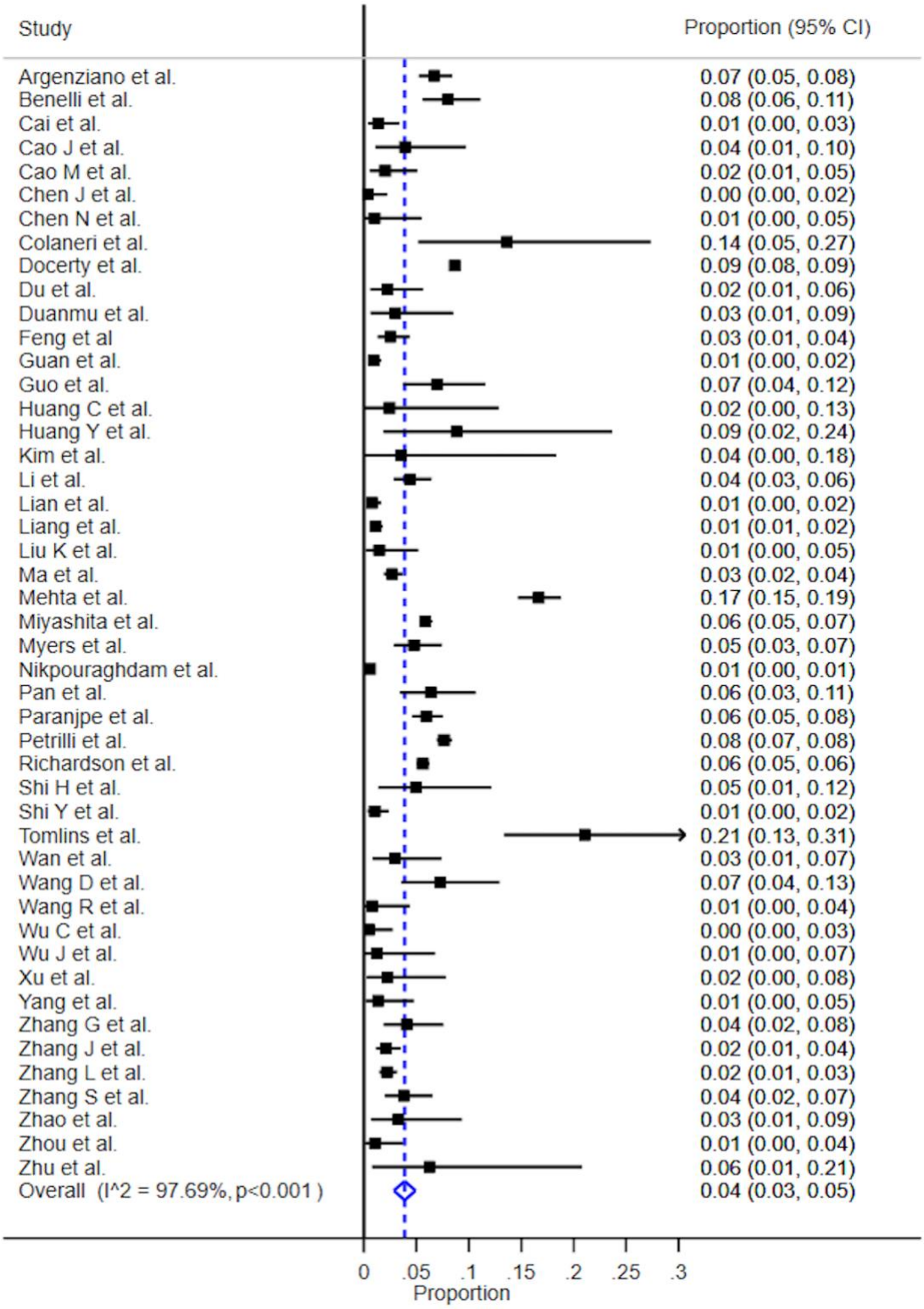
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Figure 1: Flow chart of study identification and exclusion for data analysis



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Figure 2: Random effects pooled proportion of cancer patients among patients diagnosed with COVID-19



*10 papers were excluded in the proportion calculations (did not represent overall COVID-19 populations)

Figure 3: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by median/mean age ≥ 60 vs < 60 years)

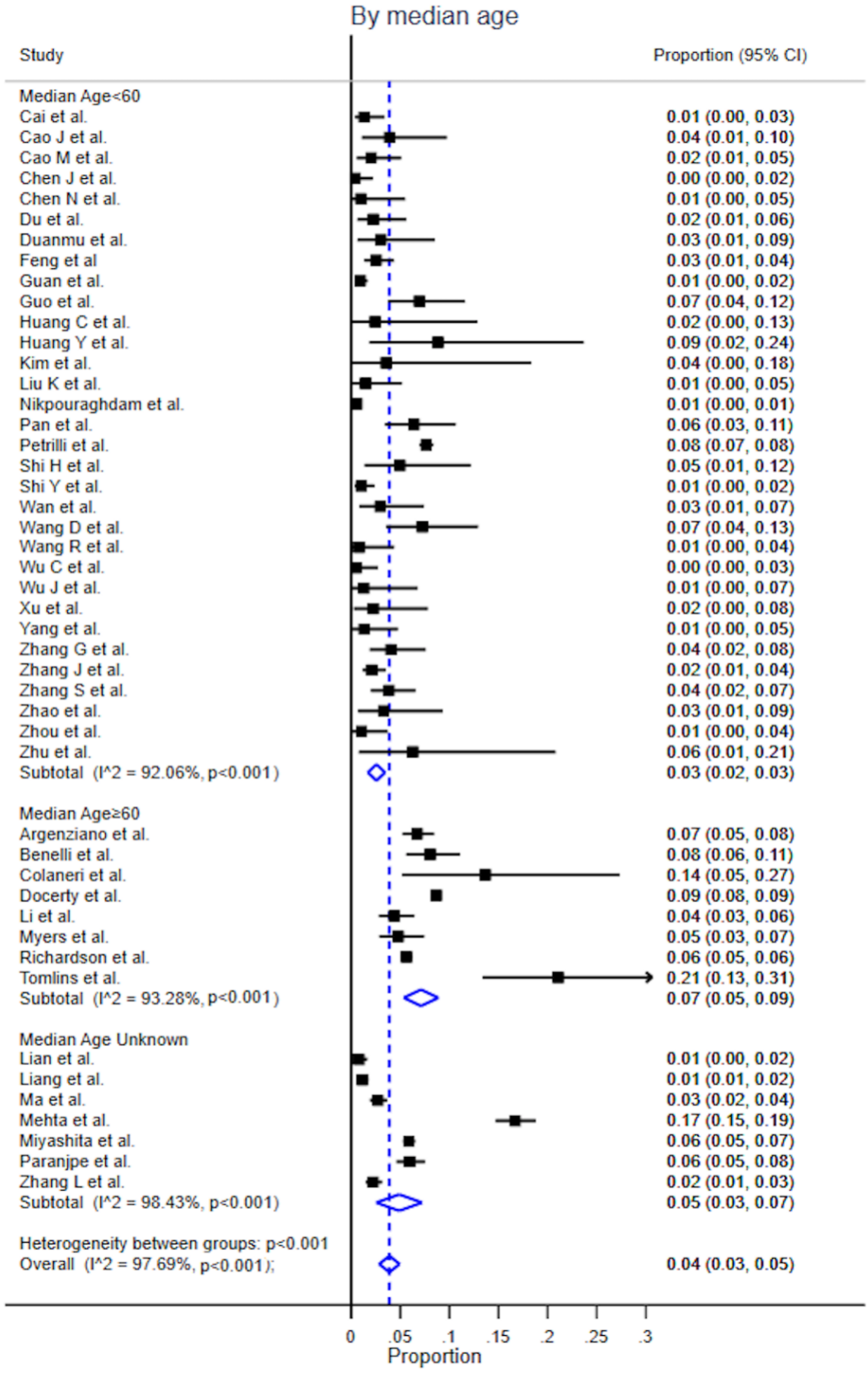


Figure 4: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by male ≥ 60% or < 60%)

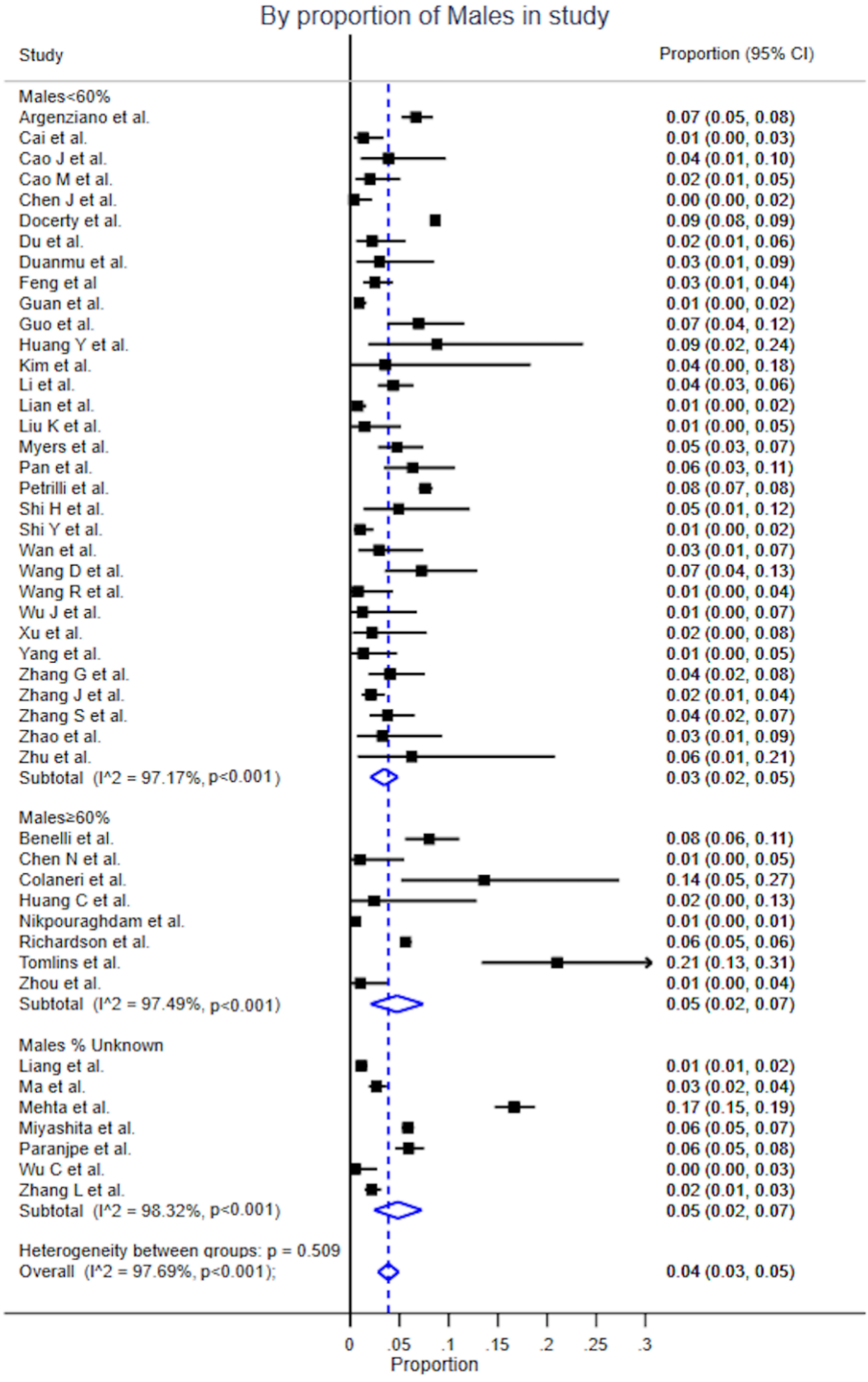


Figure 5a: Random effects pooled relative risk of death for cancer COVID-19 patients versus total population of COVID-19 patients

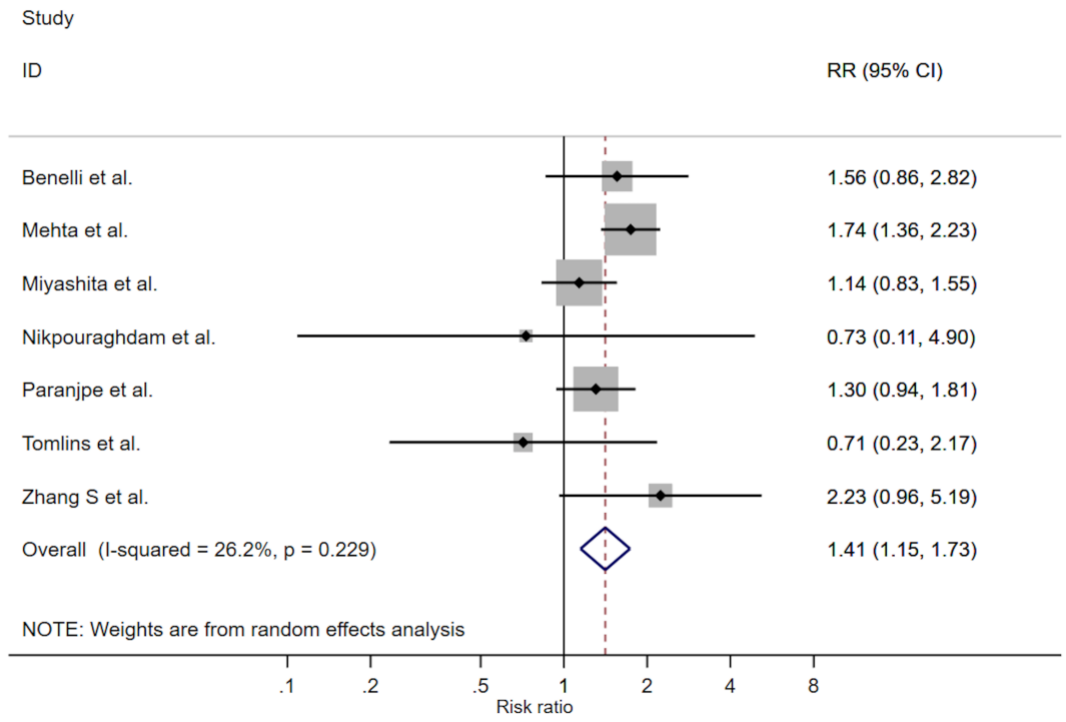


Figure 5b: Random effects pooled relative risk for severe outcome of cancer COVID-19 patients versus total population of COVID-19 patients

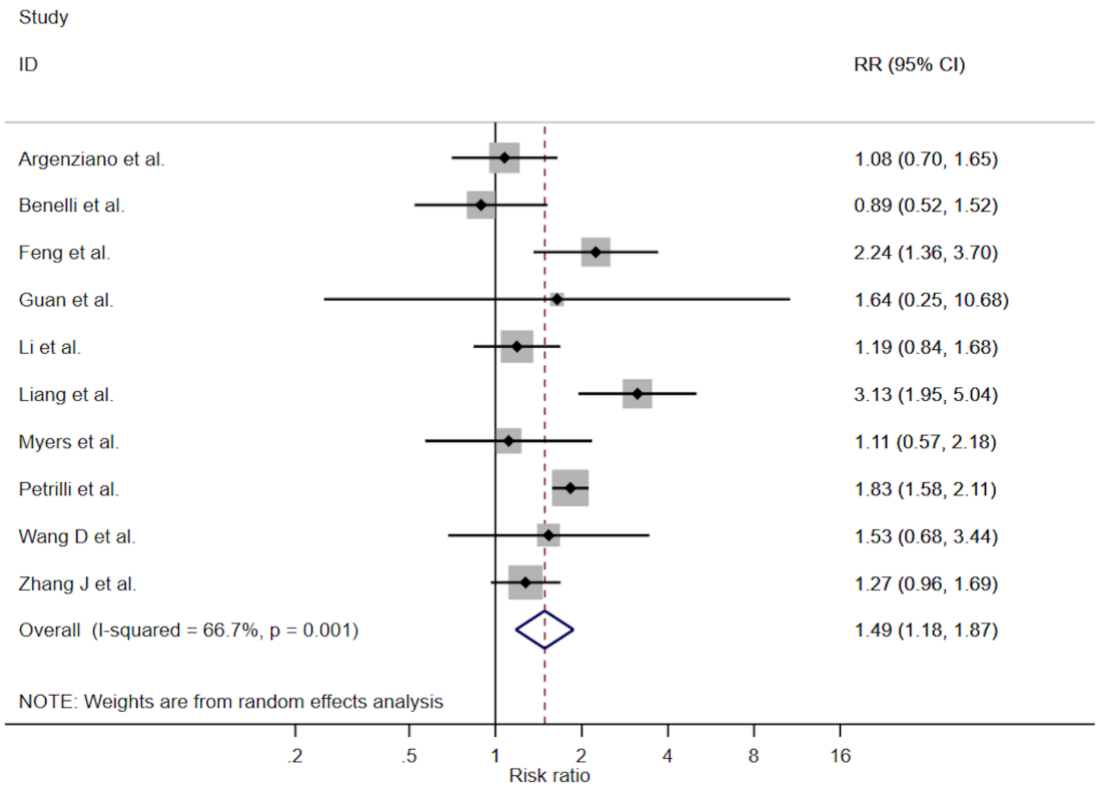


Figure 6a: Random effects pooled relative risk for severe outcome of lung cancer vs other cancer patients infected with COVID-19

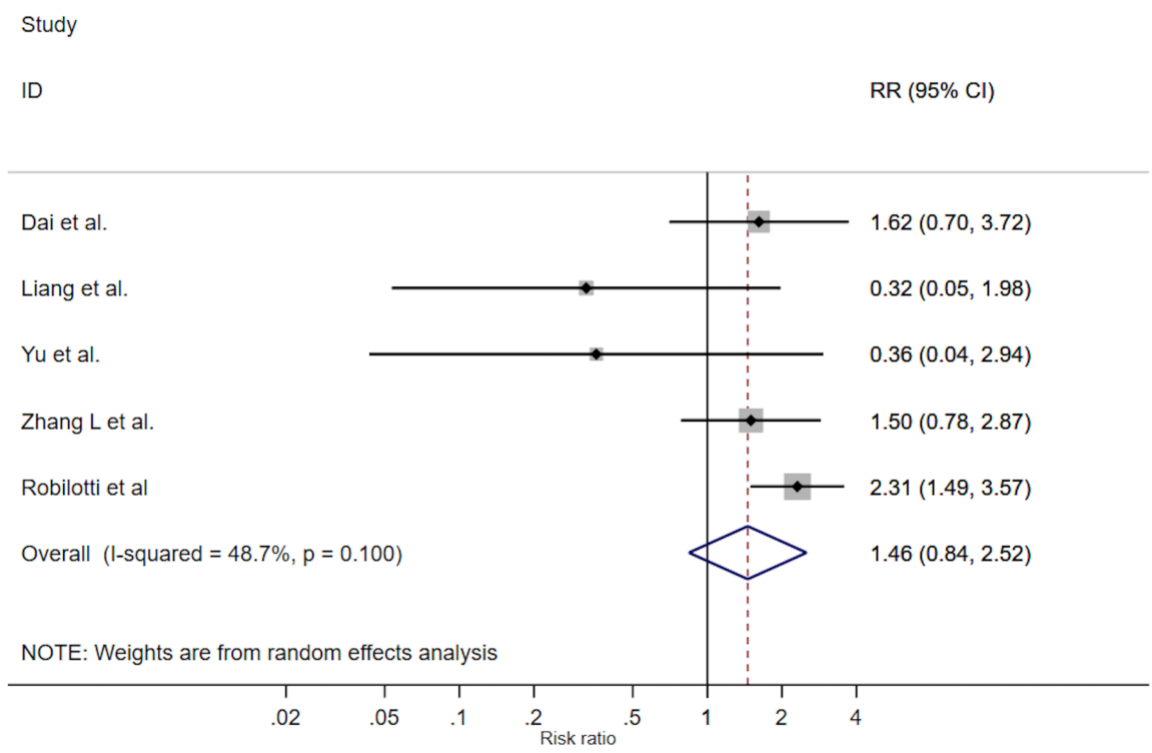
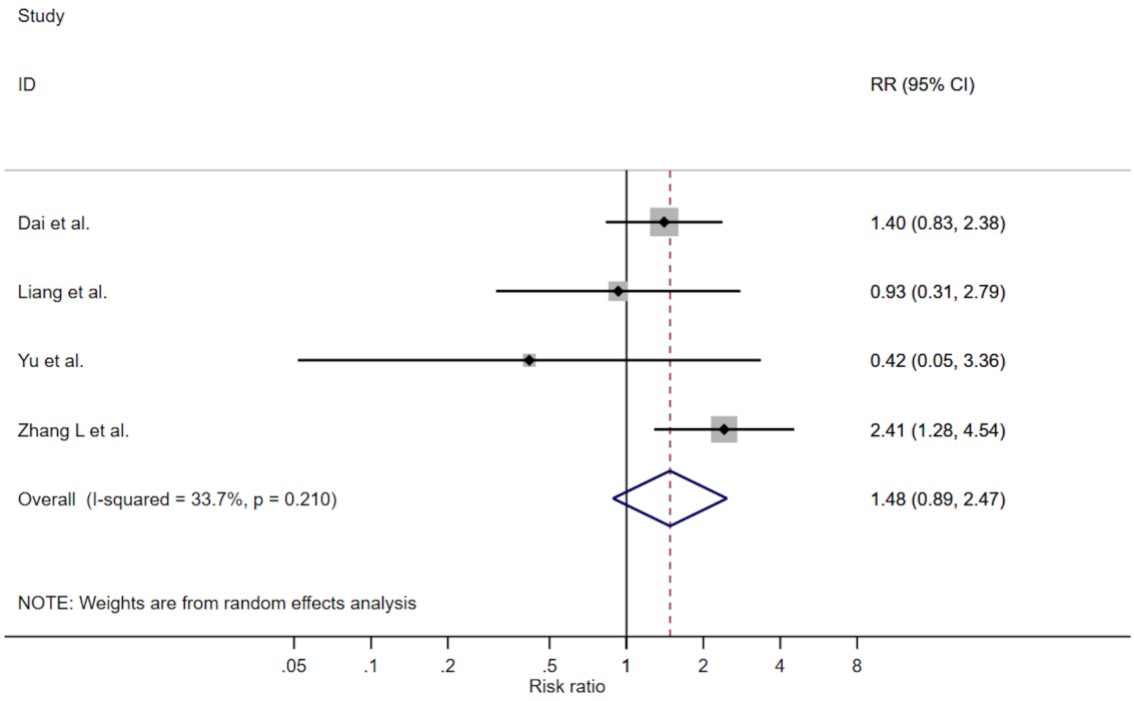


Figure 6b: Random effects pooled relative risk for severe outcome of stage IV cancer vs non-stage IV cancer patients infected with COVID-19



Appendix 1

A. 2019 Novel Coronavirus Disease (COVID-19) Diagnostic criteria 5th edition (5 Feb 2020)

Other provinces except Hubei province

1. Suspected cases by clinical criteria

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Computed tomography (CT) changes for viral pneumonia

c) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease. Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor all of the above clinical manifestations.

2. Laboratory confirmed cases

Suspected cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by real-time reverse transcriptase polymerase chain reaction (RT-PCR) assay.

2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

Hubei province

1. Suspected cases

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease.

Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor any two of clinical manifestations

2. Clinically diagnosed cases

CT chest changes that are consistent with atypical viral pneumonia.

3. Laboratory confirmed cases

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Suspected cases or clinically diagnosed cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by RT-PCR assay.
2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

B. National Health Commission Guideline on Severity Classification of COVID-19 (China)

Type	Criteria
Mild	the clinical symptoms were light, and there was no sign of pneumonia on imaging
Moderate	with fever, respiratory tract and other symptoms, imaging suggests pneumonia
Severe	coincide with any of the following: (1) respiratory distress, respiration rate (RR) ≥ 30 times / min; (2) the oxygen saturation ≤ 93% in the resting state; (3) PaO_2 / FiO_2 ≤ 300 mmHg (1mmHg = 0.133 kPa)
Critically ill	coincide with any of the following: (1) respiratory failure occurs and mechanical ventilation is required; (2) shock; (3) the patient develops other organ failure and needs ICU monitoring and treatment

(A and B released in The Diagnosis and Treatment Protocol for Novel Coronavirus Pneumonia, Version 5, National Health Commission, China)

C. IDSA/ATS Guidelines on Severe Pneumonia 2007

2007 Infectious Diseases Society of America/American Thoracic Society Criteria for Defining Severe Community-acquired Pneumonia
Validated definition includes either one major criterion or three or more minor criteria
Minor criteria
Respiratory rate ≥ 30 breaths/min
Pa_{O2}/Fi_{O2} ratio ≤ 250
Multilobar infiltrates
Confusion/disorientation
Uremia (blood urea nitrogen level ≥ 20 mg/dl)
Leukopenia* (white blood cell count < 4,000 cells/μl)
Thrombocytopenia (platelet count < 100,000/μl)
Hypothermia (core temperature < 36°C)
Hypotension requiring aggressive fluid resuscitation
Major criteria
Septic shock with need for vasopressors
Respiratory failure requiring mechanical ventilation

*Due to infection alone (i.e., not chemotherapy induced).

Metlay et al. Diagnosis and Treatment of Adults with Community-acquired Pneumonia. An Official Clnical Practice Guideline of the American Thoracic Society and Infectious Diseases Society of America. AJRCCM 2019.

D. Search criteria

MeSH terms:
COVID-19, coronavirus, nCoV, SARS-CoV-2, characteristics, comorbidity, mortality, death, risk, oncology, cancer, malignancy, neoplasm, tumor, tumour, and carcinoma.

Search strategy:
(COVID-19 OR coronavirus OR nCoV OR SARS-CoV-2) AND (characteristics OR comorbidity OR mortality OR death OR risk) AND (oncology OR cancer OR malignancy OR neoplasm OR tumor OR tumour OR carcinoma)

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E. PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N.A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4-5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4, appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Appendix 2
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4-5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4-5

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4-5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4-5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-10, Appendix 2-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	2, 6-12 Appendix 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	6-12, Appendix 3
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

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Appendix 2

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Supplementary Table 1: Case series included in meta-analysis

No.	Author	Title	No.	Type of study	Outcome measured	Quality of study	Hospital	Location
1	Argenziano et al	Characterization and clinical course of 1000 patients with coronavirus disease 2019 in New York: retrospective case series	1000	Single centre retrospective cohort study	Severity, death	7	New York-Presbyterian/Columbia University Irving Medical Center	New York, USA
2	Assaad et al	High mortality rate in cancer patients with symptoms of COVID-19 with or without detectable SARS-COV-2 on RT-PCR	302	Single centre retrospective cohort study	Death	7	Comprehensive Cancer Center of Lyon	Lyon France
3	Benelli et al. 2020	SARS-CoV-2 comorbidity and outcome in hospitalized patients in Crema, Italy.	539	Single centre prospective cohort study	Severity	7	General Hospital of Crema	Crema, Italy
4	Cai et al. 2020	COVID-19 in a designated infectious diseases hospital outside Hubei Province, China.	298	Single centre retrospective cohort study	Severity	7	Third People's Hospital of Shenzhen	Shenzhen, China
5	Cao M et al. 2020	Clinical Features of Patients Infected with the 2019 Novel Coronavirus (COVID-19) in Shanghai, China.	198	Single centre retrospective cohort study (not-peer reviewed, pre-print)	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 15 Feb 2020	Shanghai, China
6	Cao J et al. 2020	Clinical features and short-term outcomes of 102 patients with coronavirus disease 2019 in Wuhan, China.	102	Single centre retrospective cohort study	Death	7	Wuhan University Zhongnan Hospital in Wuhan, China, 3 Jan to 1 Feb 2020	Wuhan, Hubei, China
7	Chen J et al. 2020	Clinical progression of patients with COVID-19 in Shanghai, China.	249	Single centre retrospective cohort study	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 6 Feb, 2020	Shanghai, China
8	Chen N et al. 2020	Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study.	99	Single centre retrospective cohort study	Severity	7	Jinyintan Hospital in Wuhan, 1 to 20 Jan 2020	Wuhan, Hubei, China
9	Colaneri et al. 2020	Clinical characteristics of coronavirus disease (COVID-19) early findings from a teaching hospital in Pavia, North Italy, 21 to 28 February 2020.	44	Single centre retrospective cohort study	Severity	6	Fondazione IRCCS Policlinico San Matteo, 21 to 28 Feb 2020	Pavia, Italy
10	Dai et al. 2020	Patients with cancer appear more vulnerable to SARS-COV-2: a multi-center study during the COVID-19 outbreak.	641	Multicentre retrospective matched cohort study	Severity	8	14 hospitals in Wuhan, China, 1 Jan to 24 Feb 2020 (Zhongnan, Tongji, Renmin Hospital etc)	Wuhan, Hubei, China
11	Docherty et al 2020	Features of 20 133 UK patients in hospital with covid-19 using the ISARIC WHO Clinical Characterisation Protocol: prospective observational cohort study	20133	Multi centre prospective cohort study	Severity, death	7	Hospitals in England and Wales, UK	UK
12	Du et al. 2020	Predictors of Mortality for Patients with COVID-19 Pneumonia Caused by SARS-CoV-2: A Prospective Cohort Study.	179	Single centre prospective cohort study	Death	7	Wuhan Pulmonary Hospital, 25 Dec 2019 to 7 Feb 2020	Wuhan, Hubei, China
13	Duanmu et al. 2020	Characteristics of Emergency Department Patients With COVID-19 at a Single Site in Northern California: Clinical Observations and Public Health Implications.	100	Single centre observational cross-section study	Admission from ED, Death	6	Stanford University School of Medicine, 4 to 23 March 2020	California, USA
14	Feng et al. 2020	COVID-19 with Different Severity: A Multi-center Study of Clinical Features	476	Single centre retrospective cohort study	Severity and death	7	3 hospitals in Wuhan, Shanghai and Anhui, 1 Jan to 15 Feb 2020	China
15	Guan et al. 2020	Clinical Characteristics of Coronavirus Disease 2019 in China.	1099	Multicentre retrospective cohort study	Severity and death	7	552 hospitals across China, December 11, 2019, and January 29 2020	China
16	Guo et al. 2020	Cardiovascular Implications of Fatal Outcomes of Patients With Coronavirus Disease 2019 (COVID-19)	187	Single centre retrospective cohort study	Severity and death	7	Seventh Hospital of Wuhan City, from 23 Jan to 23 Feb 2020	Wuhan, Hubei, China
17	Huang C et al. 2020	Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China.	41	Single centre prospective cohort study	Severity	7	Jinyintan Hospital till 2 Jan 2020	Wuhan, Hubei, China
18	Huang Y et al. 2020	Clinical characteristics of laboratory confirmed positive cases of SARS-CoV-2 infection in Wuhan, China: A retrospective single center analysis.	34	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital, 21 Dec to 28 Jan 2020	Wuhan, Hubei, China
19	Kim et al. 2020	Clinical Course and Outcomes of Patients with Severe Acute Respiratory Syndrome Coronavirus 2 Infection: a Preliminary Report of the First 28 Patients from the Korean Cohort Study on COVID-19.	28	Multicentre retrospective cohort study	Severity	6	11 Hospitals in Korea	Korea
20	Kuderer et al. 2020	Clinical impact of COVID-19 on patients with cancer (CCC19): a cohort study	928	Multicentre retrospective cohort study	Severity, death	7	US, Canada, Spain Hospitals	US, Canada, Spain
21	Lee LY et al.	COVID-19 prevalence and mortality in patients with cancer and the effect of primary tumour subtype and patient demographics: a prospective cohort study	800	Multicentre prospective cohort study	Severity, death	7	UK Hospitals	UK
22	Li et al. 2020	Risk factors for severity and mortality in adult COVID-19 inpatients in Wuhan.	548	Single centre retrospective cohort study	Severity and death	7	Tongji Hospital, 26 Jan to 5 Feb 2020	Wuhan, Hubei, China
23	Lian et al. 2020	Analysis of Epidemiological and Clinical features in older patients with Corona Virus Disease 2019 (COVID-19) out of Wuhan.	788	Multicentre retrospective cohort study	Severity	8	Zhejiang Province, 17 Jan to 12 Feb 2020	Wuhan. Hubei, China
24	Liang et al. 2020	Cancer patients in SARS-CoV-2 infection: a nationwide analysis in China.	1590	Multicentre retrospective cohort study	Severity and death	8	575 hospitals in China as of 31 Jan 2020	China
25	Liu et al. 2020	Clinical characteristics of novel coronavirus cases in tertiary hospitals in Hubei Province.	137	Multicentre retrospective cohort study	Death	7	9 hospitals in Hubei, 30 Dec 2019 to 24 Jan 2020 (Tongji Hosp, other hospitals outside Wuhan)	Hubei, China
26	Ma et al. 2020	Clinical characteristics and prognosis in cancer patients with COVID-19: A single center's retrospective study.	1380	Single centre retrospective cohort study	Severity and death	8	Renmin Hospital of Wuhan University, 1 Jan to 30 March 2020	Wuhan, Hubei, China
27	Mehta et al. 2020	Case Fatality Rate of Cancer Patients with COVID-19 in a New York Hospital System.	218	Single centre retrospective cohort study (peer-reviewed, pre-print)	Death	7	Montefiore Medical Center and Albert Einstein College of Medicine, 18 March to 8 April 2020	New York, USA
28	Miyashita et al. 2020	Do Patients with Cancer Have a Poorer Prognosis of COVID-19? An Experience in New York City.	5688	Single centre retrospective cohort study	Severity and death	8	Mount Sinai Health System, 1 March to 6 April 2020	New York, USA
29	Myers et al. 2020	Characteristics of Hospitalized Adults With COVID-19 in an Integrated Health Care System in California.	377	Multicentre retrospective cohort study	Severity	7	Kaiser Permanente Northern California (KPNC) health system, 1 to 31 March 2020	California, USA
30	Nikpouraghdam et al. 2020	Epidemiological characteristics of coronavirus disease 2019 (COVID-19) patients in IRAN: A single center study.	2964	Single centre retrospective cohort study	Death	7	Baqiyatallah Hospital, 19 Feb to 15 April 2020	Tehran, Iran
31	Pan et al. 2020	Clinical Characteristics of COVID-19 Patients With Digestive Symptoms in Hubei, China: A Descriptive, Cross-Sectional, Multicenter Study	204	Multicentre retrospective cross sectional study	Severity and death	7	3 hospitals in Hubei province, including Wuhan Hanan Hospital, Wuhan Union Hospital, and Huanggang Central	Hubei, China

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							Hospital, 18 Jan to 28 Feb 2020	
32	Paranjpe et al. 2020	Clinical Characteristics of Hospitalized Covid-19 Patients in New York City.	1078	Multicentre retrospective cohort study (not peer reviewed, pre-print)	Death	8	Mount Sinai Health System Hospitals, 27 Feb to 2 April 2020	New York, USA
33	Petrilli et al 2020	Factors associated with hospital admission and critical illness among 5279 people with coronavirus disease 2019 in New York City: prospective cohort study	5279	Single centre prospective cohort study	Severity, death	7	NYU Langone Health	New York, USA
34	Pinato et al. 2020	Clinical Portrait of the SARS-CoV-2 Epidemic in European Patients with Cancer	890	Multicentre retrospective cohort study	Severity, death	7	Multiple Hospitals in Europe	Europe
35	Robilotti et al. 2020	Determinants of COVID-19 disease severity in patients with cancer	423	Single centre retrospective cohort study	Severity, death	7	MSKCC	New York, USA
36	Richardson et al. 2020	Presenting Characteristics, Comorbidities, and Outcomes Among 5700 Patients Hospitalized With COVID-19 in the New York City Area.	5700	Multicentre retrospective cohort study	Severity and death	8	Northwell Health Hospitals, 1 March to 4 April 2020	New York, USA
37	Shi H et al. 2020	Radiological findings from 81 patients with COVID-19 pneumonia in Wuhan, China: a descriptive study.	81	Multicentre retrospective cohort study	Not defined	6	Wuhan Jinyintan hospital or Union Hospital of Tongji Medical College, 20 Dec to 23 Jan 2020	Wuhan, Hubei, China
38	Shi Y et al. 2020	Host susceptibility to severe COVID-19 and establishment of a host risk score: findings of 487 cases outside Wuhan.	487	Multicentre retrospective cohort study	Not defined	7	Hospitals in Zhejiang Province, until 17 Feb 2020	Zhejiang, China
39	Tomlins et al. 2020	Clinical features of 95 sequential hospitalised patients with novel coronavirus 2019 disease (COVID-19), the first UK cohort	95	Single centre retrospective cohort study	Severity and death	7	North Bristol NHS Trust, 10 to 30 March 2020	UK
40	Wan et al. 2020	Clinical features and treatment of COVID-19 patients in northeast Chongqing.	135	Single centre retrospective cohort study	Not defined	7	Chongqing University Three Gorges Hospital, 23 Jan to 8 Feb 2020	Chongqing, Sichuan, China
41	Wang D et al. 2020	Clinical Characteristics of 138 Hospitalized Patients With 2019 Novel Coronavirus-Infected Pneumonia in Wuhan, China.	138	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital of Wuhan University, 1 to 28 Jan 2020	Wuhan, Hubei, China
42	Wang R et al. 2020	Epidemiological and clinical features of 125 Hospitalized Patients with COVID-19 in Fuyang, Anhui, China.	125	Single centre retrospective cohort study	Severity	7	NO.2 People's Hospital of Fuyang City 20 Jan to 9 Feb 2020.	Anhui, China
43	Wu C et al. 2020	Risk Factors Associated With Acute Respiratory Distress Syndrome and Death in Patients With Coronavirus Disease 2019 Pneumonia in Wuhan, China.	201	Single centre retrospective study	Severity and death	8	Jinyintan Hospital in Wuhan, China, 25 Dec to 26 Jan 2020	Wuhan, Hubei, China
44	Wu J et al. 2020	Clinical Characteristics of Imported Cases of COVID-19 in Jiangsu Province: A Multicenter Descriptive Study.	80	Multicentre retrospective cohort study	Severity	7	Three Grade IIIA hospitals of Jiangsu, 22 Jan to 14 Feb 2020	Jiangsu, China
45	Xu et al. 2020	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2.	90	Single centre retrospective study	Severity	7	Guangzhou Eighth People's Hospital, 23 Jan to 4 Feb 2020	Guangzhou, China
46	Yang et al. 2020	Clinical characteristics and imaging manifestations of the 2019 novel coronavirus disease (COVID-19):A multi-center study in Wenzhou city, Zhejiang, China.	149	Multicentre retrospective cohort study	Severity	7	Three tertiary hospitals of Wenzhou, 17 Jan to 10 Feb 2020	Wenzhou, Zhejiang, China
47	Yang K et al 2020	Clinical characteristics, outcomes, and risk factors for mortality in patients with cancer and COVID-19 in Hubei, China: a multicentre, retrospective, cohort stud	205	Multientre retrospective cohort study	Severity, death	7	9 Hospitals in Wuhan, 11 Jan to 18 March (overlap with other case series)	Wuhan, China
48	Yarza et al	SARS-CoV-2 infection in cancer patients undergoing active treatment: analysis of clinical features and predictive factors for severe respiratory failure and death	63	Single centre retrospective cohort study	Death	6	Hospital in Madrid	Madrid, Spain
49	Yu et al. 2020	SARS-CoV-2 Transmission in Patients With Cancer at a Tertiary Care Hospital in Wuhan, China.	1524	Single centre retrospective study	Severity and death	6	Zhongnan Hospital, 30 Dec to 17 Feb 2020	Wuhan, Hubei, China
50	Zhang G et al. 2020	Clinical features and short-term outcomes of 221 patients with COVID-19 in Wuhan. China.	221	Single centre retrospective cohort study	Severity	7	Zhongnan Hospital of Wuhan, 2 Jan to 10 Feb 2020	Wuhan, Hubei, China
51	Zhang J et al. 2020	Risk factors for disease severity, unimprovement, and mortality in COVID-19 patients in Wuhan, China.	663	Single centre retrospective cohort study	Severity	7	Renmin Hospital of Wuhan University, 11 Jan to 2 Feb 2020	Wuhan, Hubei, China
52	Zhang L et al. 2020	Clinical characteristics of COVID-19-infected cancer patients: a retrospective case study in three hospitals within Wuhan, China.	1276	Single centre retrospective cohort study	Severity and death	8	Tongji Sino-French New Town Hospital, Union Red Cross Hospital, and Union West Hospital, 13 Jan to 26 Feb 2020	Wuhan, Hubei, China
53	Zhang N et al. 2020	A multicentre study of coronavirus disease 2019 outcomes of cancer patients in Wuhan, China	1548	Multicentre retrospective observational study (peer reviewed, pre-print)	Severity, death	7	Zhongnan Hospital of Wuhan University, the Fifth hospital of Wuhamn, the seventh hospital of Wuhan and Wuhan Hankou hospital, 5 Jan to 18 Feb 2020	Wuhan, Hubei, China
54	Zhang S et al. 2020	Short Term Outcomes and Risk Factors for Mortality in Patients with COVID-19 in Wuhan, China: A Retrospective Study.	315	Single centre retrospective cohort study	Death	7	Wuhan Union Hospital West Area, 12 Jan to 3 Feb 2020	Wuhan, Hubei, China
55	Zhao et al. 2020	Clinical characteristics of patients with 2019 coronavirus disease in a non-Wuhan area of Hubei Province, China: a retrospective study.	91	Single centre retrospective cohort study	Severity	7	Jingzhou Central Hospital, 16 Jan to 10 Feb 2020	Jingzhou, Hubei, China
56	Zhou et al. 2020	Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study.	191	Multicentre retrospective cohort study	Death	7	Jinyintan Hospital and Wuhan Pulmonary Hospital, 29 Dec to 31 Jan 2020	Wuhan, Hubei, China
57	Zhu et al. 2020	Initial clinical features of suspected coronavirus disease 2019 in two emergency departments outside of Hubei, China.	32	Single centre retrospective cohort study	Not defined	6	First Affiliated Hospital of USTC, 24 Jan to 20 Feb 2020	Hefei, China

Supplementary Table 2: Meta-analyses and database reports on COVID-19 (not included in analysis)

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No.	Authors	Title:
1.	Desai et al.	COVID-19 and Cancer: Lessons From a Pooled Meta-Analysis
2.	Emami et al.	Prevalence of Underlying Diseases in Hospitalized Patients with COVID-19: a Systematic Review and Meta-Analysis
3.	Wang B et al.	Does comorbidity increase the risk of patients with COVID-19: evidence from meta-analysis
4.	Cao Y et al.	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2: A systematic review and meta-analysis
5.	Hu Y et al.	Prevalence and severity of corona virus disease 2019 (COVID-19): A systematic review and meta-analysis
6.	Yang J et al.	Prevalence of comorbidities and its effects in patients infected with SARS-CoV-2: a systematic review and meta-analysis
7.	Fu L et al.	Clinical characteristics of coronavirus disease 2019 (COVID-19) in China: A systematic review and meta-analysis
8.	Li LQ et al.	COVID-19 patients' clinical characteristics, discharge rate, and fatality rate of meta-analysis
9.	Zhu J et al.	Clinicopathological Characteristics of 8697 Patients With COVID-19 in China: A Meta-Analysis
10.	Zheng Z et al.	Risk factors of critical & mortal COVID-19 cases: A systematic literature review and meta-analysis
11.	Rodriguez-Morales et al.	Clinical, laboratory and imaging features of COVID-19: A systematic review and meta-analysis
12.	Ofori-asenso et al.	Cancer is associated with severe disease in COVID-19 patients: a systematic review and meta-analysis
13.	Khan et al.	Effects of underlying morbidities on the occurrence of deaths in COVID- 19 patients: A systematic review and meta-analysis (in-press)
14.	Moujaess et al.	Cancer patients and research during COVID-19 pandemic: A systematic review of current evidence
15.	Sun P et al.	Clinical characteristics of hospitalized patients with SARS-CoV-2 infection: A single arm meta-analysis
16.	Jin JM et al.	Gender Differences in Patients with COVID-19 Focus on Severity and Mortality
17.	Deng G et al.	Clinical determinants for fatality of 44,672 patients with COVID-19
18.	Wu and Mcgoogan	Characteristics of and Important Lessons From the Coronavirus Disease 2019 (COVID-19) Outbreak in China: Summary of a Report of 72 314 Cases From the Chinese Center for Disease Control and Prevention
19.	Onder G, Rezza G, Brusaferro S.	Case-Fatality Rate and Characteristics of Patients Dying in Relation to COVID-19 in Italy
20.	Guan et al	Comorbidity and its impact on 1590 patients with Covid-19 in China: A Nationwide Analysis
21.	Adams ML, Katz DL, Grandpre J.	Population-Based Estimates of Chronic Conditions Affecting Risk for Complications from Coronavirus Disease, United States
22.	Kang YJ et al	Mortality Rate of Infection With COVID-19 in Korea From the Perspective of Underlying Disease
23.	Barek et al	Impact of age, sex, comorbidities and clinical symptoms on the severity of COVID-19 cases: A meta-analysis with 55 studies and 10014 cases.
24.	Fuentes-Antrás J	A snapshot of COVID-19 infection in patients with solid tumors.
25.	Sng C et al	Cancer History and Systemic Anti-Cancer Therapy Independently Predict COVID-19 Mortality: A UK Tertiary Hospital Experience.
26.	Pinto et al	SARS-CoV-2 Positive Hospitalized Cancer Patients during the Italian Outbreak: The Cohort Study in Reggio Emilia. Biology (Basel).
27.	Sun L et al	Rates of COVID-19-related Outcomes in Cancer compared to non-Cancer Patients. (Pre-print)
28.	Anantharaman et al	SARS-CoV-2 Clinical Outcomes in Patients with Cancer in a Large Integrated Health Care System in Northern California.
29.	Zarifkar P et al	Clinical Characteristics and Outcomes in Patients with COVID-19 and Cancer: a Systematic Review and Meta-analysis.
30.	Jarahzadeh MH et al	Cancer and Coronavirus Disease (COVID-19): Comorbidity, Mechanical Ventilation, and Death Risk.
31.	ElGohary GM et al	The risk and prognosis of COVID-19 infection in cancer patients: A systematic review and meta-analysis.
32.	Russell B et al	Factors Affecting COVID-19 Outcomes in Cancer Patients: A First Report From Guy's Cancer Center in London.
33.	Duarte MBO et al	Outcomes of COVID-19 Patients under Cytotoxic Cancer Chemotherapy in Brazil.
34.	Gagandeep Brar et al	COVID-19 Severity and Outcomes in Patients With Cancer: A Matched Cohort Study
35.	Yang B et al	Risk of Coronavirus Disease Incidence, Severe Presentation, and Mortality in Patients with Lung Cancer.
36.	Motamedi F et al	Clinical characteristics, outcomes, and risk factors for mortality in hospitalized patients with COVID-19 and cancer history: a propensity score-matched study.
37.	Cheruiyot I et al	Cancer is associated with coronavirus disease (COVID-19) severity and mortality: A pooled analysis.
38.	Antikchi MH et al	The Risk and Prevalence of COVID-19 Infection in Colorectal Cancer Patients: a Systematic Review and Meta-analysis.

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39.	Hultcrantz M et al	COVID-19 infections and outcomes in patients with multiple myeloma in New York City: a cohort study from five academic centers.
40.	Fillmore NR et al	Prevalence and outcome of COVID-19 infection in cancer patients:a national Veterans Affairs study.
41.	Venkatesulu BP et al	A systematic review and meta-analysis of cancer patients affected by a novel coronavirus.
42.	Joharatnam-Hogan et al	Outcomes of the 2019 novel coronavirus in patients with or without a history of cancer: a multi-centre North London experience.
43.	Alpert N et al	Clinical Course of Cancer Patients With COVID-19: A Retrospective Cohort Study.
44.	Ali J et al	COVID-19-positive cancer patients undergoing active anticancer treatment: An analysis of clinical features and outcomes.
45.	Jee J et al	Chemotherapy and COVID-19 Outcomes in Patients With Cancer
46.	Liu H et al	The effect of anticancer treatment on cancer patients with COVID-19: A systematic review and meta-analysis.
47.	Park R et al	Sex-bias in COVID-19-associated illness severity and mortality in cancer patients: A systematic review and meta-analysis.
48.	Asokan I et al	The COVID-19 Pandemic and its Impact on the Cardio-Oncology Population.
49.	Rogado J et al	Characteristics and outcome of lung cancer patients and Covid-19 infection in a second-level hospital during the first 3 months of the pandemic.
50.	Park R et al	Association of active oncologic treatment and risk of death in cancer patients with COVID-19: a systematic review and meta-analysis of patient data

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The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort
   a) truly representative of the average cancer COVID-19 cases in the community *****
   b) somewhat representative of the average cancer COVID-19 cases in the community *****
   c) selected group of users eg nurses, doctors
   d) no description of the derivation of the cohort

2) Selection of the non-exposed cohort
   a) drawn from the same community as the exposed cohort *****
   b) drawn from a different source
   c) no description of the derivation of the non-exposed cohort

3) Ascertainment of exposure
   a) secure record (eg medical records) *****
   b) structured interview *****
   c) written self-report
   d) no description

4) Demonstration that outcome of interest was not present at start of study
   a) yes *****
   b) no

Comparability

5) Comparability of cohorts on the basis of the design or analysis
   a) study controls for median age *****
   b) study controls for any additional factors such as other co-morbidities *****

Outcome

6) Assessment of outcome
   a) independent blind assessment *****
   b) record linkage *****
   c) self-report
   d) no description

7) Was follow-up long enough for outcomes to occur
   a) yes (until discharge or death or at least 1 month, whichever come first) *****
   b) no

8) Adequacy of follow up of cohorts
   a) complete follow up - all subjects accounted for *****
   b) subjects lost to follow up unlikely to introduce bias - small number lost - > 80 % follow up, or description provided of those lost) *****
   c) follow up rate < 60 % and no description of those lost
   d) no statement

Case-control studies

1) Is the case definition adequate?
   a) yes, with independent validation *****
   b) yes, eg record linkage or based on self reports
   c) no description

2) Representativeness of the cases
   a) consecutive or obviously representative series of cases *****
   b) potential for selection biases or not stated

3) Selection of Controls
   a) community controls *****

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b) hospital controls
c) no description

4) Definition of Controls
a) no history of disease (endpoint) *
b) no description of source

Comparability

5) Comparability of cases and controls on the basis of the design or analysis
a) study controls for median age *
b) study controls for any additional factor * (co-morbidities)

Exposure

6) Ascertainment of exposure
a) secure record (eg surgical records) *
b) structured interview where blind to case/control status *
c) interview not blinded to case/control status
d) written self report or medical record only
e) no description

7) Same method of ascertainment for cases and controls
a) yes *
b) no

8) Non-Response rate
a) same rate for both groups *
b) non respondents described
c) rate different and no designation

Supplementary Table 3: The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

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N	Authors	Selection				Comparability		Outcome			Study quality
	Cohort studies	1 Exposed cohort truly representative	2 Non-exposed cohort drawn from same community	3 Ascertainment of exposure	4 Outcome of interest not present at start	5A Cohorts comparable ie. Median age	5B Cohorts comparable on other co-morbidities	6 Quality of outcome assessment	7 Follow-up long enough for outcomes to occur	8 Complete accounting for cohorts	
	Case control studies	1 Case definition adequate	2 Representativeness with the cases	3 Selection of controls	4 Definition of controls	5A Case and control comparable on the basis of analysis ie median age	5B Case and control comparable on other co-morbidities	6 Ascertainment of exposure	7 Same method of ascertainment for cases and controls	8 Non-Response rate	
1	Argenziano et al.	*	*	*	*			*	*	*	7
2	Assaad et al.	*	*	*	*			*	*	*	7
3	Benelli et al.	*	*	*	*			*	*	*	7
4	Cai et al.	*	*	*	*			*	*	*	7
5	Cao M et al.	*	*	*	*			*	*	*	7
6	Cao J et al.	*	*	*	*			*	*	*	7
7	Chen J et al	*	*	*	*			*	*	*	7
8	Chen N et al.	*	*	*	*			*	*	*	7
9	Colaneri et al.	*	*	*	*			*		*	6
10	Dai et al.	*	*	*	*	*		*	*	*	8
11	Docherty et al.	*	*	*	*			*	*	*	7
12	Du et al.	*	*	*	*			*	*	*	7
13	Duanmu et al.	*	*	*	*			*		*	6
14	Feng et al.	*	*	*	*			*	*	*	7
15	Guan et al.	*	*	*	*			*	*	*	7
16	Guo et al.	*	*	*	*			*	*	*	7
17	Huang C et al.	*	*	*	*			*	*	*	7
18	Huang Y et al.	*	*	*	*			*		*	6
19	Kim et al.	*	*	*	*			*		*	6
20	Kuderer et al.	*	*	*	*			*	*	*	7
21	Lee LY et al.	*	*	*	*			*	*	*	7
22	Li et al.	*	*	*	*			*	*	*	7
23	Lian et al.	*	*	*	*	*		*	*	*	8
24	Liang et al.	*	*	*	*		*	*	*	*	8
25	Liu et al.	*	*	*	*			*	*	*	7
26	Ma et al.	*	*	*	*	*		*	*	*	8
27	Mehta et al.	*	*	*	*			*	*	*	7
28	Miyashita et al.	*	*	*	*	*		*	*	*	8
29	Myers et al.	*	*	*	*			*	*	*	7
30	Nikpouraghdam et al.	*	*	*	*			*	*	*	7
31	Pan et al.	*	*	*	*			*	*	*	7
32	Paranjpe et al.	*	*	*	*		*	*	*	*	8
33	Petrilli et al.	*	*	*	*			*	*	*	7
34	Pinato et al.	*	*	*	*			*	*	*	7
35	Robilotti et al.	*	*	*	*			*	*	*	7
36	Richardson et al.	*	*	*	*	*		*	*	*	8
37	Shi H et al.	*	*	*	*			*		*	6
38	Shi Y et al.	*	*	*	*	*		*	*		7
39	Tomlins et al	*	*	*	*			*	*	*	7
40	Wan et al.	*	*	*	*			*	*	*	7
41	Wang D et al.	*	*	*	*			*	*		6
42	Wang R et al.	*	*	*	*			*	*	*	7
43	Wu C et al.	*	*	*	*	*		*	*	*	8
44	Wu J et al.	*	*	*	*			*	*	*	7
45	Xu et al.	*	*	*	*			*	*	*	7
46	Yang et al.	*	*	*	*			*	*	*	7
47	Yang K et al.	*	*	*	*			*	*	*	7
48	Yarza et al.	*	*	*	*			*	*		6
49	Yu et al.	*		*	*			*	*	*	6
50	Zhang G et al.	*	*	*	*			*	*	*	7
51	Zhang J et al.	*	*	*	*			*	*	*	7
52	Zhang L et al.	*	*	*	*		*	*	*	*	8
53	Zhang N et al.	*	*	*	*			*	*	*	7
54	Zhang S et al.	*	*	*	*			*	*	*	7
55	Zhao et al.	*	*	*	*			*	*	*	7
56	Zhou et al.	*	*	*	*			*	*	*	7
57	Zhu et al.	*	*	*	*				*	*	6

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Appendix 3

Supplementary Table 1: List of 230 patients with available individual patient data and oncological history

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No	Gender	Age	Co-morbidities	Length of dx of cancer in years	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	M	83	No	4	Bladder ca	Postoperative routine follow-up	Y	NA	N
2	M	87	DM, HTN	2	Colon ca	Postoperative routine follow-up	Y	NA	N
3	M	67	No	0	Adrenal neoplasm	3 weeks after operation	Y	NA	N
4	F	68	No	8	Breast ca	Surgical resection plus adjuvant chemotherapy 8 years ago	Y	NA	N
5	M	58	No	7	CRCC	Recurrence, on immunotherapy	Y	NA	Y
6	F	62	No	4	Breast ca	Surgical resection plus adjuvant chemotherapy 4 years ago	Y	NA	N
7	M	56	No	5	Rectal ca	Surgical resection plus adjuvant chemotherapy 5 years ago	N	NA	N
8	M	53	No	16	Colon ca	Surgical resection plus adjuvant chemotherapy 16 years ago	N	NA	N
9	F	63	No	1	Papillary thyroid ca	Postoperative, in TSH inhibition therapy	N	NA	N
10	M	47	No	1	Lung adenoca	In chemotherapy for advanced tumor	N	NA	Y
11	F	52	No	0.5	Breast ca	Postoperative, loss of chemotherapy information	N	NA	N
12	M	47	No	1	Lymphoma	N.A.	N	NA	NA
13	M	80	DM, HTN, CVS	4	Bladder ca	Postoperative, no chemotherapy information	Y	NA	N
14	M	79	COPD	5	Colorectal ca	Surgical resection plus adjuvant radiotherapy 5 years ago	Y	NA	N
15	M	63	No	1	Lung adenoca	In chemotherapy for advanced tumour	Y	NA	Y
16	F	58	CKD	2	Lung carcinoma in situ	Postoperative routine follow-up	N	NA	N
17	M	58	No	2	Lung adenoca	Postoperative, on targeted therapy	N	NA	N
18	F	55	No	6	Lung adenoca	Advanced, on targeted therapy	N	NA	Y

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No	Gender	Age	Co-morbidities	Length of dx of cancer	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	M	NA	NA	NA	NSCLC	Concurrent TKI with RT	N	Y	N
2	M	NA	NA	NA	NSCLC	Week 5 of chemoradiotherapy	Y	N	N
3	F	NA	NA	NA	NSCLC	Pleural effusion, BSC	N	Y	Y
4	M	NA	NA	NA	NSCLC	Follow up at 2 year post chemoRT	Y	N	N
5	M	NA	NA	NA	NSCLC	After first cycle of doce, cis and sintilimab	N	Y	Y
6	M	NA	NA	NA	NSCLC	After third cycle of pem/carbo and pembro	N	Y	Y
7	M	NA	NA	NA	NSCLC	Surgery (April 2019) followed by adjuvant doce/cis	N	Y	N
8	M	NA	NA	NA	Rectal ca	BSC	N	Y	Y
9	M	NA	NA	NA	Colon ca	Newly diagnosed	N	Y	NA
10	M	NA	NA	NA	Pancreatic ca	BSC	Y	N	Y
11	F	NA	NA	NA	Breast ca	Adjuvant radiotherapy	N	Y	N
12	M	NA	NA	NA	Urothelial ca	BSC	N	Y	Y

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No	Gender	Age	Co-morbidities	Length of dx of cancer in months	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	M	56	CHB, cirrhosis	26	Liver ca	Stage IV, recurrence with bone mets, bone RT with 14 days	Y	NA	Y
2	M	74	DM	3	Esophageal ca	Stage IV, chemotherapy 1 month ago	Y	NA	Y
3	M	66	HTN, CHD	11	Lung SCC	Surgical resection plus adjuvant chemoRT 10m ago, immune therapy 3m ago, gem and nivo within 14 days	Y	NA	Y
4	M	65	No	23	Lung adenoca	Stage IV, chemo for advanced tumour. Last cycle of pemetrexed was within 14 days	Y	NA	Y
5	M	65	No	4	Esophageal ca	Stage III, surgical resection plus adjuvant chemo and RT	N	NA	N
6	F	48	DM, HTN	5	Serous ca of ovary	Surgical resection plus adjuvant chemo 4m ago, 3 weeks after THBSO	N	NA	N
7	M	54	No	13	Laryngocarcinoma	4 weeks after operation	N	NA	N
8	M	80	HTN, CHD	1	Colorectal ca	3 weeks after operation and neoadjuvant chemoRT	N	NA	N
9	M	65	No	13	Esophageal ca	Surgical resection plus adjuvant chemo and RT 11m ago	Y	NA	0
10	F	70	No	124	Lung adenoca	Postoperative, recurrence. On targeted therapy	Y	NA	Y
11	F	64	No	26	Lung adenoca	Surgical resection plus adjuvant chemo, on EGFR TKI for advanced tumour	N	NA	Y
12	F	24	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 1 year ago	Y	NA	N
13	F	70	DM, HTN	148	Breast ca	Surgical resection plus adjuvant chemo and RT 12 year ago	Y	NA	N
14	F	59	No	62	Lung adenoca	Stage IV, on EGFR TKI treatment	Y	NA	Y
15	F	68	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 20m ago	N	NA	N
16	F	60	No	9	Endometrial ca	Surgical resection 7m ago, on adjuvant chemo 3 weeks ago	N	NA	N
17	M	65	No	26	Small cell lung ca	Maintenance chemo for advanced tumour	N	NA	Y
18	M	57	No	15	Lung SCC	Surgical resection plus adjuvant chemoRT 1 year ago	N	NA	N
19	F	65	No	15	Cervical ca	Surgical resection and adjuvant chemo 1 week ago	N	NA	N
20	M	47	No	51	NPC	Concurrent chemoRT 4m ago	Y	NA	N
21	M	77	No	62	Prostate ca	Postop and adjuvant RT 5 years ago, on follow up	Y	NA	N
22	M	71	DM	75	Esophageal ca	Surgical resection plus adjuvant chemo and RT 6 years ago	Y	NA	N
23	M	65	CHD	26	Prostate ca	Postoperative and adjuvant RT 2 years ago, on follow up	Y	NA	N
24	M	75	COPD	148	Rectal ca	Post-operative routine follow up	Y	NA	N
25	M	56	CHB	73	Liver ca	Surgical resection plus adjuvant chemo and RT 6 years ago, recurrence on adjuvant chemo within 4 weeks	N	NA	Y
26	M	69	No	124	Laryngocarcinoma	Postoperative and adjuvant RT 10 years ago, routine follow up	N	NA	N
27	F	52	No	8	Gastric ca	Postoperative, 4 cycles of chemo and adjuvant RT, routine follow up	N	NA	N
28	M	60	No	13	Testicular ca	Sequential chemo and RT, last cycle of chemo was within 4 weeks	Y	NA	N

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No	Gender	Age	Co-morbidities	Date of diagnosed cancer	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	M	69	Intestinal obstruction	2019.12	Peritoneal cancer	N.A	Y	Y	N
2	M	66	Jaundice, Abdominal infection, chest infection	2019.11	Gastric cancer	N.A	N	Y	Y
3	M	49	No	2020.01	Esophageal SCC	N.A	Y	Y	NA
4	M	65	No	2018.01	Gastric cancer	N.A	N	Y	N
5	M	72	Asthma	2016.09	Lung cancer	Targeted therapy (EGFR-TKI) and PD-1 immunotherapy	N	Y	Y
6	M	73	Chronic bronchitis	2019.11	Laryngeal SCC	Radiotherapy	Y	Y	Y
7	M	69	CHD, HTN	2019.08	Lung cancer	Radiotherapy	N	Y	N
8	F	55	No	2019.11	Glioblastoma	Radiotherapy and chemotherapy	N	Y	Y
9	M	74	Duodenal ulcer, COPD, HTN, CHD	2019.08	Esophageal cancer	Radiotherapy	N	Y	Y
10	M	55	Hepatic encephalopathy, GI bleeding, Chronic viral hepatitis, Anemia	2014.01	Hepatocellular carcinoma	Radiotherapy	N	Y	Y

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11	M	66	HTN, DM	2019.03	Lung SCC	Chemotherapy	Y	Y	Y
12	M	67	Chronic bronchitis, HTN, Atrial fibrillation, Peptic ulcer, CKD	2019.12	Esophageal SCC	Surgery	Y	N	N
13	M	64	Cerebral infarction	2020.01	Myelodysplastic syndrome	N.A	Y	Y	NA
14	M	83	HTN	2016	Meningioma	N.A	Y	Y	NA
15	M	73	HTN	2020.01	Lung cancer	Surgery	Y	Y	N
16	F	48	No	2019.06	Breast cancer	N.A	N	Y	N
17	M	57	Prostatic hyperplasia	2016.02	Lung cancer	Radiotherapy and Targeted therapy (EGFR-TKI)	N	Y	Y
18	M	61	No	2019.12	Lung SCC	DP chemotherapy and PD-1 immunotherapy	Y	N	Y
19	F	84	HTN, DM, lumbar compression fracture	2019.06	Lung cancer	Surgery	Y	N	N
20	M	69	Abdominal aortic aneurysm	2019.09	Lung adenocarcinoma	Radiotherapy and chemotherapy	N	Y	N
21	M	69	Prosopalgia	2019.11	Lung SCC	Chemotherapy and PD-1 immunotherapy	N	Y	Y
22	M	65	Chronic viral hepatitis	2019.04	Lung cancer	N.A	N	Y	N
23	F	76	HTN, CKD, hypothyroidism, cardiac insufficiency	Unknown	Multiple myeloma	N.A	Y	N	NA
24	M	54	Incomplete ileus	2020.01	Rectal cancer	Surgery	N	Y	N
25	F	78	HTN,, COPD, CHD, CKD	2014.06	Chronic granulocytic leukemia	N.A	Y	N	N
26	M	61	HTN, DM	2018.05	Bladder cancer	N.A	Y	Y	N
27	F	50	No	2017.05	Lung cancer	N.A	N	Y	N
28	F	87	HTN	2009	Gastric cancer	N.A	Y	Y	N
29	F	55	No	2018	Cervical cancer	N.A	N	Y	N
30	F	70	HTN	Unknown	Colon cancer	N.A	N	Y	N
31	F	69	No	Unknown	Gastric cancer	N.A	N	Y	NA
32	F	59	No	2009	Breast cancer	N.A	N	Y	N
33	F	48	No	2019.06	Breast cancer	Radiotherapy	N	Y	N
34	M	57	No	2015.12	Lung adenocarcinoma	Concurrent targeted therapy（EGFR-TKI）with radiotherapy	N	Y	Y
35	M	69	Prosopalgia	2019.11	Lung SCC	Radiotherapy and Chemotherapy	N	Y	Y
36	M	69	Chronic viral hepatitis	2019.04	Lung cancer	Chemotherapy and PD-1 immunotherapy	Y	Y	Y
37	M	69	CKD	2019.09	Lung adenocarcinoma	Chemotherapy and Radiotherapy	N	Y	N
38	M	66	No	2019.08	Lung adenocarcinoma	PD-1 immunotherapy	Y	Y	Y
39	M	65	CHD	2019.04	Bladder cancer	Radiotherapy and Chemotherapy	Y	N	Y
40	M	81	HTN	2019.09	Cancer of unknown primary	PD-1 immunotherapy	Y	N	NA
41	F	46	Bacteremia	2019.01	Rectal cancer	N.A	N	Y	N
42	M	76	Intestinal obstruction	2018.11	Rectal cancer	N.A	Y	Y	Y
43	F	58	No	2020.1.2	Ovarian cancer	Chemotherapy	N	Y	N
44	F	74	DM	2020.1.12	Cervical cancer	Chemotherapy	N	Y	N
45	M	26	No	Unknown	Testicular cancer	N.A	N	Y	N
46	F	66	HTN, HLD	Unknown	Breast cancer	N.A	Y	Y	N
47	F	52	DM	Unknown	Thyroid carcinoma	N.A	N	Y	N
48	M	50	Pulmonary tuberculosis, Abdominal infection, Biliary infection, Liver abscess	2015	Thyroid carcinoma	N.A	N	Y	N
49	F	66	No	2019	Thyroid carcinoma	N.A	Y	Y	NA
50	M	83	No	2019.05	Prostate cancer	N.A	N	Y	Y
51	F	29	No	2013.05	Thyroid carcinoma	N.A	N	Y	N
52	F	32	No	2018.01	Thyroid carcinoma	N.A	N	Y	N
53	M	76	No	2018	Prostate cancer	N.A	N	Y	NA
54	F	80	HTN, HLD, Phlebothrombosis, DM	2019.01	Lung adenocarcinoma	N.A	Y	Y	N
55	M	72	HTN	2019	Lung cancer	N.A	Y	N	Y
56	F	57	No	2019.12	Breast cancer	Chemotherapy	N	Y	N
57	F	69	Chronic Hepatitis	2018	Thyroid cancer	N.A	N	Y	N
58	F	60	No	2012	Cervical cancer	N.A	N	Y	N
59	F	72	HTN,,heart disease	2013.06	Breast cancer	N.A	N	Y	NA
60	M	75	HTN	2009	Colon cancer	N.A	N	Y	NA
61	M	67	Hypothyroidism	Unknown	Liver cancer	N.A	N	Y	NA
62	F	52	HTN, CKD	2016.03	Endometrial cancer	N.A	N	Y	N
63	M	64	No	2019.08	Gallbladder carcinoma	N.A	N	Y	NA
64	M	67	No	1994	Nasopharynx cancer	N.A	N	Y	NA
65	M	63	HTN	2010	Kidney cancer	N.A	N	Y	NA
66	M	34	No	2017	Thyroid cancer	N.A	N	Y	NA
67	F	59	No	2019.07	Colon cancer	Chemotherapy	N	Y	NA
68	M	57	Cirrhosis,diabetes	2020.01	Nasopharynx cancer	Chemotherapy	N	Y	NA
69	M	62	No	2015	Bladder cancer	N.A	N	Y	NA
70	F	66	No	2015	Thyroid cancer	N.A	N	Y	NA
71	M	60	No	2015	Multiple myeloma	N.A	N	Y	NA
72	F	64	No	Unknown	Gastric stromal tumor	N.A	N	Y	NA
73	F	66	No	2004	Ocular malignancy	N.A	N	Y	NA
74	M	61	CHD, HTN	2020.01	Multiple myeloma	N.A	N	Y	NA
75	F	53	No	2019.9	Multiple myeloma	N.A	N	Y	NA
76	M	61	DM, HTN	Unknown	Malignant lymphoma	N.A	Y	N	Y
77	F	54	No	2019.03	Lung carcinoma	N.A	Y	N	N
78	F	55	No	2016	Breast cancer	N.A	N	Y	N
79	F	38	No	2019	Breast cancer	N.A	N	Y	N
80	F	50	No	2019	Rectal carcinoma	N.A	Y	N	N
81	M	67	Bronchitis	2017.12	Esophageal carcinoma	N.A	Y	Y	N
82	F	37	No	2018	Thyroid carcinoma	N.A	N	Y	N
83	F	45	No	2019.11	Cervical carcinoma	Surgery and Radiotherapy	Y	Y	N
84	M	35	No	2019.07	Glioma	N.A	Y	Y	NA
85	F	57	No	2019.11	Cervical carcinoma	Surgery	N	Y	N
86	F	49	No	2019.11	Renal carcinoma	N.A	N	Y	N
87	M	62	HTN, Tuberculosis	2018.08	Esophageal carcinoma	N.A	N	Y	N
88	M	70	CHD, Choledochectasia	2019.06	Gastric carcinoma	N.A	N	Y	N
89	M	44	No	2019.12	Thyroid carcinoma	N.A	N	Y	N
90	F	72	Cerebrovascular disease	2017.06	Breast cancer	N.A	N	Y	N
91	F	57	HTN	2019.02	Breast cancer	N.A	Y	Y	N
92	F	56	No	2019.04	Cervical carcinoma	N.A	Y	Y	N
93	M	58	No	2019.06	Rectal carcinoma	Chemotherapy	Y	Y	Y
94	M	71	CHD, HTN	2019.1	Lung SCC	Chemotherapy	Y	Y	Y
95	F	67	DM HTN	2019.12	Breast cancer	Surgery	Y	Y	N
96	M	70	No	2019.11	gastric carcinoma	Chemotherapy	Y	Y	N
97	M	70	HTN	2014	Esophageal carcinoma	N.A	Y	Y	N
98	F	42	HTN	2020.01	Pancreatic cancer	Surgery	N	Y	N
99	M	43	No	2011	gastric carcinoma	N.A	Y	Y	N
100	M	69	No	2015	Lung cancer	N.A	Y	Y	N
101	M	62	Cerebral hemorrhage, Uremia, DM, HTN	2020.01	Lung cancer	N.A	N	Y	N
102	M	37	HLD	2018.06	Sigmoid colon cancer	N.A	Y	Y	N

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103	F	70	CHD, HTN	2002	Thyroid cancer	N.A	Y	Y	N
104	M	51	No	2015	Leukemia	Targeted therapy (Imatinib Mesylate)	N	Y	N
105	F	55	HTN	2019.07	Nasopharyngeal lymphoma	N.A	N	Y	N

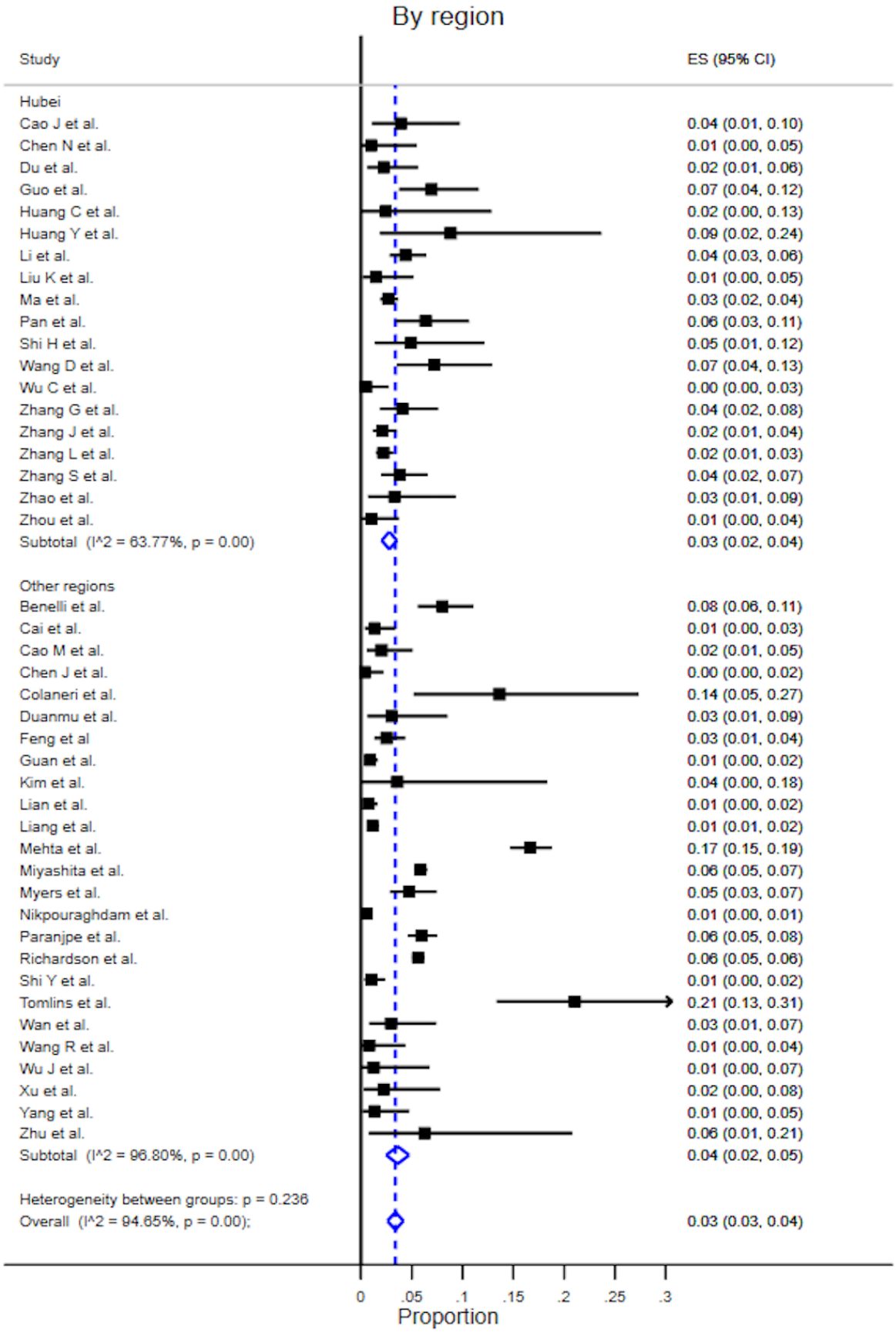
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No	Gender	Age	Co-morbidities	Length of dx of cancer in months	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	F	48	No	46	Breast	RT+endocrinotherapy	N	Y	NA
2	M	57	No	53	Lung	targeted therapy/RT	N	Y	NA
3	M	65	Liver cirrhosis	50	Lung	chemotherapy+immunotherapy	N	Y	NA
4	M	66	No	53	Lung	chemotherapy+immunotherapy+targeted therapy	N	Y	NA
5	M	69	No	48	Lung	chemotherapy+RT+immunotherapy	N	Y	NA
6	M	69	No	32	Lung	chemotherapy+RT	Y	N	NA
7	M	61	No	36	Urinary	chemotherapy	Y	N	NA
8	M	37	No	36	liver	surgery+chemotherapy	N	Y	NA
9	M	87	COPD, HTN, CHD, cerebrovascular disease/chronic nephrosis/immunological diseases	44	Gastric	follow-up	Y	N	NA
10	M	61	Disturbance of consciousness	16	Lung	chemotherapy+immunotherapy	Y	N	NA
11	F	67	DM, HTN, liver cirrhosis	8	Esophagus	perioperative	Y	N	NA
12	M	65	HTN, cerebrovascular disease	48	Lung	follow-up	N	Y	NA
13	F	50	No	54	HN	follow-up	N	Y	NA
14	F	46	HTN	50	Thyroid	follow-up	N	Y	NA
15	M	77	HTN	42	Colorectal	follow-up	Y	Y	NA
16	F	70	No	34	Breast	follow-up	Y	Y	NA
17	F	75	No	51	Colorectal	follow-up	N	Y	NA
18	F	53	No	57	HN	follow-up	N	Y	NA
19	M	48	HTN	49	CNS	follow-up	N	Y	NA
20	F	58	HTN, HLD, CHD	45	Thyroid	follow-up	N	Y	NA
21	M	54	No	44	Colorectal	perioperative	N	Y	NA
22	F	78	HLD	18	Hematology	supportive treatment	Y	N	NA
23	M	60	HTN	64	Hematology	supportive treatment	N	Y	NA
24	F	83	HTN	64	CNS	follow-up	Y	Y	NA
25	M	68	HTN	47	Urinary	follow-up	N	Y	NA
26	M	66	COPD, HTN, CHD, liver cirrhosis	48	Urinary	follow-up	N	Y	NA
27	M	63	HTN	29	Urinary	follow-up	Y	N	NA
28	M	77	No	55	Urinary	radiotherapy	Y	Y	NA
29	M	73	HTN/HLD/immune disease	53	Lung	perioperative	N	Y	NA
30	M	66	HD/DM/CHD/immune disease	22	Duodenal	supporting treatment	Y	Y	NA
31	F	84	HTN, immune disease	11	Lung	perioperative	Y	N	NA
32	M	67	DM, HTN, CHD, chronic nephrosis	42	Colorectal	follow-up	Y	Y	NA
33	F	65	DM, HTN, liver cirrhosis	41	CNS	follow-up	N	Y	NA
34	M	78	HTN	40	Colorectal	follow-up	N	Y	NA
35	M	65	No	38	Colorectal	perioperative	Y	Y	NA
36	F	71	Cerebrovascular disease	37	Colorectal	follow-up	Y	Y	NA
37	F	65	HTN, tuberculosis	44	Breast	endocrinotherapy	Y	Y	NA
38	F	40	No	41	Thyroid	follow-up	N	Y	NA
39	F	61	HTN	47	Thyroid	follow-up	N	Y	NA
40	F	70	COPD, DM, HTN, cerebrovascular disease	6	Breast	follow-up	Y	N	NA
41	M	98	No	21	Colorectal	follow-up	Y	N	NA
42	M	64	HTN	53	Esophagus	chemotherapy	N	Y	NA
43	M	67	No	14	Urinary	follow-up	Y	N	NA
44	F	76	COPD, CHD, HTN	52	Hematology	follow-up	N	Y	NA
45	F	68	Tuberculosis	53	Thyroid	follow-up	N	Y	NA
46	M	66	HTN	50	Lung	follow-up	N	Y	NA
47	M	69	No	47	Urinary	follow-up	Y	Y	NA
48	M	69	Tuberculosis	44	Colorectal	follow-up	N	Y	NA
49	M	70	DM, HTN	48	Gastric	follow-up	Y	Y	NA
50	M	72	No	19	Lung	chemotherapy	Y	N	NA
51	F	55	HTN	48	Lung	follow-up	N	Y	NA
52	F	58	No	50	CNS	follow-up	N	Y	NA
53	M	54	HTN	50	HN	follow-up	N	Y	NA
54	F	64	No	40	Breast	follow-up	N	Y	NA
55	F	51	DM, CHD	52	Thyroid	follow-up	N	Y	NA
56	M	56	HTN	50	Thyroid	follow-up	Y	Y	NA
57	M	71	No	13	HN	follow-up	Y	N	NA
58	F	65	DM, HTN	36	Ovary	follow-up	N	Y	NA
59	M	90	HTN	9	Colorectal	perioperative	Y	N	NA
60	F	64	DM, HTN	54	Breast	follow-up	N	Y	NA
61	F	69	No	54	Lung	follow-up	Y	Y	NA
62	M	52	No	39	Thyroid	follow-up	Y	N	NA
63	M	74	HTN	22	Colorectal	follow-up	Y	N	NA
64	M	63	HTN	23	HN	follow-up	Y	N	NA
65	M	70	DM, HTN	37	Lung	follow-up	N	Y	NA
66	M	67	HTN	21	Lung	immunotherapy	Y	N	NA
67	M	64	HTN	57	Urinary	follow-up	Y	Y	NA

Abbreviations: BSC: best supportive care; ca – carcinoma, CHB: chronic hepatitis b; CHD: coronary heart disease; CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; DM: diabetes mellitus; EGFR: epidermal growth factor receptor; HLD: hyperlipidemia; HTN: hypertension; NPC: nasopharyngeal cancer; NSCLC: non-small cell lung cancer; PD1: programmed death 1; RCC: renal cell cancer; RT: radiotherapy; TKI: tyrosine kinase inhibitor; SCC: squamous cell cancer; THBSO: total hysterectomy and bilateral salpingo-oophrectomy
Y: yes; N: no; NA: not applicable/available

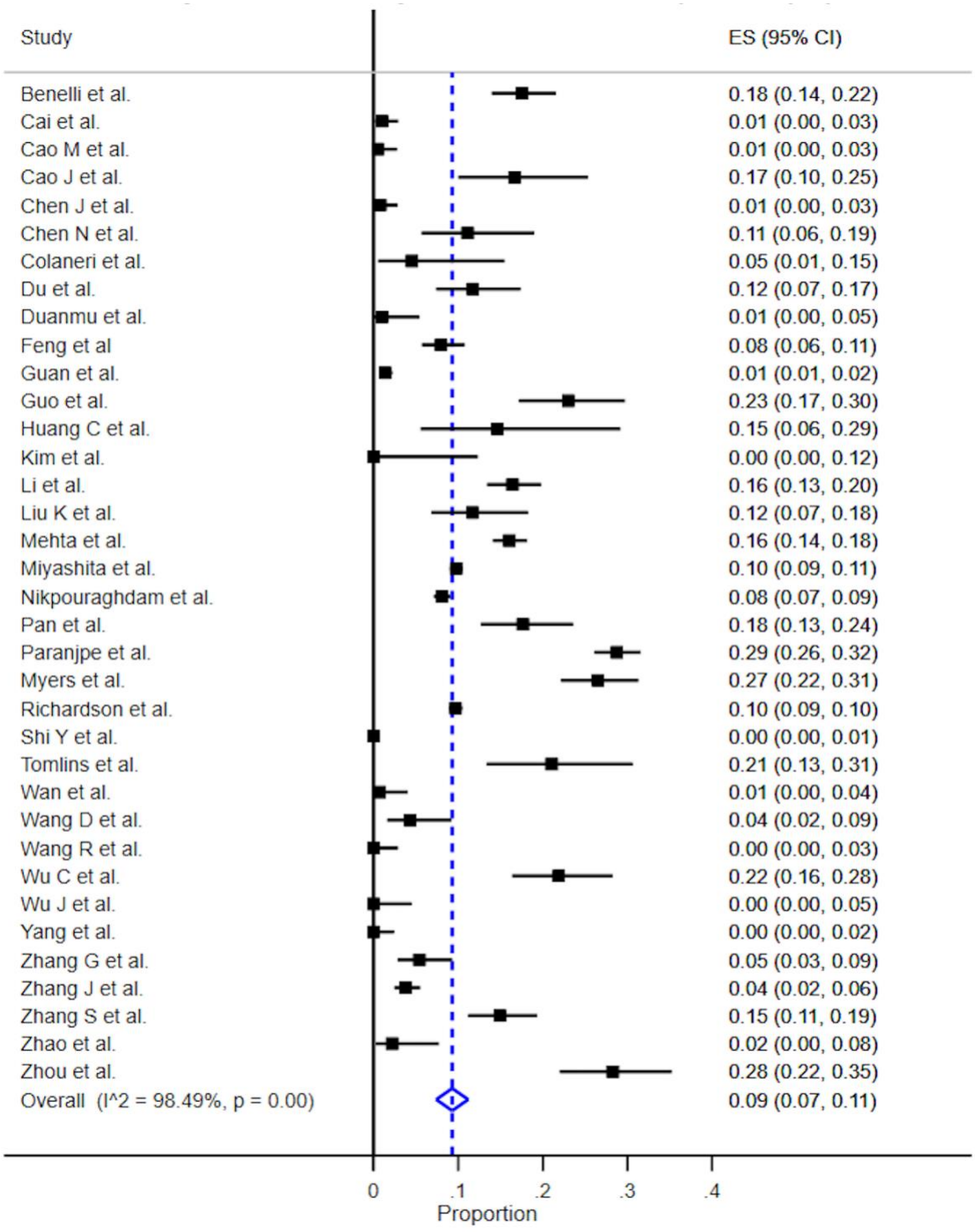
Supplementary Figure 1: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by region – within Hubei province and outside Hubei province)

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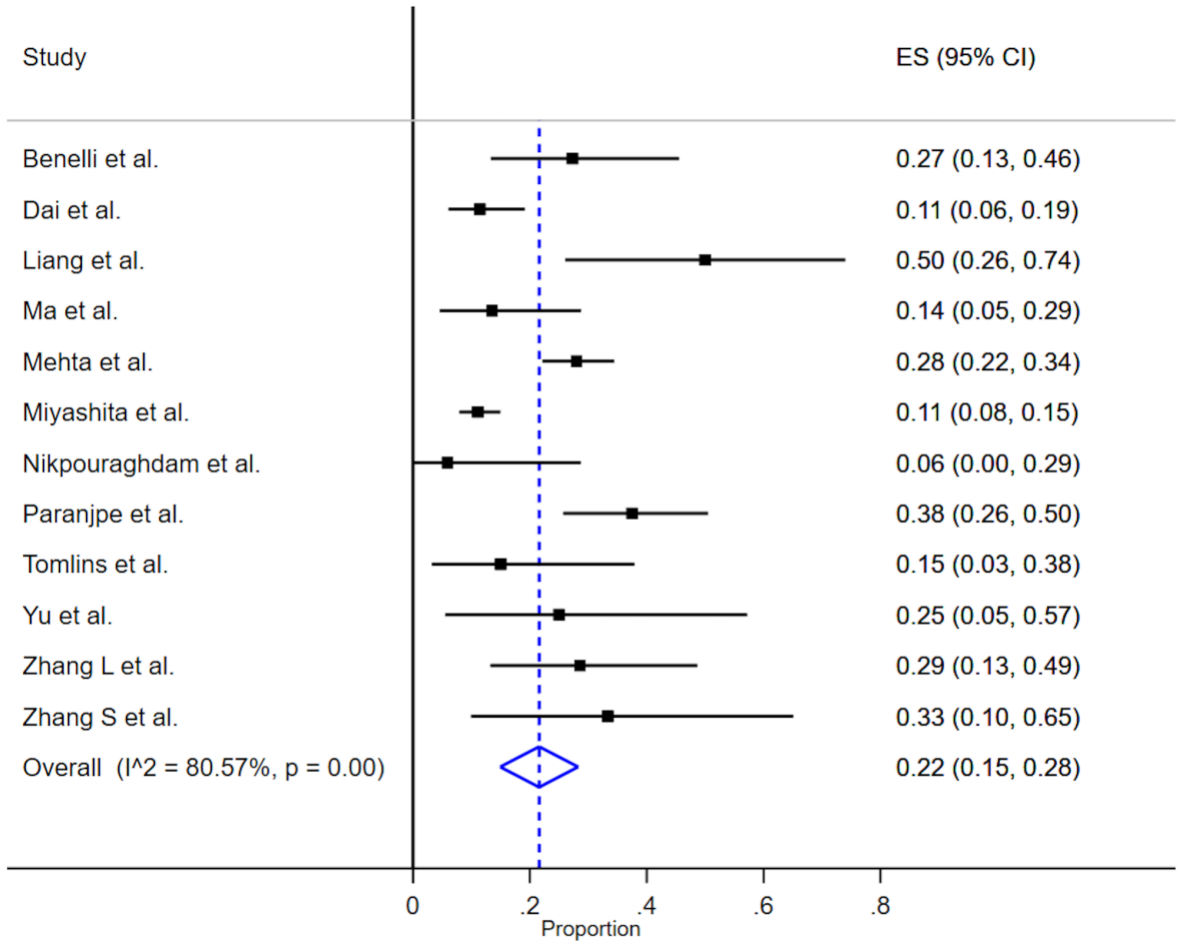
Supplementary Figure 2a: Random effects pooled proportion of deaths among patients diagnosed with COVID-19

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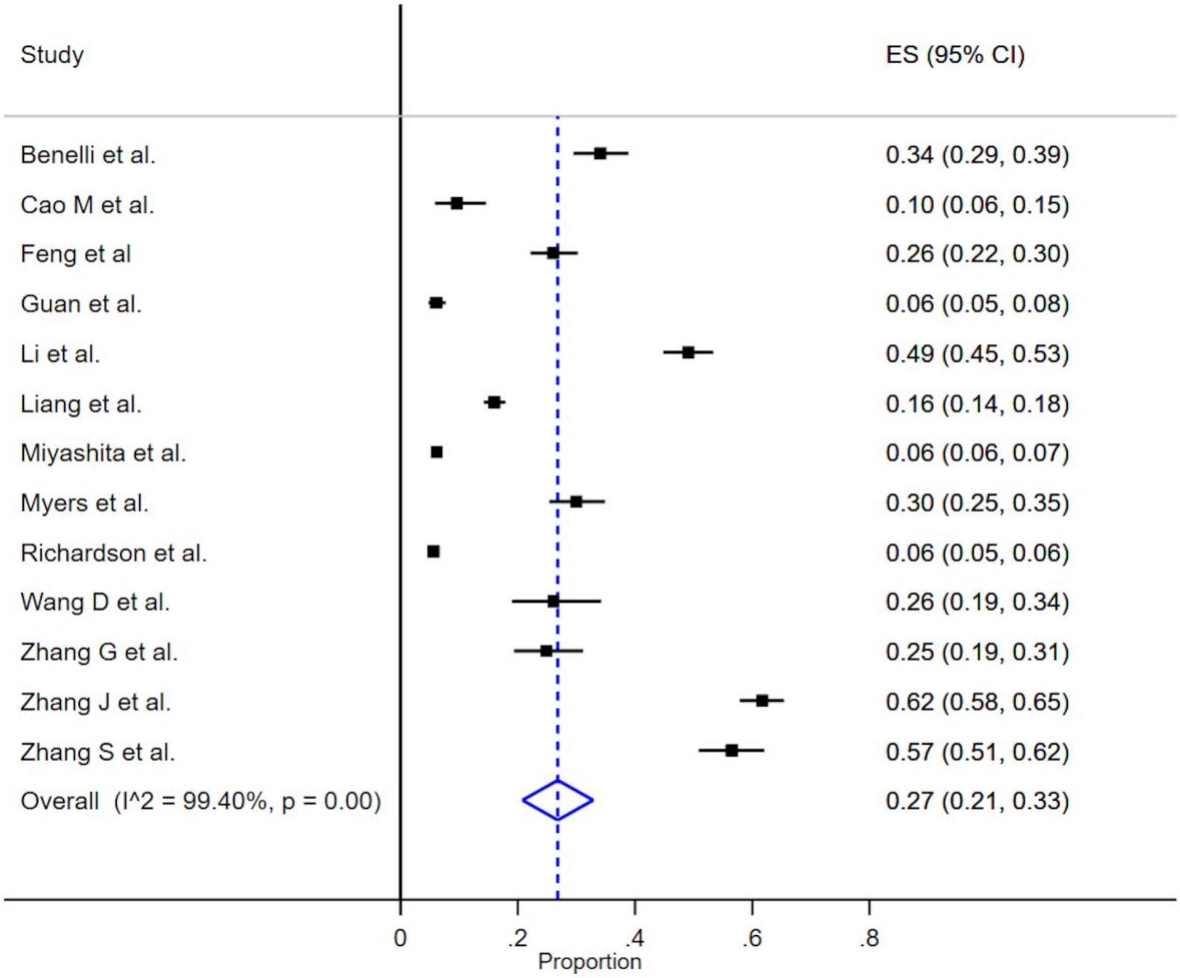
Supplementary Figure 2b: Random effects pooled proportion of deaths among cancer patients diagnosed with COVID-19

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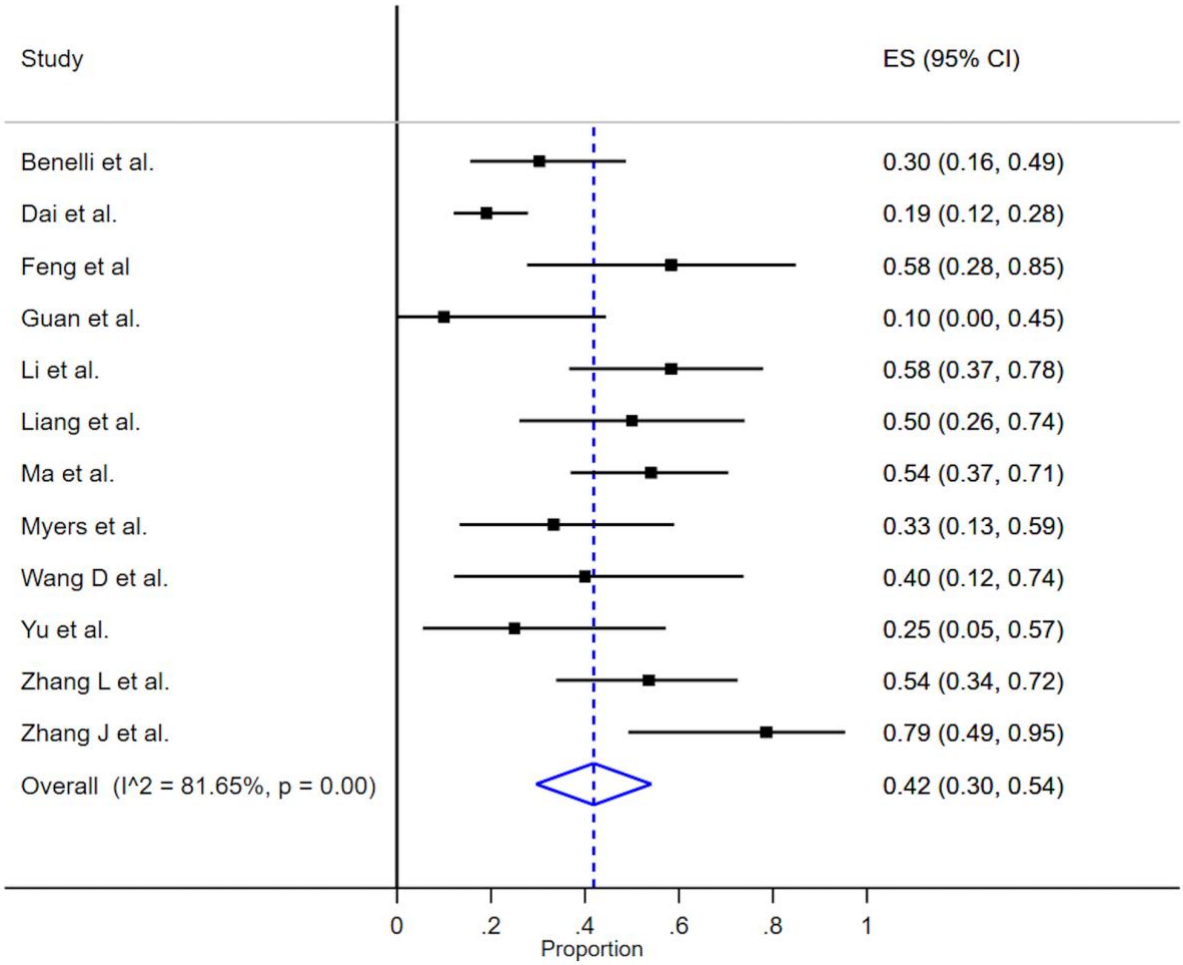
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Supplementary Figure 3a: Random effects pooled proportion of severe symptoms among patients diagnosed with COVID-19



Supplementary Figure 3b: Random effects pooled proportion of severe symptoms among cancer patients diagnosed with COVID-19

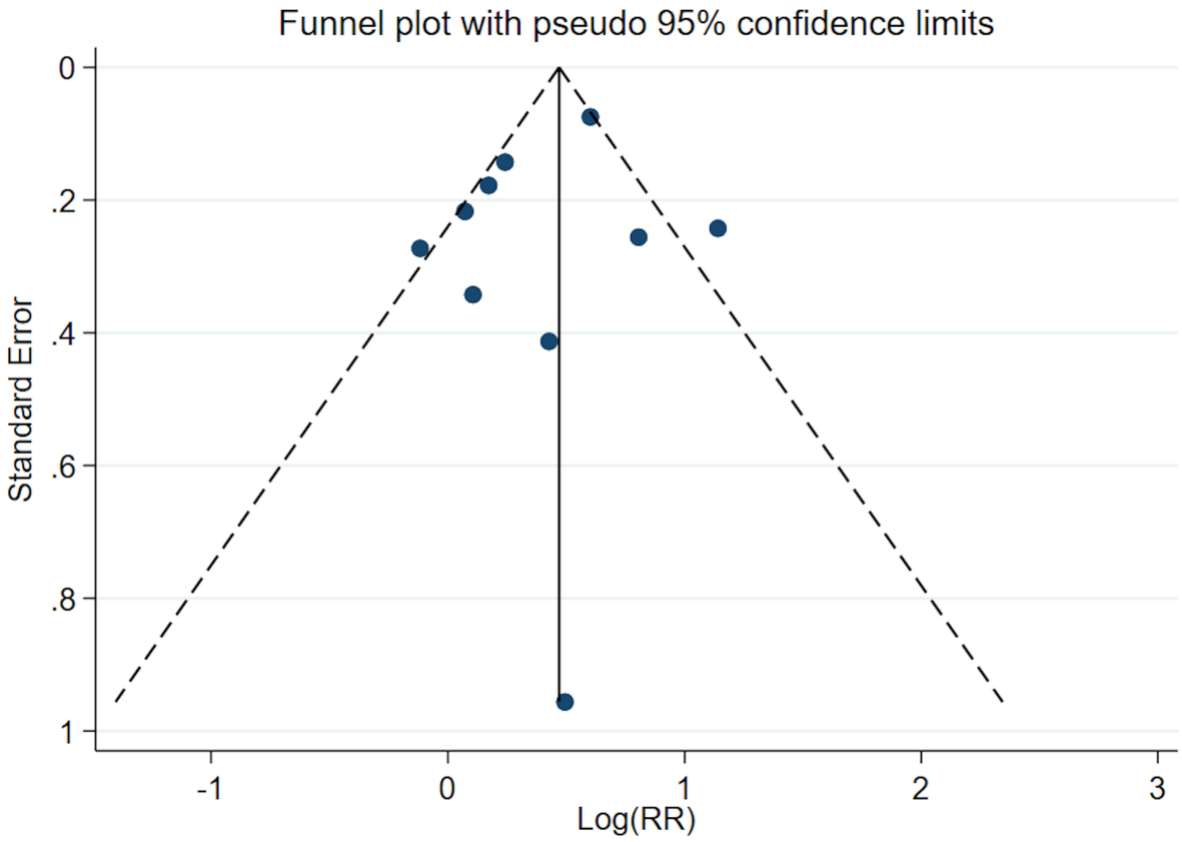
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Supplementary Figure 4a: Funnel plot of ten case series included in pooled relative risk analysis of death outcomes between cancer and general COVID-19 patients

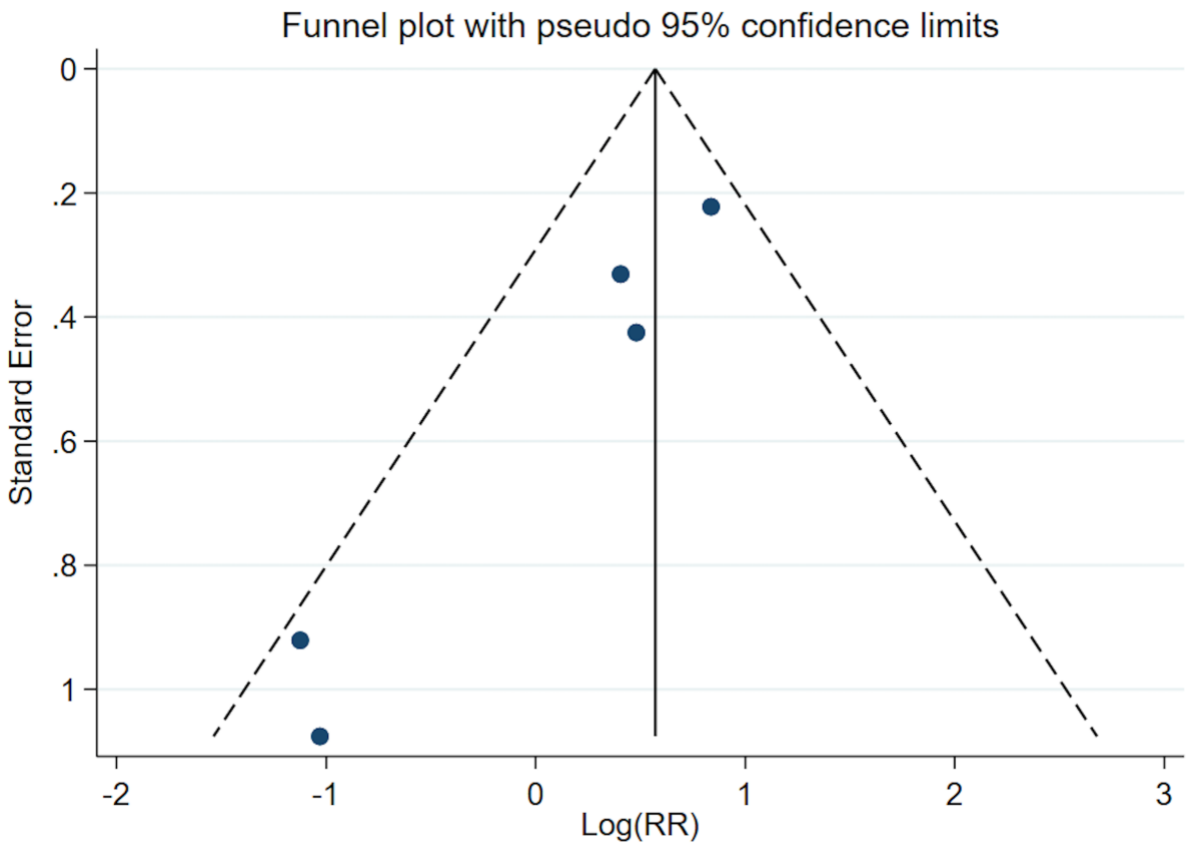
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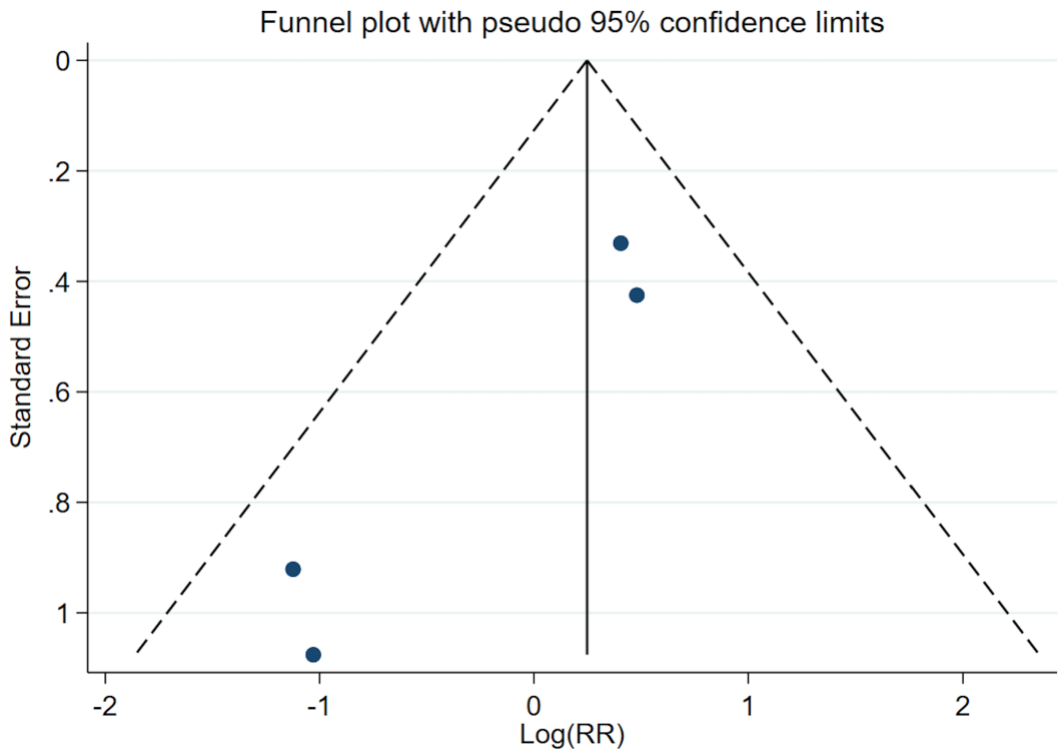
Supplementary Figure 4b: Funnel plot of five case series included in pooled relative risk analysis of severe outcomes between cancer and general COVID-19 patients

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Supplementary Figure 4c: Funnel plot of four case series included in pooled relative risk analysis of lung cancer vs non-lung cancer and stage IV vs non stage IV cancers



Supplementary Table 2: Characteristics of 230 cancer patients with COVID-19 infection with severe and non-severe outcomes

COVID-19 severity	Non-severe (N = 131)	Severe (N = 99)	P-value (Chi-square)
Gender			
Male	70 (50.0)	70 (50.0)	0.010
Female	61 (67.8)	29 (32.2)	
Co-morbidities			
≥2	20 (8.7)	25 (10.9)	0.093
0-1	102 (44.3)	71 (30.9)	
Stage			
Stage IV	20 (8.7)	20 (8.7)	0.635
Stage I-III	56 (24.3)	40 (17.4)	
Unknown	55 (23.9)	39 (17.0)	
Tumour Type			
Lung cancers	32 (13.9)	24 (10.4)	0.974
Non-lung cancers	99 (43.0)	75 (32.6)	
Age			
≥ 60	63 (27.4)	79 (34.3)	<0.001
< 60	59 (25.7)	17 (7.4)	
Treatment			
Active systemic therapy	33 (14.3)	23 (10.0)	0.758
No systemic therapy	97 (42.2)	76 (33.0)	

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N.A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4-5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4, appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Appendix 2
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4-5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4-5

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4-5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4-5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-10, Appendix 2-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	2, 6-12 Appendix 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	6-12, Appendix 3
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

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Impact of cancer diagnoses on the outcomes of patients with COVID-19 – a systematic review and meta-analysis

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Impact of cancer diagnoses on the outcomes of patients with COVID-19 – a systematic review and meta-analysis

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ABSTRACT

Background:

The coronavirus disease 2019 (COVID-19) has caused significant mortality and morbidity across the globe. Cancer patients are especially vulnerable given their immunocompromised state. We aimed to determine the proportion of COVID-19 patients with cancer, their severity and mortality outcomes through a systematic review and meta-analysis (MA).

Methods:

Systematic review was performed through online databases, PubMed, Medline and Google Scholar, with keywords listed in methods section (November 1st 2019 - Dec 31st 2020). Studies with clinical outcomes of at least 10 COVID-19 patients and at least 1 with a diagnosis of cancer were included. The studies for MA were assessed with PRISMA guidelines and appraised with Newcastle-Ottawa

Scale (NOS). The data were pooled using a random-effects model using STATA software. The main outcomes were planned before data collection, including proportion of cancer patients among COVID-19 populations, relative risk of severe outcomes and death of cancer patients compared to general COVID-19 patients.

Results:

We identified 57 case series (63 413 patients), with 230 cancer patients with individual patient data (IPD). We found that the pooled proportion of cancer among COVID-19 patients was 0.04 (0.03-0.05, I2 = 97.69%, p<0.001). The pooled relative risk (RR) of death was 1.44 (95% CI 1.19-1.76) between cancer patients and the general population with COVID-19 infection. The pooled RR of severe outcome was 1.49 (95% CI 1.18-1.87) between cancer and general COVID-19 patients. The presence of lung cancer and stage IV cancer did not result in significantly increased RR of severe outcome. Among the available IPD, only age and gender were associated with severe outcomes.

Conclusion:

Cancer patients were at a higher risk of severe and death outcomes from COVID-19 infection as compared to general COVID-19 populations. Limitations of this study include publication bias, A collaborative effort is required for a more complete database.

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STRENGTHS AND LIMITATIONS

- Largest meta-analysis on cancer and COVID-19 to date
- Specific objectives looking at proportion of cancer patients among COVID-19 cases reported, with severity and mortality outcomes presented as relative risk against general population with COVID-19
- Detailed cancer cohorts presented with breakdown of tumour types, recent systemic anti-cancer therapy and other characteristics, with analysis of lung cancer vs non-lung cancer patients and stage IV vs non-stage IV patients
- Limitations include publication biases and incomplete reporting of mortality outcomes by selected papers

  Lack of stratification for age and co-morbidities for specific analysis due to limited clinical data

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BACKGROUND

The coronavirus disease 2019 (COVID-19) pandemic, since its emergence in December 2019, has caused significant mortality and morbidity across the globe. In early 2021, more than 100 million people had been infected with more than 2,000,000 deaths. It has disrupted many aspects of healthcare delivery and overwhelmed many hospitals and their intensive care resources.

Cancer patients are an at-risk population as they are generally older, immunocompromised, and have multiple co-morbidities. However, the outcomes of cancer patients with COVID-19 remains conflicted despite multiple cohort studies and meta-analyses. A multi-centre study of 105 cancer patients in China suggested that cancer patients with COVID-19 infection had higher risk of severe disease outcomes as compared to 536 age-matched non-cancer patients[1]. Another study from the Chinese Centre for Disease Control and Prevention Group reported a case fatality rate (CFR) of 2.3% (1023/44672) among confirmed COVID-19 cases and a more than double CFR of 5.6% (6/107) among the cancer patients with COVID-19[2]. In contrast, a database report of 5,688 COVID-19 positive patients from a tertiary hospital in New York found no significant difference in risk of death between cancer and non-cancer patients (RR 1.15, 95% CI 0.84-1.57) although there was an increased risk of intubation among cancer patients (RR 1.89, 95% CI 1.37-2.61)[3], and other cross sectional studies also showed similar risk of morbidity and mortality as the general population[4,5] though with a small number of patients. A few meta-analyses had recently addressed this as well and found that there was increased risk of severe COVID-19 infection among cancer patients on active oncological treatment[6], especially chemotherapy[7].

Hence, we aimed to review the current literature and to determine firstly the proportion of cancer patients among the reported COVID-19 cases, secondly whether cancer patients with COVID-19 had more severe outcomes and higher mortality compared to the general population and lastly, if there were any specific characteristics of patients with cancer that were associated with an increased risk of severe COVID-19 infection.

METHODS

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Search Methods and Study Selection

A systematic search of the literature was conducted to identify studies that include cancer patients with COVID-19 infection. The investigators searched biomedical databases including PubMed, Medline and Google Scholar using the terms *COVID-19, coronavirus, nCoV, SARS-CoV-2, characteristics, comorbidity, mortality, death, risk, oncology, cancer, malignancy, neoplasm, tumor, tumour, and carcinoma* (Appendix 1). The search was conducted on Jan 29, 2021 for a search period from November 1, 2019 through to December 31, 2020. The references of relevant meta-analysis and published reviewers were searched for additional eligible studies. The articles included met the following criteria: (1) COVID-19 diagnosis should be confirmed by Real Time Polymerase Chain Reaction (RT-PCR), or based on 2019 Novel Coronavirus Disease (COVID-19) Diagnostic criteria, if applicable (Appendix 1), (2) studies should contain clinical data of at least ten COVID-19 patients with at least one patient with a diagnosis of cancer (including haematological malignancies), (3) severity outcomes (defined as hypoxaemia, admission to intensive care unit (ICU), and/or mechanical ventilation) and/or death outcomes should be presented and, (4) studies should be available in English and full text. Exclusion criteria included studies that reported only specific patient populations, i.e. intensive care setting, paediatric population or single tumour type cohorts. Studies published online ahead of print and selected pre-print papers were included at the discretion of the authors. For the pooled analysis on severity and death outcomes of cancer and general COVID-19 patients, we mandated that the studies included at least 10 cancer patients.

The literature search and study selection were performed independently by three of the authors (SH, QZ and VSY), and discrepancies were resolved by consensus. The study followed the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines (Appendix 1).

Data Extraction

Two investigators (SH and VSY) independently graded the studies based on Newcastle-Ottawa Scale (NOS) for assessing the quality of non-randomised studies and discrepancies were resolved by consensus (Appendix 2). Data extraction was performed by SH, and the parameters extracted

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included the location (hospital and city) of the study, type of study, number of patients, demographics including median or mean age, gender, proportion of patient with co-morbidities including cancer, and the clinical outcomes of severe disease and death (of cancer patients and overall COVID-19 population) were collected for the available studies.

Statistical analysis

All statistical analyses were performed using the STATA V.15 software. Meta-analysis using random-effects models was performed to analyse the pooled proportion of cancer among COVID-19 patients. The included studies were stratified into subgroups by region (within Hubei Province, China vs outside Hubei Province, including other countries), median/mean age (above or equal to vs below 60 years old) and gender (proportion of male more than or equal to 60% vs less than 60% in population). We further analysed the pooled relative risks (RR) of severe COVID-19 infection outcomes and related deaths for cancer patients with COVID-19 versus the general population with COVID-19. The subgroup of lung cancer and stage IV patients were also evaluated for relative risk of severe COVID-19 infection outcomes. The results were presented as proportions (decimal) with 95% confidence interval (95% CI). Heterogeneity between studies was analysed by the I^2 statistics, where $I^2 > 50\%$ was considered moderate heterogeneity and chi-square $p < 0.05$ was considered statistically significant. Chi-square tests were performed to look for associations between cancer patient characteristics (age, gender, lung cancers, stage IV cancers, co-morbidities and recent systemic therapy) and severe COVID-19 outcomes in the individual patient data (IPD).

Patient and Public Involvement

The research question was formed based on the urgent need to understand the effect of COVID-19 on cancer patients and to address the clinical decisions that follows this understanding. No patient was directly involved in this systematic review and meta-analysis. We would like to thank the authors and also the patients that were involved in the publications that we had included in our meta-analysis.

RESULTS

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Eligible studies and characteristics

We identified 3083 relevant publications after literature search and review. Additional studies were identified from the references of published meta-analyses. After screening and eligibility assessment, we included in our meta-analysis a total of 57 case series involving 63 413 patients (Figure 1; Appendix 2 Supplementary Table 1). Studies that did not meet the MA criteria were summarised in Appendix 2 Supplementary Table 2.

Among the 57 studies, 37 studies were from public hospitals in China (24 from Hubei Province) with the rest from worldwide (USA, Canada and Spain in CCC19 study) (1), Europe (1), Korea (1), Iran (1), France (1), Spain (1), Italy (2), United Kingdom (UK) (3) and USA (9) (Table 1). Most studies were retrospective cohort studies and a few observational cross-sectional studies. The quality of all the included studies ranged from 6-8 (out of 9) using the Newcastle-Ottawa Scale (Appendix 2 Supplementary Table 3). Among the 57 case series, 13 contained detailed descriptions of COVID-19 patients with cancer (Table 2). Only 5 studies provided detailed IPD (Appendix 3 Supplementary Table 1). Forty-seven studies were used for proportion analyses for cancer diagnoses among COVID-19 patients. For the pooled analysis on relative risk of severe outcomes among cancer COVID-19 patients, we identified 10 case series that contained severity data, and 7 studies that contained mortality data for analysis (studies with at least 10 cancer patients). Five studies contained severity outcomes of lung cancer versus other cancer patients with COVID-19 infection, and four studies contained severity outcomes of stage IV vs non-stage IV cancer patients.

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Table 1: Baseline characteristics, comorbidities and severity outcomes of 57 COVID-19 case series

No.	Authors	Location	Definition of severity	Total	Death n (%)	Severe n (%)	Gender – Male n (%)	Age	DM	HTN	Pulmonary disease	Smoking	CKD	CVS	CLD	Cancer total n (%)	Cancer – severe n (% of cancer total)	Cancer – death (% of cancer total)
1	Argenziano et al[8]	New York, USA	ICU, death	1000	211 (21.1)	236 (23.6)	596 (59.6)	63 (50-75	372 (37.2)	601 (60.1)	66 (6.6)	49 (4.9)	137 (13.7)	131 (13.1)	15 (1.5)	67 (6.7)	17 (25.4)	N.A
2	Assaad et al[9]	Lyon, France	Death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	302	N.A	30 (9.9)
3	Benelli et al.[10]	Crema, Italy	ICU + NIV, death	411	72 (17.5)	140 (34.1)	359 (66.6)	70.5 (1-99)	67 (16.3)	193 (47.0)	48 (11.7)	N.A	22 (5.3)	93 (22.6)	N.A	33 (8.0)	10 (30.3)	9 (27.3)
4	Cai et al.[11]	Shenzhen, China	ICU, death	298	3 (1.0)	58 (19.5)	145 (48.7)	47.5 (33-61)	18 (6.0)	47 (15.8)	N.A	N.A	N.A	25 (8.4)	28 (9.4)	4 (1.3)	2 (50.0)	N.A
5	Cao M et al.[12]	Shanghai, China	ICU, death	198	1 (0.5)	19 (9.6)	101 (51.0)	50.1 (±16.3)	15 (7.6)	42 (21.2)	N.A	11 (5.6)	N.A	12 (6.0)	6 (3.0)	4 (2.0)	0	N.A
6	Cao J et al.[12]	Hubei, China	ICU, death	102	17 (16.7)	18 (17.6)	53 (52.0)	54 (37-67)	11 (10.8)	28 (27.5)	10 (9.8)	N.A	4 (3.9)	11 (10.8)	2 (2.0)	4 (3.9)	N.A	1 (25.0)
7	Chen J et al.[13]	Shanghai, China	ICU, death	249	2 (0.8)	22 (8.8)	126 (50.6)	51 (31-64)	25 (10.0)	N.A	5 (2.0)	N.A	N.A	55 (22.1)	N.A	1 (0.4)	N.A	N.A
8	Chen N et al.[14]	Hubei, China	ICU, death	99	11 (0.11)	23 (23.2)	67 (67.7)	55.5 (±13.1)	12 (12.1)	0	1 (1.0)	N.A	N.A	40 (40.4)	N.A	1 (1.0)	N.A	N.A
9	Colaneri et al.[15]	Pavia, Italy	Require high FiO2	44	2 (4.5)	17 (38.6)	28 (63.6)	67.5 (10-94)	7 (15.9)	15 (34.1)	2 (4.5)	N.A	N.A	11 (25.0)	N.A	6 (13.6)	2 (4.5)	N.A
10	Dai et al.[16]	Hubei, China	NIV and ventilation, death	641	N.A	84 (13.1)	57 (8.9)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	105 (16.4)	20 (19.1)	12 (11.4)
11	Docherty et al[17]	UK	HD/ICU, death	20133	5165 (25.7)	3001 (14.9)	12068 (59.9)	73 (58-82)	3650 (18.1)	N.A	3128 (15.5)	852 (4.2)	2830 (14.1)	5469 (27.2)	281 (1.4)	1743 (8.7)	N.A	N.A
12	Du et al.[18]	Hubei, China	Death	179	21 (11.7)	N.A	97 (54.1)	57.6 (±13.7)	33 (18.4)	58 (32.4)	N.A	N.A	4 (2.2)	29 (16.2)	N.A	4 (2.2)	N.A	1 (25.0)
13	Duanmu et al.[19]	California, USA	Death	100	1 (1.0)	N.A	56 (56.0)	45 (32-65)	10 (10.0)	19 (19.0)	1 (1.0)	2 (2.0)	6 (6.0)	N.A	N.A	3 (3.0)	N.A	N.A
14	Feng et al.[20]	Hubei, China	See below	476	38 (8.0)	124 (26.1)	271 (56.9)	53 (40-64)	17 (3.6)	40 (8.4)	14 (2.9)	44 (9.2)	2 (0.4)	26 (5.5)	N.A	12 (2.5)	7 (58.3)	N.A
15	Guan et al.[21]	China	Composite endpoint[a]	1099	15 (!.4)	67 (6.1)	640 (58.2)	47 (35-58)	81 (7.4)	165 (15.0)	12 (1.1)	158 (14.4)	8 (0.7)	42 (3.8)	23 (2.1)	10 (0.9)	1 (10.0)	N.A
16	Guo et al.[22]	Hubei, China	Mechanical ventilation	187	43 (23.0)	45 (24.1)	91 (48.7)	58.5 (±14.7)	28 (15.0)	61 (32.6)	4 (2.1)	18 (9.6)	6 (3.2)	29 (15.5)	N.A	13 (7.0)	N.A	N.A
17	Huang C et al.[23]	Hubei, China	ICU	41	6 (14.6)	13 (31.7)	30 (73.2)	49 (41-58)	8 (19.5)	6 (14.6)	1 (2.4)	3 (7.3)	N.A	6 (14.6)	1 (2.4)	1 (2.4)	0	N.A
18	Huang Y et al.[24]	Hubei, China	ICU	34	N.A	8 (23.5)	14 (41.2)	56 (26-88)	4 (11.8)	8 (23.5)	3 (8.8)	N.A	N.A	6 (17.6)	1 (2.9)	3 (8.8)	N.A	N.A
19	Kim et al. [25]	Korea	Require high FiO2	28	0	6 (24.1)	15 (53.6)	40 (20-73)	2 (7.1)	0	0	5 (17.9)	0	0	1 (3.6)	1 (3.6)	N.A	N.A
20	Kudere r et al.[26]	Worldwide	Composite endpoint	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	928	242 (26.0)	121 (13.0)
21	Lee LY et al.[27]	UK	ICU, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	800	53 (7.0)	226 (28.3)
22	Li et al.[28]	Hubei, China	IDSA/ATS[b]	548	90 (16.4)	269 (49.1)	279 (50.9)	60 (48-69)	83 (15.1)	166 (30.3)	17 (3.1)	N.A	10 (1.8)	34 (6.2)	5	24 (0.9)	14	N.A
23	Lian et al.[29]	Zhejiang, China	Not defined	788	N.A	18 (2.3)	407 (51.6)	N.A	57 (7.2)	126 (16.0)	3	54 (6.9)	7 (0.9)	11 (1.4)	31 (3.9)	6 (0.8)	N.A	N.A
24	Liang et al.[30]	China	Composite endpoint	1590	N.A	254 (16.0)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	18 (1.1)	9 (50.0)	9 (50.0)
25	Liu K et al.[31]	Hubei, China	NIV	137	16 (11.7)	34 (24.8)	61 (44.5)	57 (20-83)	14 (10.2)	13 (9.5)	2 (1.5)	N.A	N.A	10	N.A	2 (1.5)	N.A	N.A
26	Ma et al.[32]	Hubei, China	See below[c]	1380	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	37 (2.7)	20 (54.1)	5 (13.5)
27	Mehta et al.[33]	New York, USA	Death	1308	210 (16.1)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	218 (16.7)	N.A	61 (28.0)
28	Miyashita et al.[3]	New York, USA	Intubation, death	5688	555 (9.8)	351 (6.2)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	334 (5.9)	N.A	37 (11.1)
29	Myers et al.[34]	California, USA	ICU	377	100 (26.5)	113 (30.0)	212 (56.2)	61 (50-73)	118 (31.3)	164 (43.5)	28 (7.4)	N.A	48 (12.7)	22 (5.8)	21 (5.6)	18 (4.8)	6 (33.3)	N.A
30	Nikpouraghdam et al.[35]	Iran	Death	2964	239 (8.1)	N.A	1955 (66.0)	56 (46-65)	113 (3.8)	59 (2.0)	60 (2.0)	N.A	18 (0.6)	37 (1.2)	N.A	17 (0.6)	N.A	1 (5.9%)
31	Pan et al[36]	Hubei, China	ICU, death	204	36 (17.6)	16 (7.8)	107 (52.5)	52.9 (±16·0)	44 (21.6)	N.A	9 (4.4)	N.A	N.A	44 (21.6)	N.A	13 (6.4)	N.A	N.A
32	Paranjpe et al.[37]	New York, USA	Death	1078	310 (28.8)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	64 (5.9)	N.A	24 (37.5)
33	Petrilli et al[38]	New York, USA	ICU, death	5279	665 (12.6)	990 (18.8)	2615 (49.5)	54 (38-66)	1195 (22.6)	2256 (42.7)	786 (14.9)	288 (5.5)	647 (12.3)	704 (13.3)	N.A	403 (7.6)	138 (34.2)	N.A
34	Pinato et al[39]	Europe	HD/ICU, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	890	110 (12.4)	299 (33.6)
35	Robilotti et al[40]	New York, USA	Mechanical ventilation/high flow, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	423	87 (20.6)	51 (12.1)
36	Richardson et al.[41]	New York, USA	ICU, death	5700	553 (9.7)	320 (5.6)	3437 (60.3)	63 (52-75)	1808 (31.7)	3026 (53.1)	287 (5.0)	2691 (47.2)	454 (8.0)	966 (16.9)	30 (0.5)	320 (5.6)	N.A	N.A

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37	Shi H et al.[42]	Hubei. China	Not defined	81	N.A	N.A	42 (51.9)	49·5 (±11·0)	19 (23.5)	12 (14.8)	9 (11.1)	N.A	3 (3.7)	14 (17.3)	N.A	4 (4.9)	N.A	N.A
38	Shi Y et al.[43]	Zhejiang, China	Not defined	487	0	49 (10.1)	259 (53.2)	46 (±19·0)	29 (6.0)	99 (20.3)	0	40 (8.2)	7 (1.4)	11 (2.3)	22 (4.5)	5 (1.0)	2 (40.0)	N.A
39	Tomlins et al.[44]	UK	NIV + intubation, death	95	20 (21.1)	16 (16.8)	60 (63.2)	75 (59-82)	N.A	N.A	N.A	N.A	N.A	N.A	N.A	20 (21.1)	N.A	3 (15.0)
40	Wan et al.[45]	Chongqing, China	Not defined	135	1 (0.7)	40 (29.6)	72 (53.3)	47 (36-55)	12 (8.9)	13 (9.6)	0	9 (6.7)	N.A	7 (5.2)	2 (1.5)	4 (3.0)	3 (75.0)	N.A
41	Wang D et al.[46]	Hubei, China	ICU	138	6 (4.3)	36 (26.1)	75 (54.3)	56 (42-68)	14 (10.1)	43 (31.2)	4 (2.9)	N.A	2 (1.5)	27 (19.6)	0	10 (7.2)	4 (40.0)	N.A
42	Wang R et al.[47]	Anhui. China	ICU	125	0	19 (15.2)	71 (56.8)	38.8 (±13.8)	10 (8.0)	N.A	2 (1.6)	16 (12.8)	N.A	18 (14.4)	N.A	1 (0.8)	N.A	N.A
43	Wu C[48] et al.	Hubei, China	ARDS, death	201	44 (21.9)	84 (41.8)	N.A	51 (43-60)	22 (10.9)	39 (19.4)	5 (2.5)	N.A	2 (1.0)	8 (4.0)	7 (3.5)	1 (0.5)	N.A	N.A
44	Wu J et al.[49]	Jiangsu, China	See below	80	0	3 (37.5)	39 (48.8)	46.1 (±15.4)	5 (6.3)	N.A	0	N.A	1 (1.3)	25 (31.3)	1 (1.3)	1 (1.3)	N.A	N.A
45	Xu et al.[50]	Guangzhou, China	Not defined	90	N.A	N.A	39 (43.3)	50 (18-86)	5 (55.6)	17 (18.9)	1 (1.1)	N.A	N.A	3 (3.3)	N.A	2 (2.2)	N.A	N.A
46	Yang et al.[51]	Zhejiang, China	NIV	149	0	2 (1.3)	81 (54.4)	45.1 (±13.4)	9 (6.0)	0	1 (0.7)	N.A	N.A	28 (18.8)	N.A	2 (1.3)	N.A	N.A
47	Yang K et al.[52]	Hubei, China	ICU, mechanical ventilation, death	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	205	30 (14.6)	40 (19.5)
48	Yarza et al[53]	Spain	Respi failure	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	63	34 (54.0)	16 (25.4)
49	Yu et al.[54]	Hubei, China	ARDS, death	1524	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	12 (0.8)	3 (25.0)	3 (25.0)
50	Zhang L et al.[55]	Hubei, China	Composite endpoint	1276	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	28 (2.2)	15 (53.6)	8 (28.6)
51	Zhang G et al.[56]	Hubei, China	IDSA/ATS	221	12 (5.4)	55 (24.9)	108 (48.9)	55 (39-66.5)	22 (10.0)	54 (24.4)	6 (2.7)	N.A	6 (2.7)	37 (16.7)	7 (3.2)	9 (4.1)	4 (44.4)	N.A
52	Zhang J et al.[57]	Hubei, China	See below	663	25 (3.8)	409 (61.7)	321 (48.4)	55.6 (44-69)	67 (10.1)	N.A	51 (7.7)	N.A	N.A	164 (24.7)	31 (4.7)	14 (2.1)	11 (78.6)	N.A
53	Zhang N et al.[d]	Hubei, China	See below	1548	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	67 (4.3)	32 (47.8)	18 (26.9)
54	Zhang S et al.[e]	Hubei, China	Death	315	47 (14.9)	178 (56.5)	175 (55.6)	57 (44-66)	41 (13.0)	78 (24.8)	3 (1.0)	N.A	2 (0.6)	42 (13.3)	9 (2.9)	12 (3.8)	N.A	4 (33.3)
55	Zhao et al.[58]	Hubei, China	Not defined	91	2 (2.2)	30 (33.0)	49 (53.8)	46 (Not stated)	3 (3.3)	18 (19.8)	1 (1.1)	N.A	1 (1.1)	N.A	N.A	3 (3.3)	2 (2.2)	N.A
56	Zhou et al.[59]	Hubei, China	Not defined	191	54 (28.3)	119 (62.3)	119 (62.3)	56 (46-67)	36 (18.8)	58 (30.4)	6 (3.1)	11 (5.8)	2 (1.0)	15 (7.9)	0	2 (1.0)	N.A	N.A
57	Zhu et al.[60]	Hefei, China	Not defined	32	N.A	N.A	15 (46.9)	46 (35-52)	4 (12.5)	7 (21.9)	2 (6.3)	6 (18.8)	1 (3.1)	10 (31.3)	2 (6.3)	2 (6.3)	N.A	N.A

[a]composite endpoint: Severe disease composite endpoints (ICU, mechanical ventilation and/or death)

[b]see below: Severity as defined by Chinese National Health Commission Definition of Severity (Refer to Appendix 1)

[c]IDSA/ATS: Infectious Diseases Society of America/American Thoracic Society Criteria for Defining Severe Community-acquired Pneumonia 2007 (Appendix 1)

[d]article in press (accepted, peer-reviewed), [e]article not peer-reviewed

All figures presented as n (%) – number (percentage) unless otherwise stated. Age presented as median with interquartile range (IQR): age (range) or as mean: age (± standard deviation (SD))

Abbreviations: ARDS – acute respiratory distress syndrome; CKD – chronic kidney disease; CLD (chronic liver disease); CVS – cardiovascular disease (and included cerebrovascular disease); DM – diabetes mellitus; FiO2 – supplemental oxygen; HTN – hypertension; ICU – intensive care unit; IDSA/ATS – Infectious Disease Society of America/American Thoracic Society Guidelines (Refer to Appendix 1); NIV – non-invasive ventilation; N.A – not applicable

Overall, there were more males than females affected, with a median age above 60 in all case series. This median age was notably higher than the non-cancer populations reported in previous studies[16]. Severe COVID-19 disease among cancer patients ranged from 9 to 54% in thirteen eligible studies (Table 2). Overall, 230 patients from five case series were identified with oncological history, stage, recent treatment history and clinical outcomes (Appendix 3 Supplementary Table 1). However, among these patients, there was inconsistent reporting in those on active systemic chemotherapy and some clinical data on stage of cancer were missing, limiting the usefulness of the dataset in predicting the outcomes of COVID-19 in patients with active cancer.

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Table 2: Baseline characteristics of cancer patients with COVID-19 infections from 13 case series

Characteristics	Zhang L et al. 2020 (n=28) n (%) or median (range)		Yu et al. 2020 (n=12) n (%) or median (range)		Liang et al. 2020 (n=18) n (%) or median (range)		Ma et al. 2020 (n=37) n (%) or median (range)		Dai et al. 2020 (n=105) n (%) or median (range)		Zhang N et al. 2020 (n=67) n (%) or median (range)	
Age	65.0 (56.0-70.0)		66 (48-78)		63.1 (51-75)		62 (59-70)		64 (50-78)		66 (37-98)	
Gender												
Male	17 (60.7)		10 (83.3)		12 (66.7)		20 (54.1)		57 (54.7)		41 (61.2)	
Female	11 (39.3)		2 (16.7)		6 (33.3)		17 (45.9)		48 (45.3)		26 (38.8)	
Stage												
I-III	18 (64.3)		5 (41.7)		13 (72.2)		N.A		88 (83.8)		N.A	
IV	10 (35.7)		6 (50.0)		4 (22.2)		N.A		17 (16.2)		N.A	
Unknown	N.A		1 (8.3)		1 (5.6)		N.A		N.A		N.A	
Smoking history												
Yes	N.A		N.A		4 (22.2)		N.A		36 (34.3)		9 (13.4)	
No	N.A		N.A		N.A		N.A		69 (65.7)		N.A	
Unknown	N.A		N.A		N.A		N.A		N.A		N.A	
	Lung ca	7 (25.0)	Lung ca	7 (58.3)	Lung ca	5 (27.8)	CRC	11 (29.7)	Lung ca	22 (20.1)	Lung ca	15 (22.4)
	Esophageal ca	4 (14.3)	Rectal ca	1 (8.3)	CRC	4 (22.2)	Lung ca	8 (21.6)	GI ca	13 (12.4)	CRC	11 (16.4)
	Breast ca	3 (10.7)	Colon ca	1 (8.3)	Breast ca	3 (16.7)	Breast ca	7 (18.9)	Breast ca	11 (10.5)	Thyroid ca	8 (11.9)
	Laryngoca	2 (7.1)	Pancreatic ca	1 (8.3)	Bladder ca	2 (11.1)	Gynae ca	5 (13.5)	Thyroid ca	11 (10.5)	Urinary system	8 (11.9)
	Liver ca	2 (7.1)	Breast ca	1 (8.3)	Lymphoma	1 (0.56)	Other ca	6 (16.2)	Blood ca	9 (8.6)	Breast & gynae	7 (10.4)
	Prostatic ca	2 (7.1)	Urothelial ca	1 (8.3)	Thyroid ca	1 (0.56)			Cervix ca	6 (5.7)	Head and Neck	5 (7.5)
	Cervical ca	1 (3.6)			Adrenal ca	1 (0.56)			Esophageal ca	6 (5.7)	CNS tumour	4 (5.9)
	Gastric ca	1 (3.6)			RCC	1 (0.56)					Haematological	3 (4.5)
	Colon ca	1 (3.6)									Gastric ca	3 (4.5)
	Rectum ca	1 (3.6)									Esophageal ca	2 (3.0)
	NPC	1 (3.6)									HCC	1 (1.15)
	Endometrial ca	1 (3.6)										
	Ovarian ca	1 (3.6)										
	Ca of testis	1 (3.6)										
Baseline treatment	ChemoRT	25 (89.3)	BSC	4 (33.3)	Surveillance	12 (75.0)	Antitumour therapy	13 (35.1)	Chemotherapy	17 (16.2)	Follow up	44 (65.7)
	Operation	21 (75.0)	Chemo-immuno Therapy	2 (16.7)	Chemo and surgery within 1 month	4 (25.0)			RT	13 (12.4)	On treatment	23 (34.3)
	Targeted therapy/ immunotherapy	6 (21.4)	Surveillance	2 (16.7)					Surgery	8 (7.6)		
	Chemo (<14d)	3 (10.7)	Targeted Therapy/RT	1 (8.3)					Immunotherapy	6 (5.7)		
	Targeted Therapy	2 (7.1)	Chemo/RT	1 (8.3)					Targeted Therapy	4 (3.8)		
	RT (<14d)	1 (3.6)	Adjuvant RT	1 (8.3)								
	Immunotherapy (<14d)	1 (3.6)	Not started	1 (8.3)								
Severe disease	Severe (53.6) 8/15 Non-invasive ventilation 2/15 Invasive ventilation	15	Severe (25.0)	3	Severe	9 (50.0)	Severe (54.1)	20	Severe (38.0) ICU (19.1)	40 20	Severe (47.8) Mechanical ventilation (41.8)	32 28
ARDS	8 (28.6)		N.A		N.A		N.A		N.A		14 (20.9)	
Death	8 (28.6)		3 (25.0)		9 (50.0)		5 (13.5)		12 (11.4)		18 (26.9)	

Table 2 continued

Characteristics	Kuderer et al (CCC19) (n=928) n (%) or median (range)		Lee LY et al (UKCCMP) (n=800) n (%) or median (range)		Robilotti et al (MSKCC) (n=423) n (%) or median (range)		Pinato et al (n=890) n (%) or median (range)		Assaad et al (n=302) n (%) or median (range)		Yang et al (n=205) n (%) or median (range)		Yarza et al (n=63) n (%) or median (range)	
Age	66 (57–76)		69 (59–76)		N.A		Mean: 68.0±12.8		Mean: 58.2±1.1		63 (56–70)		66 (63 - 69)	
Gender														
Male	468 (50)		449 (56)		212 (50.1)		503 (56.5)		144 (47.7)		96 (46.8)		34 (54)	
Female	459 (49)		349 (44)		211 (49.9)		387 (43.5)		158 (52.3)		109 (53.2)		29 (46)	
Stage														
I-III	N.A		227 (29)		N.A		539 (60.6)		N.A		N.A		52 (82.5)	
IV	N.A		347 (43)		238 (56.3)		351 (39.4)		161 (53.3)		N.A		11 (17.5)	
Unknown	N.A		205 (25)		N.A		0		N.A		N.A		N.A	
Smoking history														
Yes	369 (40)		N.A		167 (39.5)		380 (42.7)		N.A		N.A		34 (54.0)	
No	469 (51)		N.A		249 (58.9)		343 (38.5)		N.A		N.A		29 (46.0)	
Unknown	90 (10)		N.A		7 (1.7)		167 (18.8)		N.A		N.A		0	
	Haematological ca	204 (22.0)	Digestive organs	150 (18.8)	Others	137 (32.4)	Breast	162 (18.2)	Solid tumour	234 (77.5)	Solid tumour	183 (89.3)	Lung	17 (27.0)
	Breast ca	191 (20.6)	Other haematological	109 (13.6)	Breast	86 (20.3)	Haematological	137 (15.4)	Haematological	68 (22.5)	Breast	40 (19.5)		
	Prostate ca	152 (16.4)	Breast	102 (12.8)	Lymphoma	48 (11.3)	Genito-Urinary	132 (14.8)	Lung	42 (13.9)	Colorectal	28 (13.6)		
	Gastrointestinal ca	108 (11.6)	Respiratory and intrathoracic organs	90 (11.3)	Colorectal ca	37 (8.7)	Lung	119 (13.4)			Lung	24 (11.7)		
	Lung ca	91 (9.8)	Male genital organs	78 (9.8)	Lung	35 (8.3)	Gastrointestinal	105 (11.8)			Haematological	22 (10.7)		
	Gynaecological ca	49 (5.3)	Lymphoma	60 (7.5)	Leukemia	32 (7.6)	Gynaecological	41 (4.6)			Thyroid	16 (7.8)		
	RCC	45 (4.8)	Female genital organs	45 (5.6)	Prostate	26 (6.1)	Gastro-esophageal	40 (4.5)			Stomach	12 (5.9)		

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	Endocrine ca	39 (4.2)	Urinary tract	50 (6.3)	Melanoma	22 (5.2)	Hepatobiliary	45 (5.1)			Cervical	9 (4.4)		
	Melanoma	38 (4.1)	Other or unspecified‡	47 (5.9)			Head and Neck	29 (3.3)			Lymphoma	7 (3.4)		
	Head and neck ca	30 (3.2)	Lip, oral cavity, and pharynx	27 (3.4)			Skin	28 (3.1)			Acute lymphoblastic leukaemia	5 (2.4)		
	Sarcoma	24 (2.6)	Melanoma (skin)	27 (3.4)			Other	52 (5.8)			Chronic lymphoblastic leukaemia	4 (2.0)		
	CNS	12 (1.3)	CNS	15 (1.9)							Multiple myeloma	3 (1.5)		
	Solid tumour, not otherwise specified	43 (4.6)									Acute myelogenous leukaemia	2 (1.0)		
											MDS	1 (<1)		
Baseline treatment	None	553 (59.6)	Chemotherapy	281 (35.1)	Chemo in last 30 days	191 (45.2)	Chemotherapy	164 (69.8)	Any cancer treatment	194 (64.2)	All	54/182 (29.7)	Chemotherapy	36 (57.1)
	Chemotherapy	160 (17.2)	None	272 (34.0)	Surgery in last 30 days	31 (7.3)	Ongoing anticancer therapy at Covid-19 diagnosis	479 (53.8)	Cytotoxics	137 (45.4)	Chemotherapy	31/182 (17.0)	Endocrine therapy	10 (15.9)
	Endocrine	85 (9.2)	RT	76 (9.5)	ICI	31 (7.3)	Surgery	417 (46.8)	No cancer treatment	108 (35.8)	Targeted therapy	12/182 (6.6)	Immunotherapy	8 (12.7)
	Targeted therapy	75 (8.1)	Targeted treatment	72 (9)			Adjuvant/Neoadjuvant chemotherapy	210 (23.6)	Immunotherapy	26 (8.6)	Others	10/182 (5.5)	Targeted therapy	7 (11.1)
	Immunotherapy	38 (4.1)	Hormone therapy	64 (8.0)			Endocrine Therapy	48 (20.4)	Antiangiogenic	18 (6.0)	RT	9/182 (5.0)	No active tx	2 (3.2)
	RT	12 (1.3)	Other	60 (7.5)			Target Therapy	41 (17.4)	Anti-CD20	14 (4.6)	Surgery	4/182 (2.2)		
			Immunotherapy	44 (5.5)			Immunotherapy	38 (16.2)	Anti-HER2	12 (4.0)	Immunotherapy	4/182 (2.2)		
			Surgery	29 (3.6)					Antiproteasomes	8 (2.6)				
			No information	10 (1.3)					Everolimus (mTORi)	4 (1.3)				
Severe disease	Severe (composite) 242 (26.1) ICU 132 (14.2) Mechanical ventilation 116 (12.5)		Severe/critical 360 (45) ICU 53 (6.6)		Mechanical ventilation 40 (9.5) High flow 47 (11.1)		Admission to ICU/sub-intensive 110/760 (14.5)		N.A		ICU 30 (14.6) Mechanical ventilation 21(10.2)		Respi failure 34 (54.0)	
ARDS	NA		NA		NA		127 (22.5)		N.A		23 (11.2)		23 (36.5)	
Death	121 (13.0)		226 (28.3)		51 (12.1)		299 (33.6)		30 (9.9)		40 (19.5)		16 (25.4)	

Abbreviations: ARDS – acute respiratory distress syndrome; BSC – best supportive care; Ca – carcinoma; CNS – central nervous system; CRC – colorectal cancer; GI – gastrointestinal; ICU – intensive care unit; NPC – nasopharyngeal cancer; RCC – renal cell carcinoma; RT – radiotherapy

Proportion of cancer diagnoses

The overall pooled proportion of cancer diagnoses among COVID-19 patients from 47 studies was 0.04 (95% CI 0.03-0.05) with significant heterogeneity among the included studies, I^2 = 97.69%, chi-square p=<0.001 (Figure 2). The proportion of cancer patients was significantly lower in the studies with median or mean age less than 60 years, at 0.03 (95% CI 0.02-0.03) compared to 0.07 (95% CI 0.05-0.09) for studies with median or mean age above or equal 60 years (Figure 3). The proportion of cancer was higher in the studies with male pre-dominance (more than or at least 60% population being male), at 0.05 (95% CI 0.02-0.07) than studies with less than 60% population being male, at 0.03 (95% CI 0.02-0.05; Figure 4). The proportion of cancer in Hubei Province was 0.03 (95% CI 0.02-0.05) and outside Hubei Province 0.04 (95% CI 0.02-0.04; Appendix 3 Supplementary Figure 1).

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Mortality outcomes of cancer patients with COVID-19

Overall, the mortality rates of cancer patients with COVID-19 infection ranged widely from 5.9% to 50% across 20 studies (Table 3). The pooled relative risk of death was significantly higher in cancer patients with COVID-19 infection than in the general population with COVID-19 infection based on seven studies as shown in table 3 (RR 1.41; 95% CI 1.15-1.73), I^2 = 26.2%, chi-square p=0.229 (Figure 5a). The pooled mortality rate was estimated at 0.22 (95% CI 0.21-0.33) among cancer patients with COVID-19, which was significantly higher than that of the overall COVID-19 patients, at 0.09 (95% CI 0.07-0.11; Appendix 3 Supplementary Figure 2a-c).

Table 3: Mortality rate among cancer patients diagnosed with COVID-19 infection*

Case series	Death/cancer population	Mortality rate in cancer cohorts (%)	Death/total COVID-19 population	Mortality rate in general COVID-19 cohorts (%)	Relative risk of death (95% CI)
Liang et al.	9/18	50.0	N.A	N.A.	N.A
Zhang L et al.	8/28	28.6	N.A	N.A	N.A
Yu et al.	3/12	25	N.A	N.A	N.A
Ma et al.	5/37	13.5	N.A	N.A.	N.A
Zhang N et al.	18/67	26.9	N.A	N.A	N.A
Dai et al.	12/105	11.4	N.A	N.A	N.A
Kuderer et al.	121/928	13.0	N.A	N.A	N.A
Lee LY et al.	226/800	28.3	N.A	N.A	N.A
Robilotti et al.	51/423	12.1	N.A	N.A	N.A
Assaad et al.	30/302	9.9	N.A	N.A	N.A
Pinato et al.	299/890	33.6	N.A	N.A	N.A
Yang et al.	40/205	19.5	N.A	N.A	N.A
Yarza et al.	16/63	25.4	N.A	N.A	N.A
Benelli et al.	**9/33**	**27.3**	**72/411**	**17.5**	**1.56 (0.86-2.82)**
Mehta et al.	**61/218**	**28.0**	**210/1308**	**16.1**	**1.74 (1.36-2.23)**
Miyashita et al.	**37/334**	**11.1**	**555/5688**	**9.8**	**1.14 (0.83-1.55)**
Nikpouraghdam et al.	**1/17**	**5.9**	**239/2964**	**8.1**	**0.73 (0.11-4.90)**
Paranjpe et al.	**24/64**	**37.5**	**310/1078**	**28.8**	**1.30 (0.94-1.81)**
Tomlins et al	**3/20**	**15.0**	**20/95**	**21.1**	**0.71 (0.23-2.17)**
Zhang S et al.	**4/12**	**33.3**	**47/315**	**14.9**	**2.23 (0.96-5.19)**

***Only papers with at least 10 cancer patients were included**

Severity outcomes of cancer patients with COVID-19

Further analyses were conducted for the ten case series with available severity outcomes for both cancer and general COVID-19 patient cohorts. We found that cancer patients had higher risk of severe COVID-19 outcomes compared to the general COVID-19 population, with a pooled relative

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risk of 1.49 (95% CI 1.18-1.87) and substantial heterogeneity, I^2 = 66.7%, chi-square p=0.001 (Figure 5b). Also, we noted that the pooled proportion of severe outcomes in cancer patients was 0.42 (95% CI 0.30-0.54), which was higher than the proportion of severe cases of all COVID-19 patients at 0.27 (95% CI 0.21-0.33; Appendix 3, Supplementary Figure 3a-c). There was no statistically significant difference in relative risk of severe outcomes between those patients (available data from 5 studies identified) with lung cancers versus non lung cancers (RR at 1.46; 95% CI 1.84-2.52, I^2 = 48.1%, chi-square p=0.7; Figure 6a). There was no statistically significant difference in relative risks of severe outcomes between stage IV vs non stage IV cancers from four studies identified (RR 1.48; 95% CI 0.89-2.47; Figure 6b, Supplementary Figure 4).

The severe outcomes of COVID-19 in cancer patients was also analysed from the available 230 individual patient data and summarised in Appendix 3 Supplementary table 1. Only patient age and gender appeared to be significantly associated with severity outcomes by Chi-Square Test (Appendix 3 Supplementary Table 2).

DISCUSSION

We performed a systematic review and meta-analysis to elucidate the proportion of cancer diagnoses among COVID-19 patients, and whether cancer patients had more severe outcomes compared to the general population. We also looked at the patient characteristics that may be associated with severe clinical outcomes among these patients. To our knowledge, this is the largest meta-analysis that directly addresses the clinical outcomes of COVID-19 patients with cancer, including the relative risks of severity and death as compared to general COVID-19 patients.

Firstly, we found that the pooled proportion of cancer patients among COVID-19 cases was 4%. This result was similar to the proportion rate from an earlier meta-analysis[61] of 11 studies with a proportion of 2%, but higher than the 0.2% described by the Chinese Centre for Disease Control and Prevention Group[2]. We found that the cancer proportion was higher among older patients and in male patients with COVID-19. In the subsequent IPD analysis, it was also advanced age and male gender that were significantly associated with severity of COVID-19 infection. We also noted that the cancer proportion

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among the COVID-19 case series from Hubei Province was 3% (Appendix 3 Supplementary Figure 1), which was higher than the estimated cancer prevalence of 0.45% in the region[30]. However, cancer proportion can vary across geographical locations and specialist medical centres.

We have shown that COVID-19 patients with cancer had significantly more severe outcomes compared to the general COVID-19 patients, with a pooled RR of 1.49 (95% CI 1.18-1.87, p=0.001). The pooled estimated proportion of severe COVID-19 outcomes was higher among cancer patients in our meta-analysis compared to the largest cancer cohort data (CCC19)[62]. Overall, the proportion of cancer patients with severe COVID-19 outcomes varied across studies but remained high, highlighting the vulnerability of cancer patients to SARS-CoV-2 infection.

We also observed that lung cancer was the most common cancer among those affected by COVID-19 in Chinese studies (table 2) and this may be due to the fact that lung cancer is the most common cancer in China, rather than a direct relation to the risk of COVID-19 infection[63]. A few case series noted that patients with lung cancers had significantly worse outcomes compared to non-lung cancer cohorts[16,55], which was also supported by data from TERAVOLT[64], an international database of thoracic cancers (not included in our meta-analysis as it is a single tumour registry). TERAVOLT demonstrated 33% mortality and high morbidity among patients with thoracic malignancies, but with low ICU admission rates, presumably partly due to resource allocations and patient prognosis. However, in our meta-analysis, we did not find the relative risk of severity to be significant for lung cancers (vs other tumour types) or stage IV cancers (vs stage I-III) across four studies. This was also echoed by our IPD analysis, where lung cancer and the stage of cancer were not significantly associated with severe outcomes. This could reflect that lung cancer or stage IV cancer did not necessarily confer a higher risk of severe COVID-19 infection, but given the small number of included studies for this part of the analysis, this conclusion should be cautiously interpreted.

Separately, haematological cancers also showed severe outcomes with high mortality rates in multiple cohort studies[16,65].The available individual data in our study were not large enough to analyse the influence of modality of systemic treatment (targeted therapy, immunotherapy or chemotherapy), surgery or radiotherapy on the outcome of COVID-19 infections. Recent studies did not consistently

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find active systemic therapy to be associated with risk of severe COVID-19 infection[66] though some studies suggest that patients on recent chemotherapy[30] or immunotherapy[16] were at higher risk. These questions remain unanswered and require further collaborative efforts to identify risk factors of severe COVID-19 infection among cancer patients. One suggestion is to have standardised reporting format for cancer COVID-19 cohorts including specific tumour type, stage, date of last chemotherapy or systemic treatment, co-morbidities, active vs past cancer diagnosis and also the severity and mortality data for each subgroup.

Overall, the pooled relative risk of death among the cancer patients was 1.41 (95% CI 1.15-1.73), compared to general COVID-19 patients. However, pooled mortality risk of cancer patients with COVID-19 should be interpreted with caution due to the limited sample size. Some case series also had patients who remained hospitalised or with incomplete patient outcome at the time of reporting. CFR were lower at 5.6% among cancer COVID-19 patients in a large Chinese report[2] and 11.1% from a tertiary centre in New York, USA[3]. Notably, among selected COVID-19 mortality reports from Italy and USA, around 20-30% of deaths had a history of cancer[33,67]. We would encourage mortality data to be taken from database reports instead of small case series from single centres.

In light of the severity of COVID-19 in cancer patients, efforts should be undertaken to reduce nosocomial exposure in cancer patients, as nosocomial infections may contribute to incidence of COVID-19 in oncology patients[54]. The decision to attend clinics and continue non-urgent systemic therapy should be made judiciously, especially in regions with high burden of disease transmission. This has been echoed by various oncology bodies, including American Society of Clinical Oncology (ASCO) and European Society of Medical Oncology (ESMO)[68]. Given the higher risk of severe outcomes and mortality among cancer patients, it is important that cancer patients with COVID-19 infection should be monitored closely and decisions for ICU support discussed early. ICU outcomes should also be an important question for future retrospective studies in view of the ethical and clinical considerations of putting patients with advanced cancer on mechanical ventilation, especially in countries with significant outbreak and limited ICU resources. In light of the recent advances in COVID-19 vaccine efforts, another area of active discussion is that of the risk and benefit of vaccination for cancer patients, especially those who are deemed to be significantly

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immunocompromised. Our data shows that there is risk of severe COVID-19 infection and risk of death among cancer patients compared to general population. Hence, this factor should be strongly considered in the individualised decision for vaccination.

The limitations of this meta-analysis include the heterogeneity and retrospective nature of the case studies, including reporting and selection biases. A few studies were reported from the same Hospitals and hence there may be overlapping patients although efforts were taken to reduce this. During screening of papers by the authors, we have ensured that the analysed studies were not reported from the same hospital during the same period of admission (Appendix 2 Supplementary table 1). The small numbers and incomplete death outcomes at the time of reporting also restricted our mortality analysis. Due to the limited clinical data, we could not adjust for co-morbidities and other characteristics that may contribute to severe disease. It should also be noted that severity outcomes were defined inconsistently across some studies but efforts were taken to indicate the different definitions in table 1. Of note, some earlier studies also included clinically diagnosed COVID-19 cases.

Whilst we acknowledge other risk factors such as age[48,69,70], gender[71], cardiovascular risk factors[70,72,73] including diabetes mellitus[74,75] and hypertension[76] remain important in COVID-19 infection, this is the largest meta-analysis to show that malignancy is also a significant risk factor for severe disease and mortality. However, we did not identify other tumour or treatment-related factors associated with severe outcomes among cancer patients with COVID-19. We also did not stratify for gender, age and co-morbidities in our analysis due to limited available data for pooled analysis on severity and mortality. As more data becomes available in the literature, further analyses may be performed to discern the clinical outcomes of cancer patients with COVID-19 infection, with specific attention to underlying lung cancers, haematological cancers and the effects of various systemic therapies or other treatment modalities. CCC19, UKCCMP and TERAVOLT are some examples of collaborative efforts on the outcomes of cancer patients with COVID-19 infection, though a limitation is that there is a lack of comparison against general population. Information on COVID-19 patients with cancer should also be collected in more detail and with consistency across studies, to allow further meta-analyses to ascertain the outcomes of cancer patients with COVID-19 infection.

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CONCLUSION

Given that COVID-19 may persist in the foreseeable future, it is imperative to continue active research and data collection on the effect of SARS-CoV-2 on oncology patients. Due to the clinical burden of their disease, oncology patients will continue to require systemic chemotherapy and urgent care at medical centres[77]. As such, treating physicians need to be able to make more discerning decisions on the choice of systemic treatment and escalation of care.

Overall, we found through meta-analysis that the severity and mortality rates of COVID-19 in cancer patients were higher than that in the general population. However, the dataset remains small and should be interpreted with caution. We hope that with collaborative efforts, comprehensive patient outcomes of COVID-19 patients, and specifically, of cancer patients, can be collated to better understand the disease trajectory and outcomes of COVID-19.

Word Count: 3311

Figure legends
Figure 1: Flow chart of study identification and exclusion for data analysis
Figure 2: Random effects pooled proportion of cancer patients among patients diagnosed with COVID-19
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Figure 5a: Random effects pooled relative risk of death for cancer COVID-19 patients versus total population of COVID-19 patients
Figure 5b: Random effects pooled relative risk for severe outcome of cancer COVID-19 patients versus total population of COVID-19 patients
Figure 6a: Random effects pooled relative risk for severe outcome of lung cancer vs other cancer patients infected with COVID-19
Figure 6b: Random effects pooled relative risk for severe outcome of stage IV cancer vs non-stage IV cancer patients infected with COVID-19

Declarations

a. Contributorship statement
SH and VSY conceived the idea of the study. SH, VSY and QZ contributed to data collection. ST, SH and YYS contributed to data analysis. SH, QZ, JC, GWYC and VSY contributed to the manuscript writing. MLKC and CX contributed to clinical data. MLKC, YYS and VSY provided guidance.

b. Competing interests

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No conflicts of interest to declare.
Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

c. Funding
No funding or acknowledgement to declare.

d. Data sharing statement
All data relevant to the study are included in the article or uploaded as supplementary information.

e. Ethics statement
Ethics approval was not required as no human or animal subjects were involved. No consent taking was required.

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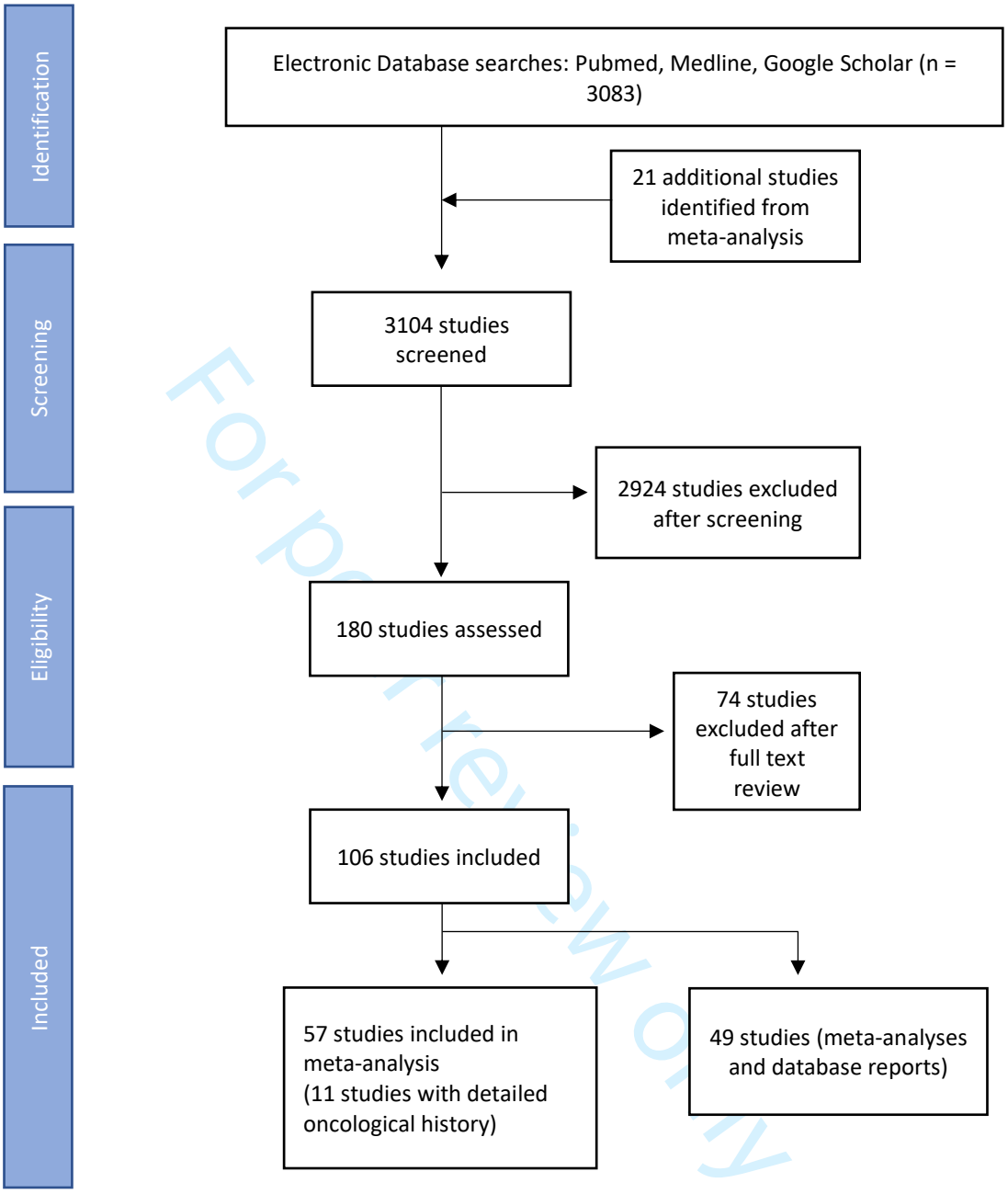
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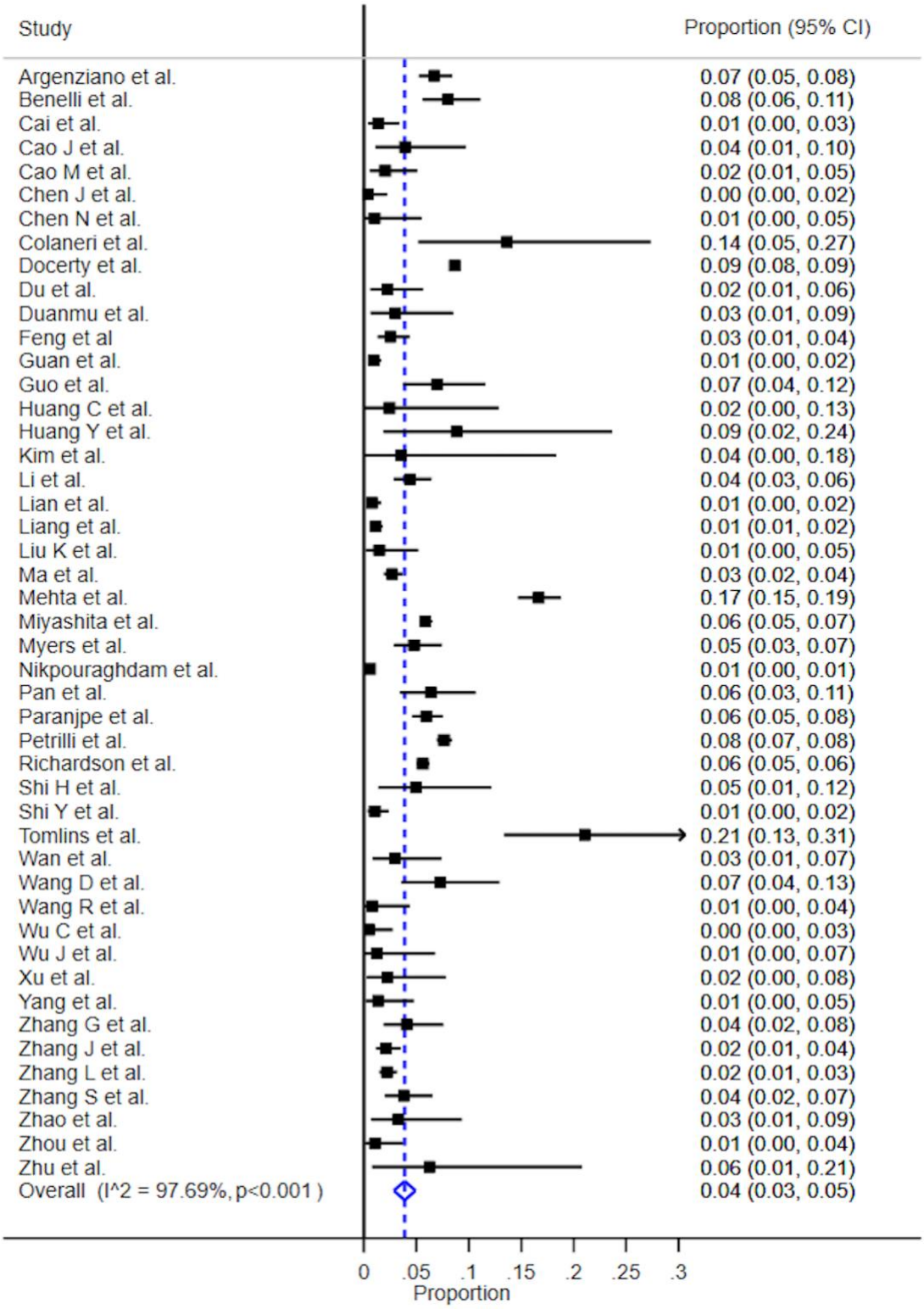
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Figure 1: Flow chart of study identification and exclusion for data analysis



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Figure 2: Random effects pooled proportion of cancer patients among patients diagnosed with COVID-19



*10 papers were excluded in the proportion calculations (did not represent overall COVID-19 populations)

Figure 3: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by median/mean age ≥ 60 vs < 60 years)

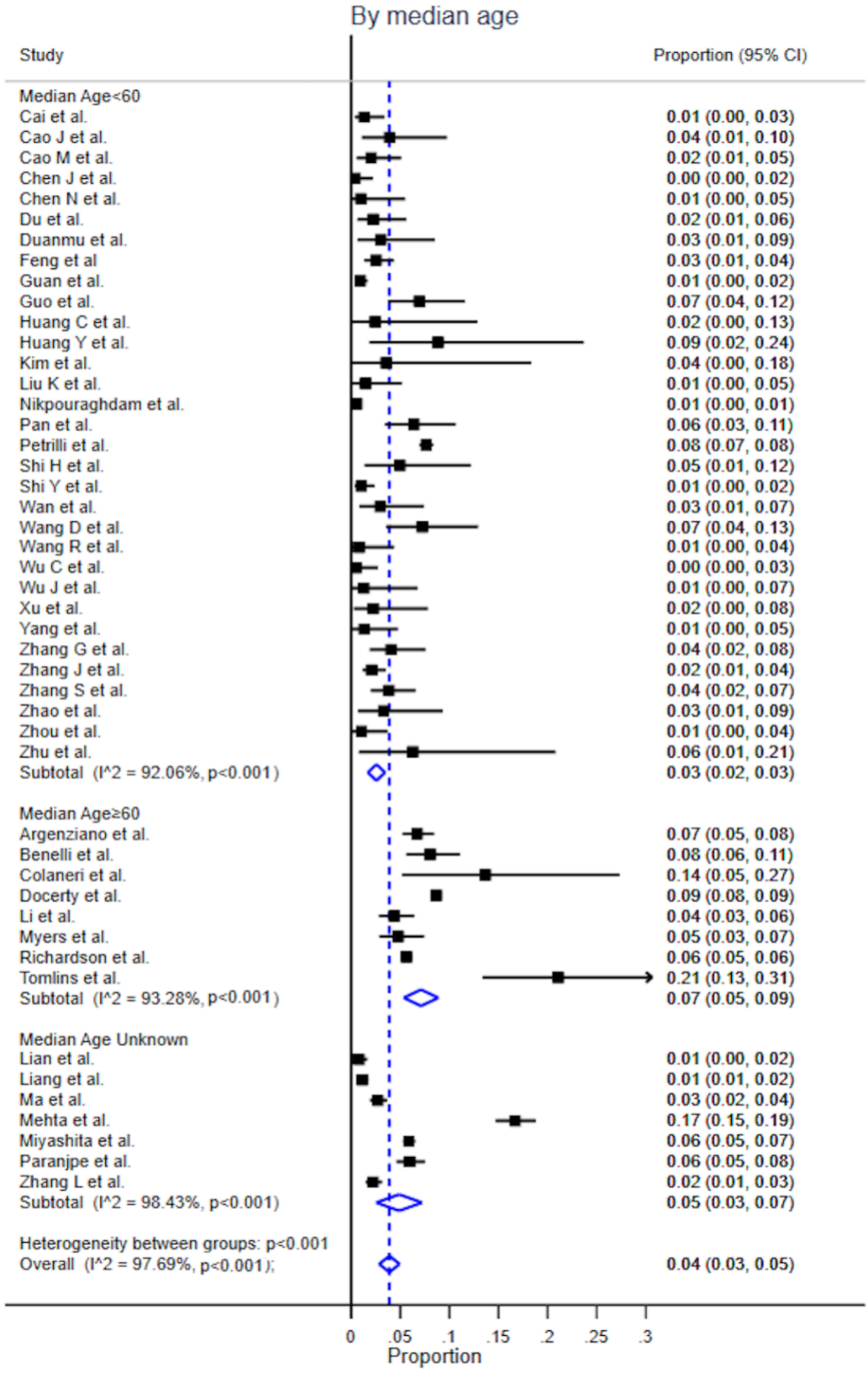


Figure 4: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by male ≥ 60% or < 60%)

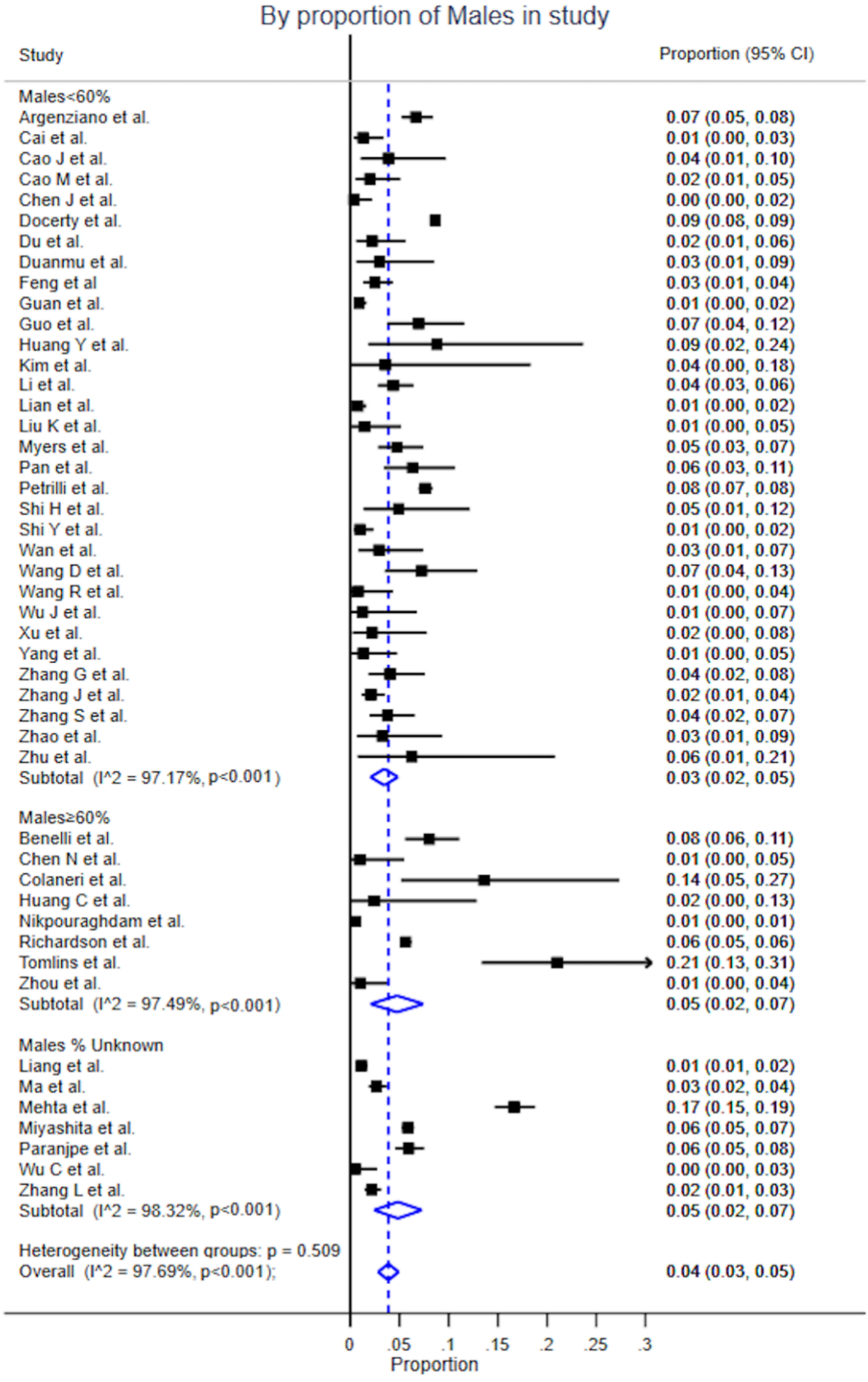


Figure 5a: Random effects pooled relative risk of death for cancer COVID-19 patients versus total population of COVID-19 patients

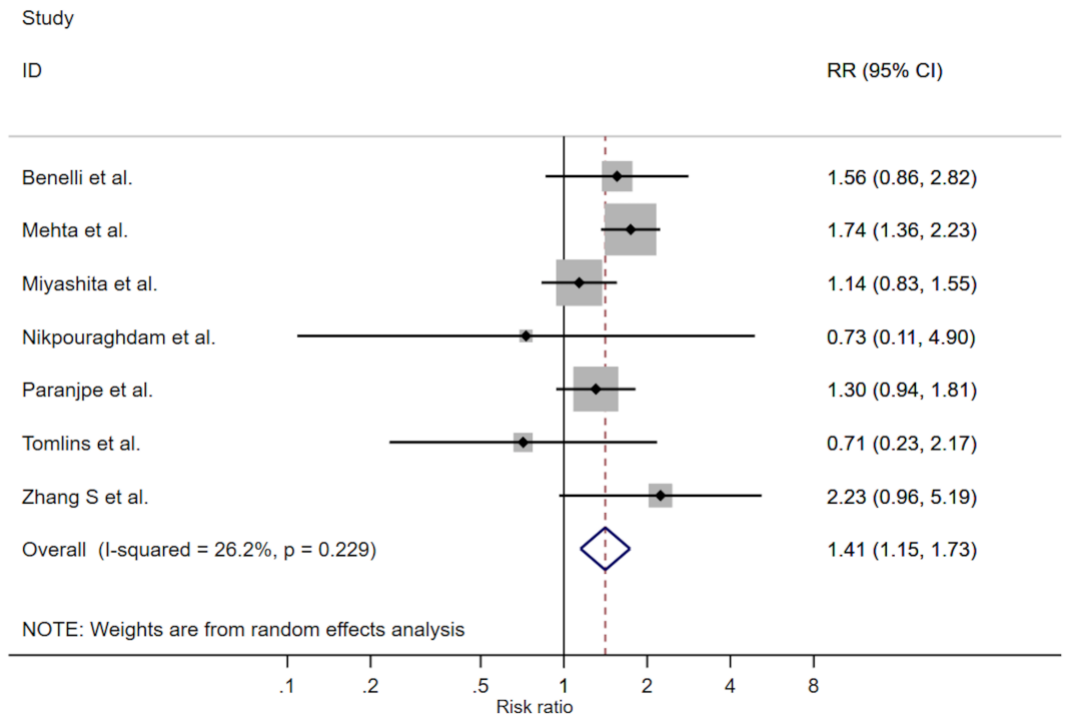


Figure 5b: Random effects pooled relative risk for severe outcome of cancer COVID-19 patients versus total population of COVID-19 patients

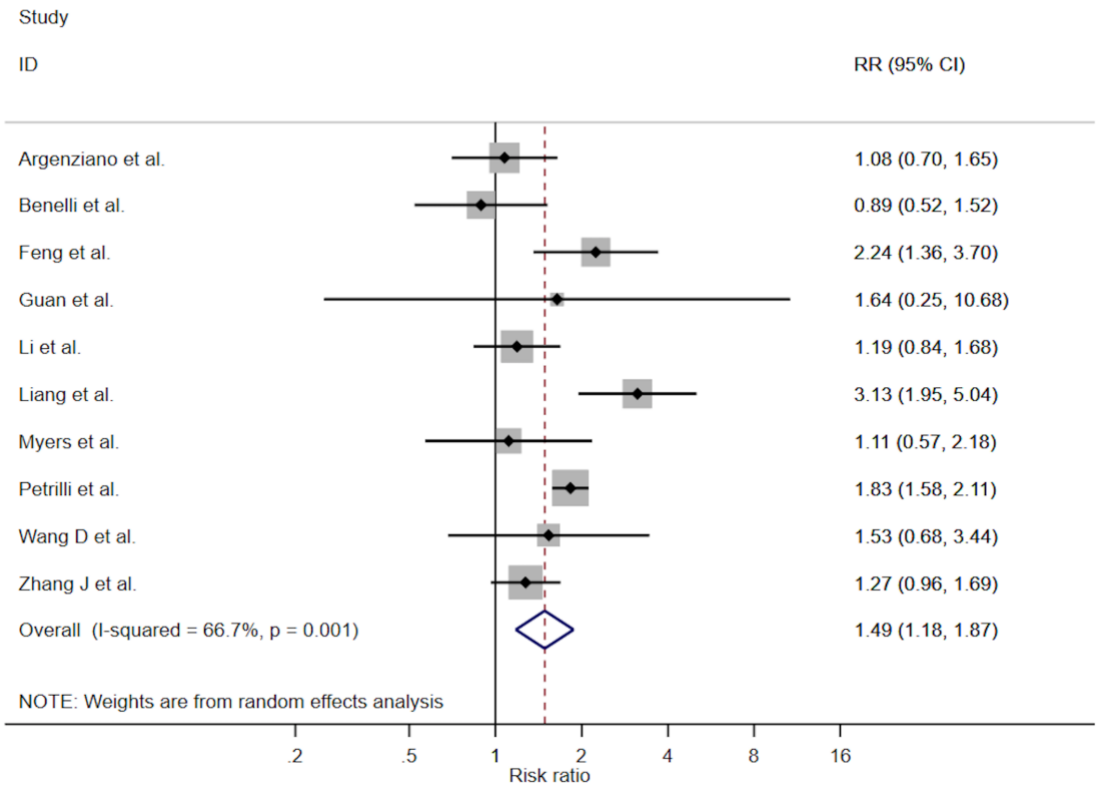


Figure 6a: Random effects pooled relative risk for severe outcome of lung cancer vs other cancer patients infected with COVID-19

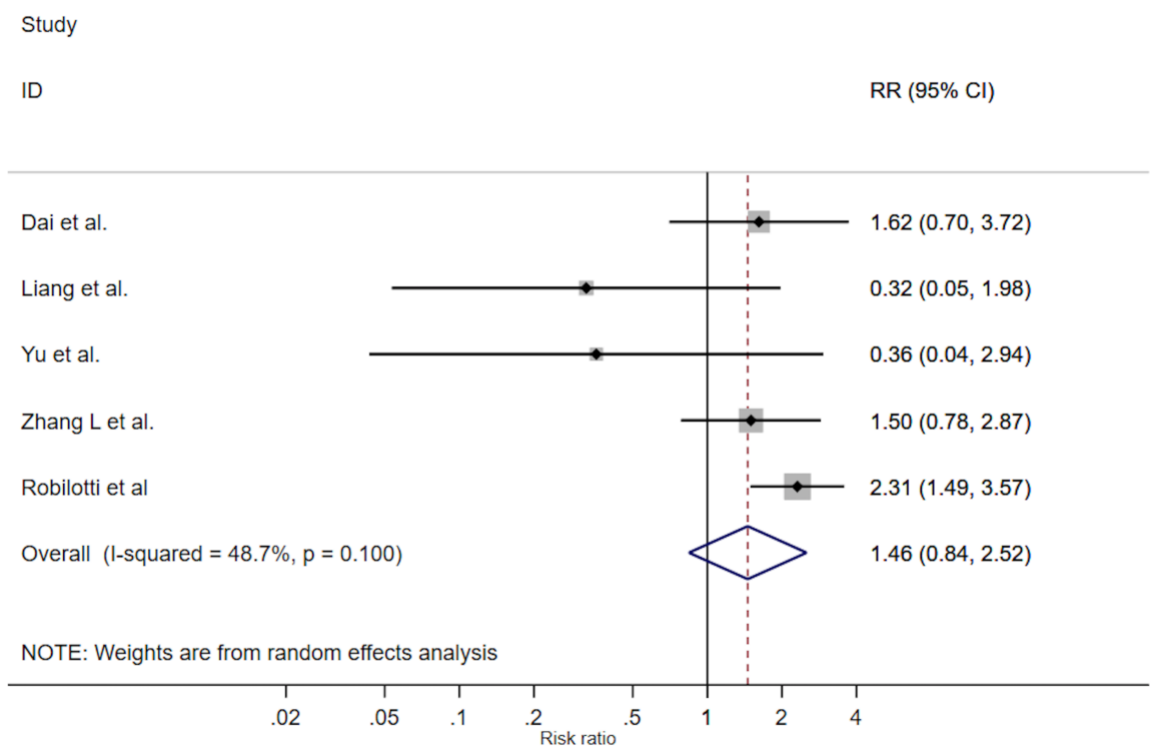
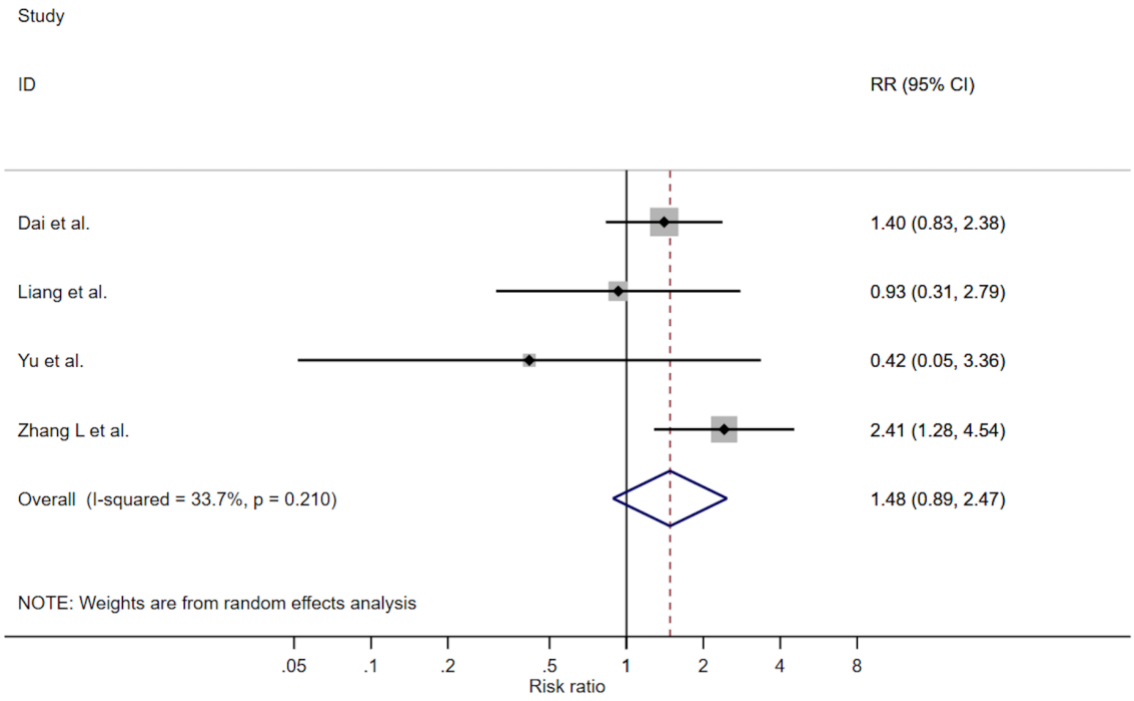


Figure 6b: Random effects pooled relative risk for severe outcome of stage IV cancer vs non-stage IV cancer patients infected with COVID-19



Appendix 1

A. 2019 Novel Coronavirus Disease (COVID-19) Diagnostic criteria 5th edition (5 Feb 2020)

Other provinces except Hubei province

1. Suspected cases by clinical criteria

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Computed tomography (CT) changes for viral pneumonia

c) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease. Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor all of the above clinical manifestations.

2. Laboratory confirmed cases

Suspected cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by real-time reverse transcriptase polymerase chain reaction (RT-PCR) assay.

2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

Hubei province

1. Suspected cases

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease.

Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor any two of clinical manifestations

2. Clinically diagnosed cases

CT chest changes that are consistent with atypical viral pneumonia.

3. Laboratory confirmed cases

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Suspected cases or clinically diagnosed cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by RT-PCR assay.
2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

B. National Health Commission Guideline on Severity Classification of COVID-19 (China)

Type	Criteria
Mild	the clinical symptoms were light, and there was no sign of pneumonia on imaging
Moderate	with fever, respiratory tract and other symptoms, imaging suggests pneumonia
Severe	coincide with any of the following: (1) respiratory distress, respiration rate (RR) ≥ 30 times / min; (2) the oxygen saturation ≤ 93% in the resting state; (3) PaO_2 / FiO_2 ≤ 300 mmHg (1mmHg = 0.133 kPa)
Critically ill	coincide with any of the following: (1) respiratory failure occurs and mechanical ventilation is required; (2) shock; (3) the patient develops other organ failure and needs ICU monitoring and treatment

(A and B released in The Diagnosis and Treatment Protocol for Novel Coronavirus Pneumonia, Version 5, National Health Commission, China)

C. IDSA/ATS Guidelines on Severe Pneumonia 2007

2007 Infectious Diseases Society of America/American Thoracic Society Criteria for Defining Severe Community-acquired Pneumonia
Validated definition includes either one major criterion or three or more minor criteria
Minor criteria
Respiratory rate ≥ 30 breaths/min
Pa_{O2}/Fi_{O2} ratio ≤ 250
Multilobar infiltrates
Confusion/disorientation
Uremia (blood urea nitrogen level ≥ 20 mg/dl)
Leukopenia* (white blood cell count < 4,000 cells/μl)
Thrombocytopenia (platelet count < 100,000/μl)
Hypothermia (core temperature < 36°C)
Hypotension requiring aggressive fluid resuscitation
Major criteria
Septic shock with need for vasopressors
Respiratory failure requiring mechanical ventilation

*Due to infection alone (i.e., not chemotherapy induced).

Metlay et al. Diagnosis and Treatment of Adults with Community-acquired Pneumonia. An Official Clnical Practice Guideline of the American Thoracic Society and Infectious Diseases Society of America. AJRCCM 2019.

D. Search criteria

MeSH terms for Pubmed/Medline (also used for google scholar search):
COVID-19, coronavirus, nCoV, SARS-CoV-2, characteristics, comorbidity, mortality, death, risk, oncology, cancer, malignancy, neoplasm, tumor, tumour, and carcinoma.

Search strategy on Pubmed:
(COVID-19 OR coronavirus OR nCoV OR SARS-CoV-2) AND (characteristics OR comorbidity OR mortality OR death OR risk) AND (oncology OR cancer OR malignancy OR neoplasm OR tumor OR tumour OR carcinoma)
Filtered for English language, full article and date range November 1, 2019 to December 31, 2020

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E. PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N.A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4-5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4, appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Appendix 2
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4-5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4-5

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4-5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4-5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-10, Appendix 2-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	2, 6-12 Appendix 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	6-12, Appendix 3
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

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Appendix 2

Supplementary Table 1: Case series included in meta-analysis

No.	Author	Title	No.	Type of study	Outcome measured	Quality of study	Hospital	Location
1	Argenziano et al	Characterization and clinical course of 1000 patients with coronavirus disease 2019 in New York: retrospective case series	1000	Single centre retrospective cohort study	Severity, death	7	New York-Presbyterian/Columbia University Irving Medical Center	New York, USA
2	Assaad et al	High mortality rate in cancer patients with symptoms of COVID-19 with or without detectable SARS-COV-2 on RT-PCR	302	Single centre retrospective cohort study	Death	7	Comprehensive Cancer Center of Lyon	Lyon France
3	Benelli et al. 2020	SARS-CoV-2 comorbidity and outcome in hospitalized patients in Crema, Italy.	539	Single centre prospective cohort study	Severity	7	General Hospital of Crema	Crema, Italy
4	Cai et al. 2020	COVID-19 in a designated infectious diseases hospital outside Hubei Province, China.	298	Single centre retrospective cohort study	Severity	7	Third People's Hospital of Shenzhen	Shenzhen, China
5	Cao M et al. 2020	Clinical Features of Patients Infected with the 2019 Novel Coronavirus (COVID-19) in Shanghai, China.	198	Single centre retrospective cohort study (not-peer reviewed, pre-print)	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 15 Feb 2020	Shanghai, China
6	Cao J et al. 2020	Clinical features and short-term outcomes of 102 patients with coronavirus disease 2019 in Wuhan, China.	102	Single centre retrospective cohort study	Death	7	Wuhan University Zhongnan Hospital in Wuhan, China, 3 Jan to 1 Feb 2020	Wuhan, Hubei, China
7	Chen J et al. 2020	Clinical progression of patients with COVID-19 in Shanghai, China.	249	Single centre retrospective cohort study	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 6 Feb, 2020	Shanghai, China
8	Chen N et al. 2020	Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study.	99	Single centre retrospective cohort study	Severity	7	Jinyintan Hospital in Wuhan, 1 to 20 Jan 2020	Wuhan, Hubei, China
9	Colaneri et al. 2020	Clinical characteristics of coronavirus disease (COVID-19) early findings from a teaching hospital in Pavia, North Italy, 21 to 28 February 2020.	44	Single centre retrospective cohort study	Severity	6	Fondazione IRCCS Policlinico San Matteo, 21 to 28 Feb 2020	Pavia, Italy
10	Dai et al. 2020	Patients with cancer appear more vulnerable to SARS-COV-2: a multi-center study during the COVID-19 outbreak.	641	Multicentre retrospective matched cohort study	Severity	8	14 hospitals in Wuhan, China, 1 Jan to 24 Feb 2020 (Zhongnan, Tongji, Renmin Hospital etc)	Wuhan, Hubei, China
11	Docherty et al 2020	Features of 20 133 UK patients in hospital with covid-19 using the ISARIC WHO Clinical Characterisation Protocol: prospective observational cohort study	20133	Multi centre prospective cohort study	Severity, death	7	Hospitals in England and Wales, UK	UK
12	Du et al. 2020	Predictors of Mortality for Patients with COVID-19 Pneumonia Caused by SARS-CoV-2: A Prospective Cohort Study.	179	Single centre prospective cohort study	Death	7	Wuhan Pulmonary Hospital, 25 Dec 2019 to 7 Feb 2020	Wuhan, Hubei, China
13	Duanmu et al. 2020	Characteristics of Emergency Department Patients With COVID-19 at a Single Site in Northern California: Clinical Observations and Public Health Implications.	100	Single centre observational cross-section study	Admission from ED, Death	6	Stanford University School of Medicine, 4 to 23 March 2020	California, USA
14	Feng et al. 2020	COVID-19 with Different Severity: A Multi-center Study of Clinical Features	476	Single centre retrospective cohort study	Severity and death	7	3 hospitals in Wuhan, Shanghai and Anhui, 1 Jan to 15 Feb 2020	China
15	Guan et al. 2020	Clinical Characteristics of Coronavirus Disease 2019 in China.	1099	Multicentre retrospective cohort study	Severity and death	7	552 hospitals across China, December 11, 2019, and January 29 2020	China
16	Guo et al. 2020	Cardiovascular Implications of Fatal Outcomes of Patients With Coronavirus Disease 2019 (COVID-19)	187	Single centre retrospective cohort study	Severity and death	7	Seventh Hospital of Wuhan City, from 23 Jan to 23 Feb 2020	Wuhan, Hubei, China
17	Huang C et al. 2020	Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China.	41	Single centre prospective cohort study	Severity	7	Jinyintan Hospital till 2 Jan 2020	Wuhan, Hubei, China
18	Huang Y et al. 2020	Clinical characteristics of laboratory confirmed positive cases of SARS-CoV-2 infection in Wuhan, China: A retrospective single center analysis.	34	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital, 21 Dec to 28 Jan 2020	Wuhan, Hubei, China
19	Kim et al. 2020	Clinical Course and Outcomes of Patients with Severe Acute Respiratory Syndrome Coronavirus 2 Infection: a Preliminary Report of the First 28 Patients from the Korean Cohort Study on COVID-19.	28	Multicentre retrospective cohort study	Severity	6	11 Hospitals in Korea	Korea
20	Kuderer et al. 2020	Clinical impact of COVID-19 on patients with cancer (CCC19): a cohort study	928	Multicentre retrospective cohort study	Severity, death	7	US, Canada, Spain Hospitals	US, Canada, Spain
21	Lee LY et al.	COVID-19 prevalence and mortality in patients with cancer and the effect of primary tumour subtype and patient demographics: a prospective cohort study	800	Multicentre prospective cohort study	Severity, death	7	UK Hospitals	UK
22	Li et al. 2020	Risk factors for severity and mortality in adult COVID-19 inpatients in Wuhan.	548	Single centre retrospective cohort study	Severity and death	7	Tongji Hospital, 26 Jan to 5 Feb 2020	Wuhan, Hubei, China
23	Lian et al. 2020	Analysis of Epidemiological and Clinical features in older patients with Corona Virus Disease 2019 (COVID-19) out of Wuhan.	788	Multicentre retrospective cohort study	Severity	8	Zhejiang Province, 17 Jan to 12 Feb 2020	Wuhan. Hubei, China
24	Liang et al. 2020	Cancer patients in SARS-CoV-2 infection: a nationwide analysis in China.	1590	Multicentre retrospective cohort study	Severity and death	8	575 hospitals in China as of 31 Jan 2020	China
25	Liu et al. 2020	Clinical characteristics of novel coronavirus cases in tertiary hospitals in Hubei Province.	137	Multicentre retrospective cohort study	Death	7	9 hospitals in Hubei, 30 Dec 2019 to 24 Jan 2020 (Tongji Hosp, other hospitals outside Wuhan)	Hubei, China
26	Ma et al. 2020	Clinical characteristics and prognosis in cancer patients with COVID-19: A single center's retrospective study.	1380	Single centre retrospective cohort study	Severity and death	8	Renmin Hospital of Wuhan University, 1 Jan to 30 March 2020	Wuhan, Hubei, China
27	Mehta et al. 2020	Case Fatality Rate of Cancer Patients with COVID-19 in a New York Hospital System.	218	Single centre retrospective cohort study (peer-reviewed, pre-print)	Death	7	Montefiore Medical Center and Albert Einstein College of Medicine, 18 March to 8 April 2020	New York, USA
28	Miyashita et al. 2020	Do Patients with Cancer Have a Poorer Prognosis of COVID-19? An Experience in New York City.	5688	Single centre retrospective cohort study	Severity and death	8	Mount Sinai Health System, 1 March to 6 April 2020	New York, USA
29	Myers et al. 2020	Characteristics of Hospitalized Adults With COVID-19 in an Integrated Health Care System in California.	377	Multicentre retrospective cohort study	Severity	7	Kaiser Permanente Northern California (KPNC) health system, 1 to 31 March 2020	California, USA
30	Nikpouraghdam et al. 2020	Epidemiological characteristics of coronavirus disease 2019 (COVID-19) patients in IRAN: A single center study.	2964	Single centre retrospective cohort study	Death	7	Baqiyatallah Hospital, 19 Feb to 15 April 2020	Tehran, Iran
31	Pan et al. 2020	Clinical Characteristics of COVID-19 Patients With Digestive Symptoms in Hubei, China: A Descriptive, Cross-Sectional, Multicenter Study	204	Multicentre retrospective cross sectional study	Severity and death	7	3 hospitals in Hubei province, including Wuhan Hanan Hospital, Wuhan	Hubei, China

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							Union Hospital, and Huanggang Central Hospital, 18 Jan to 28 Feb 2020	
32	Paranjpe et al. 2020	Clinical Characteristics of Hospitalized Covid-19 Patients in New York City.	1078	Multicentre retrospective cohort study (not peer reviewed, pre-print)	Death	8	Mount Sinai Health System Hospitals, 27 Feb to 2 April 2020	New York, USA
33	Petrilli et al 2020	Factors associated with hospital admission and critical illness among 5279 people with coronavirus disease 2019 in New York City: prospective cohort study	5279	Single centre prospective cohort study	Severity, death	7	NYU Langone Health	New York, USA
34	Pinato et al. 2020	Clinical Portrait of the SARS-CoV-2 Epidemic in European Patients with Cancer	890	Multicentre retrospective cohort study	Severity, death	7	Multiple Hospitals in Europe	Europe
35	Robilotti et al. 2020	Determinants of COVID-19 disease severity in patients with cancer	423	Single centre retrospective cohort study	Severity, death	7	MSKCC	New York, USA
36	Richardson et al. 2020	Presenting Characteristics, Comorbidities, and Outcomes Among 5700 Patients Hospitalized With COVID-19 in the New York City Area.	5700	Multicentre retrospective cohort study	Severity and death	8	Northwell Health Hospitals, 1 March to 4 April 2020	New York, USA
37	Shi H et al. 2020	Radiological findings from 81 patients with COVID-19 pneumonia in Wuhan, China: a descriptive study.	81	Multicentre retrospective cohort study	Not defined	6	Wuhan Jinyintan hospital or Union Hospital of Tongji Medical College, 20 Dec to 23 Jan 2020	Wuhan, Hubei, China
38	Shi Y et al. 2020	Host susceptibility to severe COVID-19 and establishment of a host risk score: findings of 487 cases outside Wuhan.	487	Multicentre retrospective cohort study	Not defined	7	Hospitals in Zhejiang Province, until 17 Feb 2020	Zhejiang, China
39	Tomlins et al. 2020	Clinical features of 95 sequential hospitalised patients with novel coronavirus 2019 disease (COVID-19), the first UK cohort	95	Single centre retrospective cohort study	Severity and death	7	North Bristol NHS Trust, 10 to 30 March 2020	UK
40	Wan et al. 2020	Clinical features and treatment of COVID-19 patients in northeast Chongqing.	135	Single centre retrospective cohort study	Not defined	7	Chongqing University Three Gorges Hospital, 23 Jan to 8 Feb 2020	Chongqing, Sichuan, China
41	Wang D et al. 2020	Clinical Characteristics of 138 Hospitalized Patients With 2019 Novel Coronavirus-Infected Pneumonia in Wuhan, China.	138	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital of Wuhan University, 1 to 28 Jan 2020	Wuhan, Hubei, China
42	Wang R et al. 2020	Epidemiological and clinical features of 125 Hospitalized Patients with COVID-19 in Fuyang, Anhui, China.	125	Single centre retrospective cohort study	Severity	7	NO.2 People's Hospital of Fuyang City 20 Jan to 9 Feb 2020.	Anhui, China
43	Wu C et al. 2020	Risk Factors Associated With Acute Respiratory Distress Syndrome and Death in Patients With Coronavirus Disease 2019 Pneumonia in Wuhan, China.	201	Single centre retrospective study	Severity and death	8	Jinyintan Hospital in Wuhan, China, 25 Dec to 26 Jan 2020	Wuhan, Hubei, China
44	Wu J et al. 2020	Clinical Characteristics of Imported Cases of COVID-19 in Jiangsu Province: A Multicenter Descriptive Study.	80	Multicentre retrospective cohort study	Severity	7	Three Grade IIIA hospitals of Jiangsu, 22 Jan to 14 Feb 2020	Jiangsu, China
45	Xu et al. 2020	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2.	90	Single centre retrospective study	Severity	7	Guangzhou Eighth People's Hospital, 23 Jan to 4 Feb 2020	Guangzhou, China
46	Yang et al. 2020	Clinical characteristics and imaging manifestations of the 2019 novel coronavirus disease (COVID-19):A multi-center study in Wenzhou city, Zhejiang, China.	149	Multicentre retrospective cohort study	Severity	7	Three tertiary hospitals of Wenzhou, 17 Jan to 10 Feb 2020	Wenzhou, Zhejiang, China
47	Yang K et al 2020	Clinical characteristics, outcomes, and risk factors for mortality in patients with cancer and COVID-19 in Hubei, China: a multicentre, retrospective, cohort stud	205	Multientre retrospective cohort study	Severity, death	7	9 Hospitals in Wuhan, 11 Jan to 18 March (overlap with other case series)	Wuhan, China
48	Yarza et al	SARS-CoV-2 infection in cancer patients undergoing active treatment: analysis of clinical features and predictive factors for severe respiratory failure and death	63	Single centre retrospective cohort study	Death	6	Hospital in Madrid	Madrid, Spain
49	Yu et al. 2020	SARS-CoV-2 Transmission in Patients With Cancer at a Tertiary Care Hospital in Wuhan, China.	1524	Single centre retrospective study	Severity and death	6	Zhongnan Hospital, 30 Dec to 17 Feb 2020	Wuhan, Hubei, China
50	Zhang G et al. 2020	Clinical features and short-term outcomes of 221 patients with COVID-19 in Wuhan. China.	221	Single centre retrospective cohort study	Severity	7	Zhongnan Hospital of Wuhan, 2 Jan to 10 Feb 2020	Wuhan, Hubei, China
51	Zhang J et al. 2020	Risk factors for disease severity, unimprovement, and mortality in COVID-19 patients in Wuhan, China.	663	Single centre retrospective cohort study	Severity	7	Renmin Hospital of Wuhan University, 11 Jan to 2 Feb 2020	Wuhan, Hubei, China
52	Zhang L et al. 2020	Clinical characteristics of COVID-19-infected cancer patients: a retrospective case study in three hospitals within Wuhan, China.	1276	Single centre retrospective cohort study	Severity and death	8	Tongji Sino-French New Town Hospital, Union Red Cross Hospital, and Union West Hospital, 13 Jan to 26 Feb 2020	Wuhan, Hubei, China
53	Zhang N et al. 2020	A multicentre study of coronavirus disease 2019 outcomes of cancer patients in Wuhan, China	1548	Multicentre retrospective observational study (peer reviewed, pre-print)	Severity, death	7	Zhongnan Hospital of Wuhan University, the Fifth hospital of Wuhamn, the seventh hospital of Wuhan and Wuhan Hankou hospital, 5 Jan to 18 Feb 2020	Wuhan, Hubei, China
54	Zhang S et al. 2020	Short Term Outcomes and Risk Factors for Mortality in Patients with COVID-19 in Wuhan, China: A Retrospective Study.	315	Single centre retrospective cohort study	Death	7	Wuhan Union Hospital West Area, 12 Jan to 3 Feb 2020	Wuhan, Hubei, China
55	Zhao et al. 2020	Clinical characteristics of patients with 2019 coronavirus disease in a non-Wuhan area of Hubei Province, China: a retrospective study.	91	Single centre retrospective cohort study	Severity	7	Jingzhou Central Hospital, 16 Jan to 10 Feb 2020	Jingzhou, Hubei, China
56	Zhou et al. 2020	Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study.	191	Multicentre retrospective cohort study	Death	7	Jinyintan Hospital and Wuhan Pulmonary Hospital, 29 Dec to 31 Jan 2020	Wuhan, Hubei, China
57	Zhu et al. 2020	Initial clinical features of suspected coronavirus disease 2019 in two emergency departments outside of Hubei, China.	32	Single centre retrospective cohort study	Not defined	6	First Affiliated Hospital of USTC, 24 Jan to 20 Feb 2020	Hefei, China

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Supplementary Table 2: Meta-analyses and database reports on COVID-19 (not included in analysis)

No.	Authors	Title:
1.	Desai et al.	COVID-19 and Cancer: Lessons From a Pooled Meta-Analysis
2.	Emami et al.	Prevalence of Underlying Diseases in Hospitalized Patients with COVID-19: a Systematic Review and Meta-Analysis
3.	Wang B et al.	Does comorbidity increase the risk of patients with COVID-19: evidence from meta-analysis
4.	Cao Y et al.	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2: A systematic review and meta-analysis
5.	Hu Y et al.	Prevalence and severity of corona virus disease 2019 (COVID-19): A systematic review and meta-analysis
6.	Yang J et al.	Prevalence of comorbidities and its effects in patients infected with SARS-CoV-2: a systematic review and meta-analysis
7.	Fu L et al.	Clinical characteristics of coronavirus disease 2019 (COVID-19) in China: A systematic review and meta-analysis
8.	Li LQ et al.	COVID-19 patients' clinical characteristics, discharge rate, and fatality rate of meta-analysis
9.	Zhu J et al.	Clinicopathological Characteristics of 8697 Patients With COVID-19 in China: A Meta-Analysis
10.	Zheng Z et al.	Risk factors of critical & mortal COVID-19 cases: A systematic literature review and meta-analysis
11.	Rodriguez-Morales et al.	Clinical, laboratory and imaging features of COVID-19: A systematic review and meta-analysis
12.	Ofori-asenso et al.	Cancer is associated with severe disease in COVID-19 patients: a systematic review and meta-analysis
13.	Khan et al.	Effects of underlying morbidities on the occurrence of deaths in COVID- 19 patients: A systematic review and meta-analysis (in-press)
14.	Moujaess et al.	Cancer patients and research during COVID-19 pandemic: A systematic review of current evidence
15.	Sun P et al.	Clinical characteristics of hospitalized patients with SARS-CoV-2 infection: A single arm meta-analysis
16.	Jin JM et al.	Gender Differences in Patients with COVID-19 Focus on Severity and Mortality
17.	Deng G et al.	Clinical determinants for fatality of 44,672 patients with COVID-19
18.	Wu and Mcgoogan	Characteristics of and Important Lessons From the Coronavirus Disease 2019 (COVID-19) Outbreak in China: Summary of a Report of 72 314 Cases From the Chinese Center for Disease Control and Prevention
19.	Onder G, Rezza G, Brusaferro S.	Case-Fatality Rate and Characteristics of Patients Dying in Relation to COVID-19 in Italy
20.	Guan et al	Comorbidity and its impact on 1590 patients with Covid-19 in China: A Nationwide Analysis
21.	Adams ML, Katz DL, Grandpre J.	Population-Based Estimates of Chronic Conditions Affecting Risk for Complications from Coronavirus Disease, United States
22.	Kang YJ et al	Mortality Rate of Infection With COVID-19 in Korea From the Perspective of Underlying Disease
23.	Barek et al	Impact of age, sex, comorbidities and clinical symptoms on the severity of COVID-19 cases: A meta-analysis with 55 studies and 10014 cases.
24.	Fuentes-Antrás J	A snapshot of COVID-19 infection in patients with solid tumors.
25.	Sng C et al	Cancer History and Systemic Anti-Cancer Therapy Independently Predict COVID-19 Mortality: A UK Tertiary Hospital Experience.
26.	Pinto et al	SARS-CoV-2 Positive Hospitalized Cancer Patients during the Italian Outbreak: The Cohort Study in Reggio Emilia. Biology (Basel).
27.	Sun L et al	Rates of COVID-19-related Outcomes in Cancer compared to non-Cancer Patients. (Pre-print)
28.	Anantharaman et al	SARS-CoV-2 Clinical Outcomes in Patients with Cancer in a Large Integrated Health Care System in Northern California.
29.	Zarifkar P et al	Clinical Characteristics and Outcomes in Patients with COVID-19 and Cancer: a Systematic Review and Meta-analysis.
30.	Jarahzadeh MH et al	Cancer and Coronavirus Disease (COVID-19): Comorbidity, Mechanical Ventilation, and Death Risk.
31.	ElGohary GM et al	The risk and prognosis of COVID-19 infection in cancer patients: A systematic review and meta-analysis.
32.	Russell B et al	Factors Affecting COVID-19 Outcomes in Cancer Patients: A First Report From Guy's Cancer Center in London.
33.	Duarte MBO et al	Outcomes of COVID-19 Patients under Cytotoxic Cancer Chemotherapy in Brazil.
34.	Gagandeep Brar et al	COVID-19 Severity and Outcomes in Patients With Cancer: A Matched Cohort Study
35.	Yang B et al	Risk of Coronavirus Disease Incidence, Severe Presentation, and Mortality in Patients with Lung Cancer.
36.	Motamedi F et al	Clinical characteristics, outcomes, and risk factors for mortality in hospitalized patients with COVID-19 and cancer history: a propensity score-matched study.
37.	Cheruiyot I et al	Cancer is associated with coronavirus disease (COVID-19) severity and mortality: A pooled analysis.

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38.	Antikchi MH et al	The Risk and Prevalence of COVID-19 Infection in Colorectal Cancer Patients: a Systematic Review and Meta-analysis.
39.	Hultcrantz M et al	COVID-19 infections and outcomes in patients with multiple myeloma in New York City: a cohort study from five academic centers.
40.	Fillmore NR et al	Prevalence and outcome of COVID-19 infection in cancer patients:a national Veterans Affairs study.
41.	Venkatesulu BP et al	A systematic review and meta-analysis of cancer patients affected by a novel coronavirus.
42.	Joharatnam-Hogan et al	Outcomes of the 2019 novel coronavirus in patients with or without a history of cancer: a multi-centre North London experience.
43.	Alpert N et al	Clinical Course of Cancer Patients With COVID-19: A Retrospective Cohort Study.
44.	Ali J et al	COVID-19-positive cancer patients undergoing active anticancer treatment: An analysis of clinical features and outcomes.
45.	Jee J et al	Chemotherapy and COVID-19 Outcomes in Patients With Cancer
46.	Liu H et al	The effect of anticancer treatment on cancer patients with COVID-19: A systematic review and meta-analysis.
47.	Park R et al	Sex-bias in COVID-19-associated illness severity and mortality in cancer patients: A systematic review and meta-analysis.
48.	Asokan I et al	The COVID-19 Pandemic and its Impact on the Cardio-Oncology Population.
49.	Rogado J et al	Characteristics and outcome of lung cancer patients and Covid-19 infection in a second-level hospital during the first 3 months of the pandemic.
50.	Park R et al	Association of active oncologic treatment and risk of death in cancer patients with COVID-19: a systematic review and meta-analysis of patient data

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The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort
   a) truly representative of the average cancer COVID-19 cases in the community *****
   b) somewhat representative of the average cancer COVID-19 cases in the community *****
   c) selected group of users eg nurses, doctors
   d) no description of the derivation of the cohort

2) Selection of the non-exposed cohort
   a) drawn from the same community as the exposed cohort *****
   b) drawn from a different source
   c) no description of the derivation of the non-exposed cohort

3) Ascertainment of exposure
   a) secure record (eg medical records) *****
   b) structured interview *****
   c) written self-report
   d) no description

4) Demonstration that outcome of interest was not present at start of study
   a) yes *****

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b) no

Comparability

5) Comparability of cohorts on the basis of the design or analysis
a) study controls for median age *****
b) study controls for any additional factors such as other co-morbidities *****

Outcome

6) Assessment of outcome
a) independent blind assessment *****
b) record linkage *****
c) self-report
d) no description

7) Was follow-up long enough for outcomes to occur
a) yes (until discharge or death or at least 1 month, whichever come first) *****
b) no

8) Adequacy of follow up of cohorts
a) complete follow up - all subjects accounted for *****
b) subjects lost to follow up unlikely to introduce bias - small number lost - > 80 % follow up, or description provided of those lost) *****
c) follow up rate < 60 % and no description of those lost
d) no statement

Case-control studies

1) Is the case definition adequate?
a) yes, with independent validation *****
b) yes, eg record linkage or based on self reports
c) no description

2) Representativeness of the cases
a) consecutive or obviously representative series of cases *****
b) potential for selection biases or not stated

3) Selection of Controls
a) community controls *****
b) hospital controls
c) no description

4) Definition of Controls
a) no history of disease (endpoint) *****
b) no description of source

Comparability

5) Comparability of cases and controls on the basis of the design or analysis
a) study controls for median age *****
b) study controls for any additional factor ***** (co-morbidities)

Exposure

6) Ascertainment of exposure
a) secure record (eg surgical records) *****
b) structured interview where blind to case/control status *****
c) interview not blinded to case/control status
d) written self report or medical record only
e) no description

7) Same method of ascertainment for cases and controls
a) yes *****

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b) no

8) Non-Response rate
a) same rate for both groups *
b) non respondents described
c) rate different and no designation

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Supplementary Table 3: The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

N	Authors	Selection				Comparability		Outcome			Study quality
	Cohort studies	1 Exposed cohort truly representative	2 Non-exposed cohort drawn from same community	3 Ascertainment of exposure	4 Outcome of interest not present at start	5A Cohorts comparable ie. Median age	5B Cohorts comparable on other co-morbidities	6 Quality of outcome assessment	7 Follow-up long enough for outcomes to occur	8 Complete accounting for cohorts	
	Case control studies	1 Case definition adequate	2 Representativeness with the cases	3 Selection of controls	4 Definition of controls	5A Case and control comparable on the basis of analysis ie median age	5B Case and control comparable on other co-morbidities	6 Ascertainment of exposure	7 Same method of ascertainment for cases and controls	8 Non-Response rate	
1	Argenziano et al.	*	*	*	*			*	*	*	7
2	Assaad et al.	*	*	*	*			*	*	*	7
3	Benelli et al.	*	*	*	*			*	*	*	7
4	Cai et al.	*	*	*	*			*	*	*	7
5	Cao M et al.	*	*	*	*			*	*	*	7
6	Cao J et al.	*	*	*	*			*	*	*	7
7	Chen J et al	*	*	*	*			*	*	*	7
8	Chen N et al.	*	*	*	*			*	*	*	7
9	Colaneri et al.	*	*	*	*			*		*	6
10	Dai et al.	*	*	*	*	*		*	*	*	8
11	Docherty et al.	*	*	*	*			*	*	*	7
12	Du et al.	*	*	*	*			*	*	*	7
13	Duanmu et al.	*	*	*	*			*		*	6
14	Feng et al.	*	*	*	*			*	*	*	7
15	Guan et al.	*	*	*	*			*	*	*	7
16	Guo et al.	*	*	*	*			*	*	*	7
17	Huang C et al.	*	*	*	*			*	*	*	7
18	Huang Y et al.	*	*	*	*			*		*	6
19	Kim et al.	*	*	*	*			*		*	6
20	Kuderer et al.	*	*	*	*			*	*	*	7
21	Lee LY et al.	*	*	*	*			*	*	*	7
22	Li et al.	*	*	*	*			*	*	*	7
23	Lian et al.	*	*	*	*	*		*	*	*	8
24	Liang et al.	*	*	*	*		*	*	*	*	8
25	Liu et al.	*	*	*	*			*	*	*	7
26	Ma et al.	*	*	*	*	*		*	*	*	8
27	Mehta et al.	*	*	*	*			*	*	*	7
28	Miyashita et al.	*	*	*	*	*		*	*	*	8
29	Myers et al.	*	*	*	*			*	*	*	7
30	Nikpouraghdam et al.	*	*	*	*			*	*	*	7
31	Pan et al.	*	*	*	*			*	*	*	7
32	Paranjpe et al.	*	*	*	*		*	*	*	*	8
33	Petrilli et al.	*	*	*	*			*	*	*	7
34	Pinato et al.	*	*	*	*			*	*	*	7
35	Robilotti et al.	*	*	*	*			*	*	*	7
36	Richardson et al.	*	*	*	*	*		*	*	*	8
37	Shi H et al.	*	*	*	*			*		*	6
38	Shi Y et al.	*	*	*	*	*		*	*		7
39	Tomlins et al	*	*	*	*			*	*	*	7
40	Wan et al.	*	*	*	*			*	*	*	7
41	Wang D et al.	*	*	*	*			*	*		6
42	Wang R et al.	*	*	*	*			*	*	*	7
43	Wu C et al.	*	*	*	*	*		*	*	*	8
44	Wu J et al.	*	*	*	*			*	*	*	7
45	Xu et al.	*	*	*	*			*	*	*	7
46	Yang et al.	*	*	*	*			*	*	*	7
47	Yang K et al.	*	*	*	*			*	*	*	7
48	Yarza et al.	*	*	*	*			*	*		6
49	Yu et al.	*		*	*			*	*	*	6
50	Zhang G et al.	*	*	*	*			*	*	*	7
51	Zhang J et al.	*	*	*	*			*	*	*	7
52	Zhang L et al.	*	*	*	*		*	*	*	*	8
53	Zhang N et al.	*	*	*	*			*	*	*	7
54	Zhang S et al.	*	*	*	*			*	*	*	7
55	Zhao et al.	*	*	*	*			*	*	*	7
56	Zhou et al.	*	*	*	*			*	*	*	7
57	Zhu et al.	*	*	*	*				*	*	6

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Appendix 3

Supplementary Table 1: List of 230 patients with available individual patient data and oncological history

Liang et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer in years**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	83	No	4	Bladder ca	Postoperative routine follow-up	Y	NA	N
2	M	87	DM, HTN	2	Colon ca	Postoperative routine follow-up	Y	NA	N
3	M	67	No	0	Adrenal neoplasm	3 weeks after operation	Y	NA	N
4	F	68	No	8	Breast ca	Surgical resection plus adjuvant chemotherapy 8 years ago	Y	NA	N
5	M	58	No	7	CRCC	Recurrence, on immunotherapy	Y	NA	Y
6	F	62	No	4	Breast ca	Surgical resection plus adjuvant chemotherapy 4 years ago	Y	NA	N
7	M	56	No	5	Rectal ca	Surgical resection plus adjuvant chemotherapy 5 years ago	N	NA	N
8	M	53	No	16	Colon ca	Surgical resection plus adjuvant chemotherapy 16 years ago	N	NA	N
9	F	63	No	1	Papillary thyroid ca	Postoperative, in TSH inhibition therapy	N	NA	N
10	M	47	No	1	Lung adenoca	In chemotherapy for advanced tumor	N	NA	Y
11	F	52	No	0.5	Breast ca	Postoperative, loss of chemotherapy information	N	NA	N
12	M	47	No	1	Lymphoma	N.A.	N	NA	NA
13	M	80	DM, HTN, CVS	4	Bladder ca	Postoperative, no chemotherapy information	Y	NA	N
14	M	79	COPD	5	Colorectal ca	Surgical resection plus adjuvant radiotherapy 5 years ago	Y	NA	N
15	M	63	No	1	Lung adenoca	In chemotherapy for advanced tumour	Y	NA	Y
16	F	58	CKD	2	Lung carcinoma in situ	Postoperative routine follow-up	N	NA	N
17	M	58	No	2	Lung adenoca	Postoperative, on targeted therapy	N	NA	N
18	F	55	No	6	Lung adenoca	Advanced, on targeted therapy	N	NA	Y
Yu et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	NA	NA	NA	NSCLC	Concurrent TKI with RT	N	Y	N
2	M	NA	NA	NA	NSCLC	Week 5 of chemoradiotherapy	Y	N	N
3	F	NA	NA	NA	NSCLC	Pleural effusion, BSC	N	Y	Y
4	M	NA	NA	NA	NSCLC	Follow up at 2 year post chemoRT	Y	N	N
5	M	NA	NA	NA	NSCLC	After first cycle of doce, cis and sintilimab	N	Y	Y
6	M	NA	NA	NA	NSCLC	After third cycle of pem/carbo and pembro	N	Y	Y
7	M	NA	NA	NA	NSCLC	Surgery (April 2019) followed by adjuvant doce/cis	N	Y	N
8	M	NA	NA	NA	Rectal ca	BSC	N	Y	Y
9	M	NA	NA	NA	Colon ca	Newly diagnosed	N	Y	NA
10	M	NA	NA	NA	Pancreatic ca	BSC	Y	N	Y
11	F	NA	NA	NA	Breast ca	Adjuvant radiotherapy	N	Y	N
12	M	NA	NA	NA	Urothelial ca	BSC	N	Y	Y
Zhang L et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer in months**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	56	CHB, cirrhosis	26	Liver ca	Stage IV, recurrence with bone mets, bone RT with 14 days	Y	NA	Y
2	M	74	DM	3	Esophageal ca	Stage IV, chemotherapy 1 month ago	Y	NA	Y
3	M	66	HTN, CHD	11	Lung SCC	Surgical resection plus adjuvant chemoRT 10m ago, immune therapy 3m ago, gem and nivo within 14 days	Y	NA	Y
4	M	65	No	23	Lung adenoca	Stage IV, chemo for advanced tumour. Last cycle of pemetrexed was within 14 days	Y	NA	Y
5	M	65	No	4	Esophageal ca	Stage III, surgical resection plus adjuvant chemo and RT	N	NA	N
6	F	48	DM, HTN	5	Serous ca of ovary	Surgical resection plus adjuvant chemo 4m ago, 3 weeks after THBSO	N	NA	N
7	M	54	No	13	Laryngocarcinoma	4 weeks after operation	N	NA	N
8	M	80	HTN, CHD	1	Colorectal ca	3 weeks after operation and neoadjuvant chemoRT	N	NA	N
9	M	65	No	13	Esophageal ca	Surgical resection plus adjuvant chemo and RT 11m ago	Y	NA	0
10	F	70	No	124	Lung adenoca	Postoperative, recurrence. On targeted therapy	Y	NA	Y
11	F	64	No	26	Lung adenoca	Surgical resection plus adjuvant chemo, on EGFR TKI for advanced tumour	N	NA	Y
12	F	24	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 1 year ago	Y	NA	N
13	F	70	DM, HTN	148	Breast ca	Surgical resection plus adjuvant chemo and RT 12 year ago	Y	NA	N
14	F	59	No	62	Lung adenoca	Stage IV, on EGFR TKI treatment	Y	NA	Y
15	F	68	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 20m ago	N	NA	N
16	F	60	No	9	Endometrial ca	Surgical resection 7m ago, on adjuvant chemo 3 weeks ago	N	NA	N
17	M	65	No	26	Small cell lung ca	Maintenance chemo for advanced tumour	N	NA	Y
18	M	57	No	15	Lung SCC	Surgical resection plus adjuvant chemoRT 1 year ago	N	NA	N
19	F	65	No	15	Cervical ca	Surgical resection and adjuvant chemo 1 week ago	N	NA	N
20	M	47	No	51	NPC	Concurrent chemoRT 4m ago	Y	NA	N
21	M	77	No	62	Prostate ca	Postop and adjuvant RT 5 years ago, on follow up	Y	NA	N
22	M	71	DM	75	Esophageal ca	Surgical resection plus adjuvant chemo and RT 6 years ago	Y	NA	N
23	M	65	CHD	26	Prostate ca	Postoperative and adjuvant RT 2 years ago, on follow up	Y	NA	N
24	M	75	COPD	148	Rectal ca	Post-operative routine follow up	Y	NA	N
25	M	56	CHB	73	Liver ca	Surgical resection plus adjuvant chemo and RT 6 years ago, recurrence on adjuvant chemo within 4 weeks	N	NA	Y
26	M	69	No	124	Laryngocarcinoma	Postoperative and adjuvant RT 10 years ago, routine follow up	N	NA	N
27	F	52	No	8	Gastric ca	Postoperative, 4 cycles of chemo and adjuvant RT, routine follow up	N	NA	N
28	M	60	No	13	Testicular ca	Sequential chemo and RT, last cycle of chemo was within 4 weeks	Y	NA	N
Dai et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Date of diagnosed cancer**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	69	Intestinal obstruction	2019.12	Peritoneal cancer	N.A	Y	Y	N
2	M	66	Jaundice, Abdominal infection, chest infection	2019.11	Gastric cancer	N.A	N	Y	Y
3	M	49	No	2020.01	Esophageal SCC	N.A	Y	Y	NA
4	M	65	No	2018.01	Gastric cancer	N.A	N	Y	N
5	M	72	Asthma	2016.09	Lung cancer	Targeted therapy (EGFR-TKI) and PD-1 immunotherapy	N	Y	Y
6	M	73	Chronic bronchitis	2019.11	Laryngeal SCC	Radiotherapy	Y	Y	Y
7	M	69	CHD, HTN	2019.08	Lung cancer	Radiotherapy	N	Y	N

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8	F	55	No	2019.11	Glioblastoma	Radiotherapy and chemotherapy	N	Y	Y
9	M	74	Duodenal ulcer, COPD, HTN, CHD	2019.08	Esophageal cancer	Radiotherapy	N	Y	Y
10	M	55	Hepatic encephalopathy, GI bleeding, Chronic viral hepatitis, Anemia	2014.01	Hepatocellular carcinoma	Radiotherapy	N	Y	Y
11	M	66	HTN, DM	2019.03	Lung SCC	Chemotherapy	Y	Y	Y
12	M	67	Chronic bronchitis, HTN, Atrial fibrillation, Peptic ulcer, CKD	2019.12	Esophageal SCC	Surgery	Y	N	N
13	M	64	Cerebral infarction	2020.01	Myelodysplastic syndrome	N.A	Y	Y	NA
14	M	83	HTN	2016	Meningioma	N.A	Y	Y	NA
15	M	73	HTN	2020.01	Lung cancer	Surgery	Y	Y	N
16	F	48	No	2019.06	Breast cancer	N.A	N	Y	N
17	M	57	Prostatic hyperplasia	2016.02	Lung cancer	Radiotherapy and Targeted therapy (EGFR-TKI)	N	Y	Y
18	M	61	No	2019.12	Lung SCC	DP chemotherapy and PD-1 immunotherapy	Y	N	Y
19	F	84	HTN, DM, lumbar compression fracture	2019.06	Lung cancer	Surgery	Y	N	N
20	M	69	Abdominal aortic aneurysm	2019.09	Lung adenocarcinoma	Radiotherapy and chemotherapy	N	Y	N
21	M	69	Prosopalgia	2019.11	Lung SCC	Chemotherapy and PD-1 immunotherapy	N	Y	Y
22	M	65	Chronic viral hepatitis	2019.04	Lung cancer	N.A	N	Y	N
23	F	76	HTN, CKD, hypothyroidism, cardiac insufficiency	Unknown	Multiple myeloma	N.A	Y	N	NA
24	M	54	Incomplete ileus	2020.01	Rectal cancer	Surgery	N	Y	N
25	F	78	HTN,, COPD, CHD, CKD	2014.06	Chronic granulocytic leukemia	N.A	Y	N	N
26	M	61	HTN, DM	2018.05	Bladder cancer	N.A	Y	Y	N
27	F	50	No	2017.05	Lung cancer	N.A	N	Y	N
28	F	87	HTN	2009	Gastric cancer	N.A	Y	Y	N
29	F	55	No	2018	Cervical cancer	N.A	N	Y	N
30	F	70	HTN	Unknown	Colon cancer	N.A	N	Y	N
31	F	69	No	Unknown	Gastric cancer	N.A	N	Y	NA
32	F	59	No	2009	Breast cancer	N.A	N	Y	N
33	F	48	No	2019.06	Breast cancer	Radiotherapy	N	Y	N
34	M	57	No	2015.12	Lung adenocarcinoma	Concurrent targeted therapy（EGFR-TKI）with radiotherapy	N	Y	Y
35	M	69	Prosopalgia	2019.11	Lung SCC	Radiotherapy and Chemotherapy	N	Y	Y
36	M	69	Chronic viral hepatitis	2019.04	Lung cancer	Chemotherapy and PD-1 immunotherapy	Y	Y	Y
37	M	69	CKD	2019.09	Lung adenocarcinoma	Chemotherapy and Radiotherapy	N	Y	N
38	M	66	No	2019.08	Lung adenocarcinoma	PD-1 immunotherapy	Y	Y	Y
39	M	65	CHD	2019.04	Bladder cancer	Radiotherapy and Chemotherapy	Y	N	Y
40	M	81	HTN	2019.09	Cancer of unknown primary	PD-1 immunotherapy	Y	N	NA
41	F	46	Bacteremia	2019.01	Rectal cancer	N.A	N	Y	N
42	M	76	Intestinal obstruction	2018.11	Rectal cancer	N.A	Y	Y	Y
43	F	58	No	2020.1.2	Ovarian cancer	Chemotherapy	N	Y	N
44	F	74	DM	2020.1.12	Cervical cancer	Chemotherapy	N	Y	N
45	M	26	No	Unknown	Testicular cancer	N.A	N	Y	N
46	F	66	HTN, HLD	Unknown	Breast cancer	N.A	Y	Y	N
47	F	52	DM	Unknown	Thyroid carcinoma	N.A	N	Y	N
48	M	50	Pulmonary tuberculosis, Abdominal infection, Biliary infection, Liver abscess	2015	Thyroid carcinoma	N.A	N	Y	N
49	F	66	No	2019	Thyroid carcinoma	N.A	Y	Y	NA
50	M	83	No	2019.05	Prostate cancer	N.A	N	Y	Y
51	F	29	No	2013.05	Thyroid carcinoma	N.A	N	Y	N
52	F	32	No	2018.01	Thyroid carcinoma	N.A	N	Y	N
53	M	76	No	2018	Prostate cancer	N.A	N	Y	NA
54	F	80	HTN, HLD, Phlebothrombosis, DM	2019.01	Lung adenocarcinoma	N.A	Y	Y	N
55	M	72	HTN	2019	Lung cancer	N.A	Y	N	Y
56	F	57	No	2019.12	Breast cancer	Chemotherapy	N	Y	N
57	F	69	Chronic Hepatitis	2018	Thyroid cancer	N.A	N	Y	N
58	F	60	No	2012	Cervical cancer	N.A	N	Y	N
59	F	72	HTN,,heart disease	2013.06	Breast cancer	N.A	N	Y	NA
60	M	75	HTN	2009	Colon cancer	N.A	N	Y	NA
61	M	67	Hypothyroidism	Unknown	Liver cancer	N.A	N	Y	NA
62	F	52	HTN, CKD	2016.03	Endometrial cancer	N.A	N	Y	N
63	M	64	No	2019.08	Gallbladder carcinoma	N.A	N	Y	NA
64	M	67	No	1994	Nasopharynx cancer	N.A	N	Y	NA
65	M	63	HTN	2010	Kidney cancer	N.A	N	Y	NA
66	M	34	No	2017	Thyroid cancer	N.A	N	Y	NA
67	F	59	No	2019.07	Colon cancer	Chemotherapy	N	Y	NA
68	M	57	Cirrhosis,diabetes	2020.01	Nasopharynx cancer	Chemotherapy	N	Y	NA
69	M	62	No	2015	Bladder cancer	N.A	N	Y	NA
70	F	66	No	2015	Thyroid cancer	N.A	N	Y	NA
71	M	60	No	2015	Multiple myeloma	N.A	N	Y	NA
72	F	64	No	Unknown	Gastric stromal tumor	N.A	N	Y	NA
73	F	66	No	2004	Ocular malignancy	N.A	N	Y	NA
74	M	61	CHD, HTN	2020.01	Multiple myeloma	N.A	N	Y	NA
75	F	53	No	2019.9	Multiple myeloma	N.A	N	Y	NA
76	M	61	DM, HTN	Unknown	Malignant lymphoma	N.A	Y	N	Y
77	F	54	No	2019.03	Lung carcinoma	N.A	Y	N	N
78	F	55	No	2016	Breast cancer	N.A	N	Y	N
79	F	38	No	2019	Breast cancer	N.A	N	Y	N
80	F	50	No	2019	Rectal carcinoma	N.A	Y	N	N
81	M	67	Bronchitis	2017.12	Esophageal carcinoma	N.A	Y	Y	N
82	F	37	No	2018	Thyroid carcinoma	N.A	N	Y	N
83	F	45	No	2019.11	Cervical carcinoma	Surgery and Radiotherapy	Y	Y	N
84	M	35	No	2019.07	Glioma	N.A	Y	Y	NA
85	F	57	No	2019.11	Cervical carcinoma	Surgery	N	Y	N
86	F	49	No	2019.11	Renal carcinoma	N.A	N	Y	N
87	M	62	HTN, Tuberculosis	2018.08	Esophageal carcinoma	N.A	N	Y	N
88	M	70	CHD, Choledochectasia	2019.06	Gastric carcinoma	N.A	N	Y	N
89	M	44	No	2019.12	Thyroid carcinoma	N.A	N	Y	N
90	F	72	Cerebrovascular disease	2017.06	Breast cancer	N.A	N	Y	N
91	F	57	HTN	2019.02	Breast cancer	N.A	Y	Y	N
92	F	56	No	2019.04	Cervical carcinoma	N.A	Y	Y	N
93	M	58	No	2019.06	Rectal carcinoma	Chemotherapy	Y	Y	Y
94	M	71	CHD, HTN	2019.1	Lung SCC	Chemotherapy	Y	Y	Y
95	F	67	DM HTN	2019.12	Breast cancer	Surgery	Y	Y	N
96	M	70	No	2019.11	gastric carcinoma	Chemotherapy	Y	Y	N
97	M	70	HTN	2014	Esophageal carcinoma	N.A	Y	Y	N
98	F	42	HTN	2020.01	Pancreatic cancer	Surgery	N	Y	N

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99	M	43	No	2011	gastric carcinoma	N.A	Y	Y	N
100	M	69	No	2015	Lung cancer	N.A	Y	Y	N
101	M	62	Cerebral hemorrhage, Uremia, DM, HTN	2020.01	Lung cancer	N.A	N	Y	N
102	M	37	HLD	2018.06	Sigmoid colon cancer	N.A	Y	Y	N
103	F	70	CHD, HTN	2002	Thyroid cancer	N.A	Y	Y	N
104	M	51	No	2015	Leukemia	Targeted therapy (Imatinib Mesylate)	N	Y	N
105	F	55	HTN	2019.07	Nasopharyngeal lymphoma	N.A	N	Y	N

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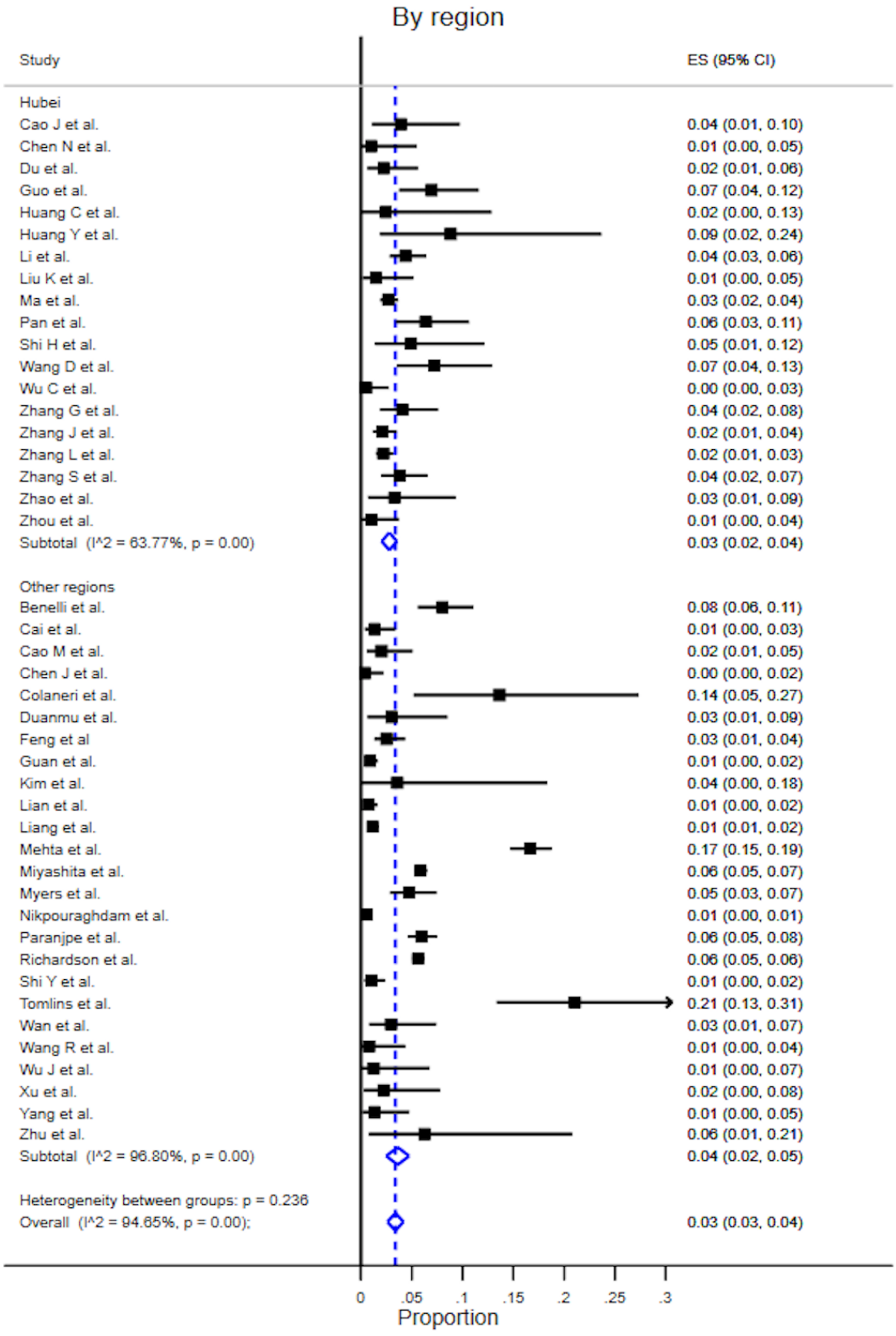
No	Gender	Age	Co-morbidities	Length of dx of cancer in months	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	F	48	No	46	Breast	RT+endocrinotherapy	N	Y	NA
2	M	57	No	53	Lung	targeted therapy/RT	N	Y	NA
3	M	65	Liver cirrhosis	50	Lung	chemotherapy+immunotherapy	N	Y	NA
4	M	66	No	53	Lung	chemotherapy+immunotherapy+targeted therapy	N	Y	NA
5	M	69	No	48	Lung	chemotherapy+RT+immunotherapy	N	Y	NA
6	M	69	No	32	Lung	chemotherapy+RT	Y	N	NA
7	M	61	No	36	Urinary	chemotherapy	Y	N	NA
8	M	37	No	36	liver	surgery+chemotherapy	N	Y	NA
9	M	87	COPD, HTN, CHD, cerebrovascular disease/chronic nephrosis/immunological diseases	44	Gastric	follow-up	Y	N	NA
10	M	61	Disturbance of consciousness	16	Lung	chemotherapy+immunotherapy	Y	N	NA
11	F	67	DM, HTN, liver cirrhosis	8	Esophagus	perioperative	Y	N	NA
12	M	65	HTN, cerebrovascular disease	48	Lung	follow-up	N	Y	NA
13	F	50	No	54	HN	follow-up	N	Y	NA
14	F	46	HTN	50	Thyroid	follow-up	N	Y	NA
15	M	77	HTN	42	Colorectal	follow-up	Y	Y	NA
16	F	70	No	34	Breast	follow-up	Y	Y	NA
17	F	75	No	51	Colorectal	follow-up	N	Y	NA
18	F	53	No	57	HN	follow-up	N	Y	NA
19	M	48	HTN	49	CNS	follow-up	N	Y	NA
20	F	58	HTN, HLD, CHD	45	Thyroid	follow-up	N	Y	NA
21	M	54	No	44	Colorectal	perioperative	N	Y	NA
22	F	78	HLD	18	Hematology	supportive treatment	Y	N	NA
23	M	60	HTN	64	Hematology	supportive treatment	N	Y	NA
24	F	83	HTN	64	CNS	follow-up	Y	Y	NA
25	M	68	HTN	47	Urinary	follow-up	N	Y	NA
26	M	66	COPD, HTN, CHD, liver cirrhosis	48	Urinary	follow-up	N	Y	NA
27	M	63	HTN	29	Urinary	follow-up	Y	N	NA
28	M	77	No	55	Urinary	radiotherapy	Y	Y	NA
29	M	73	HTN/HLD/immune disease	53	Lung	perioperative	N	Y	NA
30	M	66	HD/DM/CHD/immune disease	22	Duodenal	supporting treatment	Y	Y	NA
31	F	84	HTN, immune disease	11	Lung	perioperative	Y	N	NA
32	M	67	DM, HTN, CHD, chronic nephrosis	42	Colorectal	follow-up	Y	Y	NA
33	F	65	DM, HTN, liver cirrhosis	41	CNS	follow-up	N	Y	NA
34	M	78	HTN	40	Colorectal	follow-up	N	Y	NA
35	M	65	No	38	Colorectal	perioperative	Y	Y	NA
36	F	71	Cerebrovascular disease	37	Colorectal	follow-up	Y	Y	NA
37	F	65	HTN, tuberculosis	44	Breast	endocrinotherapy	Y	Y	NA
38	F	40	No	41	Thyroid	follow-up	N	Y	NA
39	F	61	HTN	47	Thyroid	follow-up	N	Y	NA
40	F	70	COPD, DM, HTN, cerebrovascular disease	6	Breast	follow-up	Y	N	NA
41	M	98	No	21	Colorectal	follow-up	Y	N	NA
42	M	64	HTN	53	Esophagus	chemotherapy	N	Y	NA
43	M	67	No	14	Urinary	follow-up	Y	N	NA
44	F	76	COPD, CHD, HTN	52	Hematology	follow-up	N	Y	NA
45	F	68	Tuberculosis	53	Thyroid	follow-up	N	Y	NA
46	M	66	HTN	50	Lung	follow-up	N	Y	NA
47	M	69	No	47	Urinary	follow-up	Y	Y	NA
48	M	69	Tuberculosis	44	Colorectal	follow-up	N	Y	NA
49	M	70	DM, HTN	48	Gastric	follow-up	Y	Y	NA
50	M	72	No	19	Lung	chemotherapy	Y	N	NA
51	F	55	HTN	48	Lung	follow-up	N	Y	NA
52	F	58	No	50	CNS	follow-up	N	Y	NA
53	M	54	HTN	50	HN	follow-up	N	Y	NA
54	F	64	No	40	Breast	follow-up	N	Y	NA
55	F	51	DM, CHD	52	Thyroid	follow-up	N	Y	NA
56	M	56	HTN	50	Thyroid	follow-up	Y	Y	NA
57	M	71	No	13	HN	follow-up	Y	N	NA
58	F	65	DM, HTN	36	Ovary	follow-up	N	Y	NA
59	M	90	HTN	9	Colorectal	perioperative	Y	N	NA
60	F	64	DM, HTN	54	Breast	follow-up	N	Y	NA
61	F	69	No	54	Lung	follow-up	Y	Y	NA
62	M	52	No	39	Thyroid	follow-up	Y	N	NA
63	M	74	HTN	22	Colorectal	follow-up	Y	N	NA
64	M	63	HTN	23	HN	follow-up	Y	N	NA
65	M	70	DM, HTN	37	Lung	follow-up	N	Y	NA
66	M	67	HTN	21	Lung	immunotherapy	Y	N	NA
67	M	64	HTN	57	Urinary	follow-up	Y	Y	NA

Abbreviations: BSC: best supportive care; ca – carcinoma, CHB: chronic hepatitis b; CHD: coronary heart disease; CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; DM: diabetes mellitus; EGFR: epidermal growth factor receptor; HLD: hyperlipidemia; HTN: hypertension; NPC: nasopharyngeal cancer; NSCLC: non-small cell lung cancer; PD1: programmed death 1; RCC: renal cell cancer; RT: radiotherapy; TKI: tyrosine kinase inhibitor; SCC: squamous cell cancer; THBSO: total hysterectomy and bilateral salpingo-oophrectomy
Y: yes; N: no; NA: not applicable/available

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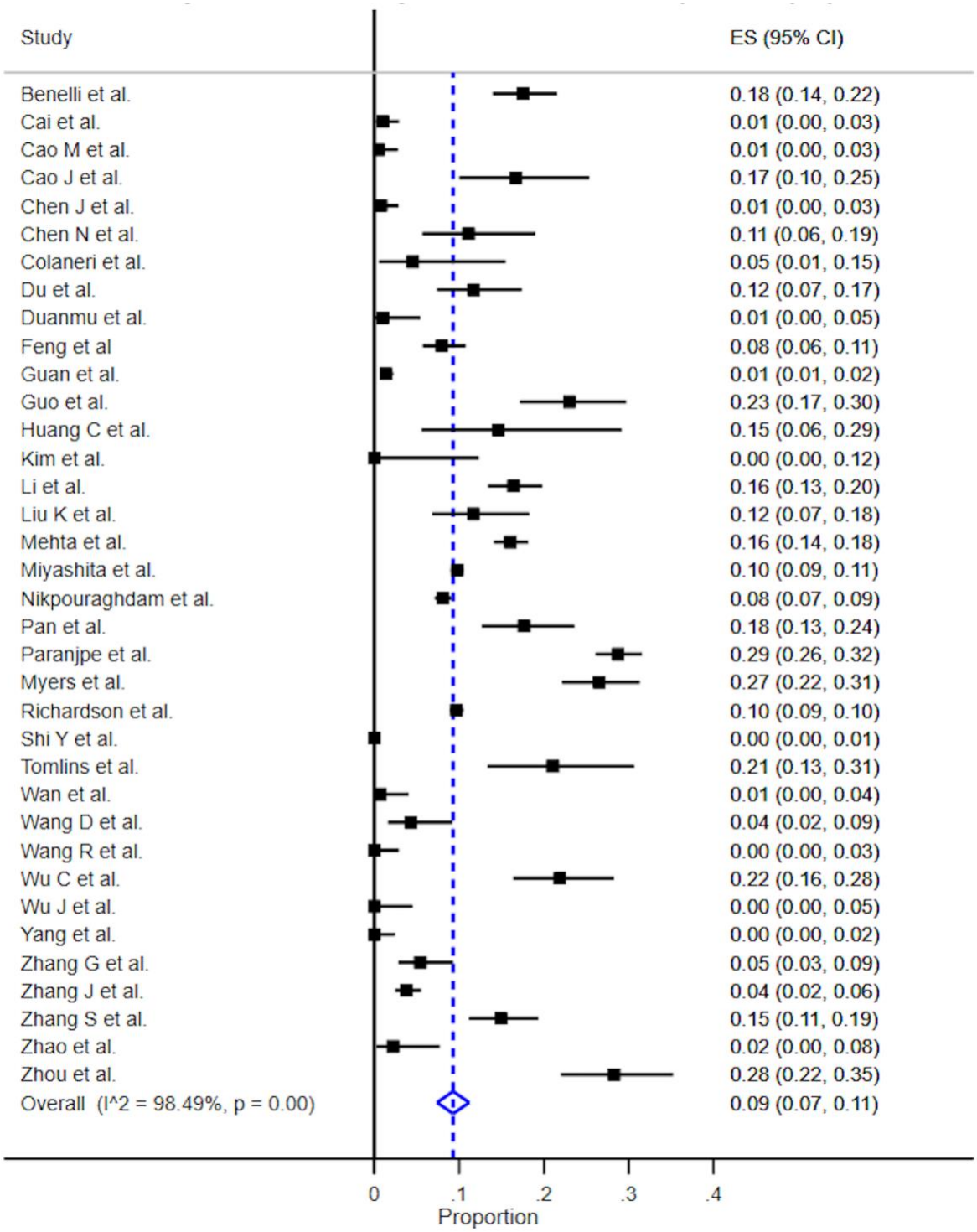
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Supplementary Figure 1: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by region – within Hubei province and outside Hubei province)



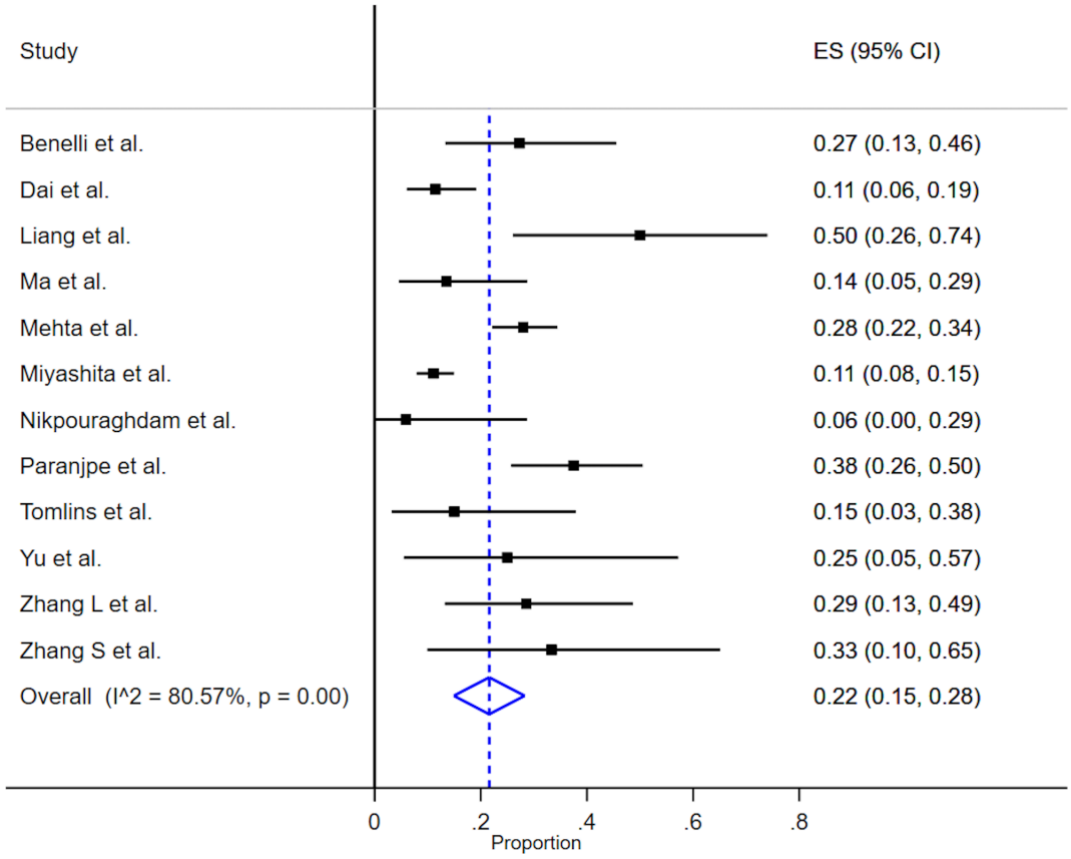
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Supplementary Figure 2a: Random effects pooled proportion of deaths among patients diagnosed with COVID-19

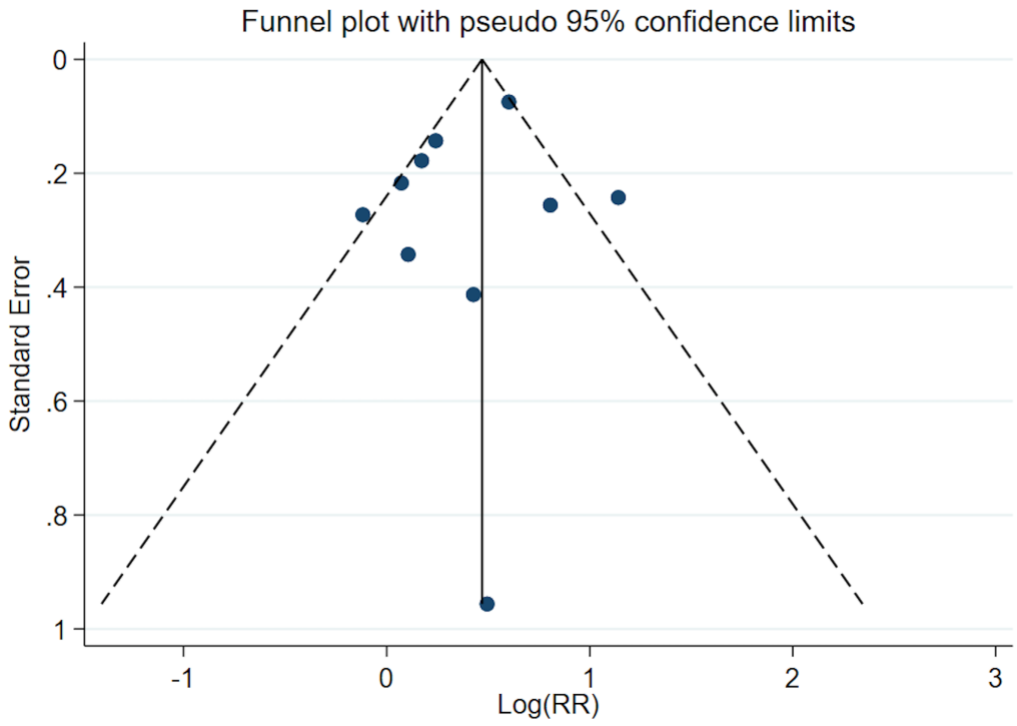


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Supplementary Figure 2b: Random effects pooled proportion of deaths among cancer patients diagnosed with COVID-19

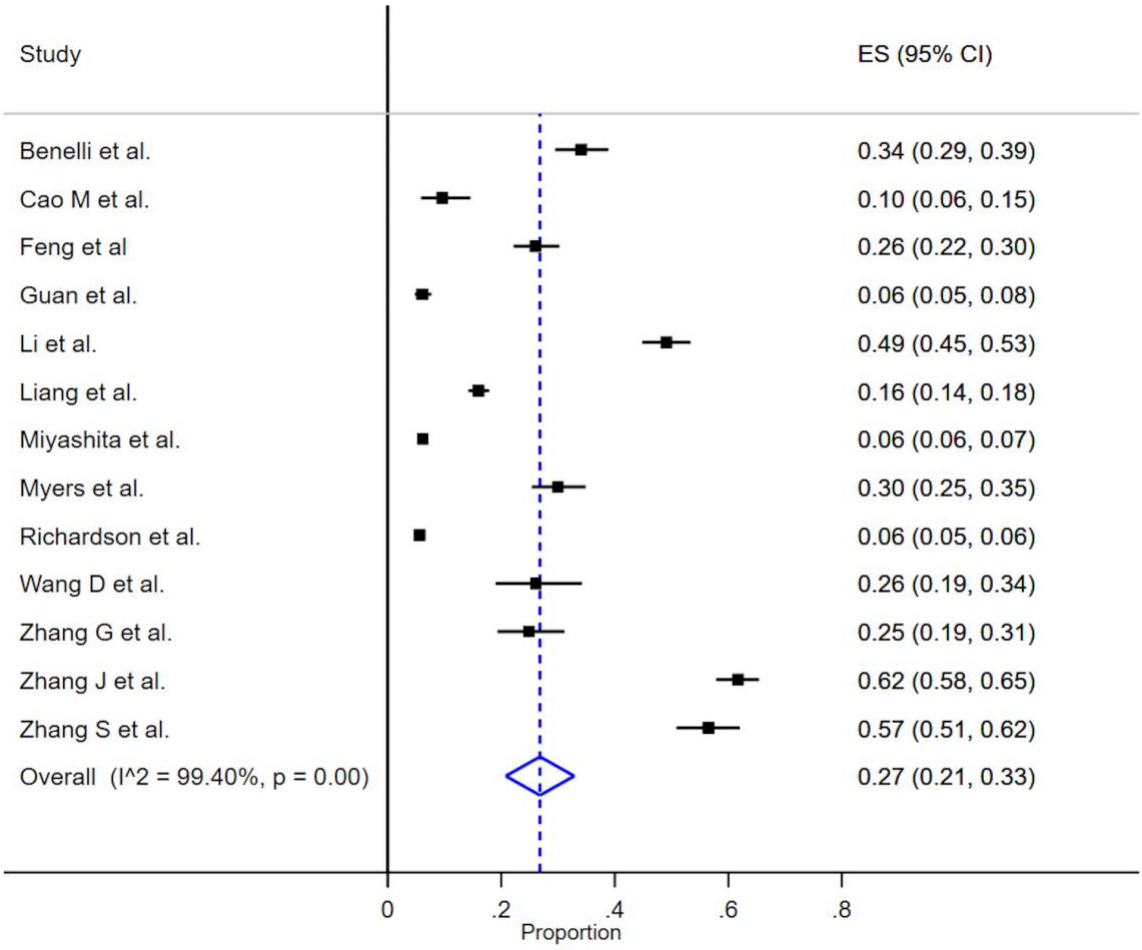


Supplementary Figure 2c: Funnel plot of ten case series included in pooled relative risk analysis of death outcomes between cancer and general COVID-19 patients



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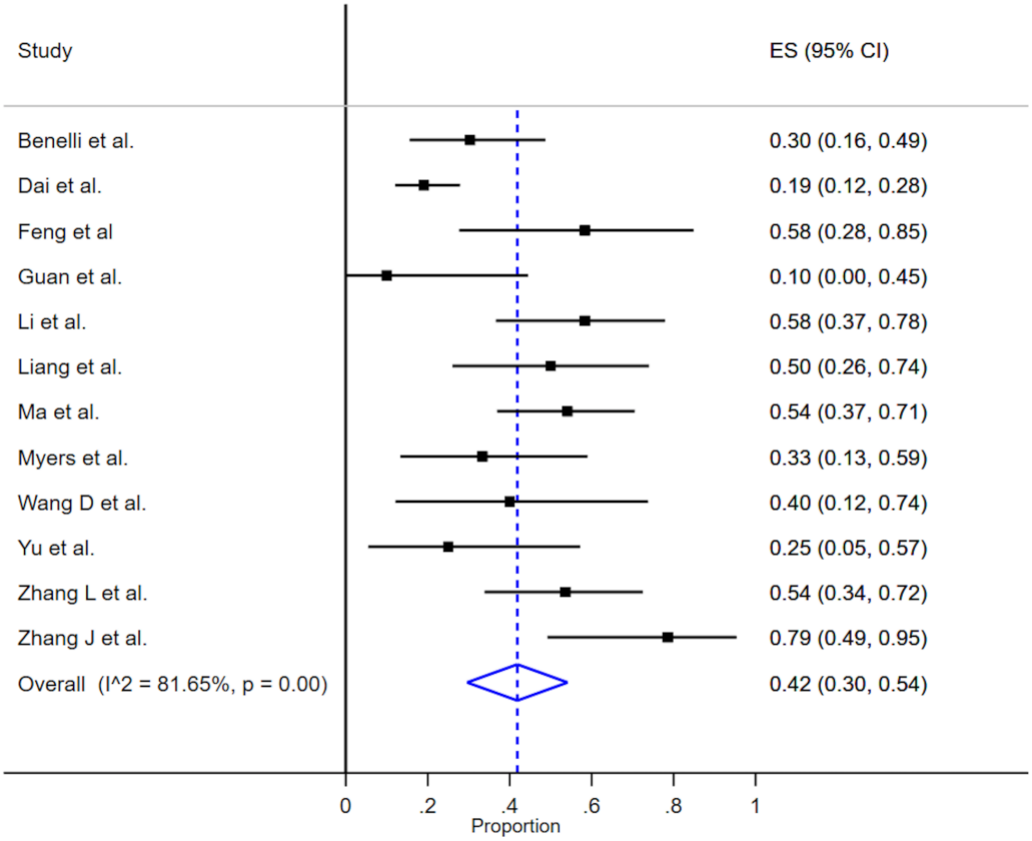
Supplementary Figure 3a: Random effects pooled proportion of severe symptoms among patients diagnosed with COVID-19



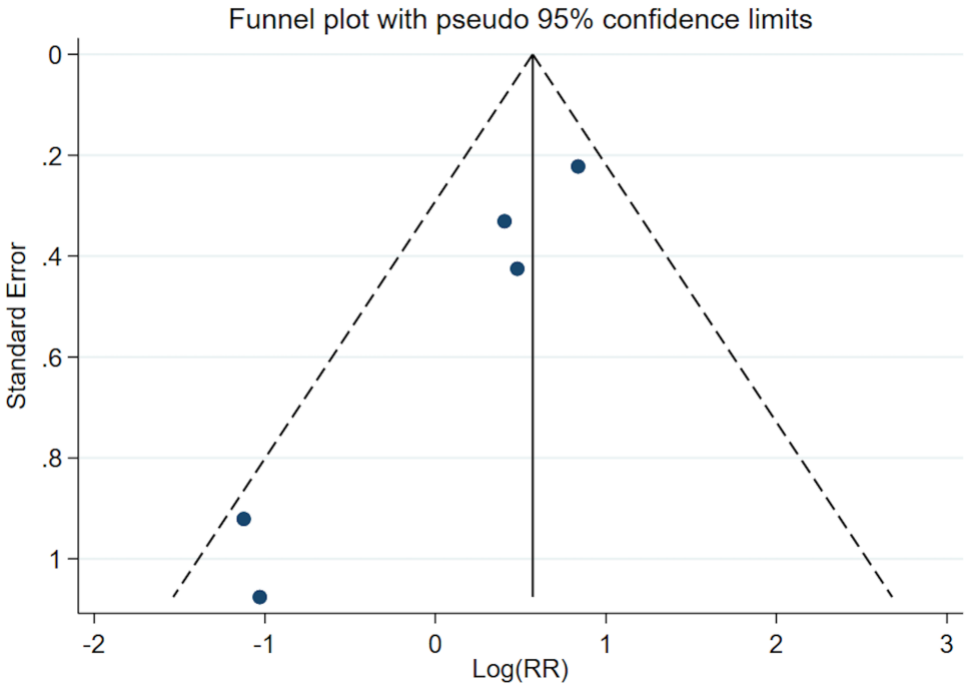
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Supplementary Figure 3b: Random effects pooled proportion of severe symptoms among cancer patients diagnosed with COVID-19

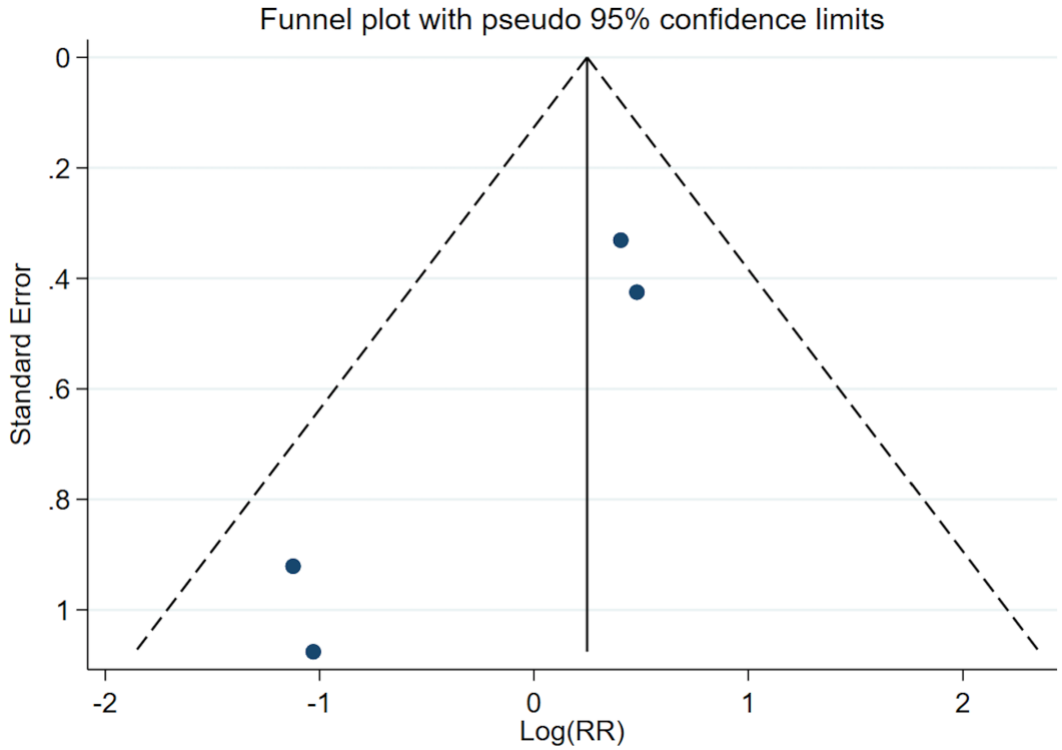


Supplementary Figure 3c: Funnel plot of five case series included in pooled relative risk analysis of severe outcomes between cancer and general COVID-19 patients



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Supplementary Figure 4: Funnel plot of four case series included in pooled relative risk analysis of lung cancer vs non-lung cancer and stage IV vs non stage IV cancers



Supplementary Table 2: Characteristics of 230 cancer patients with COVID-19 infection with severe and non-severe outcomes

COVID-19 severity	Non-severe (N = 131)	Severe (N = 99)	P-value (Chi-square)
Gender			
Male	70 (50.0)	70 (50.0)	0.010
Female	61 (67.8)	29 (32.2)	
Co-morbidities			
≥2	20 (8.7)	25 (10.9)	0.093
0-1	102 (44.3)	71 (30.9)	
Stage			
Stage IV	20 (8.7)	20 (8.7)	0.635
Stage I-III	56 (24.3)	40 (17.4)	
Unknown	55 (23.9)	39 (17.0)	
Tumour Type			
Lung cancers	32 (13.9)	24 (10.4)	0.974
Non-lung cancers	99 (43.0)	75 (32.6)	
Age			
≥ 60	63 (27.4)	79 (34.3)	<0.001
< 60	59 (25.7)	17 (7.4)	
Treatment			
Active systemic therapy	33 (14.3)	23 (10.0)	0.758
No systemic therapy	97 (42.2)	76 (33.0)	

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PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N.A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4-5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4, appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Appendix 2
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4-5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4-5

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4-5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4-5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-10, Appendix 2-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	2, 6-12 Appendix 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	6-12, Appendix 3
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

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